GENERAL APTITUDE

Previous Year Questions

GATE — ALL BRANCHES

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

General Aptitude

Previous Year Questions

General Aptitude (all GATE branches)

Authors: Himanshu Agarwal & Anish Saha

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Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in General Aptitude (all GATE branches), gathered from the original papers and arranged year by year and session by session. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by year and by session, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q24.5.3

24	GATE exam year — 2024 (two digits; 98 = 1998)
5	Session number of that year's exam
3	Question number within that session

So **Q24.5.3** is the 3rd aptitude question of Session 5, GATE 2024. Numbers are therefore not sequential through the book — they tell you the year and session at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a session to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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YEAR

2026

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
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Session 2	AE, BT, CH, CY, GE, PH, XH	10
Session 3	CS-1, ST	10
Session 4	CS-2, EY, NM, PE	10
Session 5	CE-1, EE, PI	10
Session 6	BM, CE-2, ME, MT	10
Session 7	EC	10
Session 8	AR, DA	10

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Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 07/02/26

Q26.1.1 MCQ 2026 1M

Suresh said, "I did it yesterday."
Which one of the following options is the correct form of this sentence in indirect speech?

A. Suresh said that I did it yesterday.

B. Suresh says I did it yesterday.

C. Suresh says that he did it the day before.

D. Suresh said that he had done it the day before.

Q26.1.2 MCQ 2026 1M

To continue the sequence of tiles shown, the tile indicated by the question mark should be

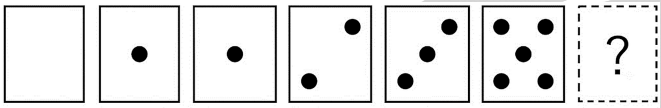
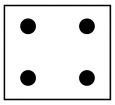
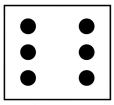
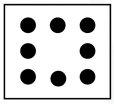
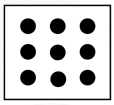


Fig. Q26.1.2

A.  **B.**  **C.**  **D.** 

Q26.1.3 MCQ 2026 1M

Consider an art gallery whose walkways are shown as lines in the diagram. A black dot represents a junction of two walkways. A guard may be placed at a junction to watch over the walkways that join at that junction. The minimum number of guards needed to watch all the walkways is ____.

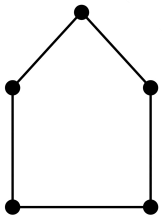


Fig. Q26.1.3

A. 2 **B.** 3 **C.** 4 **D.** 5

Q26.1.4 MCQ 2026 1M

The 2nd of June is a Thursday in a certain year. Which day of the week is the 3rd of July in that year?

A. Thursday **B.** Friday **C.** Saturday **D.** Sunday

Q26.1.5 MCQ 2026 1M

A coin with heads facing up is shown as (H) and a coin with tails facing up is shown as (T).
Six coins are placed in the Starting Arrangement, as shown in the figure below. A "step" is defined as interchanging a pair of adjacent coins without flipping them. The minimum number of steps needed to go from the Starting Arrangement to the Final Arrangement, as shown in the figure, is

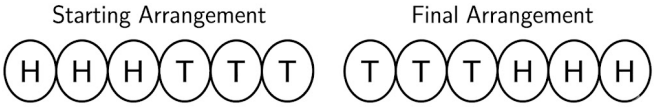


Fig. Q26.1.5

A. 3 **B.** 6 **C.** 9 **D.** 12

Q26.1.6 MCQ 2026 2M

Exacerbate : Mitigate :: _____
Choose the option with the correct pair of words to fill the blank.

A. Aggravate : Alleviate **B.** Alleviate : Precipitate

C. Aggravate : Precipitate **D.** Emancipate : Exonerate

Q26.1.7 MCQ 2026 2M

A paper shown in Panel I is folded along the dashed lines (- - -) to construct a cube. The shaded regions shown in Panel I appear on the outer surface of the cube. Referring to cubes shown in Panel II, which one of the options is correct?

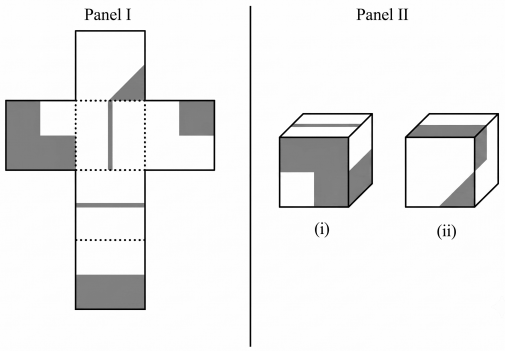


Fig. Q26.1.7

A. Only (i) can correspond to the unfolded cube in Panel I.

B. Only (ii) can correspond to the unfolded cube in Panel I.

C. Both (i) and (ii) can correspond to the unfolded cube in Panel I.

D. Neither (i) nor (ii) can correspond to the unfolded cube in Panel I.

Q26.1.8 MCQ | 2026 | 2M

In a population, patients who have high cholesterol also have high blood-pressure (BP). Some patients with high BP also have diabetes. There are no patients who have both high cholesterol and diabetes. Furthermore,

1. the total number of patients with at least one of these conditions is 75,
2. the number of patients with high cholesterol is 10,
3. the number of patients with high BP is 45, and
4. the number of patients with only high BP and no other conditions is 20.

Then the number of patients who have both diabetes and high BP is _____.

A. 0 **B.** 15 **C.** 20 **D.** 10

Q26.1.9 MCQ | 2026 | 2M

Four people P, Q, R, and S, of different ages, make the following observations.

P – I am younger than S.

Q – I am neither the youngest nor the oldest.

R – P is older than me.

Based on these observations, the youngest person is _____.

A. P **B.** Q **C.** R **D.** S

Q26.1.10 MCQ | 2026 | 2M

Circles C_1, C_2, and C_3, with centers O_1, O_2, and O_3, and radii r_1, r_2, and r_3, respectively, touch each other as shown in the following figure.

Given $r_1 = 2$ cm, $r_2 = 1$ cm and the angle $\angle O_1O_3O_2$ is $90°$, r_3 = _____ cm.

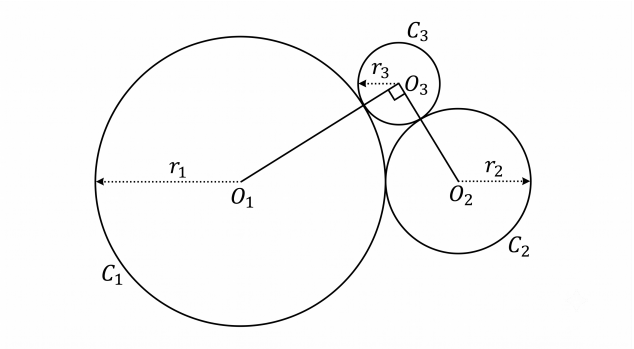


Fig. Q26.1.10

A. $\frac{1}{2}\left(-3+\sqrt{17}\right)$ **B.** $\frac{1}{2}\left(3+\sqrt{17}\right)$

C. $\frac{1}{2}\left(-2+\sqrt{17}\right)$ **D.** $\frac{1}{2}\left(-3+2\sqrt{17}\right)$

Session : 2 | Afternoon | 07/02/26

Q26.2.1 MCQ | 2026 | 1M

"He often _____ the numbers. False claims are not going to help. Honesty _____ trust", said the manager.

Choose the option with the correct order of words to fill the blanks.

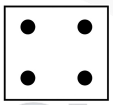
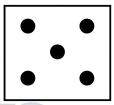
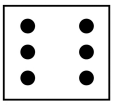
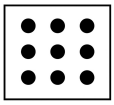
A. exaggerates; engenders **B.** excels; encourages

C. aggravates; alleviates **D.** diminishes; eliminates

Q26.2.2 MCQ | 2026 | 1M

In the sequence of tiles shown below, the missing tile indicated by the question mark should be

Fig. Q26.2.2

A.  **B.**  **C.**  **D.** 

Q26.2.3 MCQ | 2026 | 1M

A school has 100 students distributed among ten standards, numbered 1 to 10.

Based on this, which one of the following statements is always correct?

A. There are at least 10 students who belong to the same standard.

B. There is at least one student in each standard.

C. There are at most 10 students in standard 10.

D. The total number of students from standards 1 to 5 is at least 50.

Q26.2.4 MCQ 2026 1M

How many 3-digit numbers can be formed using three distinct single digit prime numbers?

A. 64 **B.** 24 **C.** 12 **D.** 4

Q26.2.5 MCQ 2026 1M

In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____.

A. 18 **B.** 20 **C.** 24 **D.** 32

Q26.2.6 MCQ 2026 2M

Charity : P :: Retaliation : Q

Choose the appropriate pair of words P and Q that fit the analogy.

A. P = Parsimonious; Q = Vengeful

B. P = Altruistic; Q = Amicable

C. P = Resentful; Q = Spiteful

D. P = Magnanimous; Q = Vindictive

Q26.2.7 MCQ 2026 2M

A paper shown in Panel I is folded along the dashed lines (- - - - -) to construct a cube. The shaded regions shown in Panel I appear on the outer surface of the cube. Referring to cubes shown in Panel II, which one of the options is correct?

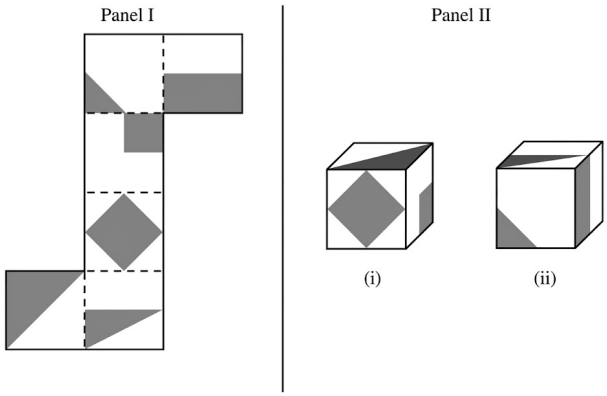


Fig. Q26.2.7

A. Only (i) can correspond to the unfolded cube in Panel I.

B. Only (ii) can correspond to the unfolded cube in Panel I.

C. Both (i) and (ii) can correspond to the unfolded cube in Panel I.

D. Neither (i) nor (ii) can correspond to the unfolded cube in Panel I.

Q26.2.8 MCQ 2026 2M

Consider the cube shown below with its 8 corners labelled a, b, c, d, e, f, g, and h. The figure is representative. All corners are to be colored such that any two corners that are connected by an edge must be of different colors. The minimum number of colors required to achieve this is ___.

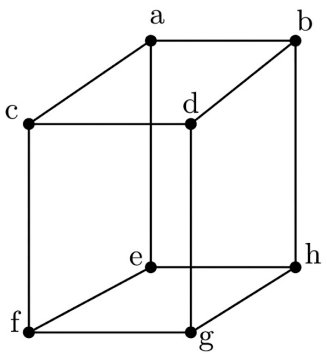


Fig. Q26.2.8

A. 8 **B.** 4 **C.** 3 **D.** 2

Q26.2.9 MCQ 2026 2M

Four hills H1, H2, H3, and H4 are present in an area. The following observations are made about them:

(i) Neither H2 nor H3 is the easternmost hill.
(ii) Neither H2 nor H3 is the westernmost hill.
(iii) Neither the easternmost hill nor the westernmost hill is the southernmost hill.
(iv) Two hills are located to the west of H2.
(v) The southernmost hill has at least two hills to its east.

The southernmost hill is _____.

A. H1 **B.** H2 **C.** H3 **D.** H4

Q26.2.10 MCQ 2026 2M

As shown in the figure, circle C_1 with center O_1 and radius r_1 touches the square $VWXY$ at points P and Q while circle C_2 with center O_2 and radius r_2 touches the square $VWXY$ at points R and S. The two circles touch each other at T. Given $r_1 = 1$ cm and $\overline{VY} = \overline{VW} = 4$ cm, r_2 = _____ cm.

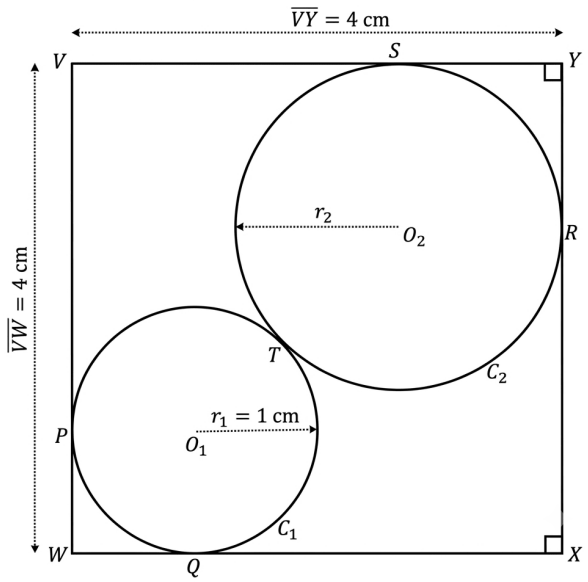


Fig. Q26.2.10

A. $4 - 3\sqrt{2}$ **B.** $1 + 2\sqrt{2}$ **C.** $7 - 4\sqrt{2}$ **D.** $5 + 3\sqrt{2}$

Session : 3 | Forenoon | 08/02/26

Q26.3.1 MCQ | 2026 | 1M

The antonym of the word protagonist is_____.

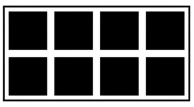
A. agnostic **B.** antagonist **C.** arsonist **D.** anarchist

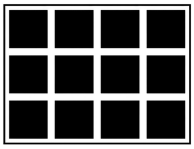
Q26.3.2 MCQ | 2026 | 1M

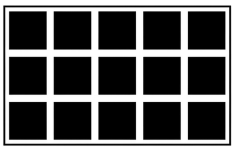
The figure shows two 4-tile patterns.

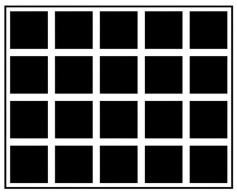
Fig. Q26.3.2

Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below **cannot** be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?

A. 

B. 

C. 

D. 

Q26.3.3 MCQ | 2026 | 1M

Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?

A. 127 **B.** 64 **C.** 63 **D.** 32

Q26.3.4 MCQ | 2026 | 1M

A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization courses. The maximum credits of elective courses that a student can enroll for is_____.

A. 10 **B.** 15 **C.** 20 **D.** 25

Q26.3.5 MCQ | 2026 | 1M

'When the teacher is in the room, all students stand silently.' If the above statement is true, which one of the following statements is **not necessarily** true?

A. If any student is not standing silently, then the teacher is not in the room.

B. When the teacher is in the room, all students are silent.

C. If all students are standing, then the teacher is in the room.

D. When the teacher is in the room, all students are standing.

Q26.3.6 MCQ 2026 2M

Combinatorics deals with problems involving counting. For example, "How many distinct arrangements of N distinct objects in M spaces on a circle are possible?" is a typical problem in combinatorics. This kind of counting is sometimes used in the modeling of several physical phenomena. Often, in such models, the different combinatorial possibilities are assigned probability values. Assigning probabilities enables the computation of the average values of physical quantities.

Consider the following statements:

P: Combinatorics is always invoked in the modeling of physical phenomena.

Q: Modeling some physical phenomena involves assigning probabilities to combinatorial possibilities in order to compute average values of physical quantities.

Based on the passage above, what can be inferred about statements P and Q?

A. P is False and Q is False **B.** P is False and Q is True

C. P is True and Q is False **D.** P is True and Q is True

Q26.3.7 MCQ 2026 2M

In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I?

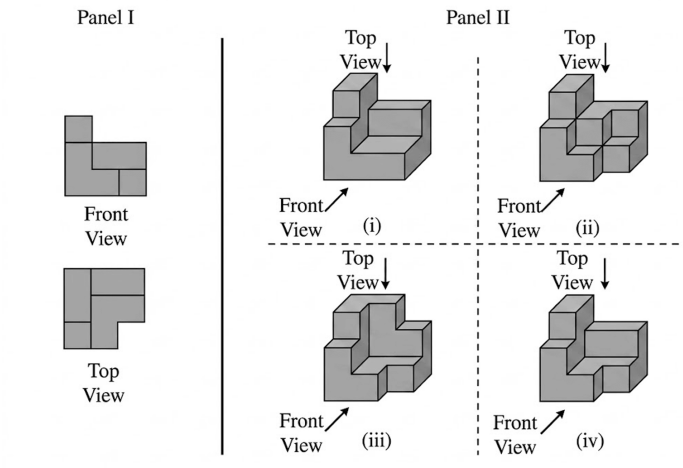


Fig. Q26.3.7

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.3.8 MCQ 2026 2M

For positive real numbers S and K, the function $H_K(S)$ is defined as: $H_K(S) = \max(S - K, 0)$. The max function is defined as:

$$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$

The graph below shows the plot of a function $N(S)$ versus S.

$N(S)$ can be expressed as _____.

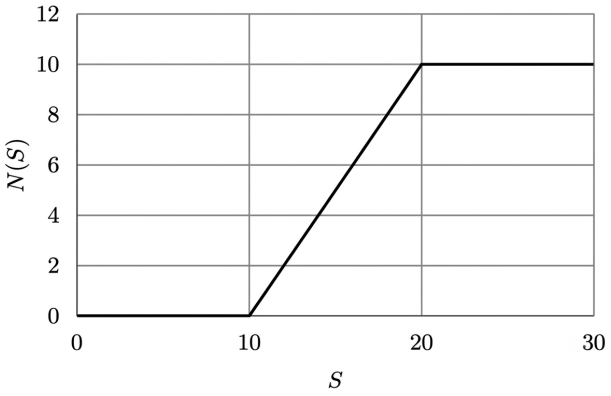


Fig. Q26.3.8

A. $H_{10}(S) - H_{20}(S)$ **B.** $H_{10}(S) - 2H_{20}(S)$

C. $-H_{10}(S) + H_{20}(S)$ **D.** $H_{15}(S) - H_{20}(S)$

Q26.3.9 MCQ 2026 2M

In the 2020 summer Olympics' Javelin throw finals, Neeraj Chopra exhibited a spectacular performance to win the gold medal. The silver medal was won by Jakub Vadlejch and the bronze medal was won by Vitezlav Vesely. There were six rounds of throws with each athlete having one throw per round. The best of all the throws of each athlete is considered for the medal. Following were the observations about the throws:

i. The first and second rounds were dominated by Neeraj Chopra with a gold medal performance in his second throw, while the other two athletes did not have any medal winning throws in these rounds.

ii. The throws in the last round by both Jakub Vadlejch and Vitezlav Vesely were fouls and were not considered for scoring.

iii. After four rounds, Vitezlav Vesely was in the second position and could not improve upon his best throw in the succeeding rounds.

iv. In the fourth round, the throw by Jakub Vadlejch was the best in that round.

In which round did Vitezlav Vesely have his best throw?

A. Third **B.** Fourth **C.** Fifth **D.** Sixth

Q26.3.10 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____.

A. $\frac{1}{6}$ **B.** $\frac{5}{18}$ **C.** $\frac{7}{18}$ **D.** $\frac{5}{6}$

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Q26.4.1 MCQ 2026 1M

Expedite, Hasten, Hurry, _____
Fill the blank by choosing a word with a meaning similar to that of the words given above.

A. Accelerate **B.** Retard **C.** Provide **D.** Disable

Q26.4.2 MCQ 2026 1M

A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part?

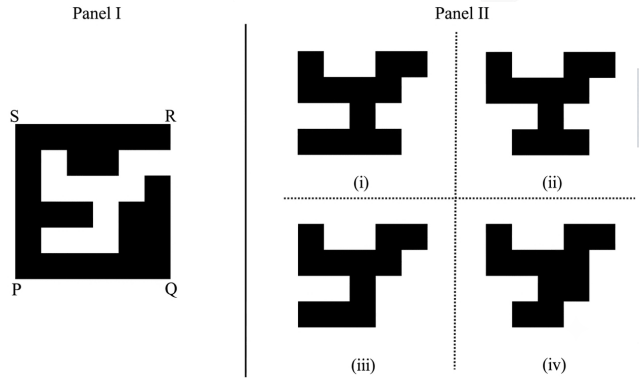


Fig. Q26.4.2

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.3 MCQ 2026 1M

A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

A. 1/4 **B.** 3/4 **C.** 2/3 **D.** 3/8

Q26.4.4 MCQ 2026 1M

The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. ___.

A. 0 **B.** 5 **C.** 10 **D.** 20

Q26.4.5 MCQ 2026 1M

'When it is raining, peacocks dance.'
Based only on this sentence, which one of the following options is necessarily true?

A. Peacocks dance only when it is raining.

B. When peacocks dance, it is raining.

C. When peacocks are not dancing, it is not raining.

D. When it is not raining, peacocks do not dance.

Q26.4.6 MCQ 2026 2M

Water : P :: Food : Q
Choose the P and Q combination from the options below to form a meaningful analogy.

A. P = Thirst; Q = Hunger **B.** P = Drink; Q = Hunger

C. P = Thirst; Q = Satiated **D.** P = Wet; Q = Critic

Q26.4.7 MCQ 2026 2M

Two tiles are missing in Panel I. Which one of the options in Panel II is the appropriate choice for the missing tiles?

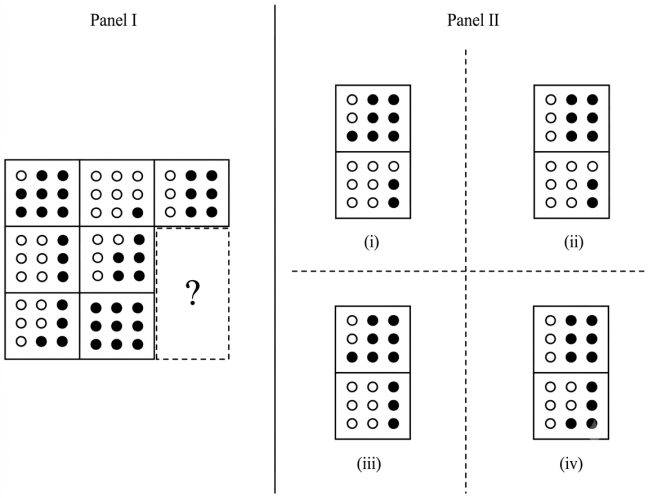


Fig. Q26.4.7

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.8 MCQ 2026 2M

Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways. A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started.
Which one of the following options is then true?

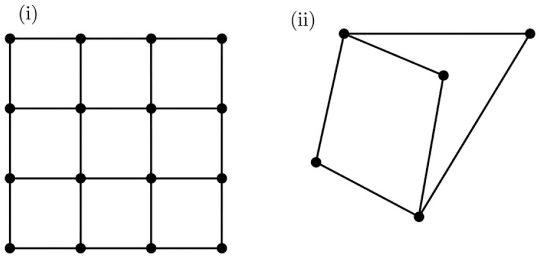


Fig. Q26.4.8

A. Such a trip is possible for (i), but not for (ii).

B. Such a trip is possible for (ii), but not for (i).

C. Such a trip is possible for both (i) and (ii).

D. Such a trip is possible neither for (i) nor for (ii).

Q26.4.9 MCQ 2026 2M

The figure in Panel I below is a grid of cells with four rows and four columns. The numbers on the top and on the left represent the number of cells that are to be shaded in that column and row, respectively. Which one of the options shown in Panel II below represents the grid shaded correctly?

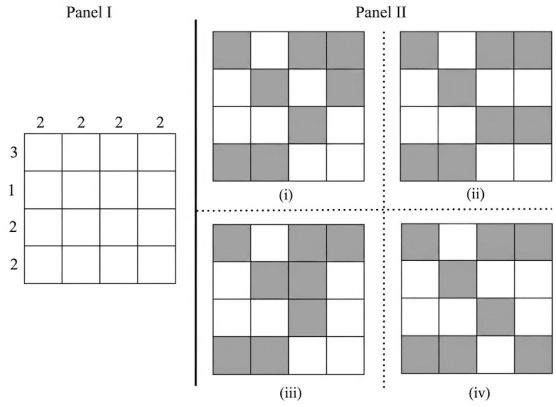


Fig. Q26.4.9

A. (i) **B.** (ii) **C.** (iii) **D.** (iv)

Q26.4.10 MCQ 2026 2M

An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____

A. $\frac{3}{36}$ **B.** $\frac{13}{36}$ **C.** $\frac{15}{36}$ **D.** $\frac{19}{36}$

Session : 5 | Forenoon | 14/02/26

Q26.5.1 MCQ 2026 1M

'The shopkeeper sells lemons.'
In this sentence, the word 'lemons' is the _____.

A. object **B.** subject **C.** predicate **D.** verb

Q26.5.2 MCQ 2026 1M

The figure below is supposed to show three non-overlapping shapes – one oval and two triangles. Which one of the following figures P, Q, R, or S fits the missing portion indicated by '?' and completes the oval and the two triangles?

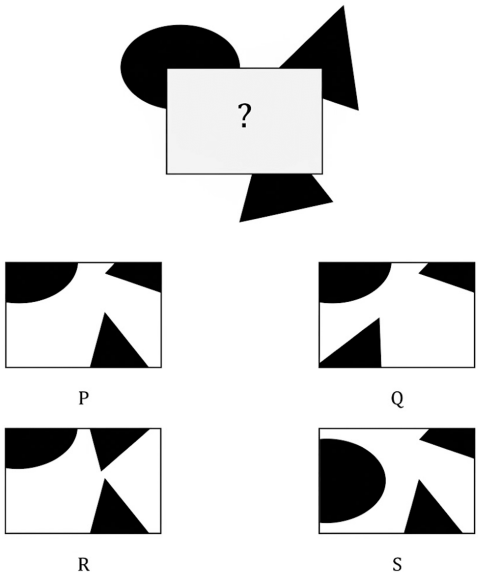


Fig. Q26.5.2

A. P **B.** Q **C.** R **D.** S

Q26.5.3 MCQ 2026 1M

At how many points will the curves $y = x^2$ and $y = -x^2 - 2x - 1$ intersect in the real (x, y) plane?

A. 0 **B.** 1 **C.** 2 **D.** 3

Q26.5.4 MCQ 2026 1M

'If Anish had scored hundred runs in today's match, he would have been made the captain of his team. He would have then become the youngest captain in his team's history. Unfortunately, he got out without scoring any runs. Hence, there won't be any change in the captaincy for now.'
Based on the paragraph above, which one of the following statements is true?

A. Anish made hundred runs but was denied captaincy.

B. Anish was the captain of his team before the game today.

C. The current captain is older than Anish.

D. Anish is the youngest player in his team.

Q26.5.5 MCQ 2026 1M

Which one of the following figures P, Q, R, or S, correctly shows the 45° clockwise-rotated version of figure (I)?

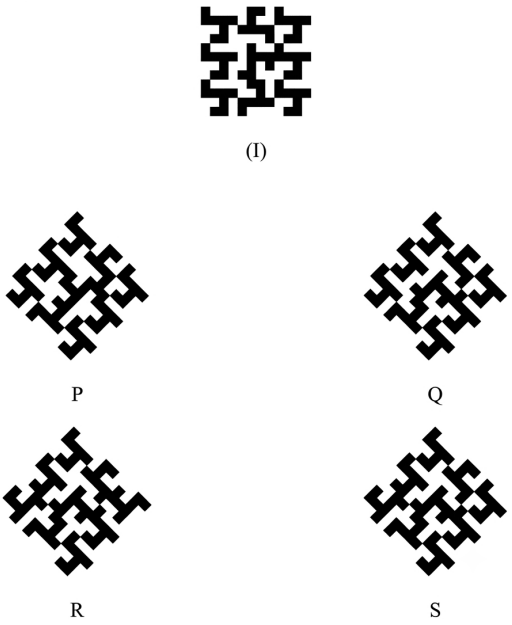


Fig. Q26.5.5

A. P **B.** Q **C.** R **D.** S

Q26.5.6 MCQ 2026 2M

Match the words in Column I with their synonyms in Column II.

Column I	Column II
(i) Lonely	(p) Verbatim
(ii) Literal	(q) Solitary
(iii) Lousy	(r) Deadly
(iv) Lethal	(s) Terrible

A. (i)-(q); (ii)-(p); (iii)-(s); (iv)-(r)

B. (i)-(q); (ii)-(s); (iii)-(r); (iv)-(p)

C. (i)-(s); (ii)-(p); (iii)-(q); (iv)-(r)

D. (i)-(r); (ii)-(s); (iii)-(p); (iv)-(q)

Q26.5.7 MCQ 2026 2M

In the given figure, $\overline{PQ}$ is the diameter of a circle with center O. Two points R and S are chosen on the circle such that $\angle ROS = 80°$. When $\overline{PR}$ and $\overline{QS}$ are extended, they meet at T. The value of $\angle RTS$ is _____.

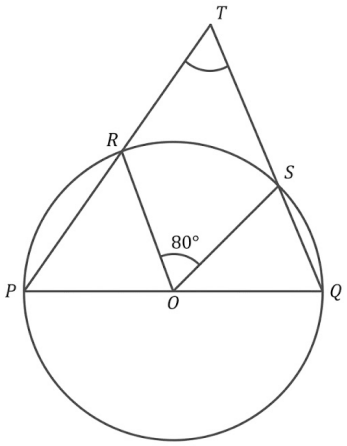


Fig. Q26.5.7

A. 40° **B.** 50° **C.** 60° **D.** 80°

Q26.5.8 MCQ 2026 2M

Based on the relationship between each polygon and the number inside it, the value of 'X' is _____

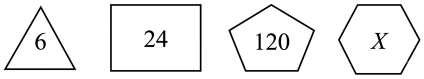


Fig. Q26.5.8

A. 720 **B.** 596 **C.** 24 **D.** 240

Q26.5.9 MCQ 2026 2M

Consider a linear arrangement of seven bulbs, each of which can be in the ON or OFF states. The initial configuration of the bulbs is shown in the figure. In every Step, the states of the bulbs are changed based on the following rules:

- Any OFF bulb with exactly one ON neighbor at the end of the previous Step is turned ON.

- Any ON bulb with both neighbors ON at the end of the previous Step is turned OFF.
- The state of any bulb not meeting the conditions above is left unchanged.

The state of bulbs at the end of Step 1 and Step 2 are also shown in the figure.
The number of bulbs which are ON at the end of Step 8 is

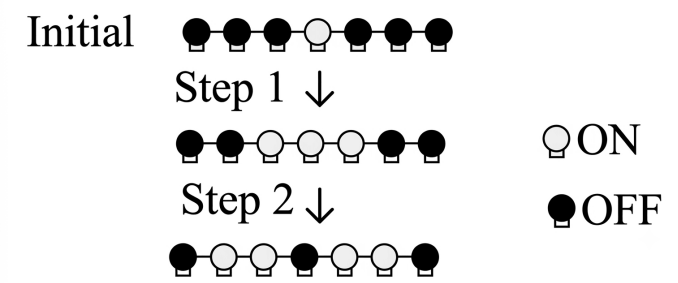


Fig. Q26.5.9

A. 5 **B.** 4 **C.** 3 **D.** 0

Q26.5.10 MCQ | 2026 | 2M

P and Q are two positive integers such that $P^2 = Q^2 + 13$. The product of the numbers P and Q is _____

A. 13 **B.** 26 **C.** 39 **D.** 42

Session : 6 | Afternoon | 14/02/26

Q26.6.1 MCQ | 2026 | 1M

'The team _____ more than 300 runs in 20 overs _____ rains. However, some players needed to improve their batting skills.'
Choose the option with the correct sequence of words to fill the blanks.

A. score; despite **B.** scoring; instead of

C. scored; despite **D.** scoring; in spite of

Q26.6.2 MCQ | 2026 | 1M

If a positive real x satisfies the following equation

$$\log_2 x + \log_{\sqrt{2}} x = 48,$$

then the value of x is_____

A. 2^{16} **B.** 4^{16} **C.** 2^{14} **D.** 4^{14}

Q26.6.3 MCQ | 2026 | 1M

The next figure (indicated by '?') in the sequence is

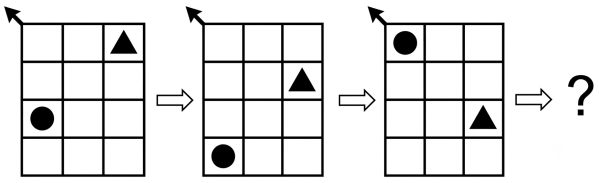
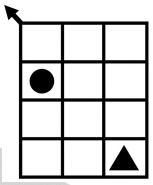
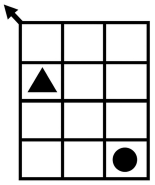
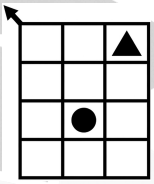
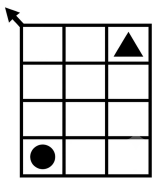


Fig. Q26.6.3

A.


B.


C.


D.


Q26.6.4 MCQ | 2026 | 1M

'All the mangoes in the basket are good.'
If the above statement is **false**, then which one of the following statements is necessarily true?

A. All the mangoes in the basket are not good.

B. No mango in the basket is good.

C. In the basket, some of the mangoes are good and some are not good.

D. There exists at least one mango in the basket that is not good.

Q26.6.5 MCQ | 2026 | 1M

Consider the following statements about four numbers:

(S1) The average of the four numbers is 25

(S2) Each number is at most 40

(S3) Each number is at least 20

Choose the option that is necessarily correct.

A. (S1) and (S2) together imply (S3)

B. (S2) and (S3) together imply (S1)

C. (S1) and (S3) together imply (S2)

D. (S1) implies (S3)

Q26.6.6 MCQ 2026 2M

'People are crowding around ____ pit into which ____ elephant has fallen. I have never seen an elephant looking more bewildered ____ miserable. Here it is in a most undignified position, thrust into a pit and made to look up ____ a vast, curiosity-stricken crowd.'

Choose the option with the correct sequence of words to fill the blanks.

A. an; a; at; and **B.** a; an; and; at

C. and; a; an; at **D.** at; a; an; and

Q26.6.7 MCQ 2026 2M

The table lists the unit selling price of five products P, Q, R, S, and T. On a particular day, 250 items were sold with the average selling price of Rs. 60.

The following observations were made:

(i) The quantity of S sold was twice that of T.
(ii) The quantity of R sold was thrice that of T.
(iii) The quantity of Q sold was four times that of T.

Product	P	Q	R	S	T
Unit selling price (Rs.)	100	50	40	60	60

What is the quantity of product P sold on that day?

A. 40 **B.** 50 **C.** 60 **D.** 70

Q26.6.8 MCQ 2026 2M

Consider a string P of length l that is laid out as a straight-line segment. Another string K is laid out as a semicircular arc with string P as its diameter, as represented in Figure (i). When both the strings are shortened by a length x they can be re-arranged such that the shortened string K forms a full circle with the shortened string P as its diameter, as represented in Figure (ii). The value of x/l is _____

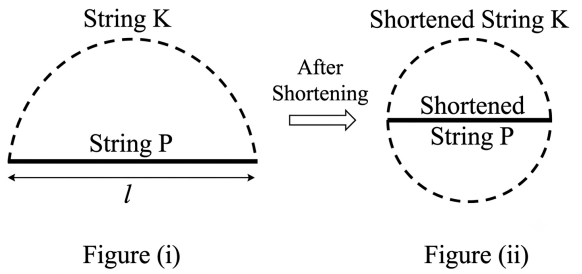


Fig. Q26.6.8

A. π **B.** $\frac{\pi - 1}{2\pi}$ **C.** $\frac{\pi}{2(\pi - 1)}$ **D.** $\frac{\pi}{\pi - 1}$

Q26.6.9 MCQ 2026 2M

The Roman senator Meritorius, his brother, his son, and his daughter have varying oratory skill levels. They are seated in rows and columns as shown in the figure with exactly one person sitting in each box. It is known that:

(i) Meritorius' daughter and his brother are seated in the same column.
(ii) His son is seated diagonally across the sibling of the worst orator.
(iii) The best and worst orators are seated in the same row.

Who is the best orator?

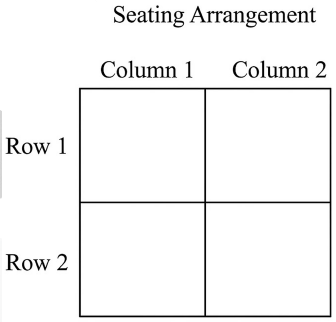


Fig. Q26.6.9

A. Meritorius **B.** Meritorius' brother

C. Meritorius' son **D.** Meritorius' daughter

Q26.6.10 MCQ 2026 2M

Which one of the patterns labelled P, Q, R, and S is used to generate the following figure?

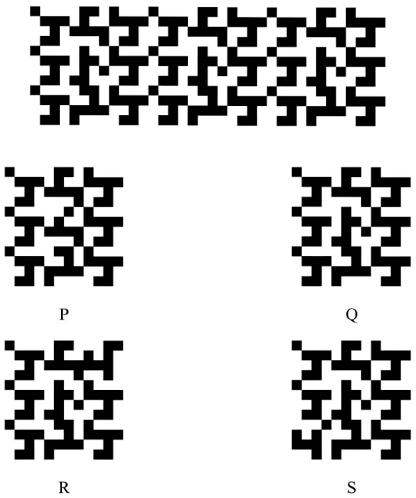


Fig. Q26.6.10

A. P **B.** Q **C.** R **D.** S

Session : 7 | Forenoon | 15/02/26

Q26.7.1 MCQ 2026 1M

Among the following options, the antonym of the word 'nocturnal' is _____.

A. normal **B.** diurnal **C.** abnormal **D.** exceptional

Q26.7.2 MCQ 2026 1M

The statements (S1), (S2), and (S3) pertain to the scores obtained by students in an exam. The maximum possible marks in the exam is 150.

(S1) The highest score is 100.
(S2) The fourth highest score is 76.
(S3) There are at least four students whose scores are within 25 of each other.

Which one of the following options is necessarily correct?

A. (S1) and (S2) together imply (S3)

B. (S1) and (S3) together imply (S2)

C. (S2) and (S3) together imply (S1)

D. (S1) implies (S3)

Q26.7.3 MCQ 2026 1M

The figure below has exactly three intersecting line segments with a rectangular portion WXYZ missing. Which one of the following options P, Q, R, and S is the missing portion WXYZ?

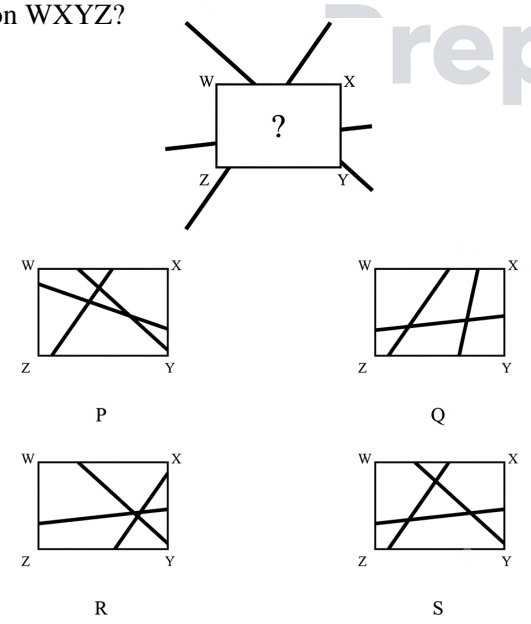


Fig. Q26.7.3

A. P **B.** Q **C.** R **D.** S

Q26.7.4 MCQ 2026 1M

Real numbers y, p, and n (all greater than 1) satisfy

$$(\log_{p^{1/n}} y)(\log_{y^{1/n}} p) = 16.$$

where the logarithms are taken to the bases $p^{1/n}$ and $y^{1/n}$. The value of n is _____

A. 2 **B.** 4 **C.** 8 **D.** 16

Q26.7.5 MCQ 2026 1M

The following observation is made about the scores obtained by 100 students in an exam:
'For each student, there exists another student in the class such that their scores are at most ten marks away.'
If the above statement is **false**, which one of the following statements is necessarily true?

A. For each student, the scores of all the other students are more than 10 marks away.

B. There exists at least one student in the class for whom the scores of all the other students are more than 10 marks away.

C. There is exactly one student in the class for whom the scores of some students are more than 10 marks away.

D. For each student, the score of exactly one other student is more than 10 marks away.

Q26.7.6 MCQ 2026 2M

Each one of the following clues contains a keyword that is partially filled.
Clue 1: Synonym of recognize (8 letters): _D_NT_FY
Clue 2: A story long enough to fill a book (5 letters): _ _ _ EL
Clue 3: Two of something (6 letters): _ _ _ PLE
Clue 4: A fraction of something, split equally into two parts (4 letters): _ _ _ F
The first letter of each of the keywords can be rearranged to form a four-letter word.
Which one of the options below is a possible choice for the four-letter word?

A. CHIN **B.** COIN **C.** ITCH **D.** NOSE

Q26.7.7 MCQ 2026 2M

Three children P, Q, R and two grown-ups X, Y play a badminton doubles tournament. X and Y are parents to two of the children playing. The child of X is not the same as the child of Y. Exactly one of the children does not have a parent playing in the tournament. The following rules are followed:

(i) A parent and his/her child cannot be on the same team.

(ii) A match can feature at most one parent and his/her child, that is, a maximum of one parent-child pair can play in a match.

The following matches were played:

	TEAM 1	TEAM 2
MATCH 1	P and X	Q and R
MATCH 2	P and R	X and Y
MATCH 3	R and X	Q and Y

Which one of the following options is correct?

A. P does not have any parent playing

B. Q does not have any parent playing

C. R does not have any parent playing

D. X does not have a child playing

Q26.7.8 MCQ 2026 2M

Let P_k represent the perimeter of a square with sides of length k. The value of the expression $P_1+P_2+P_3+\cdots+P_{10}$ is ____

A. 55 **B.** 110 **C.** 220 **D.** 440

Q26.7.9 MCQ 2026 2M

The city of Atlantis was crafted by the God of the seas, Poseidon. It was made of alternating concentric circular rings of land (shaded) and water (not shaded) as represented in the figure (not to scale). The radius of Inner Island was 2.5 stades (a unit of length used in ancient Greece). The water surrounding Inner Island was one stade wide (length AB). This was surrounded by two pairs of alternating rings of land and water. The first pair of land and water was two stades wide each (lengths BC and CD), and the outer pair is three stades wide each (lengths DE and EF).

The ratio of the surface area of the land to that of the water in the city of Atlantis is _____(round off to two decimal places).

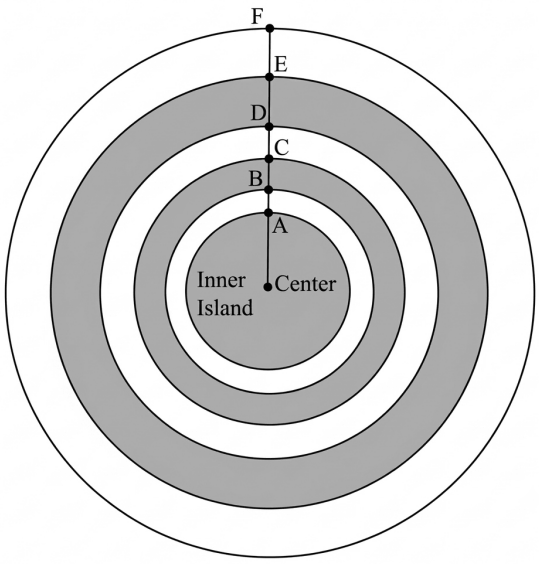


Fig. Q26.7.9

A. 0.45 **B.** 0.60 **C.** 0.75 **D.** 0.90

Q26.7.10 MCQ 2026 2M

If

Ref Fig 12 34 → ꟻ⅁ 34 ꟻ⅁

Then

GATE - 2026 Aptitude Test → ?

P: ɐqnʇıʇdV ʇsǝT / ƆAT∃ - 2026

Q: Aptitude Test / GATE - 2026

R: ɐqnʇıʇdV ʇsǝT / ƆAT∃ - 2026

S: Aptitude Test / GATE - 2026

A. P **B.** Q **C.** R **D.** S

Session : 8 | Afternoon | 15/02/26

Q26.8.1 MCQ 2026 1M

Verbosity: Brevity :: Insolence: _____
Choose the word that best fills the blank.

A. Innocence **B.** Respect **C.** Solace **D.** Wealth

Q26.8.2 MCQ 2026 1M

The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _____

A. 12 **B.** 14 **C.** 16 **D.** 18

Q26.8.3 MCQ 2026 1M

The four pieces of a puzzle are shown in the figure below.

Fig. Q26.8.3

Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?

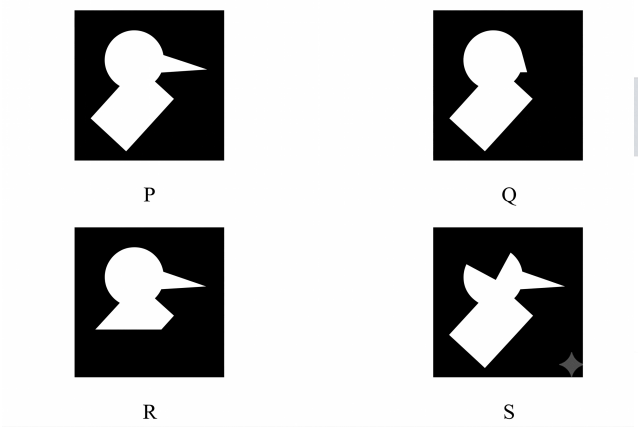


Fig. Q26.8.3

A. P **B.** Q **C.** R **D.** S

Q26.8.4 MCQ 2026 1M

Consider two distinct positive real numbers m, n, with $m > n$. Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is _____.

A. $x > y$ **B.** $x < y$ **C.** $x = y$ **D.** $x = \log_{10}(y)$

Q26.8.5 MCQ 2026 1M

'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.'
Based only on the above sentence, which one of the following statements is true?

A. The actor will certainly sponsor his next movie.

B. His latest movie was a commercial success.

C. The actor made enough money from his latest movie.

D. His latest movie was not commercially successful.

Q26.8.6 MCQ 2026 2M

'My friend and I parted ____ the door ____ the cabin that I had rented ____ the night.'
Choose the option with the correct sequence of words to fill the blanks.

A. at; of; for **B.** for; at; of **C.** of; for; in **D.** in; of; for

Q26.8.7 MCQ 2026 2M

Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20.
Ignoring permutations, the number of ways to pick these five integers is _____

A. 0 **B.** 1 **C.** 2 **D.** 3

Q26.8.8 MCQ 2026 2M

Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that:

(i) the first game was the only game where two students of the same class played against each other,
(ii) the students of Class 5 won more games than the students of Class 4, and
(iii) the boys won two games and the girls won one game.

The student who did not lose any game is _____.

A. Pavan **B.** Rishi **C.** Swathi **D.** Tanvi

Q26.8.9 MCQ 2026 2M

P, Q, R, S, X, and Y are distinct single-digit whole numbers taking values from 0 to 9.
PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number.
It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number.
The value of Y is _____

A. 4 **B.** 5 **C.** 6 **D.** 7

Q26.8.10 MCQ 2026 2M

In the given figure, P, Q, and R are three points on a circle of radius 10 cm with O as its center, $\overline{PQ} = \overline{RQ}$, and $\angle PQR = 45°$. The figure is representative.
The area of the shaded region $PQRO$ is _____ cm^2.

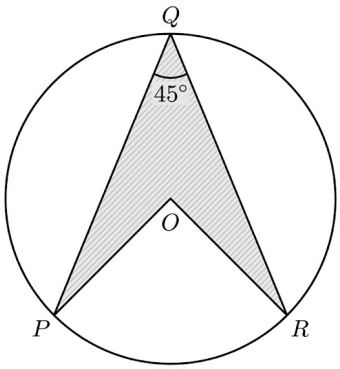


Fig. Q26.8.10

A. 50 **B.** $25\sqrt{2}$ **C.** $50\sqrt{2}$ **D.** 100

Answer Key

Session : 1					
Q26.1.1 D	Q26.1.2 C	Q26.1.3 B	Q26.1.4 D	Q26.1.5 C	Q26.1.6 A
Q26.1.7 B	Q26.1.8 B	Q26.1.9 C	Q26.1.10 A		
Session : 2					
Q26.2.1 A	Q26.2.2 B	Q26.2.3 A	Q26.2.4 B	Q26.2.5 C	Q26.2.6 D
Q26.2.7 A	Q26.2.8 D	Q26.2.9 C	Q26.2.10 C		
Session : 3					
Q26.3.1 B	Q26.3.2 C	Q26.3.3 C	Q26.3.4 D	Q26.3.5 C	Q26.3.6 B
Q26.3.7 D	Q26.3.8 A	Q26.3.9 A	Q26.3.10 C		
Session : 4					
Q26.4.1 A	Q26.4.2 C	Q26.4.3 A	Q26.4.4 A	Q26.4.5 C	Q26.4.6 A
Q26.4.7 A	Q26.4.8 A	Q26.4.9 B	Q26.4.10 C		
Session : 5					
Q26.5.1 A	Q26.5.2 A	Q26.5.3 A	Q26.5.4 C	Q26.5.5 B	Q26.5.6 A
Q26.5.7 B	Q26.5.8 A	Q26.5.9 B	Q26.5.10 D		
Session : 6					
Q26.6.1 C	Q26.6.2 A	Q26.6.3 A	Q26.6.4 D	Q26.6.5 C	Q26.6.6 B
Q26.6.7 B	Q26.6.8 C	Q26.6.9 B	Q26.6.10 B		
Session : 7					
Q26.7.1 B	Q26.7.2 A	Q26.7.3 D	Q26.7.4 B	Q26.7.5 B	Q26.7.6 A
Q26.7.7 C	Q26.7.8 C	Q26.7.9 C	Q26.7.10 C		
Session : 8					
Q26.8.1 B	Q26.8.2 B	Q26.8.3 A	Q26.8.4 C	Q26.8.5 D	Q26.8.6 A
Q26.8.7 B	Q26.8.8 D	Q26.8.9 C	Q26.8.10 C		

YEAR

2025

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS1, AG, MA	10
Session 2	CS2, NM, MT, TF, IN	10
Session 3	ME, PE, AR	10
Session 4	EE	10
Session 5	CY, AE, DA, ES, PI	10
Session 6	EC, GE, XH, BM, EY	10
Session 7	CE1, GG, CH, PH, BT	10
Session 8	CE2, ST, XE, XL, MN	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 01/02/25

Q25.1.1 MCQ 2025 1M

Ravi had ____ younger brother who taught at ____ university. He was widely regarded as ____ honorable man.
Select the option with the correct sequence of articles to fill in the blanks.

A. a; a; an **B.** the; an; a **C.** a; an; a **D.** an; an; a

Q25.1.2 MCQ 2025 1M

The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains.
Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.

A. visionary **B.** shortsighted

C. progressive **D.** innovative

Q25.1.3 MCQ 2025 1M

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

A. 25 **B.** 28 **C.** 30 **D.** 32

Q25.1.4 MCQ 2025 1M

Consider the relationships among P, Q, R, S, and T:

- P is the brother of Q.
- S is the daughter of Q.
- T is the sister of S.
- R is the mother of Q.

The following statements are made based on the relationships given above.

(1) R is the grandmother of S.
(2) P is the uncle of S and T.
(3) R has only one son.
(4) Q has only one daughter.

Which one of the following options is correct?

A. Both (1) and (2) are true. **B.** Both (1) and (3) are true.

C. Only (3) is true. **D.** Only (4) is true.

Q25.1.5 MCQ 2025 1M

According to the map shown in the figure, which one of the following statements is correct?
Note: The figure shown is representative.

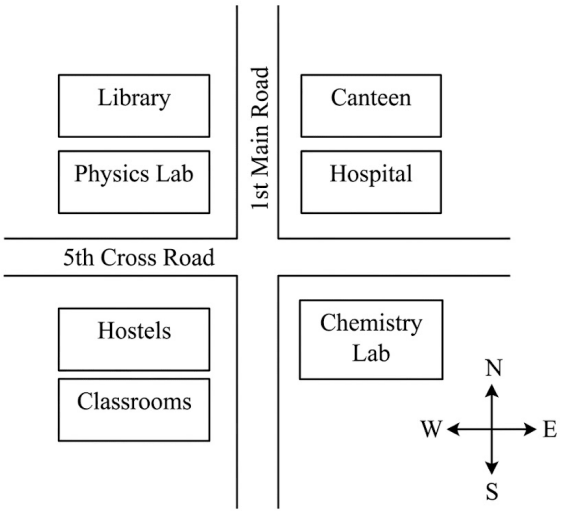


Fig. Q25.1.5

A. The library is located to the northwest of the canteen.

B. The hospital is located to the east of the chemistry lab.

C. The chemistry lab is to the southeast of physics lab.

D. The classrooms and canteen are next to each other.

Q25.1.6 MCQ 2025 2M

"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past."

(From G.K. Chesterton's "A Piece of Chalk")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The author of the passage carries a mirror in his pocket to reflect upon things.

B. The author of the passage had decided to write a poem on epics.

C. The pocket-knife is described as the infant of the sword.

D. Epics are described as too inconvenient to write.

Q25.1.7 MCQ | 2025 | 2M

In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

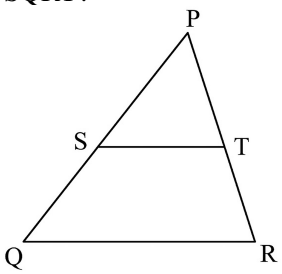


Fig. Q25.1.7

A. $\frac{1}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{2}{5}$ **D.** $\frac{1}{2}$

Q25.1.8 MCQ | 2025 | 2M

A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

A. $\frac{75}{216}$ **B.** $\frac{1}{6}$ **C.** $\frac{1}{18}$ **D.** $\frac{25}{216}$

Q25.1.9 MCQ | 2025 | 2M

A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

Note: The figures shown are representative

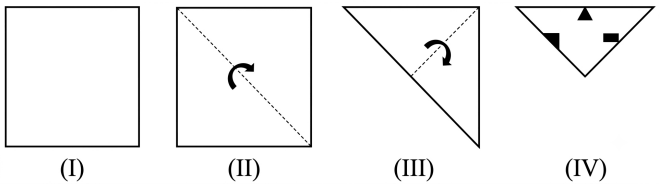
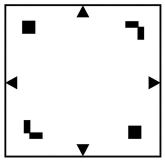
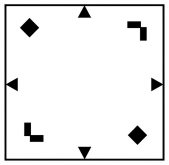


Fig. Q25.1.9

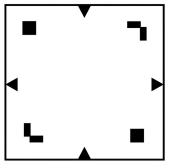
A.



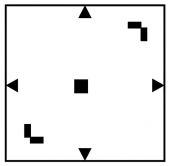
B.



C.



D.



Q25.1.10 MCQ | 2025 | 2M

A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. **The order in which the scoops are purchased is inconsequential**. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

A. 4 **B.** 20 **C.** 24 **D.** 48

Session : 2 | Afternoon | 01/02/25

Q25.2.1 MCQ | 2025 | 1M

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered.
Select the most appropriate option to complete the above sentence.

A. ambivalence **B.** satisfaction **C.** resolve **D.** revolve

Q25.2.2 MCQ | 2025 | 1M

Bird : Nest :: Bee : _____
Select the correct option to complete the analogy.

A. Kennel **B.** Hammock **C.** Hive **D.** Lair

Q25.2.3 MCQ | 2025 | 1M

If $Pe^{x} = Qe^{-x}$ for all real values of x, which one of the following statements is true?

A. $P = Q = 0$ **B.** $P = Q = 1$
C. $P = 1; Q = -1$ **D.** $\frac{P}{Q} = 0$

Q25.2.4 MCQ | 2025 | 1M

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.

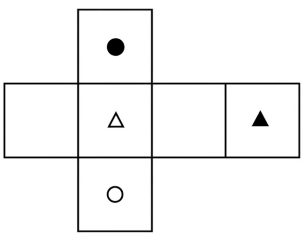
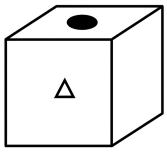
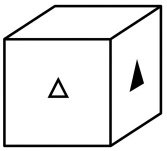


Fig. Q25.2.4

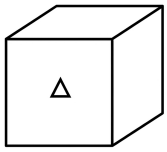
A.



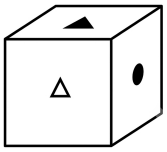
B.



C.



D.



Q25.2.5 MCQ | 2025 | 1M

Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2?

A. $p_1 + p_2$ is not a prime number.

B. $p_1 p_2$ is not a prime number.

C. $p_1 + p_2 + 1$ is a prime number.

D. $p_1 p_2 + 1$ is a prime number.

Q25.2.6 MCQ | 2025 | 2M

Based only on the conversation below, identify the logically correct inference:
"Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

A. Ramya knew that Josephine was in the hospital.

B. Ramya did not know that Josephine was in the hospital.

C. Ramya and Josephine were once close friends; but now, they are not.

D. Josephine was in the hospital due to an injury to her leg.

Q25.2.7 MCQ | 2025 | 2M

If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH?

A. CEADP **B.** IDBFC **C.** JGIBC **D.** IBCEC

Q25.2.8 MCQ | 2025 | 2M

Which one of the following options is correct for the given data in the table?

Iteration (i)	0	1	2	3
Input (I)	20	−4	10	15
Output (X)	20	16	26	41
Output (Y)	20	−80	−800	−12000

A. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$

B. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1) + I(i);\ i > 0$

C. $X(i) = X(i-1)I(i);\ Y(i) = Y(i-1)I(i);\ i > 0$

D. $X(i) = X(i-1) + I(i);\ Y(i) = Y(i-1)I(i-1);\ i > 0$

Q25.2.9 MCQ | 2025 | 2M

In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square.
What is the area (in cm^2) of the rectangle PLMN?

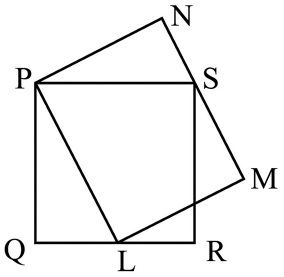


Fig. Q25.2.9

A. $2\sqrt{2}$ **B.** 2 **C.** 8 **D.** 4

Q25.2.10 MCQ | 2025 | 2M

The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct?

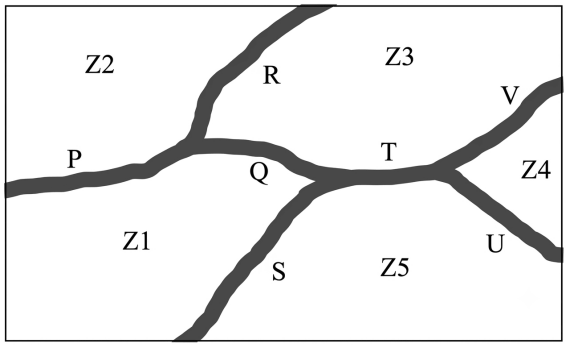


Fig. Q25.2.10

A. Bridges on P, Q, and T

B. Bridges on P, Q, S, and T

C. Bridges on Q, R, T, and V

D. Bridges on P, Q, S, U, and V

Session : 3 | Forenoon | 02/02/25

Q25.3.1 MCQ 2025 1M

Fish: Shoal :: Lion: _____

Select the correct option to complete the analogy.

A. Pride **B.** School **C.** Forest **D.** Series

Q25.3.2 MCQ 2025 1M

Identify the grammatically correct sentence:

A. It is I who am responsible for this fiasco.

B. It is myself who is responsible for this fiasco.

C. It is I who is responsible for this fiasco.

D. It is I who are responsible for this fiasco.

Q25.3.3 MCQ 2025 1M

Two cars, P and Q, start from a point X in India at 10 AM. Car P travels North with a speed of 25 km/h and car Q travels East with a speed of 30 km/h. Car P travels continuously but car Q stops for some time after travelling for one hour. If both the cars are at the same distance from X at 11:30 AM, for how long (in minutes) did car Q stop?

A. 10 **B.** 12 **C.** 15 **D.** 18

Q25.3.4 MCQ 2025 1M

The ceiling function of a real number x, denoted by $ce(x)$, is defined as the smallest integer that is greater than or equal to x. Similarly, the floor function, denoted by $fl(x)$, is defined as the largest integer that is smaller than or equal to x. Which one of the following statements is NOT correct for all possible values of x?

A. $ce(x) \geq x$ **B.** $fl(x) \leq x$

C. $ce(x) \geq fl(x)$ **D.** $fl(x) < ce(x)$

Q25.3.5 MCQ 2025 1M

P and Q play chess frequently against each other. Of these matches, P has won 80% of the matches, drawn 15% of the matches and lost 5% of the matches.

If they play 3 more matches, what is the probability of P winning exactly 2 of these 3 matches?

A. $\frac{48}{125}$ **B.** $\frac{16}{125}$ **C.** $\frac{16}{25}$ **D.** $\frac{25}{48}$

Q25.3.6 MCQ 2025 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. At once, without thinking much, people rushed towards the city in hordes with the sole aim of grabbing as much gold as they could.

Q. However, little did they realize about the impending hardships they would have to face on their way to the city: miles of mud, unfriendly forests, hungry beasts and inimical local lords all of which would reduce their chances of getting gold to almost zero.

R. All of them thought that easily they could lay their hands on gold and become wealthy overnight.

S. About a hundred years ago, the news that gold had been discovered in Kolar spread like wildfire and the whole State was in raptures.

A. $P \rightarrow Q \rightarrow R \rightarrow S$ **B.** $Q \rightarrow S \rightarrow R \rightarrow P$

C. $S \rightarrow Q \rightarrow P \rightarrow R$ **D.** $S \rightarrow P \rightarrow R \rightarrow Q$

Q25.3.7 MCQ 2025 2M

If HIDE and CAGE are coded as 19-23-7-11 and 5-2-17-11 respectively, then what is the code for HIGH?

A. 5-17-1-2 **B.** 17-19-13-17

C. 13-3-1-2 **D.** 19-23-17-19

Q25.3.8 MCQ 2025 2M

The given figure is reflected about the horizontal dashed line and then rotated clockwise by 90° about an axis perpendicular to the plane of the figure.

Which one of the following options correctly shows the resultant figure?

Note: The figures shown are representative.

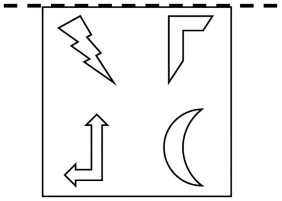
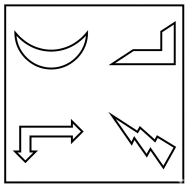
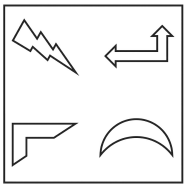
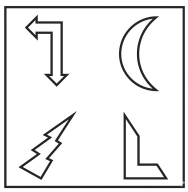
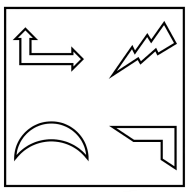


Fig. Q25.3.8

A. 

B. 

C. 

D. 

Q25.3.9 MCQ | 2025 | 2M

Which one of the following options has the correct sequence of objects arranged in the increasing number of mirror lines (lines of symmetry)?

A. Circle; Square; Equilateral triangle; Isosceles triangle

B. Isosceles triangle; Equilateral triangle; Square; Circle

C. Equilateral triangle; Isosceles triangle; Square; Circle

D. Isosceles triangle; Square; Equilateral triangle; Circle

Q25.3.10 MCQ | 2025 | 2M

A final year student appears for placement interview in two companies, S and T. Based on her interview performance, she estimates the probability of receiving job offers from companies S and T to be 0.8 and 0.6, respectively. Let p be the probability that she receives job offers from both the companies. Select the most appropriate option.

A. $0 \le p \le 0.2$

B. $0.4 \le p \le 0.6$

C. $0.2 \le p \le 0.4$

D. $0.6 \le p \le 1.0$

Session : 4 | Afternoon | 02/02/25

Q25.4.1 MCQ | 2025 | 1M

Kavya _____ go to work yesterday as she _____ feeling well.

Select the most appropriate option to complete the above sentence.

A. didn't; isn't

B. wouldn't; wasn't

C. wasn't; wasn't

D. couldn't; wasn't

Q25.4.2 MCQ | 2025 | 1M

Good : Evil :: Genuine : _____

Select the most appropriate option to complete the analogy.

A. Counterfeit

B. Contraband

C. Counterfoil

D. Counterpart

Q25.4.3 MCQ | 2025 | 1M

The relationship between two variables x and y is given by $x + py + q = 0$ and is shown in the figure. Find the values of p and q.

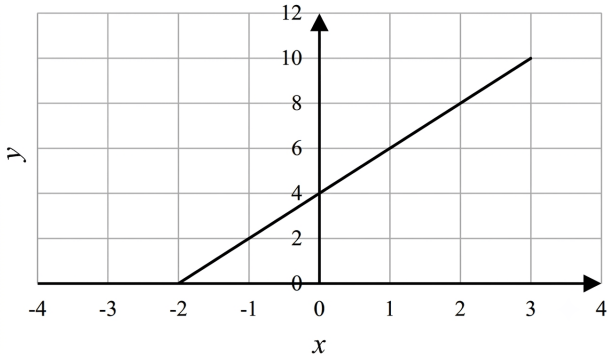


Fig. Q25.4.3

A. $p = -\frac{1}{2}; q = 2$

B. $p = 2; q = -2$

C. $p = \frac{1}{2}; q = 4$

D. $p = 2; q = 4$

Q25.4.4 MCQ | 2025 | 1M

Each row of Column-I has three items and each item is represented by a circle in Column-II. The arrangement of circles in Column-II represents the relationship among the items in Column-I. Identify the option that has the most appropriate match between Column-I and Column-II.

Column-I		Column-II	
(1)	Animals, Zebra, Giraffe	(P)	
(2)	Director, Producer, Actor	(Q)	
(3)	Word, Sentence, Novel	(R)	
(4)	Pianist, Guitarist, Instrumentalist	(S)	

Fig. Q25.4.4

A. (1)-(Q); (2)-(R); (3)-(S); (4)-(P)

B. (1)-(S); (2)-(P); (3)-(R); (4)-(Q)

C. (1)-(Q); (2)-(P); (3)-(S); (4)-(R)

D. (1)-(R); (2)-(S); (3)-(Q); (4)-(P)

Q25.4.5 MCQ | 2025 | 1M

What is the value of $\left(\frac{3^{81}}{27^4}\right)^{\frac{1}{3}}$?

A. 3^{13} **B.** 3^{96} **C.** 3^{23} **D.** 3^{69}

Q25.4.6 MCQ | 2025 | 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. It is because deer, like most of the animals that tigers normally prey on, run much faster! It simply means, another day of empty stomach for the big cats.

Q. Tigers spend most of their life searching for food.

R. If they trace the scent of deer, tigers follow the trail, chase the deer for a mile or two in the dark, and yet may not catch them.

S. For several nights, they relentlessly prowl through the forest, hunting for a trail that may lead to their prey.

A. $S \rightarrow P \rightarrow R \rightarrow Q$ **B.** $R \rightarrow P \rightarrow S \rightarrow Q$

C. $Q \rightarrow S \rightarrow R \rightarrow P$ **D.** $P \rightarrow Q \rightarrow S \rightarrow R$

Q25.4.7 MCQ | 2025 | 2M

In the given figure, EF and HJ are coded as 30 and 80, respectively. Which one among the given options is most appropriate for the entries marked (i) and (ii) ?

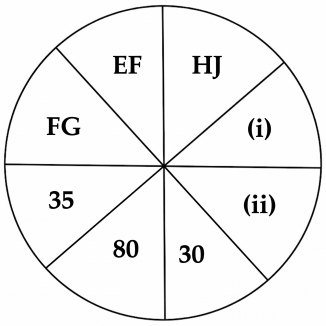


Fig. Q25.4.7

A. (i) EH; (ii) 40 **B.** (i) JK; (ii) 36

C. (i) EG; (ii) 42 **D.** (i) PS; (ii) 14

Q25.4.8 MCQ | 2025 | 2M

Scores obtained by two students P and Q in seven courses are given in the table below. Based on the information given in the table, which one of the statements is INCORRECT?

P	22	89	50	45	78	60	39
Q	35	65	60	56	81	45	50

A. Average score of P is less than the average score of Q.

B. Median score of P is same as the median score of Q.

C. Difference between the maximum and minimum scores of P is greater than the difference between the maximum and minimum scores of Q.

D. Median score and the average score of Q are same.

Q25.4.9 MCQ | 2025 | 2M

Spheres of unit diameter are centered at (l, m, n) where l, m, and n take every possible integer values.
The distance between two spheres is computed from the center of one sphere to the center of the other sphere. For a given sphere, x is the distance to its nearest sphere and y is the distance to its next nearest sphere. The value of $\frac{y}{x}$ is:

A. $2\sqrt{2}$ **B.** $\frac{1}{\sqrt{2}}$ **C.** $\sqrt{2}$ **D.** 2

Q25.4.10 MCQ | 2025 | 2M

In the triangle PQR, the lengths of PT and TR are in the ratio of 3:2.
ST is parallel to QR. Two semicircles are drawn with PS and PQ as diameters, as shown in the figure.
Which one of the following statements is true about the shaded area PQS?

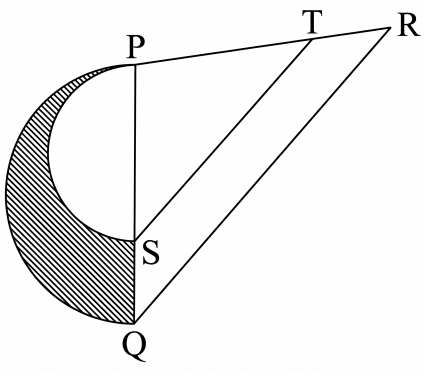


Fig. Q25.4.10

A. The shaded area is $\frac{16}{9}$ times the area of the semicircle with the diameter PS.

B. The shaded area is equal to the area of the semicircle with the diameter PS.

C. The shaded area is $\frac{14}{9}$ times the area of the semicircle with the diameter PS.

D. The shaded area is $\frac{14}{25}$ times the area of the semicircle with the diameter PQ.

Session : 5 | Forenoon | 15/02/25

Q25.5.1 MCQ 2025 1M

Courage : Bravery :: Yearning : _____

Select the most appropriate option to complete the analogy.

A. Longing **B.** Yelling **C.** Yawning **D.** Glaring

Q25.5.2 MCQ 2025 1M

We _____ tennis in the lawn when it suddenly started to rain.

Select the most appropriate option to complete the above sentence.

A. have been playing **B.** had been playing

C. could be playing **D.** would have been playing

Q25.5.3 MCQ 2025 1M

A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

A. 3 **B.** 8 **C.** 11 **D.** 9

Q25.5.4 MCQ 2025 1M

In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R?

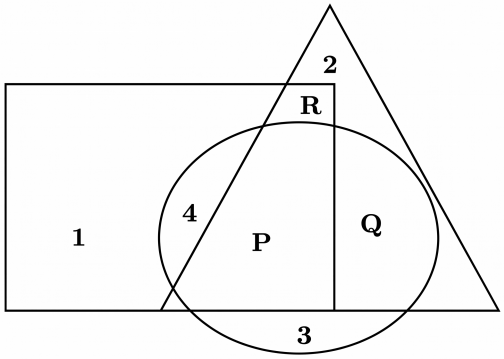


Fig. Q25.5.4

A. $P = 6; Q = 5; R = 3$ **B.** $P = 5; Q = 6; R = 3$

C. $P = 3; Q = 6; R = 6$ **D.** $P = 5; Q = 3; R = 6$

Q25.5.5 MCQ 2025 1M

A rectangle has a length L and a width W, where $L > W$. If the width, W, is increased by 10%, which one of the following statements is correct for all values of L and W?

A. Perimeter increases by 10%.

B. Length of the diagonals increases by 10%.

C. Area increases by 10%.

D. The rectangle becomes a square.

Q25.5.6 MCQ 2025 2M

Column-I has statements made by Shanthala; and, Column-II has responses given by Kanishk.

Column-I		Column-II	
P.	This house is in a mess.	1.	Alright, I won't bring it up during our conversations.
Q.	I am not happy with the marks given to me.	2.	Well, you can easily look it up.
R.	Politics is a subject I avoid talking about.	3.	No problem, let me clear it up for you.
S.	I don't know what this word means.	4.	Don't worry, I will take it up with your teacher.

Identify the option that has the correct match between Column-I and Column-II.

A. $P-2; Q-3; R-1; S-4$

B. $P-3; Q-4; R-1; S-2$

C. $P-4; Q-1; R-2; S-3$

D. $P-1; Q-2; R-4; S-3$

Q25.5.7 MCQ 2025 2M

Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$.

Which one of the following statements is correct about their age on the day when they attain the same weight?

A. $a_1 < a_2 < 50$ **B.** $a_1 < 50 < a_2$

C. $50 < a_1 < a_2$ **D.** Either $a_1 = 50$ or $a_2 = 50$

Q25.5.8 MCQ 2025 2M

A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square.

The number of squares that can be formed and the number of triangles required to form each square, respectively, are:

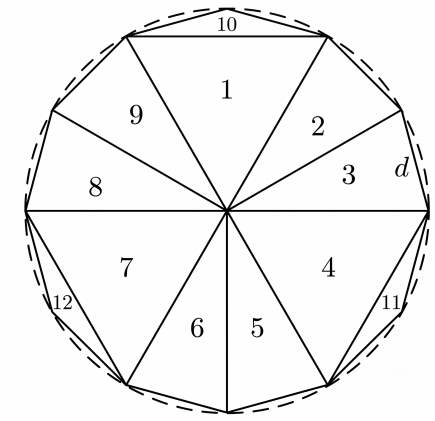


Fig. Q25.5.8

A. 3; 4 **B.** 4; 3 **C.** 3; 3 **D.** 3; 2

Q25.5.9 MCQ 2025 2M

If a real variable x satisfies $3^{x^2} = 27 \times 9^x$ then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:

A. 2^{-1} **B.** 2^0 **C.** 2^3 **D.** 2^{15}

Q25.5.10 MCQ 2025 2M

The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is Rs. $1000(X-0.2)$, what is the average amount (in Rs.) she has earned per shift in the past 100 shifts?

Note: The figure shown is representative.

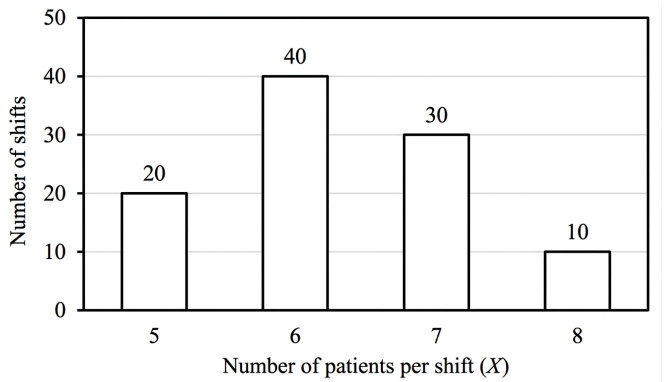


Fig. Q25.5.10

A. 6,100 **B.** 6,300 **C.** 5,800 **D.** 6,500

Session : 6 | Afternoon | 15/02/25

Q25.6.1 MCQ 2025 1M

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II.

Group-I: Abuse → Insult → Ridicule

Group-II: _____ → Praise → Appreciate

A. Extol **B.** Prize **C.** Appropriate **D.** Espouse

Q25.6.2 MCQ 2025 1M

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

A. will be **B.** can be

C. am going to be **D.** could have been

Q25.6.3 MCQ 2025 1M

The 12 musical notes are given as C, $C^{\#}$, D, $D^{\#}$, E, F, $F^{\#}$, G, $G^{\#}$, A, $A^{\#}$, and B.

Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^{\#}$ and C is:

A. $\sqrt[6]{2}$ **B.** $\sqrt{2}$ **C.** $\sqrt[4]{2}$ **D.** 2

Q25.6.4 MCQ 2025 1M

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n' is:

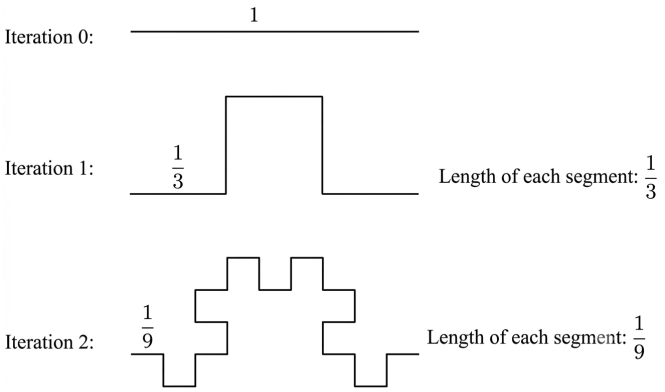


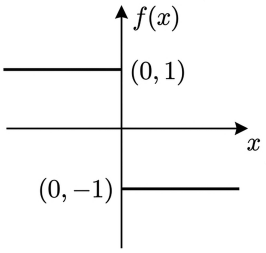
Fig. Q25.6.4

A. $\left(\frac{5}{3}\right)^{\frac{n}{2}}$ **B.** $\left(\frac{5}{3}\right)^{n}$ **C.** $\left(\frac{5}{3}\right)$ **D.** $\left(\frac{5}{3}\right)^{n(2n-1)}$

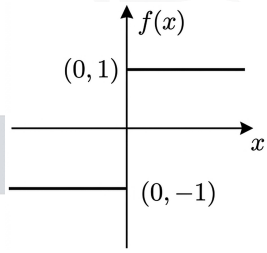
Q25.6.5 MCQ 2025 1M

Which one of the following plots represents $f(x) = -\frac{|x|}{x}$, where x is a non-zero real number?

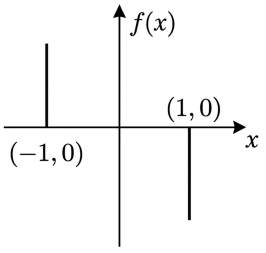
A.



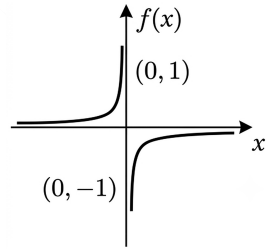
B.



C.



D.



Q25.6.6 MCQ 2025 2M

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

P. Over time, such adaptations lead to significant evolutionary changes with the potential to shape the development of new species.

Q. In natural world, organisms constantly adapt to their environments in response to challenges and opportunities.

R. This process of adaptation is driven by the principle of natural selection, where favorable traits increase an organism's chances of survival and reproduction.

S. As environments change, organisms that can adapt their behavior, structure and physiology to such changes are more likely to survive.

A. $P \rightarrow Q \rightarrow R \rightarrow S$ **B.** $Q \rightarrow S \rightarrow R \rightarrow P$

C. $R \rightarrow S \rightarrow Q \rightarrow P$ **D.** $S \rightarrow P \rightarrow R \rightarrow Q$

Q25.6.7 MCQ 2025 2M

A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

Note: All lengths are in meter. The figure shown is representative.

Fig. Q25.6.7

A. $b_1 < 0.5$ **B.** $b_2 > 0.5$

C. $b_2 < b_1 + 0.5$ **D.** $b_1 + b_2 < 1$

Q25.6.8 MCQ 2025 2M

Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1st round, 4th student behind P leaves the game.
After 2nd round, 5th student behind Q leaves the game.
After 3rd round, 3rd student behind V leaves the game.
After 4th round, 4th student behind U leaves the game.
Who all are left in the game after the 4th round?

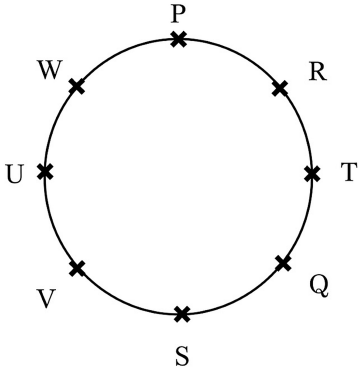


Fig. Q25.6.8

A. P; T; Q; S **B.** V; P; T; Q

C. W; R; Q; U **D.** Q; T; V; W

Q25.6.9 MCQ 2025 2M

The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

A. France will occupy the third place if the list were made on the basis of the total number of medals won.

B. The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

C. USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

D. Canada has won twice as many total medals as Japan.

Q25.6.10 MCQ 2025 2M

An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to ₹ 5,00,000. For higher fees, the overhead is ₹ 1,00,000 plus 10% of the amount by which the fee exceeds ₹ 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than ₹ 10,00,000, what is the maximum consulting fee that the employee can charge?

A. ₹ 7,01,438 **B.** ₹ 7,24,961

C. ₹ 7,51,232 **D.** ₹ 7,75,784

Session : 7 | Forenoon | 16/02/25

Q25.7.1 MCQ 2025 1M

Is there any good show _____ television tonight?
Select the most appropriate option to complete the above sentence.

A. in **B.** at **C.** within **D.** on

Q25.7.2 MCQ 2025 1M

As the police officer was found guilty of embezzlement, he was _____ dismissed from the service in accordance with the Service Rules.
Select the most appropriate option to complete the above sentence.

A. sumptuously **B.** brazenly

C. unintentionally **D.** summarily

Q25.7.3 MCQ 2025 1M

The sum of the following infinite series is:

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\cdots$$

A. π **B.** $1+e$ **C.** $e-1$ **D.** e

Q25.7.4 MCQ 2025 1M

A thin wire is used to construct all the edges of a cube of 1 m side by bending, cutting and soldering the wire. If the wire is 12 m long, what is the minimum number of cuts required to construct the wire frame to form the cube?

A. 3 **B.** 4 **C.** 6 **D.** 12

Q25.7.5 MCQ 2025 1M

The figures I, II and III are parts of a sequence. Which one of the following options comes next in the sequence at IV?

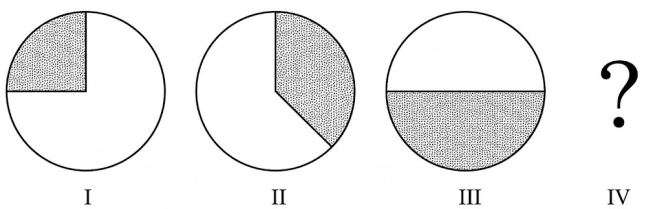
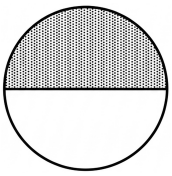
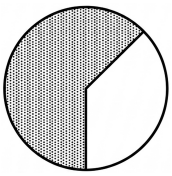
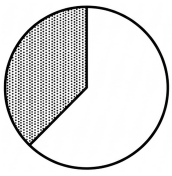
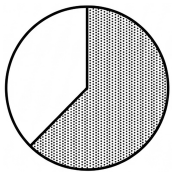


Fig. Q25.7.5

A. 

B. 

C. 

D. 

Q25.7.6 MCQ | 2025 | 2M

"Why do they pull down and do away with crooked streets, I wonder, which are my delight, and hurt no man living? Every day the wealthier nations are pulling down one or another in their capitals and their great towns: they do not know why they do it; neither do I. It ought to be enough, surely, to drive the great broad ways which commerce needs and which are the life-channels of a modern city, without destroying all history and all the humanity in between: the islands of the past."

(From Hilaire Belloc's "The Crooked Streets")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The author of the passage takes delight in wondering.

B. The wealthier nations are pulling down the crooked streets in their capitals.

C. In the past, crooked streets were only built on islands.

D. Great broad ways are needed to protect commerce and history.

Q25.7.7 MCQ | 2025 | 2M

Rohit goes to a restaurant for lunch at about 1 PM. When he enters the restaurant, he notices that the hour and minute hands on the wall clock are exactly coinciding. After about an hour, when he leaves the restaurant, he notices that the clock hands are again exactly coinciding. How much time (in minutes) did Rohit spend at the restaurant?

A. $64\frac{6}{11}$ **B.** $66\frac{5}{13}$ **C.** $65\frac{5}{11}$ **D.** $66\frac{6}{13}$

Q25.7.8 MCQ | 2025 | 2M

A color model is shown in the figure with color codes: Yellow (Y), Magenta (M), Cyan (Cy), Red (R), Blue (Bl), Green (G), and Black (K).

Which one of the following options displays the color codes that are consistent with the color model?

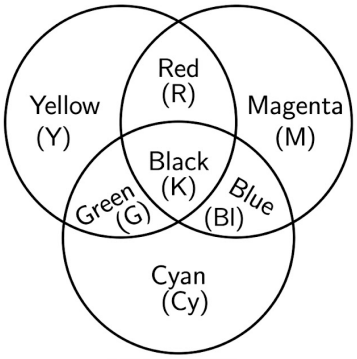
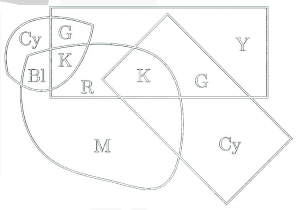
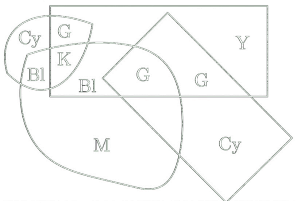
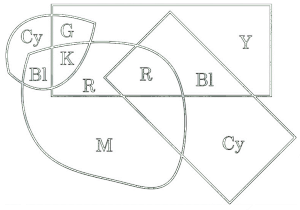
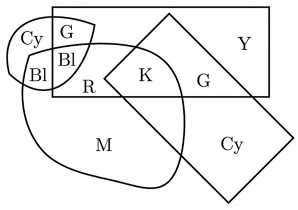


Fig. Q25.7.8

A. 

B. 

C. 

D. 

Q25.7.9 MCQ | 2025 | 2M

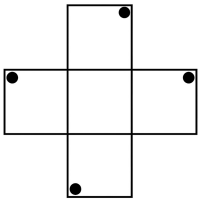
A circle with center at $(x, y) = (0.5, 0)$ and radius = 0.5 intersects with another circle with center at $(x, y) = (1, 1)$ and radius = 1 at two points. One of the points of intersection (x, y) is:

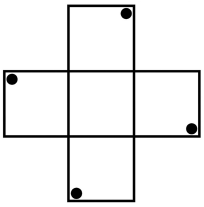
A. (0,0) **B.** (0.2, 0.4) **C.** (0.5, 0.5) **D.** (1,2)

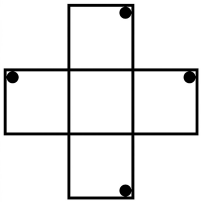
Q25.7.10 MCQ | 2025 | 2M

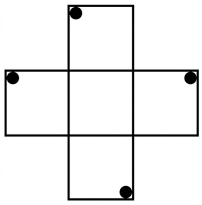
An object is said to have an n-fold rotational symmetry if the object, rotated by an angle of $\frac{2\pi}{n}$, is identical to the original.

Which one of the following objects exhibits 4-fold rotational symmetry about an axis perpendicular to the plane of the screen?

A. 

B. 

C. 

D. 

Session : 8 | Afternoon | 16/02/25

Q25.8.1 MCQ 2025 1M

Even though I had planned to go skiing with my friends, I had to _____ at the last moment because of an injury. Select the most appropriate option to complete the above sentence.

A. back up **B.** back of **C.** back on **D.** back out

Q25.8.2 MCQ 2025 1M

The President, along with the Council of Ministers, _____ to visit India next week. Select the most appropriate option to complete the above sentence.

A. wish **B.** wishes **C.** will wish **D.** is wishing

Q25.8.3 MCQ 2025 1M

An electricity utility company charges ₹ 7 per kWh (kilo watt-hour). If a 40-watt desk light is left on for 10 hours each night for 180 days, what would be the cost of energy consumption? If the desk light is on for 2 more hours each night for the 180 days, what would be the percentage-increase in the cost of energy consumption?

A. ₹ 604.8; 10% **B.** ₹ 504; 20%

C. ₹ 604.8; 12% **D.** ₹ 720; 15%

Q25.8.4 MCQ 2025 1M

In the context of the given figure, which one of the following options correctly represents the entries in the blocks labelled (i), (ii), (iii), and (iv), respectively?

N	U	F	(i)
21	14	9	6
H	L	(ii)	O
12	(iv)	15	(iii)

Fig. Q25.8.4

A. Q, M, 12, and 8 **B.** K, L, 10 and 14

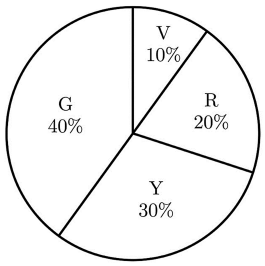
C. I, J, 10, and 8 **D.** L, K, 12 and 8

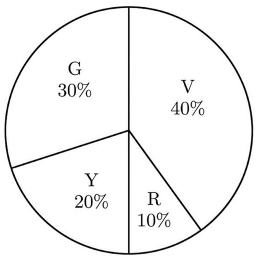
Q25.8.5 MCQ 2025 1M

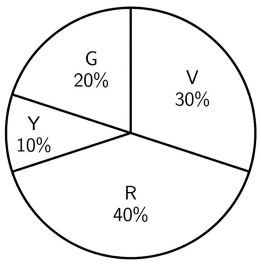
A bag contains Violet (V), Yellow (Y), Red (R), and Green (G) balls. On counting them, the following results are obtained:

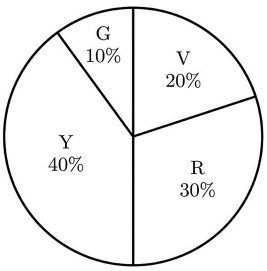
(i) The sum of Yellow balls and twice the number of Violet balls is 50.
(ii) The sum of Violet and Green balls is 50.
(iii) The sum of Yellow and Red balls is 50.
(iv) The sum of Violet and twice the number of Red balls is 50.

Which one of the following Pie charts correctly represents the balls in the bag?

A. 

B. 

C. 

D. 

Q25.8.6 MCQ 2025 2M

"His life was divided between the books, his friends, and long walks. A solitary man, he worked at all hours without much method, and probably courted his fatal illness in this way. To his own name there is not much to show; but such was his liberality that he was continually helping others, and fruits of his erudition are widely scattered, and have gone to increase many a comparative stranger's reputation."

(From E.V. Lucas's "A Funeral")

Based only on the information provided in the above passage, which one of the following statements is true?

A. The solitary man described in the passage is dead.

B. Strangers helped create a grand reputation for the solitary man described in the passage.

C. The solitary man described in the passage found joy in scattering fruits.

D. The solitary man worked in a court where he fell ill.

Q25.8.7 MCQ 2025 2M

For the clock shown in the figure, if

O^* = O Q S Z P R T, and

X^* = X Z P W O Q

then which one among the given options is most appropriate for P^*?

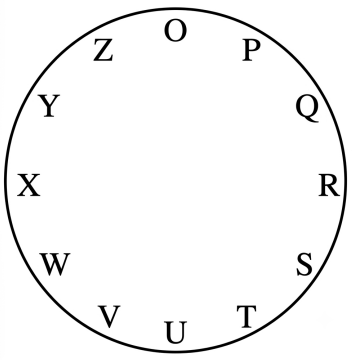


Fig. Q25.8.7

A. P U W R T V W

B. P R T O Q S U

C. P T V Q S U W

D. P S U P R T V

Q25.8.8 MCQ 2025 2M

Consider a five-digit number PQRST that has distinct digits P, Q, R, S, and T, and satisfies the following conditions:

$$P < Q$$

$$S > P > T$$

$$R < T$$

If integers 1 through 5 are used to construct such a number, the value of P is:

A. 1 **B.** 2 **C.** 3 **D.** 4

Q25.8.9 MCQ 2025 2M

A business person buys potatoes of two different varieties P and Q, mixes them in a certain ratio and sells them at ₹192 per kg. The cost of the variety P is ₹800 for 5 kg. The cost of the variety Q is ₹800 for 4 kg. If the person gets 8% profit, what is the P:Q ratio (by weight)?

A. 5:4 **B.** 3:4 **C.** 3:2 **D.** 1:1

Q25.8.10 MCQ 2025 2M

Three villages P, Q, and R are located in such a way that the distance PQ = 13 km, QR = 14 km, and RP = 15 km, as shown in the figure. A straight road joins Q and R. It is proposed to connect P to this road QR by constructing another road. What is the minimum possible length (in km) of this connecting road?

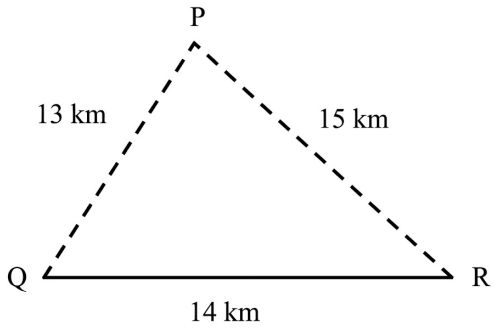


Fig. Q25.8.10

A. 10.5 **B.** 11.0 **C.** 12.0 **D.** 12.5

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q25.1.1 A	Q25.1.2 B	Q25.1.3 C	Q25.1.4 A	Q25.1.5 C	Q25.1.6 C
Q25.1.7 A	Q25.1.8 A	Q25.1.9 A	Q25.1.10 B		
Session : 2					
Q25.2.1 C	Q25.2.2 C	Q25.2.3 A	Q25.2.4 A	Q25.2.5 B	Q25.2.6 B
Q25.2.7 B	Q25.2.8 A	Q25.2.9 D	Q25.2.10 C		
Session : 3					
Q25.3.1 A	Q25.3.2 A	Q25.3.3 C	Q25.3.4 D	Q25.3.5 A	Q25.3.6 D
Q25.3.7 D	Q25.3.8 B	Q25.3.9 B	Q25.3.10 B		
Session : 4					
Q25.4.1 D	Q25.4.2 A	Q25.4.3 A	Q25.4.4 A	Q25.4.5 C	Q25.4.6 C
Q25.4.7 C	Q25.4.8 B	Q25.4.9 C	Q25.4.10 A		
Session : 5					
Q25.5.1 A	Q25.5.2 B	Q25.5.3 C	Q25.5.4 A	Q25.5.5 C	Q25.5.6 B
Q25.5.7 B	Q25.5.8 A	Q25.5.9 C	Q25.5.10 A		
Session : 6					
Q25.6.1 A	Q25.6.2 D	Q25.6.3 B	Q25.6.4 B	Q25.6.5 A	Q25.6.6 B
Q25.6.7 D	Q25.6.8 MTA	Q25.6.9 C	Q25.6.10 B		
Session : 7					
Q25.7.1 D	Q25.7.2 D	Q25.7.3 C	Q25.7.4 A	Q25.7.5 B	Q25.7.6 B
Q25.7.7 C	Q25.7.8 A	Q25.7.9 B	Q25.7.10 B		
Session : 8					
Q25.8.1 D	Q25.8.2 B	Q25.8.3 B	Q25.8.4 C	Q25.8.5 A	Q25.8.6 A
Q25.8.7 B	Q25.8.8 C	Q25.8.9 A	Q25.8.10 C		

YEAR

2024

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, CY, DA, ES, PI	10
Session 2	GE, ME, PH	10
Session 3	BM, CE1, CH, EY, GG, XH	10
Session 4	CE2, IN, MA, MN, PE	10
Session 5	AE, AG, BT, CS1, MT, NM, TF	10
Session 6	CS2, ST, XE, XL	10
Session 7	EC	10
Session 8	EE	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 03/02/24

Q24.1.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[sick → infirm → moribund] is analogous to [silly → _____ → daft].
Which one of the given options is appropriate to fill the blank?

A. frown **B.** fawn **C.** vein **D.** vain

Q24.1.2 MCQ 2024 1M

The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is

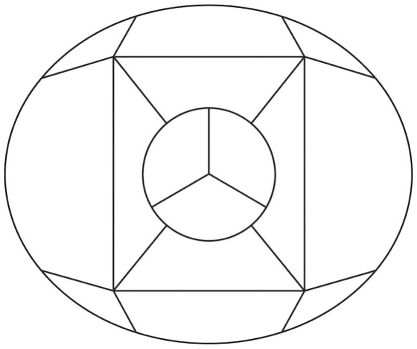


Fig. Q24.1.2

A. 4 **B.** 3 **C.** 5 **D.** 6

Q24.1.3 MCQ 2024 1M

How many 4-digit positive integers divisible by 3 can be formed using only the digits $\{1, 3, 4, 6, 7\}$, such that no digit appears more than once in a number?

A. 24 **B.** 48 **C.** 72 **D.** 12

Q24.1.4 MCQ 2024 1M

The sum of the following infinite series is

$$2 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{8} + \frac{1}{9} + \frac{1}{16} + \frac{1}{27} + \cdots$$

A. 11/3 **B.** 7/2 **C.** 13/4 **D.** 9/2

Q24.1.5 MCQ 2024 1M

In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid.

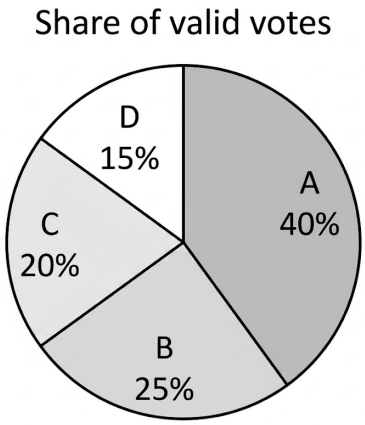


Fig. Q24.1.5

Based on the data provided, the total number of valid votes received by the candidates B and C is .

A. 45,000 **B.** 49,500 **C.** 51,750 **D.** 54,000

Q24.1.6 MCQ 2024 2M

Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk.
Based on the given passage, which of the following can be inferred?

A. All human beings can digest dairy milk.

B. No human being can digest dairy milk.

C. Digestion of dairy milk is essential for human beings.

D. In human beings, digestion of dairy milk resulted from a mutated gene.

Q24.1.7 MCQ 2024 2M

The probability of a boy or a girl being born is 1/2. For a family having only three children, what is the probability of having two girls and one boy?

A. 3/8 **B.** 1/8 **C.** 1/4 **D.** 1/2

Q24.1.8 MCQ 2024 2M

Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund		
	A	B	C
Person 1	₹10,000	₹20,000	₹20,000
Person 2	₹20,000	₹15,000	₹15,000

At the end of one year, the total amount that Person 1 gets is Rs 500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

A. 7.5% **B.** 10% **C.** 15% **D.** 20%

Q24.1.9 MCQ 2024 2M

Three different views of a dice are shown in the figure below.

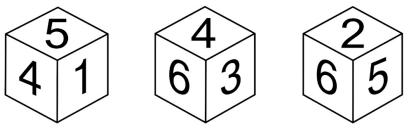


Fig. Q24.1.9

The piece of paper that can be folded to make this dice is .

A.
5	1
	4
	6
2	3

B.
5	1
	4
	2
6	3

C.
5	1
	3
	2
4	6

D.
5	1
	4
	6
3	2

Q24.1.10 MCQ 2024 2M

Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

A. A rhombus **B.** A triangle

C. An ellipse **D.** A hexagon

Session : 2 | Afternoon | 03/02/24

Q24.2.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[smile → giggle → laugh] is analogous to [disapprove → _____ → chide].
Which one of the given options is appropriate to fill the blank?

A. reprove **B.** praise **C.** reprise **D.** grieve

Q24.2.2 MCQ 2024 1M

Find the odd one out in the set:
{19, 37, 21, 17, 23, 29, 31, 11}

A. 21 **B.** 29 **C.** 37 **D.** 23

Q24.2.3 MCQ 2024 1M

In the following series, identify the number that needs to be changed to form the Fibonacci series.
1, 1, 2, 3, 6, 8, 13, 21,...

A. 8 **B.** 21 **C.** 6 **D.** 13

Q24.2.4 MCQ 2024 1M

The real variables x, y, z, and the real constants p, q, r satisfy

$$\frac{x}{pq - r^2} = \frac{y}{qr - p^2} = \frac{z}{rp - q^2}$$

Given that the denominators are non-zero, the value of $px + qy + rz$ is

A. 0 **B.** 1 **C.** pqr **D.** $p^2 + q^2 + r^2$

Q24.2.5 MCQ 2024 1M

Take two long dice (rectangular parallelepiped), each having four rectangular faces labelled as 2, 3, 5, and 7. If thrown, the long dice cannot land on the square faces and has 1/4 probability of landing on any of the four rectangular faces. The label on the top face of the dice is the score of the throw.
If thrown together, what is the probability of getting the sum of the two long dice scores greater than 11?

A. 3/8 **B.** 1/8 **C.** 1/16 **D.** 3/16

Q24.2.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

Prof. P (i) merely a man who narrated funny stories (ii) in his blackest moments he was capable of self-deprecating humor.

Prof. Q (iii) a man who hardly narrated funny stories (iv) in his blackest moments was he able to find humor.

A. (i) was, (ii) Only, (iii) wasn't, (iv) Even

B. (i) wasn't, (ii) Even, (iii) was, (iv) Only

C. (i) was, (ii) Even, (iii) wasn't, (iv) Only

D. (i) wasn't, (ii) Only, (iii) was, (iv) Even

Q24.2.7 MCQ 2024 2M

How many combinations of non-null sets A, B, C are possible from the subsets of {2, 3, 5} satisfying the conditions: (i) A is a subset of B, and (ii) B is a subset of C?

A. 28 **B.** 27 **C.** 18 **D.** 19

Q24.2.8 MCQ 2024 2M

The bar chart gives the batting averages of VK and RS for 11 calendar years from 2012 to 2022. Considering that 2015 and 2019 are world cup years, which one of the following options is true?

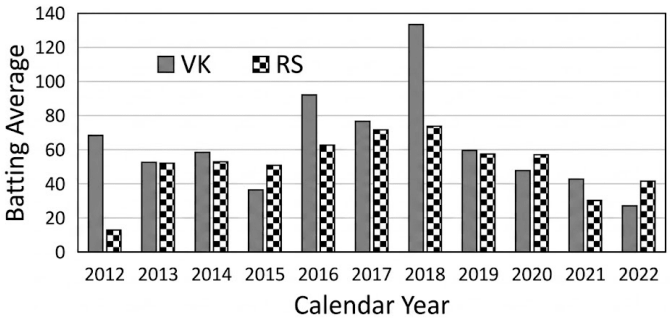


Fig. Q24.2.8

A. RS has a higher yearly batting average than that of VK in every world cup year.

B. VK has a higher yearly batting average than that of RS in every world cup year.

C. VK's yearly batting average is consistently higher than that of RS between the two world cup years.

D. RS's yearly batting average is consistently higher than that of VK in the last three calendar years.

Q24.2.9 MCQ 2024 2M

A planar rectangular paper has two V-shaped pieces attached as shown below.

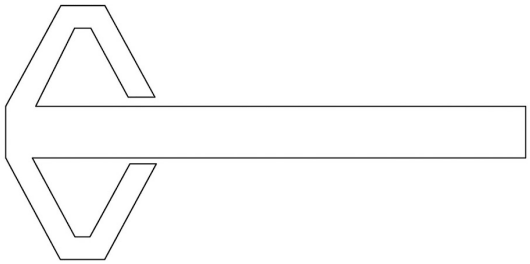


Fig. Q24.2.9

This piece of paper is folded to make the following closed three-dimensional object.

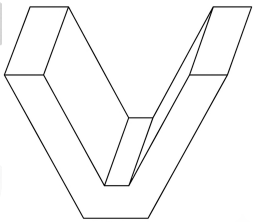


Fig. Q24.2.9

The number of folds required to form the above object is

A. 9 **B.** 7 **C.** 11 **D.** 8

Q24.2.10 MCQ 2024 2M

Four equilateral triangles are used to form a regular closed three-dimensional object by joining along the edges. The angle between any two faces is

A. 30° **B.** 60° **C.** 45° **D.** 90°

Session : 3 | Forenoon | 04/02/24

Q24.3.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[simmer → seethe → smolder] is analogous to [break → _____ → raze].

Which one of the given options is appropriate to fill the blank?

A. obfuscate **B.** obliterate **C.** fracture **D.** fissure

Q24.3.2 MCQ 2024 1M

In a locality, the houses are numbered in the following way: The house-numbers on one side of a road are consecutive odd integers starting from 301, while the house-numbers on the other side of the road are consecutive even numbers starting from 302. The total number of houses is the same on both sides of the road.

If the difference of the sum of the house-numbers between the two sides of the road is 27, then the number of houses on each side of the road is

A. 27 **B.** 52 **C.** 54 **D.** 26

Q24.3.3 MCQ 2024 1M

For positive integers p and q, with $\frac{p}{q} \neq 1$, $\left(\frac{p}{q}\right)^{\frac{p}{q}} = p^{\left(\frac{p}{q}-1\right)}$. Then,

A. $q^p = p^q$ **B.** $q^p = p^{2q}$

C. $\sqrt{q} = \sqrt{p}$ **D.** $\sqrt[q]{q} = \sqrt[q]{p}$

Q24.3.4 MCQ 2024 1M

Which one of the given options is a possible value of x in the following sequence?

3, 7, 15, x, 63, 127, 255

A. 35 **B.** 40 **C.** 45 **D.** 31

Q24.3.5 MCQ 2024 1M

On a given day, how many times will the second-hand and the minute-hand of a clock cross each other during the clock time 12:05:00 hours to 12:55:00 hours?

A. 51 **B.** 49 **C.** 50 **D.** 55

Q24.3.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

From the ancient Athenian arena to the modern Olympic stadiums, athletics (i) the potential for a spectacle. The crowd (ii) with bated breath as the Olympian artist twists his body, stretching the javelin behind him. Twelve strides in, he begins to cross-step. Six cross-steps (iii) in an abrupt stop on his left foot. As his body (iv) like a door turning on a hinge, the javelin is launched skyward at a precise angle.

A. (i) hold, (ii) waits, (iii) culminates, (iv) pivot

B. (i) holds, (ii) wait, (iii) culminates, (iv) pivot

C. (i) hold, (ii) wait, (iii) culminate, (iv) pivots

D. (i) holds, (ii) waits, (iii) culminate, (iv) pivots

Q24.3.7 MCQ 2024 2M

Three distinct sets of indistinguishable twins are to be seated at a circular table that has 8 identical chairs. Unique seating arrangements are defined by the relative positions of the people.

How many unique seating arrangements are possible such that each person is sitting next to their twin?

A. 12 **B.** 14 **C.** 10 **D.** 28

Q24.3.8 MCQ 2024 2M

The chart given below compares the Installed Capacity (MW) of four power generation technologies, T1, T2, T3, and T4, and their Electricity Generation (MWh) in a time of 1000 hours (h).

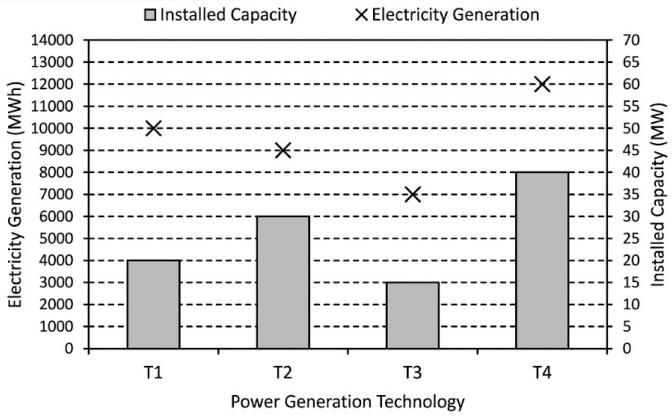


Fig. Q24.3.8

The Capacity Factor of a power generation technology is:

$$\text{Capacity Factor} = \frac{\text{Electricity Generation (MWh)}}{\text{Installed Capacity (MW)} \times 1000\ \text{(h)}}$$

Which one of the given technologies has the highest Capacity Factor?

A. T1 **B.** T2 **C.** T3 **D.** T4

Q24.3.9 MCQ 2024 2M

In the 4×4 array shown below, each cell of the first three columns has either a cross (X) or a number, as per the given rule.

1	1	2	
2	X	3	
2	X	4	
1	2	X	

Fig. Q24.3.9

Rule: The number in a cell represents the count of crosses around its immediate neighboring cells (left, right, top, bottom, diagonals).
As per this rule, the **maximum** number of crosses possible in the empty column is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q24.3.10 MCQ 2024 2M

During a half-moon phase, the Earth-Moon-Sun form a right triangle. If the Moon-Earth-Sun angle at this half-moon phase is measured to be 89.85°, the ratio of the Earth-Sun and Earth-Moon distances is closest to

A. 328 **B.** 382 **C.** 238 **D.** 283

Session : 4 | Afternoon | 04/02/24

Q24.4.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[drizzle → rain → downpour] is analogous to [_____ → quarrel → feud].
Which one of the given options is appropriate to fill the blank?

A. bicker **B.** bog **C.** dither **D.** dodge

Q24.4.2 MCQ 2024 1M

Statements:

1. All heroes are winners.
2. All winners are lucky people.

Inferences:

I. All lucky people are heroes.
II. Some lucky people are heroes.
III. Some winners are heroes.

Which of the above inferences can be logically deduced from statements 1 and 2?

A. Only I and II **B.** Only II and III
C. Only I and III **D.** Only III

Q24.4.3 MCQ 2024 1M

A student was supposed to **multiply** a positive real number p with another positive real number q. Instead, the student **divided** p by q. If the percentage error in the student's answer is 80%, the value of q is

A. 5 **B.** $\sqrt{2}$ **C.** 2 **D.** $\sqrt{5}$

Q24.4.4 MCQ 2024 1M

If the sum of the first 20 consecutive positive odd numbers is divided by 20^2, the result is

A. 1 **B.** 20 **C.** 2 **D.** 1/2

Q24.4.5 MCQ 2024 1M

The ratio of the number of girls to boys in class VIII is the same as the ratio of the number of boys to girls in class IX. The total number of students (boys and girls) in classes VIII and IX is 450 and 360, respectively. If the number of girls in classes VIII and IX is the same, then the number of girls in each class is

A. 150 **B.** 200 **C.** 250 **D.** 175

Q24.4.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.
Yoko Roi stands (i) as an author for standing (ii) as an honorary fellow, after she stood (iii) her writings that stand (iv) the freedom of speech.

A. (i) out, (ii) down, (iii) in, (iv) for

B. (i) down, (ii) out, (iii) by, (iv) in

C. (i) down, (ii) out, (iii) for, (iv) in

D. (i) out, (ii) down, (iii) by, (iv) for

Q24.4.7 MCQ 2024 2M

Seven identical cylindrical chalk-sticks are fitted tightly in a cylindrical container. The figure below shows the arrangement of the chalk-sticks inside the cylinder.

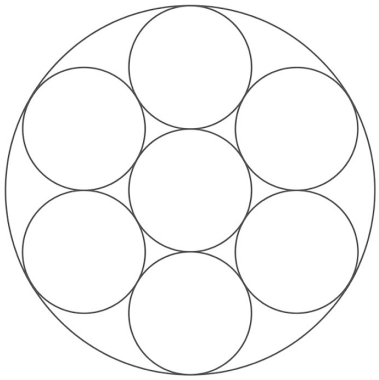


Fig. Q24.4.7

The length of the container is equal to the length of the chalk-sticks. The ratio of the occupied space to the empty space of the container is

A. 5/2 **B.** 7/2 **C.** 9/2 **D.** 3

Q24.4.8 MCQ 2024 2M

The plot below shows the relationship between the mortality risk of cardiovascular disease and the number of steps a person walks per day. Based on the data, which one of the following options is true?

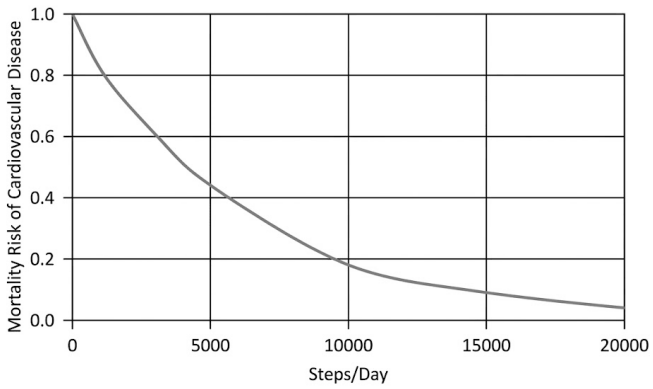


Fig. Q24.4.8

A. The risk reduction on increasing the steps/day from 0 to 10000 is less than the risk reduction on increasing the steps/day from 10000 to 20000.

B. The risk reduction on increasing the steps/day from 0 to 5000 is less than the risk reduction on increasing the steps/day from 15000 to 20000.

C. For any 5000 increment in steps/day the largest risk reduction occurs on going from 0 to 5000.

D. For any 5000 increment in steps/day the largest risk reduction occurs on going from 15000 to 20000.

Q24.4.9 MCQ 2024 2M

Five cubes of identical size and another smaller cube are assembled as shown in Figure A. If viewed from direction X, the planar image of the assembly appears as Figure B.

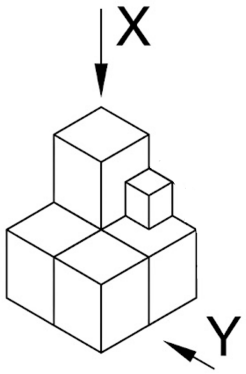


Figure A

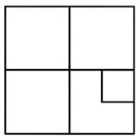
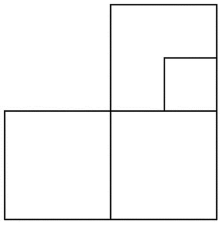


Figure B

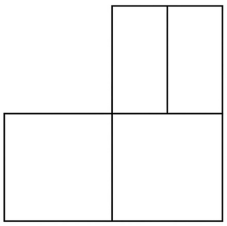
Fig. Q24.4.9

If viewed from direction Y, the planar image of the assembly appears as

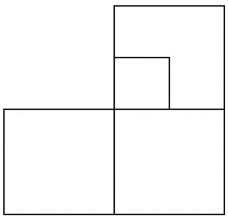
A.



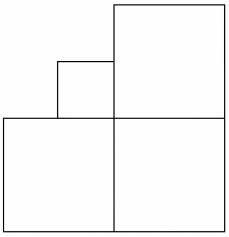
B.



C.



D.



Q24.4.10 MCQ 2024 2M

Visualize a cube that is held with one of the four body diagonals aligned to the vertical axis. Rotate the cube about this axis such that its view remains unchanged. The magnitude of the minimum angle of rotation is

A. 120° **B.** 60° **C.** 90° **D.** 180°

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Q24.5.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words
[dry → arid → parched] is analogous to [diet → fast → _____].
Which one of the given options is appropriate to fill the blank?

A. starve **B.** reject **C.** feast **D.** deny

Q24.5.2 MCQ 2024 1M

If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$

A. depends on xy

B. depends only on x and not on y

C. depends only on y and not on x

D. is a constant

Q24.5.3 MCQ 2024 1M

Consider the following sample of numbers:

$$9, 18, 11, 14, 15, 17, 10, 69, 11, 13$$

The median of the sample is

A. 13.5 **B.** 14 **C.** 11 **D.** 18.7

Q24.5.4 MCQ 2024 1M

The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is _____.

A. 21% **B.** $14\frac{2}{7}\%$ **C.** 10% **D.** 30%

Q24.5.5 MCQ 2024 1M

For positive non-zero real variables p and q, if

$$\log(p^2 + q^2) = \log p + \log q + 2\log 3$$

then, the value of $\frac{p^4 + q^4}{p^2 q^2}$ is _____.

A. 79 **B.** 81 **C.** 9 **D.** 83

Q24.5.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.
Steve was advised to keep his head __(i)__ before heading __(ii)__ to bat;for, while he had a head __(iii)__ batting, he could only do so with a cool head __(iv)__ his shoulders.

A. (i) down, (ii) down, (iii) on, (iv) for

B. (i) on, (ii) down, (iii) for, (iv) on

C. (i) down, (ii) out, (iii) for, (iv) on

D. (i) on, (ii) out, (iii) on, (iv) for

Q24.5.7 MCQ 2024 2M

A rectangular paper sheet of dimensions 54 cm x 4 cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.
Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

A. $1/\pi$ **B.** $2/\pi$ **C.** $3/\pi$ **D.** $4/\pi$

Q24.5.8 MCQ 2024 2M

The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.

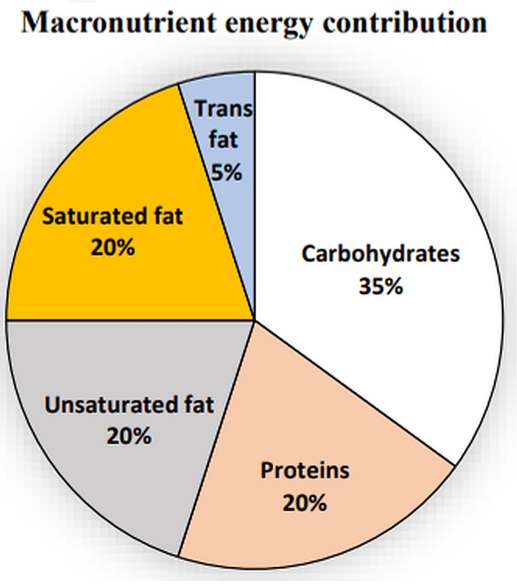


Fig. Q24.5.8

The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

A. 44.4 **B.** 77.8 **C.** 100 **D.** 3,600

Q24.5.9 MCQ 2024 2M

A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

A. 18 **B.** 24 **C.** 20 **D.** 21

Q24.5.10 MCQ 2024 2M

The least number of squares to be added in the figure to make AB a line of symmetry is

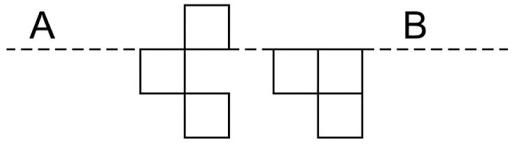


Fig. Q24.5.10

A. 6 **B.** 4 **C.** 5 **D.** 7

Session : 6 | Afternoon | 10/02/24

Q24.6.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted].
Which one of the given options is appropriate to fill the blank?

A. phased **B.** phrased **C.** fazed **D.** fused

Q24.6.2 MCQ 2024 1M

Two wizards try to create a spell using all the four elements, *water, air, fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.
How many attempts does each wizard make before coming to this conclusion, independently?

A. 24 **B.** 48 **C.** 16 **D.** 12

Q24.6.3 MCQ 2024 1M

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $1/4^{th}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.
The number of students who like their core branches is

A. 1,800 **B.** 3,500 **C.** 1,600 **D.** 1,500

Q24.6.4 MCQ 2024 1M

For positive non-zero real variables x and y, if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is .

A. 1 **B.** 1/2 **C.** 2 **D.** 4

Q24.6.5 MCQ 2024 1M

In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

A. 25 **B.** 21 **C.** 18 **D.** 20

Q24.6.6 MCQ 2024 2M

Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.
Q: Antarctic Plate collided with the massive Eurasian Plate.
R: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.
S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

A. QPSR **B.** QSPR **C.** SPRQ **D.** SRPQ

Q24.6.7 MCQ 2024 2M

A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

A. 1% profit **B.** 2% profit **C.** 1% loss **D.** 2% loss

Q24.6.8 MCQ 2024 2M

The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.

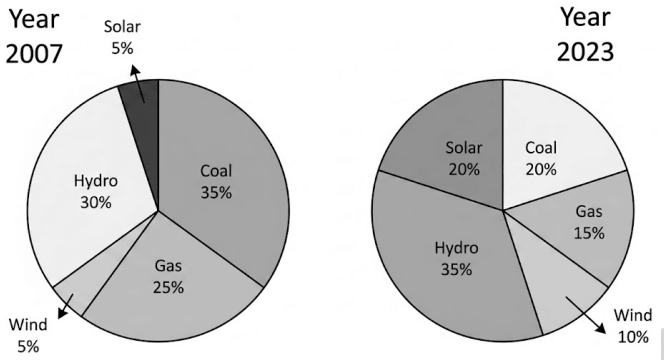


Fig. Q24.6.8

The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

A. 25% **B.** 50% **C.** 77.5% **D.** 62.5%

Q24.6.9 MCQ 2024 2M

A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

A. 3 **B.** 4 **C.** 7 **D.** 8

Q24.6.10 MCQ 2024 2M

In the 4x4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

Fig. Q24.6.10

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X).

Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

A. 11 **B.** 10 **C.** 12 **D.** 9

Session : 7 | Forenoon | 11/02/24

Q24.7.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[charm → enamor → bewitch] is analogous to [bored → _____ → weary].

Which one of the given options is appropriate to fill the blank?

A. jaded **B.** baffled **C.** dead **D.** worsted

Q24.7.2 MCQ 2024 1M

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room.

If S and Q are single children, and the room allocations shown below are acceptable to all:

P R	T S	Q

P Q	R T	S

Fig. Q24.7.2

then, which one of the given options is the siblings?

A. P and T **B.** P and S **C.** T and Q **D.** T and R

Q24.7.3 MCQ 2024 1M

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

A. 28 years **B.** 30 years **C.** 35 years **D.** 32 years

Q24.7.4 MCQ 2024 1M

For a real number $x > 1$,

$$\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$$

The value of x is .

A. 4 **B.** 12 **C.** 24 **D.** 36

Q24.7.5 MCQ 2024 1M

The greatest prime factor of $(3^{199} - 3^{196})$ is

A. 13 **B.** 17 **C.** 3 **D.** 11

Q24.7.6 MCQ 2024 2M

Sequence the following sentences (P, Q, R, S) in a coherent passage:

P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?"
Q: Shifu said, "Surely my running away from the bull is also an illusion."
R: Shifu once proclaimed that all life is illusion.
S: One day, when a bull gave him chase, Shifu began running for his life.

A. SPRQ **B.** SRPQ **C.** RSPQ **D.** RPQS

Q24.7.7 MCQ 2024 2M

Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown.

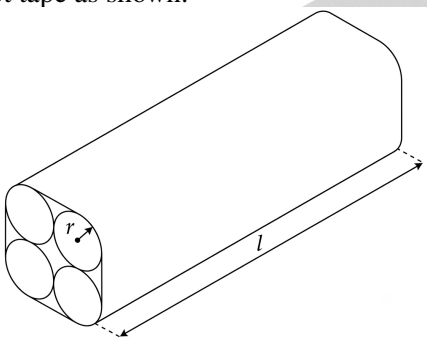


Fig. Q24.7.7

The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once is

A. $20(4+\pi)$ **B.** $20(8+\pi)$ **C.** $10(8+\pi)$ **D.** $10(4+\pi)$

Q24.7.8 MCQ 2024 2M

The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023.

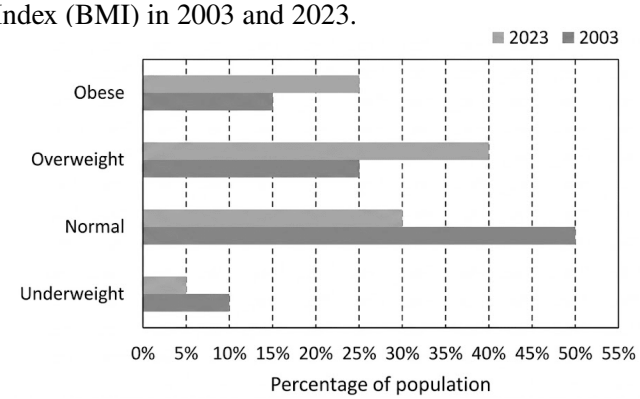


Fig. Q24.7.8

Based on the data provided, which one of the following options is INCORRECT?

A. The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.

B. The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.

C. The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.

D. The percentage of population falling into normal category has decreased in 20 years.

Q24.7.9 MCQ 2024 2M

Examples of mirror and water reflections are shown in the figures below:

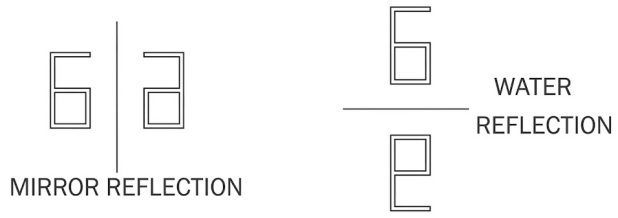


Fig. Q24.7.9

An object appears as the following image after first reflecting in a mirror and then reflecting on water.

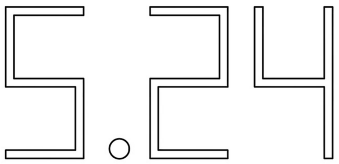
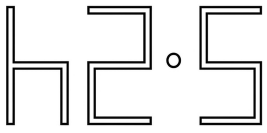
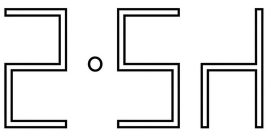
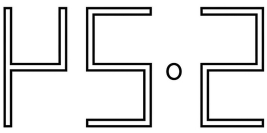


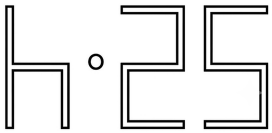
Fig. Q24.7.9

The original object is:

A. 

B. 

C. 

D. 

Q24.7.10 MCQ 2024 2M

Two identical sheets A and B, of dimensions 24 cm×16 cm, can be folded into half using two distinct operations, FO1 or FO2.

In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge.

If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

A. 14:11 **B.** 11:14 **C.** 18:11 **D.** 11:18

Session : 8 | Afternoon | 11/02/24

Q24.8.1 MCQ 2024 1M

If '→' denotes increasing order of intensity, then the meaning of the words

[talk → shout → scream] is analogous to [please → _____ → pander].

Which one of the given options is appropriate to fill the blank?

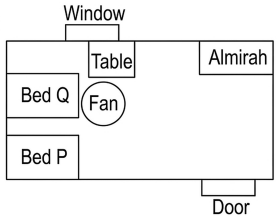
A. flatter **B.** flutter **C.** fritter **D.** frizzle

Q24.8.2 MCQ 2024 1M

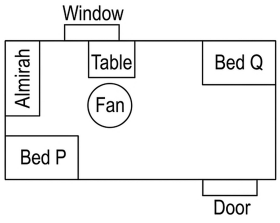
P and Q have been allotted a hostel room with two beds, a study table, and an almirah. P is an avid bird-watcher and wants to sit at the table and watch birds outside the window. Q does not mind that as long as his bed is close to the ceiling fan.

Which one of the following arrangements suits them the most?

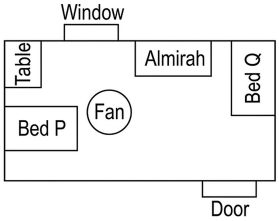
A.



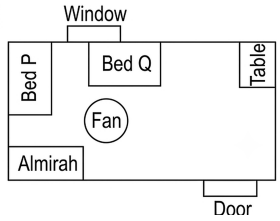
B.



C.



D.



Q24.8.3 MCQ 2024 1M

The decimal number system uses the characters 0, 1, 2, .., 8, 9, and the octal number system uses the characters 0, 1, 2, ..., 6, 7.

For example, the decimal number 12 $(= 1 \times 10^1 + 2 \times 10^0)$ is expressed as 14 $(= 1 \times 8^1 + 4 \times 8^0)$ in the octal number system.

The decimal number 108 in the octal number system is

A. 168 **B.** 108 **C.** 150 **D.** 154

Q24.8.4 MCQ 2024 1M

A shopkeeper buys shirts from a producer and sells them at 20% profit. A customer has to pay ₹3,186.00 including 18% taxes, per shirt. At what price did the shopkeeper buy each shirt from the producer?

A. ₹2,500.00 **B.** ₹1,975.40

C. ₹2,250.00 **D.** ₹2,548.80

Q24.8.5 MCQ 2024 1M

If, for non-zero real variables x, y, and real parameter $a > 1$,

$$x : y = (a + 1) : (a - 1),$$

then the ratio $(x^2 - y^2) : (x^2 + y^2)$ is

A. $2a : (a^2 + 1)$ **B.** $a : (a^2 + 1)$

C. $2a : (a^2 - 1)$ **D.** $a : (a^2 - 1)$

Q24.8.6 MCQ 2024 2M

In the given text, the blanks are numbered (i)-(iv). Select the best match for all the blanks.

Following a row (i) the shopkeeper (ii) the price of a frying pan, the cook stood (iii) a row to withdraw cash (iv) the ATM booth.

A. (i) with, (ii) over, (iii) at, (iv) with

B. (i) at, (ii) over, (iii) over, (iv) at

C. (i) with, (ii) over, (iii) at, (iv) at

D. (i) over, (ii) with, (iii) over, (iv) at

Q24.8.7 MCQ 2024 6M

In the following figure,

$$CD = 5 \text{ cm}, BE = 10 \text{ cm}, AE = 12 \text{ cm},$$
$$\angle DAB = \angle DCB, \text{ and } \angle DAE = \angle DBC = 90°.$$

Points AFCD create a rhombus.

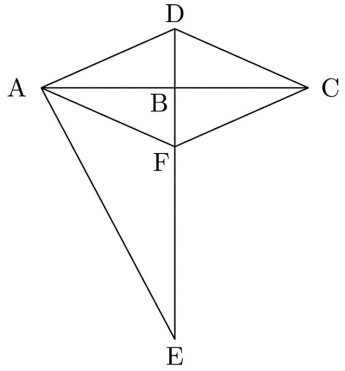


Fig. Q24.8.7

The length of BF (in cm) is

A. 3 **B.** 2 **C.** 4 **D.** 6

Q24.8.8 MCQ 2024 2M

The chart below shows the data of the number of cars bought by Millennials and Gen X people in a country from the year 2010 to 2020 as well as the yearly fuel consumption of the country (in Million liters)

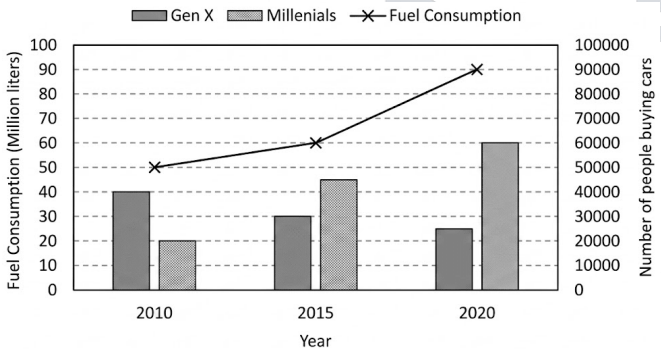


Fig. Q24.8.8

Considering the data presented in the chart, which one of the following options is true?

A. The percentage increase in fuel consumption from 2010 to 2015 is more than the percentage increase in fuel consumption from 2015 to 2020.

B. The increase in the number of Millennial car buyers from 2015 to 2020 is less than the decrease in the number of Gen X car buyers from 2010 to 2015.

C. The increase in the number of Millennial car buyers from 2010 to 2015 is more than the decrease in the number of Gen X car buyers from 2010 to 2015.

D. The decrease in the number of Gen X car buyers from 2015 to 2020 is more than the increase in the number of Millennial car buyers from 2010 to 2015.

Q24.8.9 MCQ 2024 2M

The assembly shown below has three teethed circular objects (Pinions) and two teethed flat objects (Racks), which are perfectly mating with each other. Pinions can only rotate clockwise or anti-clockwise staying at its own center. Racks can translate towards the left (←) or the right (→) direction.

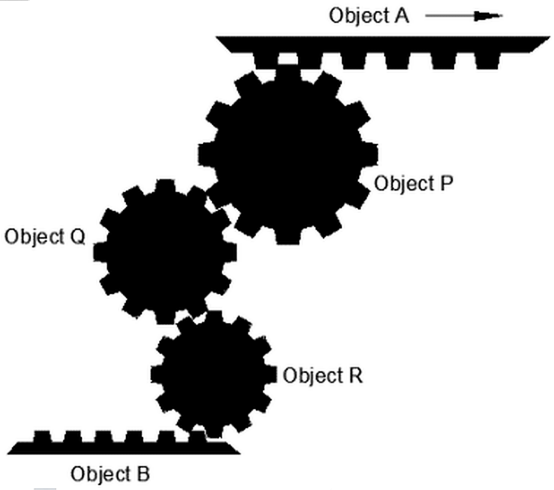


Fig. Q24.8.9

If the object A (Rack) is translating towards the right (→) direction, the correct statement among the following is

A. Object B translates towards the right direction.

B. Object B translates towards the left direction.

C. Object R rotates in the anticlockwise direction.

D. Object Q rotates in the clockwise direction.

Q24.8.10 MCQ 2024 2M

A surveyor has to measure the horizontal distance from her position to a distant reference point C. Using her position as the center, a 200 m horizontal line segment is drawn with the two endpoints A and B. Points A, B, and C are not collinear. Each of the angles $\angle CAB$ and $\angle CBA$ are measured as 87.8°. The distance (in m) of the reference point C from her position is nearest to

A. 2603 **B.** 2606 **C.** 2306 **D.** 2063

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q24.1.1 D	Q24.1.2 A	Q24.1.3 B	Q24.1.4 B	Q24.1.5 B	Q24.1.6 D
Q24.1.7 A	Q24.1.8 B	Q24.1.9 A	Q24.1.10 A		
Session : 2					
Q24.2.1 A	Q24.2.2 A	Q24.2.3 C	Q24.2.4 A	Q24.2.5 D	Q24.2.6 B
Q24.2.7 MTA	Q24.2.8 C	Q24.2.9 A	Q24.2.10 MTA		
Session : 3					
Q24.3.1 B	Q24.3.2 A	Q24.3.3 A	Q24.3.4 D	Q24.3.5 C	Q24.3.6 D
Q24.3.7 A	Q24.3.8 A	Q24.3.9 C	Q24.3.10 B		
Session : 4					
Q24.4.1 A	Q24.4.2 B	Q24.4.3 D	Q24.4.4 A	Q24.4.5 B	Q24.4.6 D
Q24.4.7 B	Q24.4.8 C	Q24.4.9 A	Q24.4.10 A		
Session : 5					
Q24.5.1 A	Q24.5.2 D	Q24.5.3 A	Q24.5.4 C	Q24.5.5 A	Q24.5.6 C
Q24.5.7 A	Q24.5.8 C	Q24.5.9 A	Q24.5.10 A		
Session : 6					
Q24.6.1 C	Q24.6.2 A	Q24.6.3 A	Q24.6.4 C	Q24.6.5 B	Q24.6.6 C
Q24.6.7 C	Q24.6.8 D	Q24.6.9 A	Q24.6.10 A		
Session : 7					
Q24.7.1 A	Q24.7.2 A	Q24.7.3 B	Q24.7.4 C	Q24.7.5 A	Q24.7.6 C
Q24.7.7 D	Q24.7.8 C	Q24.7.9 A	Q24.7.10 A		
Session : 8					
Q24.8.1 A	Q24.8.2 A	Q24.8.3 D	Q24.8.4 C	Q24.8.5 A	Q24.8.6 C
Q24.8.7 MTA	Q24.8.8 C	Q24.8.9 B	Q24.8.10 A		

YEAR

2023

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS	10
Session 2	AR, ME	10
Session 3	EE, ES, XH	10
Session 4	BM, CY, EC	10
Session 5	GG, IN, MA, PE, XE, XL	10
Session 6	AE, AG, BT, CH, EY, GE, MT, NM, PH, PI, TF	10
Session 7	CE1, ST	10
Session 8	CE2, MN	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 04/02/23

Q23.1.1 MCQ 2023 1M

We reached the station late, and _____ missed the train.

A. near **B.** nearly **C.** utterly **D.** mostly

Q23.1.2 MCQ 2023 1M

Kind: _____ :: Often: Frequently
(By word meaning)

A. Mean **B.** Type **C.** Cruel **D.** Kindly

Q23.1.3 MCQ 2023 1M

A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \ldots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$.
If $F_6 = 37$ and $F_7 = 60$, then what is F_1?

A. 4 **B.** 5 **C.** 8 **D.** 9

Q23.1.4 MCQ 2023 1M

A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers.
Given this information, which one of the following statements can be inferred with *certainty*?

A. More than half of the pregnant women received medical care at least once from a doctor.

B. Less than half of the pregnant women received medical care at least once from a doctor.

C. More than half of the pregnant women received medical care at most once from a doctor.

D. Less than half of the pregnant women received medical care at most once from a doctor.

Q23.1.5 MCQ 2023 1M

Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?

A. The surface of the object must be concave everywhere.

B. The surface of the object must be convex everywhere.

C. The surface of the object may be concave in some places and convex in other places.

D. The object can have edges, but no corners.

Q23.1.6 MCQ 2023 2M

The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.
Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. All the working population of Zombieland will henceforth ride bicycles to work.

B. Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.

C. The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.

D. The Government of Zombieland believes that riding bicycles is a form of physical exercise.

Q23.1.7 MCQ 2023 2M

Consider two functions of time (t),

$$f(t) = 0.01t^2$$
$$g(t) = 4t$$

where $0 < t < \infty$.
Now consider the following two statements:

(i) For some $t > 0$, $g(t) > f(t)$.
(ii) There exists a T, such that $f(t) > g(t)$ for all $t > T$.

Which one of the following options is TRUE?

A. only (i) is correct

B. only (ii) is correct

C. both (i) and (ii) are correct

D. neither (i) nor (ii) is correct

Q23.1.8 MCQ 2023 2M

Which one of the following sentence sequences creates a coherent narrative?

(i) Once on the terrace, on her way to her small room in the corner, she notices the man right away.

(ii) She begins to pant by the time she has climbed all the stairs.
(iii) Mina has bought vegetables and rice at the market, so her bags are heavy.
(iv) He was leaning against the parapet, watching the traffic below.

A. (i), (ii), (iv), (iii) **B.** (ii), (iii), (i), (iv)

C. (iv), (ii), (i), (iii) **D.** (iii), (ii), (i), (iv)

Q23.1.9 MCQ | 2023 | 2M

$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x) = g(y)$ for all real values of x and y. Which one of the following options is *necessarily* TRUE for all x and y?

A. $f(x) = 0$ and $g(y) = 0$

B. $f(x) = g(y) =$ constant

C. $f(x) \neq$ constant and $g(y) \neq$ constant

D. $f(x) + g(y) = f(x) - g(y)$

Q23.1.10 MCQ | 2023 | 2M

Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?

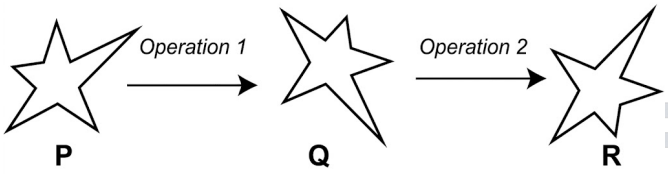


Fig. Q23.1.10

A. *Operation 1:* A clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a horizontal line

B. *Operation 1:* A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a horizontal line

C. *Operation 1:* A clockwise rotation by 90° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a vertical line

D. *Operation 1:* A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure;
Operation 2: A reflection along a vertical line

Session : 2 | Afternoon | 04/02/23

Q23.2.1 MCQ | 2023 | 1M

He did not manage to fix the car himself, so he _____ in the garage.

A. got it fixed **B.** getting it fixed

C. gets fixed **D.** got fixed

Q23.2.2 MCQ | 2023 | 1M

Planting: Seed :: Raising: _____
(By word meaning)

A. Child **B.** Temperature **C.** Height **D.** Lift

Q23.2.3 MCQ | 2023 | 1M

A certain country has 504 universities and 25951 colleges. These are categorised into Grades I, II, and III as shown in the given pie charts.
What is the percentage, correct to one decimal place, of higher education institutions (colleges and universities) that fall into Grade III?

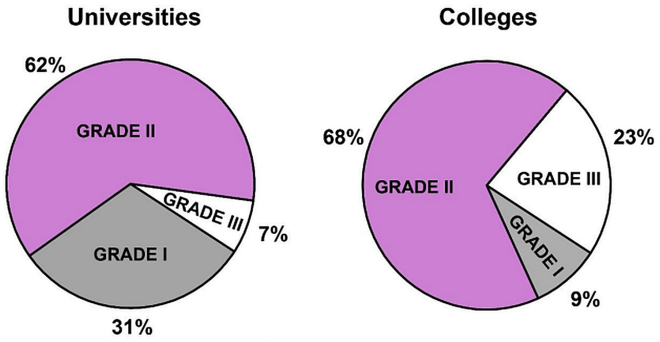


Fig. Q23.2.3

A. 22.7 **B.** 23.7 **C.** 15.0 **D.** 66.8

Q23.2.4 MCQ | 2023 | 1M

The minute-hand and second-hand of a clock cross each other ____ times between 09:15:00 AM and 09:45:00 AM on a day.

A. 30 **B.** 15 **C.** 29 **D.** 31

Q23.2.5 MCQ 2023 1M

The symbols ◯, ∗, Δ and ☐ are to be filled, one in each box, as shown below. The rules for filling in the four symbols are as follows.

1) Every row and every column must contain each of the four symbols.
2) Every 2×2 square delineated by bold lines must contain each of the four symbols.

Which symbol will occupy the box marked with '**?**' in the partially filled figure?

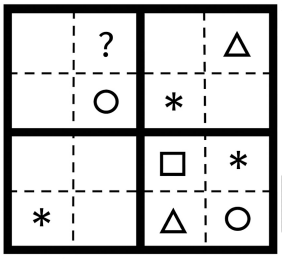


Fig. Q23.2.5

A. ◯ **B.** ∗ **C.** Δ **D.** ☐

Q23.2.6 MCQ 2023 2M

In a recently held parent-teacher meeting, the teachers had very few complaints about Ravi. After all, Ravi was a hardworking and kind student. Incidentally, almost all of Ravi's friends at school were hardworking and kind too. But the teachers drew attention to Ravi's complete lack of interest in sports. The teachers believed that, along with some of his friends who showed similar disinterest in sports, Ravi needed to engage in some sports for his overall development.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. All of Ravi's friends are hardworking and kind.

B. No one who is not a friend of Ravi is hardworking and kind.

C. None of Ravi's friends are interested in sports.

D. Some of Ravi's friends are hardworking and kind.

Q23.2.7 MCQ 2023 2M

Consider the following inequalities:

$$p^2 - 4q < 4$$
$$3p + 2q < 6$$

where p and q are positive integers.

The value of $(p + q)$ is _____.

A. 2 **B.** 1 **C.** 3 **D.** 4

Q23.2.8 MCQ 2023 2M

Which one of the sentence sequences in the given options creates a coherent narrative?

(i) I could not bring myself to knock.
(ii) There was a murmur of unfamiliar voices coming from the big drawing room and the door was firmly shut.
(iii) The passage was dark for a bit, but then it suddenly opened into a bright kitchen.
(iv) I decided I would rather wander down the passage.

A. (iv), (i), (iii), (ii) **B.** (iii), (i), (ii), (iv)

C. (ii), (i), (iv), (iii) **D.** (i), (iii), (ii), (iv)

Q23.2.9 MCQ 2023 2M

How many pairs of sets (S, T) are possible among the subsets of {1, 2, 3, 4, 5, 6} that satisfy the condition that S is a subset of T?

A. 729 **B.** 728 **C.** 665 **D.** 664

Q23.2.10 MCQ 2023 2M

An opaque pyramid shown below, with a square base and isosceles faces, is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The pyramid can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P, Q, R** and **S** is NOT possible?

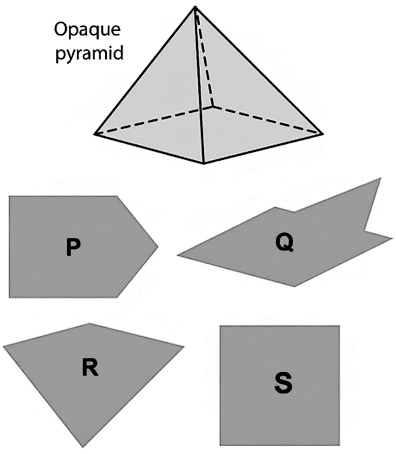


Fig. Q23.2.10

A. P **B.** Q **C.** R **D.** S

Session : 3 | Forenoon | 05/02/23

Q23.3.1 MCQ 2023 1M

Rafi told Mary, "I am thinking of watching a film this weekend."
The following reports the above statement in indirect speech:
Rafi told Mary that he _____ of watching a film that weekend.

A. thought **B.** is thinking
C. am thinking **D.** was thinking

Q23.3.2 MCQ 2023 1M

Permit: _____ :: Enforce: Relax
(By word meaning)

A. Allow **B.** Forbid **C.** License **D.** Reinforce

Q23.3.3 MCQ 2023 1M

Given a fair six-faced dice where the faces are labelled '1', '2', '3', '4', '5', and '6', what is the probability of getting a '1' on the first roll of the dice and a '4' on the second roll?

A. $\frac{1}{36}$ **B.** $\frac{1}{6}$ **C.** $\frac{5}{6}$ **D.** $\frac{1}{3}$

Q23.3.4 MCQ 2023 1M

A recent survey shows that 65% of tobacco users were advised to stop consuming tobacco. The survey also shows that 3 out of 10 tobacco users attempted to stop using tobacco.
Based only on the information in the above passage, which one of the following options can be logically inferred with *certainty*?

A. A majority of tobacco users who were advised to stop consuming tobacco made an attempt to do so.

B. A majority of tobacco users who were advised to stop consuming tobacco did not attempt to do so.

C. Approximately 30% of tobacco users successfully stopped consuming tobacco.

D. Approximately 65% of tobacco users successfully stopped consuming tobacco.

Q23.3.5 MCQ 2023 1M

How many triangles are present in the given figure?

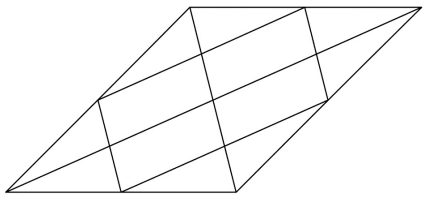


Fig. Q23.3.5

A. 12 **B.** 16 **C.** 20 **D.** 24

Q23.3.6 MCQ 2023 2M

Students of all the departments of a college who have successfully completed the registration process are eligible to vote in the upcoming college elections. However, by the time the due date for registration was over, it was found that surprisingly none of the students from the Department of Human Sciences had completed the registration process.
Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with *certainty*?

(i) All those students who would not be eligible to vote in the college elections would certainly belong to the Department of Human Sciences.
(ii) None of the students from departments other than Human Sciences failed to complete the registration process within the due time.
(iii) All the eligible voters would certainly be students who are not from the Department of Human Sciences.

A. (i) and (ii) **B.** (i) and (iii) **C.** only (i) **D.** only (iii)

Q23.3.7 MCQ 2023 2M

Which one of the following options represents the given graph?

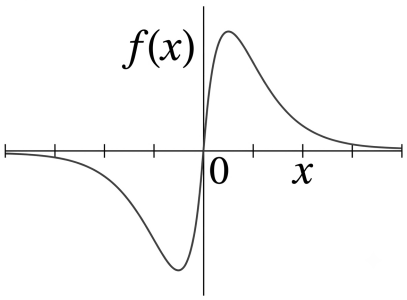


Fig. Q23.3.7

A. $f(x) = x^2 2^{-|x|}$ **B.** $f(x) = x2^{-|x|}$
C. $f(x) = |x|2^{-x}$ **D.** $f(x) = x2^{-x}$

Q23.3.8 MCQ 2023 2M

Which one of the options does NOT describe the passage below or follow from it?

We tend to think of cancer as a 'modern' illness because its metaphors are so modern. It is a disease of overproduction, of sudden growth, a growth that is unstoppable, tipped into the abyss of no control. Modern cell biology encourages us to imagine the cell as a molecular machine. Cancer is that machine unable to quench its initial command (to grow) and thus transform into an indestructible, self-propelled automaton.

[Adapted from *The Emperor of All Maladies* by Siddhartha Mukherjee]

A. It is a reflection of why cancer seems so modern to most of us.

B. It tells us that modern cell biology uses and promotes metaphors of machinery.

C. Modern cell biology encourages metaphors of machinery, and cancer is often imagined as a machine.

D. Modern cell biology never uses figurative language, such as metaphors, to describe or explain anything.

Q23.3.9 MCQ 2023 2M

The digit in the unit's place of the product $3^{999} \times 7^{1000}$ is

A. 7 **B.** 1 **C.** 3 **D.** 9

Q23.3.10 MCQ 2023 2M

A square with sides of length 6 cm is given. The boundary of the shaded region is defined by two semi-circles whose diameters are the sides of the square, as shown.

The area of the shaded region is _____ cm^2.

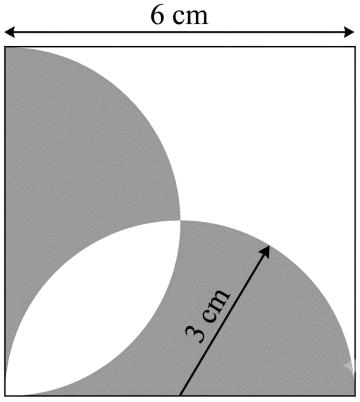


Fig. Q23.3.10

A. 6π **B.** 18 **C.** 20 **D.** 9π

Session : 4 | Afternoon | 05/02/23

Q23.4.1 MCQ 2023 1M

"I cannot support this proposal. My _____ will not permit it."

A. conscious **B.** consensus **C.** conscience **D.** consent

Q23.4.2 MCQ 2023 1M

Courts: _____ :: Parliament: Legislature

(By word meaning)

A. Judiciary **B.** Executive **C.** Governmental **D.** Legal

Q23.4.3 MCQ 2023 1M

What is the smallest number with distinct digits whose digits add up to 45?

A. 123555789 **B.** 123457869

C. 123456789 **D.** 99999

Q23.4.4 MCQ 2023 1M

In a class of 100 students,

(i) there are 30 students who neither like romantic movies nor comedy movies,

(ii) the number of students who like romantic movies is twice the number of students who like comedy movies, and

(iii) the number of students who like both romantic movies and comedy movies is 20.

How many students in the class like romantic movies?

A. 40 **B.** 20 **C.** 60 **D.** 30

Q23.4.5 MCQ 2023 1M

How many rectangles are present in the given figure?

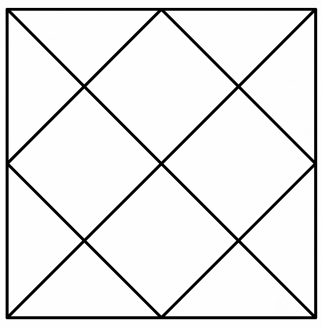


Fig. Q23.4.5

A. 8 **B.** 9 **C.** 10 **D.** 12

Q23.4.6 MCQ 2023 2M

Forestland is a planet inhabited by different kinds of creatures. Among other creatures, it is populated by animals all of whom are ferocious. There are also creatures that have claws, and some that do not. All creatures that have claws are ferocious.

Based only on the information provided above, which one of the following options can be logically inferred with *certainty*?

A. All creatures with claws are animals.

B. Some creatures with claws are non-ferocious.

C. Some non-ferocious creatures have claws.

D. Some ferocious creatures are creatures with claws.

Q23.4.7 MCQ 2023 2M

Which one of the following options represents the given graph?

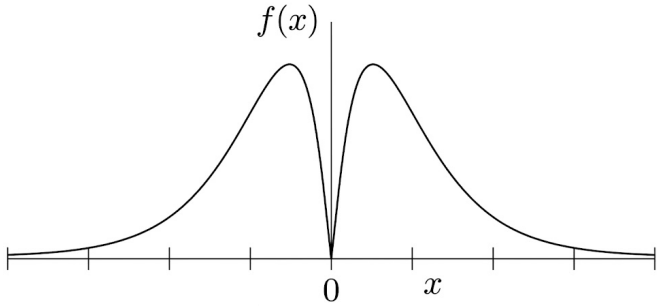


Fig. Q23.4.7

A. $f(x) = x^2 2^{-|x|}$ **B.** $f(x) = x2^{-|x|}$

C. $f(x) = |x|2^{-x}$ **D.** $f(x) = x2^{-x}$

Q23.4.8 MCQ 2023 2M

Which one of the following options can be inferred from the given passage alone?

When I was a kid, I was partial to stories about other worlds and interplanetary travel. I used to imagine that I could just gaze off into space and be whisked to another planet.

[Excerpt from The Truth about Stories by T. King]

A. It is a child's description of what he or she likes.

B. It is an adult's memory of what he or she liked as a child.

C. The child in the passage read stories about interplanetary travel only in parts.

D. It teaches us that stories are good for children.

Q23.4.9 MCQ 2023 2M

Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to:

(i) Collect saliva samples from all 1000 individuals and randomly group them into sets of 5.
(ii) Mix the samples within each set and test the mixed sample for covid.
(iii) If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative.
(iv) If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid.

Given this strategy, no more than _____ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.

A. 700 **B.** 600 **C.** 800 **D.** 1000

Q23.4.10 MCQ 2023 2M

A 100 cm × 32 cm rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions 100 cm × 1 cm.

The total number of creases visible when the sheet is unfolded is _____.

A. 32 **B.** 5 **C.** 31 **D.** 63

Session : 5 | Forenoon | 11/02/23

Q23.5.1 MCQ 2023 1M

The village was nestled in a green spot, _____ the ocean and the hills.

A. through **B.** in **C.** at **D.** between

Q23.5.2 MCQ 2023 1M

Disagree: Protest :: Agree: _____
(By word meaning)

A. Refuse **B.** Pretext **C.** Recommend **D.** Refute

Q23.5.3 MCQ 2023 1M

A 'frabjous' number is defined as a 3 digit number with all digits odd, and no two adjacent digits being the same. For example, 137 is a frabjous number, while 133 is not. How many such frabjous numbers exist?

A. 125 **B.** 720 **C.** 60 **D.** 80

Q23.5.4 MCQ 2023 1M

Which one among the following statements must be TRUE about the mean and the median of the scores of all candidates appearing for GATE 2023?

A. The median is at least as large as the mean.

B. The mean is at least as large as the median.

C. At most half the candidates have a score that is larger than the median.

D. At most half the candidates have a score that is larger than the mean.

Q23.5.5 MCQ 2023 1M

In the given diagram, ovals are marked at different heights (h) of a hill. Which one of the following options **P**, **Q**, **R**, and **S** depicts the top view of the hill?

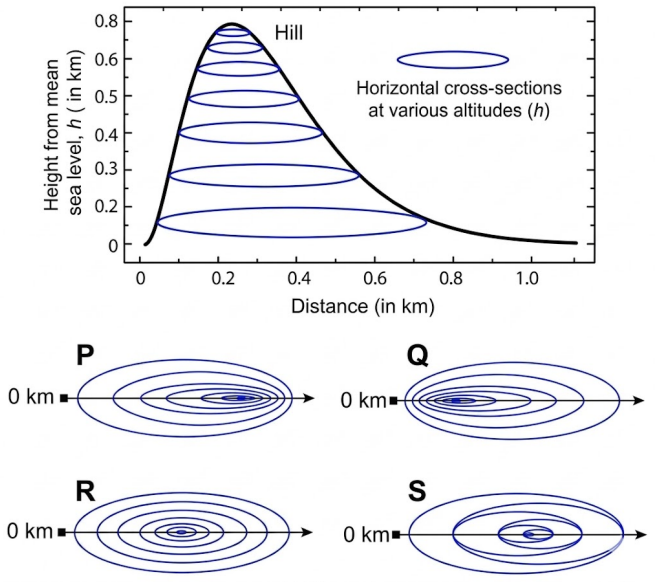


Fig. Q23.5.5

A. P **B.** Q **C.** R **D.** S

Q23.5.6 MCQ 2023 2M

Residency is a famous housing complex with many well-established individuals among its residents. A recent survey conducted among the residents of the complex revealed that all of those residents who are well established in their respective fields happen to be academicians. The survey also revealed that most of these academicians are authors of some best-selling books.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

A. Some residents of the complex who are well established in their fields are also authors of some best-selling books.

B. All academicians residing in the complex are well established in their fields.

C. Some authors of best-selling books are residents of the complex who are well established in their fields.

D. Some academicians residing in the complex are well established in their fields.

Q23.5.7 MCQ 2023 2M

Ankita has to climb 5 stairs starting at the ground, while respecting the following rules:

1. At any stage, Ankita can move either one or two stairs up.
2. At any stage, Ankita cannot move to a lower step.

Let $F(N)$ denote the number of possible ways in which Ankita can reach the N^{th} stair.

For example, $F(1) = 1$, $F(2) = 2$, $F(3) = 3$.

The value of $F(5)$ is _____.

A. 8 **B.** 7 **C.** 6 **D.** 5

Q23.5.8 MCQ 2023 2M

The information contained in DNA is used to synthesize proteins that are necessary for the functioning of life. DNA is composed of four nucleotides: Adenine (A), Thymine (T), Cytosine (C), and Guanine (G). The information contained in DNA can then be thought of as a sequence of these four nucleotides: A, T, C, and G. DNA has coding and non-coding regions. Coding regions—where the sequence of these nucleotides are read in groups of three to produce individual amino acids—constitute only about 2% of human DNA. For example, the triplet of nucleotides CCG codes for the amino acid glycine, while the triplet GGA codes for the amino acid proline. Multiple amino acids are then assembled to form a protein.

Based only on the information provided above, which of the following statements can be logically inferred with certainty?

(i) The majority of human DNA has no role in the synthesis of proteins.

(ii) The function of about 98% of human DNA is not understood.

A. only (i) **B.** only (ii)

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q23.5.9 MCQ | 2023 | 2M

Which one of the given figures P, Q, R and S represents the graph of the following function?

$$f(x) = ||x + 2| - |x - 1||$$

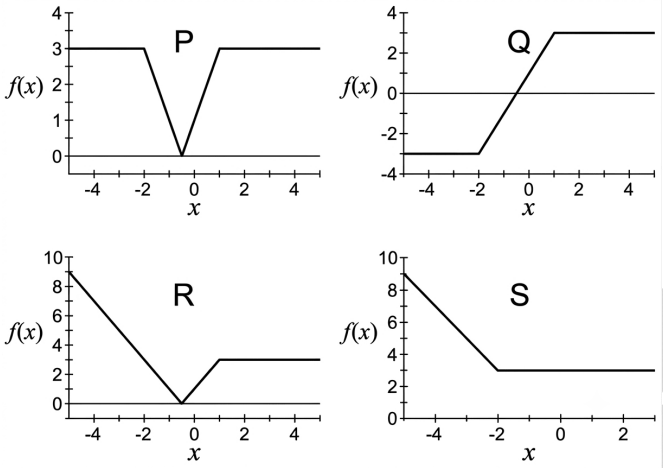


Fig. Q23.5.9

A. P **B.** Q **C.** R **D.** S

Q23.5.10 MCQ | 2023 | 2M

An opaque cylinder is suspended in the path of a parallel beam of light, such that its shadow is cast on a screen oriented perpendicular to the direction of the light beam. The cylinder can be reoriented in any direction within the light beam. Under these conditions, which one of the shadows **P, Q, R**, and **S** is NOT possible?

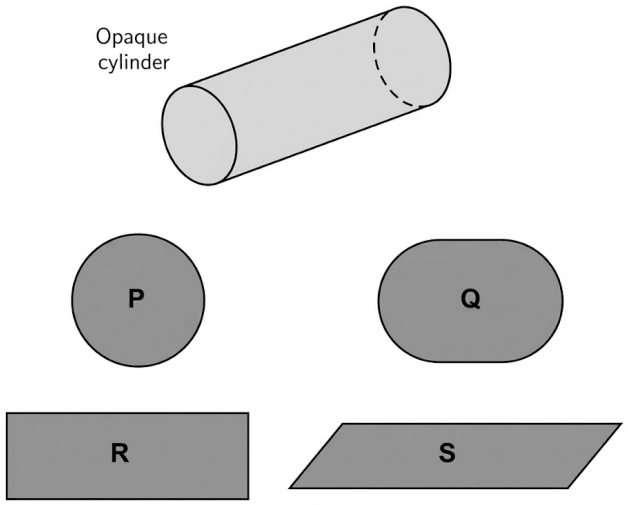


Fig. Q23.5.10

A. P **B.** Q **C.** R **D.** S

Session : 6 | Afternoon | 11/02/23

Q23.6.1 MCQ | 2023 | 1M

"You are delaying the completion of the task. Send _____ contributions at the earliest."

A. you are **B.** your **C.** you're **D.** yore

Q23.6.2 MCQ | 2023 | 1M

References:_____ :: Guidelines: Implement
(By word meaning)

A. Sight **B.** Site **C.** Cite **D.** Plagiarise

Q23.6.3 MCQ | 2023 | 1M

In the given figure, PQRS is a parallelogram with PS=7 cm, PT=4 cm and PV=5 cm. What is the length of RS in cm? (The diagram is representative.)

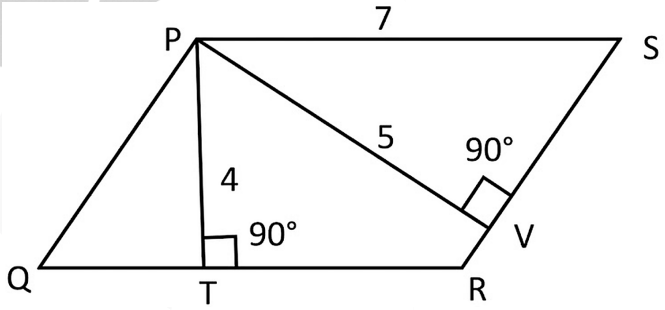


Fig. Q23.6.3

A. $\frac{20}{7}$ **B.** $\frac{28}{5}$ **C.** $\frac{9}{2}$ **D.** $\frac{35}{4}$

Q23.6.4 MCQ | 2023 | 1M

In 2022, June Huh was awarded the Fields medal, which is the highest prize in Mathematics.
When he was younger, he was also a poet. He did not win any medals in the International Mathematics Olympiads. He dropped out of college.
Based only on the above information, which one of the following statements can be logically inferred with *certainty*?

A. Every Fields medalist has won a medal in an International Mathematics Olympiad.

B. Everyone who has dropped out of college has won the Fields medal.

C. All Fields medalists are part-time poets.

D. Some Fields medalists have dropped out of college.

Q23.6.5 MCQ 2023 1M

A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.

The given figure consists of 16 unit squares arranged as shown. In addition to the three black squares, what is the minimum number of squares that must be coloured black, such that both PQ and MN form lines of symmetry? (The figure is representative)

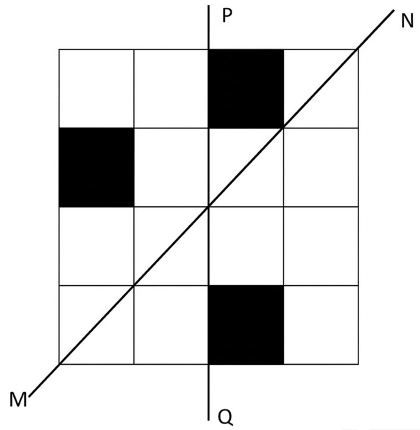


Fig. Q23.6.5

A. 3 **B.** 4 **C.** 5 **D.** 6

Q23.6.6 MCQ 2023 2M

Human beings are one among many creatures that inhabit an imagined world. In this imagined world, some creatures are cruel. If in this imagined world, it is given that the statement "Some human beings are not cruel creatures" is FALSE, then which of the following set of statement(s) can be logically inferred with *certainty*?

(i) All human beings are cruel creatures.
(ii) Some human beings are cruel creatures.
(iii) Some creatures that are cruel are human beings.
(iv) No human beings are cruel creatures.

A. only (i) **B.** only (iii) and (iv)

C. only (i) and (ii) **D.** (i), (ii) and (iii)

Q23.6.7 MCQ 2023 2M

To construct a wall, sand and cement are mixed in the ratio of 3:1. The cost of sand and that of cement are in the ratio of 1:2.

If the total cost of sand and cement to construct the wall is 1000 rupees, then what is the cost (in rupees) of cement used?

A. 400 **B.** 600 **C.** 800 **D.** 200

Q23.6.8 MCQ 2023 2M

The World Bank has declared that it does not plan to offer new financing to Sri Lanka, which is battling its worst economic crisis in decades, until the country has an adequate macroeconomic policy framework in place. In a statement, the World Bank said Sri Lanka needed to adopt structural reforms that focus on economic stabilisation and tackle the root causes of its crisis. The latter has starved it of foreign exchange and led to shortages of food, fuel, and medicines. The bank is repurposing resources under existing loans to help alleviate shortages of essential items such as medicine, cooking gas, fertiliser, meals for children, and cash for vulnerable households.

Based only on the above passage, which one of the following statements can be inferred with *certainty*?

A. According to the World Bank, the root cause of Sri Lanka's economic crisis is that it does not have enough foreign exchange.

B. The World Bank has stated that it will advise the Sri Lankan government about how to tackle the root causes of its economic crisis.

C. According to the World Bank, Sri Lanka does not yet have an adequate macroeconomic policy framework.

D. The World Bank has stated that it will provide Sri Lanka with additional funds for essentials such as food, fuel, and medicines.

Q23.6.9 MCQ 2023 2M

The coefficient of x^4 in the polynomial $(x-1)^3(x-2)^3$ is equal to _____.

A. 33 **B.** -3 **C.** 30 **D.** 21

Q23.6.10 MCQ 2023 2M

Which one of the following shapes can be used to tile (completely cover by repeating) a flat plane, extending to infinity in all directions, without leaving any empty spaces in between them? The copies of the shape used to tile are identical and are not allowed to overlap.

A. circle **B.** regular octagon

C. regular pentagon **D.** rhombus

Session : 7 | Forenoon | 12/02/23

Q23.7.1 MCQ | 2023 | 1M

"I have not yet decided what I will do this evening; I _____ visit a friend."

A. mite **B.** would **C.** might **D.** didn't

Q23.7.2 MCQ | 2023 | 1M

Eject: Insert :: Advance: _____
(By word meaning)

A. Advent **B.** Progress **C.** Retreat **D.** Loan

Q23.7.3 MCQ | 2023 | 1M

In the given figure, PQRSTV is a regular hexagon with each side of length 5 cm. A circle is drawn with its centre at V such that it passes through P. What is the area (in cm^2) of the shaded region?

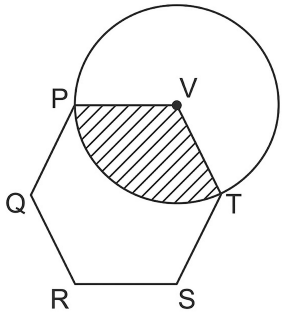


Fig. Q23.7.3

A. $\frac{25\pi}{3}$ **B.** $\frac{20\pi}{3}$ **C.** 6π **D.** 7π

Q23.7.4 MCQ | 2023 | 1M

A duck named Donald Duck says "All ducks always lie." Based only on the information above, which one of the following statements can be logically inferred with *certainty*?

A. Donald Duck always lies.

B. Donald Duck always tells the truth.

C. Donald Duck's statement is true.

D. Donald Duck's statement is false.

Q23.7.5 MCQ | 2023 | 1M

A line of symmetry is defined as a line that divides a figure into two parts in a way such that each part is a mirror image of the other part about that line.
The figure below consists of 20 unit squares arranged as shown. In addition to the given black squares, upto 5 more may be coloured black. Which one among the following options depicts the minimum number of boxes that must be coloured black to achieve two lines of symmetry?

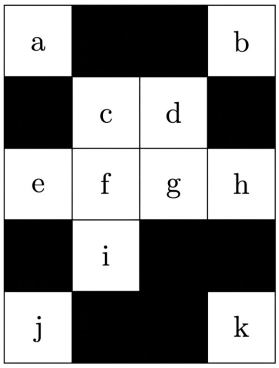


Fig. Q23.7.5

A. d **B.** c, d, i **C.** c, i **D.** c, d, i, f, g

Q23.7.6 MCQ | 2023 | 2M

Based only on the truth of the statement 'Some humans are intelligent', which one of the following options can be logically inferred with *certainty*?

A. No human is intelligent.

B. All humans are intelligent.

C. Some non-humans are intelligent.

D. Some intelligent beings are humans.

Q23.7.7 MCQ | 2023 | 2M

Which one of the options can be inferred about the mean, median, and mode for the given probability distribution (i.e. probability mass function), $P(x)$ of a variable x?

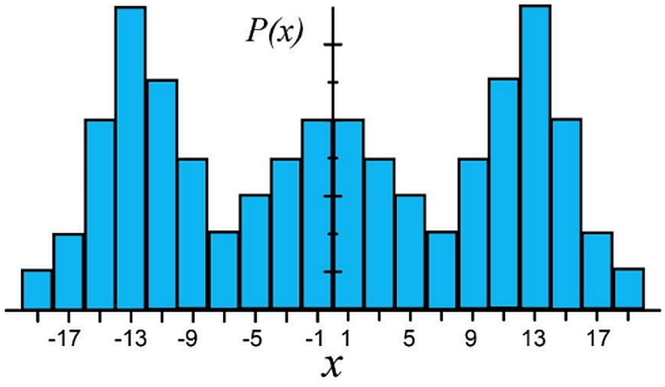


Fig. Q23.7.7

A. mean = median ≠ mode **B.** mean = median = mode

C. mean ≠ median = mode **D.** mean ≠ mode = median

Q23.7.8 MCQ | 2023 | 2M

The James Webb telescope, recently launched in space, is giving humankind unprecedented access to the depths of time by imaging very old stars formed almost 13 billion years ago. Astrophysicists and cosmologists believe that this odyssey in space may even shed light on the existence of dark matter. Dark matter is supposed to interact only via the gravitational interaction and not through the electromagnetic-, the weak- or the strong-interaction. This may justify the epithet "dark" in dark matter.
Based on the above paragraph, which one of the following statements is FALSE?

A. No other telescope has captured images of stars older than those captured by the James Webb telescope.

B. People other than astrophysicists and cosmologists may also believe in the existence of dark matter.

C. The James Webb telescope could be of use in the research on dark matter.

D. If dark matter was known to interact via the strong-interaction, then the epithet "dark" would be justified.

Q23.7.9 MCQ | 2023 | 2M

Let $a = 30!$, $b = 50!$ and $c = 100!$. Consider the following numbers:

$$log_a c, \quad log_c a, \quad log_b a, \quad log_a b.$$

Which one of the following inequalities is CORRECT?

A. $log_c a < log_b a < log_a b < log_a c$

B. $log_c a < log_a b < log_b a < log_b c$

C. $log_c a < log_b a < log_a c < log_a b$

D. $log_b a < log_c a < log_a b < log_a c$

Q23.7.10 MCQ | 2023 | 2M

A square of side length 4 cm is given. The boundary of the shaded region is defined by one semi-circle on the top and two circular arcs at the bottom, each of radius 2 cm, as shown.

The area of the shaded region is _____ cm^2.

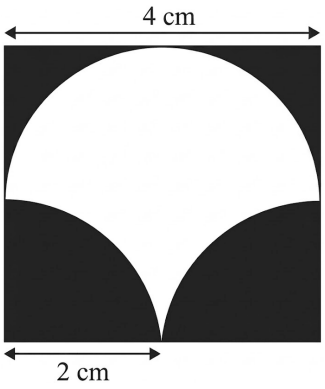


Fig. Q23.7.10

A. 8 **B.** 4 **C.** 12 **D.** 10

Session : 8 | Afternoon | 12/02/23

Q23.8.1 MCQ | 2023 | 1M

The line ran _____ the page, right through the centre, and divided the page into two.

A. across **B.** of **C.** between **D.** about

Q23.8.2 MCQ | 2023 | 1M

Kind: _____ :: Often: Seldom
(By word meaning)

A. Cruel **B.** Variety **C.** Type **D.** Kindred

Q23.8.3 MCQ | 2023 | 1M

In how many ways can cells in a 3×3 grid be shaded, such that each row and each column have exactly one shaded cell? An example of one valid shading is shown.

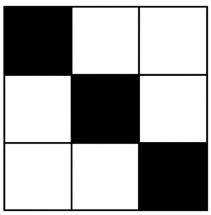


Fig. Q23.8.3

A. 2 **B.** 9 **C.** 3 **D.** 6

Q23.8.4 MCQ | 2023 | 1M

There are 4 red, 5 green, and 6 blue balls inside a box. If N number of balls are picked simultaneously, what is the smallest value of N that guarantees there will be at least two balls of the same colour?
One cannot see the colour of the balls until they are picked.

A. 4 **B.** 15 **C.** 5 **D.** 2

Q23.8.5 MCQ 2023 1M

Consider a circle with its centre at the origin (O), as shown. Two operations are allowed on the circle.

Operation 1: Scale independently along the x and y axes.
Operation 2: Rotation in any direction about the origin.

Which figure among the options can be achieved through a combination of these two operations on the given circle?

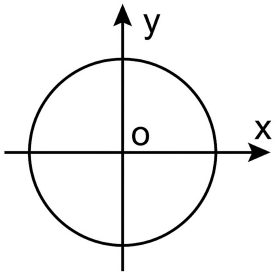
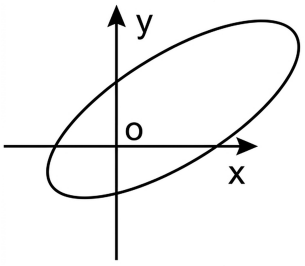
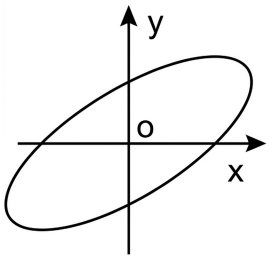
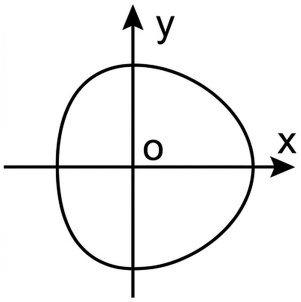
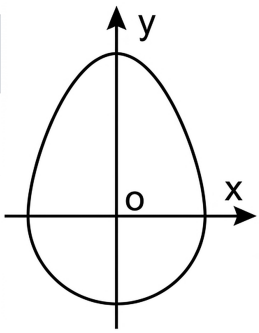


Fig. Q23.8.5

A.


B.


C.


D.


Q23.8.6 MCQ 2023 2M

Elvesland is a country that has peculiar beliefs and practices. They express almost all their emotions by gifting flowers. For instance, if anyone gifts a white flower to someone, then it is always taken to be a declaration of one's love for that person.In a similar manner, the gifting of a yellow flower to someone often means that one is angry with that person.
Based only on the information provided above, which one of the following sets of statement(s) can be logically inferred with certainty?

(i) In Elvesland, one always declares one's love by gifting a white flower.
(ii) In Elvesland, all emotions are declared by gifting flowers.
(iii) In Elvesland, sometimes one expresses one's anger by gifting a flower that is not yellow.
(iv) In Elvesland, sometimes one expresses one's love by gifting a white flower.

A. only (ii)
B. (i), (ii) and (iii)
C. (i), (iii) and (iv)
D. only (iv)

Q23.8.7 MCQ 2023 2M

Three husband-wife pairs are to be seated at a circular table that has six identical chairs. Seating arrangements are defined only by the relative position of the people. How many seating arrangements are possible such that every husband sits next to his wife?

A. 16 **B.** 4 **C.** 120 **D.** 720

Q23.8.8 MCQ 2023 2M

Based only on the following passage, which one of the options can be inferred with *certainty*? When the congregation sang together, Apenyo would also join, though her little screams were not quite audible because of the group singing. But whenever there was a special number, trouble would begin; Apenyo would try singing along, much to the embarrassment of her mother. After two or three such mortifying Sunday evenings, the mother stopped going to church altogether until Apenyo became older and learnt to behave.
At home too, Apenyo never kept quiet; she hummed or made up silly songs to sing by herself, which annoyed her mother at times but most often made her become pensive. She was by now convinced that her daughter had inherited her love of singing from her father who had died unexpectedly away from home.

[Excerpt from *These Hills Called Home* by Temsula Ao]

A. The mother was embarrassed about her daughter's singing at home.

B. The mother's feelings about her daughter's singing at home were only of annoyance.

C. The mother was not sure if Apenyo had inherited her love of singing from her father.

D. When Apenyo hummed at home, her mother tended to become thoughtful.

Q23.8.9 MCQ 2023 2M

If x satisfies the equation $4^{8^x} = 256$, then x is equal to

A. 1/2 **B.** $log_{16}8$ **C.** 2/3 **D.** $log_4 8$

Q23.8.10 MCQ 2023 2M

Consider a spherical globe rotating about an axis passing through its poles. There are three points *P, Q,* and *R* situated respectively on the equator, the north pole, and midway between the equator and the north pole in the northern hemisphere. Let *P, Q,* and *R* move with speeds v_P, v_Q, and v_R, respectively.
Which one of the following options is CORRECT?

A. $v_P < v_R < v_Q$ **B.** $v_P < v_Q < v_R$

C. $v_P > v_R > v_Q$ **D.** $v_P = v_R \neq v_Q$

PrepFusion

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q23.1.1 B	Q23.1.2 MTA	Q23.1.3 A	Q23.1.4 A	Q23.1.5 C	Q23.1.6 D
Q23.1.7 C	Q23.1.8 D	Q23.1.9 B	Q23.1.10 A		
Session : 2					
Q23.2.1 A	Q23.2.2 A	Q23.2.3 A	Q23.2.4 A	Q23.2.5 B	Q23.2.6 D
Q23.2.7 A	Q23.2.8 C	Q23.2.9 A	Q23.2.10 B		
Session : 3					
Q23.3.1 D	Q23.3.2 B	Q23.3.3 A	Q23.3.4 B	Q23.3.5 C	Q23.3.6 D
Q23.3.7 B	Q23.3.8 D	Q23.3.9 A	Q23.3.10 B		
Session : 4					
Q23.4.1 C	Q23.4.2 A	Q23.4.3 C	Q23.4.4 C	Q23.4.5 C	Q23.4.6 D
Q23.4.7 A	Q23.4.8 B	Q23.4.9 A	Q23.4.10 C		
Session : 5					
Q23.5.1 D	Q23.5.2 C	Q23.5.3 D	Q23.5.4 C	Q23.5.5 B	Q23.5.6 MTA
Q23.5.7 A	Q23.5.8 D	Q23.5.9 A	Q23.5.10 D		
Session : 6					
Q23.6.1 B	Q23.6.2 C	Q23.6.3 B	Q23.6.4 D	Q23.6.5 C	Q23.6.6 D
Q23.6.7 A	Q23.6.8 C	Q23.6.9 A	Q23.6.10 D		
Session : 7					
Q23.7.1 C	Q23.7.2 C	Q23.7.3 A	Q23.7.4 D	Q23.7.5 B	Q23.7.6 D
Q23.7.7 A	Q23.7.8 D	Q23.7.9 A	Q23.7.10 A		
Session : 8					
Q23.8.1 A	Q23.8.2 A	Q23.8.3 D	Q23.8.4 A	Q23.8.5 B	Q23.8.6 D
Q23.8.7 A	Q23.8.8 D	Q23.8.9 C	Q23.8.10 C		

YEAR

2022

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CS, BM	10
Session 2	EE, MA T	10
Session 3	EC, ES, ST, NM, MT, MN	10
Session 4	CY, CH, PI, XH, IN, AG, GG, TF	10
Session 5	CE1, BT, PH, EY	10
Session 6	CE2, XE, XL	10
Session 7	ME1, PE, AR	10
Session 8	ME2, GE, AE	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 05/02/22

Q22.1.1 MCQ 2022 1M

The _____ is too high for it to be considered _____.

A. fair / fare **B.** faer / fair

C. fare / fare **D.** fare / fair

Q22.1.2 MCQ 2022 1M

A function $y(x)$ is defined in the interval $[0, 1]$ on the x-axis as

$$y(x) = \begin{cases} 2 & \text{if } 0 \le x < \frac{1}{3} \\ 3 & \text{if } \frac{1}{3} \le x < \frac{3}{4} \\ 1 & \text{if } \frac{3}{4} \le x \le 1 \end{cases}$$

Which one of the following is the area under the curve for the interval $[0, 1]$ on the x-axis?

A. $\frac{5}{6}$ **B.** $\frac{6}{5}$ **C.** $\frac{13}{6}$ **D.** $\frac{6}{13}$

Q22.1.3 MCQ 2022 1M

Let r be a root of the equation $x^2 + 2x + 6 = 0$.
Then the value of the expression $(r+2)(r+3)(r+4)(r+5)$ is

A. 51 **B.** -51 **C.** 126 **D.** -126

Q22.1.4 MCQ 2022 1M

Given below are four statements.

Statement 1: All students are inquisitive.
Statement 2: Some students are inquisitive.
Statement 3: No student is inquisitive.
Statement 4: Some students are not inquisitive.

From the given four statements, find the two statements that **CANNOT BE TRUE** simultaneously, assuming that there is at least one student in the class.

A. Statement 1 and Statement 3

B. Statement 1 and Statement 2

C. Statement 2 and Statement 4

D. Statement 3 and Statement 4

Q22.1.5 MCQ 2022 1M

A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters:

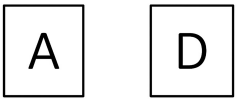
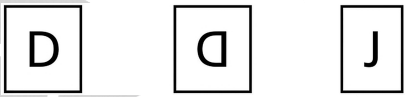


Fig. Q22.1.5

From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome? The player should use all the five plates exactly once. The plates can be rotated in their plane.

A. 

B.

C.

D.

Q22.1.6 MCQ 2022 2M

Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture.
Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Metrics are useful in organizations with poor work culture

B. Metrics are useful in organizations with good work culture

C. Metrics are always counterproductive in organizations with good work culture

D. Metrics are never useful in organizations with good work culture

Q22.1.7 MCQ 2022 2M

In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.
Which one of the following is the correct logical inference based on the information provided in the above passage?

A. Equal number of boys and girls qualified

B. Equal number of boys and girls appeared for the test

C. The number of boys who appeared for the test is less than the number of girls who appeared

D. The number of boys who qualified the test is less than the number of girls who qualified

Q22.1.8 MCQ 2022 2M

A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.
First ball is drawn. What is the probability of getting an orange ball in the next draw?

A. $\frac{1}{2}$ **B.** $\frac{8}{25}$ **C.** $\frac{19}{50}$ **D.** $\frac{23}{50}$

Q22.1.9 MCQ 2022 2M

The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to the corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

A. P cannot be placed at a corner

B. S cannot be placed at a corner

C. U cannot be placed at a mid-point

D. R cannot be placed at a corner

Q22.1.10 MCQ 2022 2M

A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved.
What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.

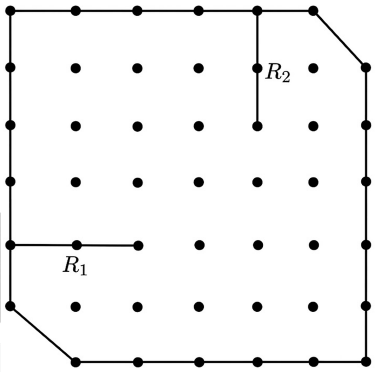


Fig. Q22.1.10

A. 2 **B.** 4 **C.** 5 **D.** 3

Session : 2 | Afternoon | 05/02/22

Q22.2.1 MCQ 2022 1M

As you grow older, an injury to your _____ may take longer to _____.

A. heel / heel **B.** heal / heel

C. heal / heal **D.** heel / heal

Q22.2.2 MCQ 2022 1M

In a 500 m race, P and Q have speeds in the ratio of 3 : 4. Q starts the race when P has already covered 140 m.
What is the distance between P and Q (in m) when P wins the race?

A. 20 **B.** 40 **C.** 60 **D.** 140

Q22.2.3 MCQ 2022 1M

Three bells P, Q, and R are rung periodically in a school. P is rung every 20 minutes; Q is rung every 30 minutes and R is rung every 50 minutes.
If all the three bells are rung at 12:00 PM, when will the three bells ring together again the next time?

A. 5:00 PM **B.** 5:30 PM **C.** 6:00 PM **D.** 6:30 PM

Q22.2.4 MCQ 2022 1M

Given below are two statements and four conclusions drawn based on the statements.

Statement 1: Some bottles are cups.
Statement 2: All cups are knives.

Conclusion I: Some bottles are knives.
Conclusion II: Some knives are cups.
Conclusion III: All cups are bottles.
Conclusion IV: All knives are cups.

Which one of the following options can be logically inferred?

A. Only conclusion I and conclusion II are correct

B. Only conclusion II and conclusion III are correct

C. Only conclusion II and conclusion IV are correct

D. Only conclusion III and conclusion IV are correct

Q22.2.5 MCQ 2022 1M

The figure below shows the front and rear view of a disc, which is shaded with identical patterns. The disc is flipped once with respect to any one of the fixed axes 1-1, 2-2 or 3-3 chosen uniformly at random.

What is the probability that the disc **DOES NOT** retain the same front and rear views after the flipping operation?

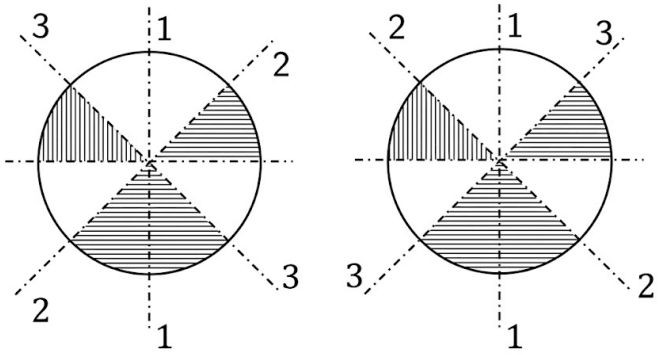


Fig. Q22.2.5

A. 0 **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** 1

Q22.2.6 MCQ 2022 2M

Altruism is the human concern for the wellbeing of others. Altruism has been shown to be motivated more by social bonding, familiarity and identification of belongingness to a group. The notion that altruism may be attributed to empathy or guilt has now been rejected.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Humans engage in altruism due to guilt but not empathy

B. Humans engage in altruism due to empathy but not guilt

C. Humans engage in altruism due to group identification but not empathy

D. Humans engage in altruism due to empathy but not familiarity

Q22.2.7 MCQ 2022 2M

There are two identical dice with a single letter on each of the faces. The following six letters: Q, R, S, T, U, and V, one on each of the faces. Any of the six outcomes are equally likely.

The two dice are thrown once independently at random.

What is the probability that the outcomes on the dice were composed only of any combination of the following possible outcomes: Q, U and V?

A. $\frac{1}{4}$ **B.** $\frac{3}{4}$ **C.** $\frac{1}{6}$ **D.** $\frac{5}{36}$

Q22.2.8 MCQ 2022 2M

The price of an item is 10% cheaper in an online store S compared to the price at another online store M. Store S charges ₹150 for delivery. There are no delivery charges for orders from the store M. A person bought the item from the store S and saved ₹100.

What is the price of the item at the online store S (in ₹) if there are no other charges than what is described above?

A. 2500 **B.** 2250 **C.** 1750 **D.** 1500

Q22.2.9 MCQ 2022 2M

The letters P, Q, R, S, T and U are to be placed one per vertex on a regular convex hexagon, but not necessarily in the same order.

Consider the following statements:

- The line segment joining R and S is longer than the line segment joining P and Q.

- The line segment joining R and S is perpendicular to the line segment joining P and Q.
- The line segment joining R and U is parallel to the line segment joining T and Q.

Based on the above statements, which one of the following options is CORRECT?

A. The line segment joining R and T is parallel to the line segment joining Q and S

B. The line segment joining T and Q is parallel to the line joining P and U

C. The line segment joining R and P is perpendicular to the line segment joining U and Q

D. The line segment joining Q and S is perpendicular to the line segment joining R and P

Q22.2.10 MCQ | 2022 | 2M

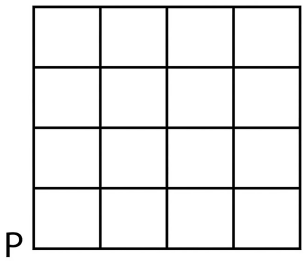
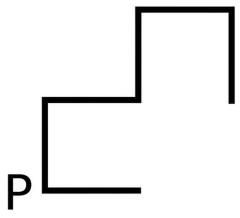


Fig. Q22.2.10

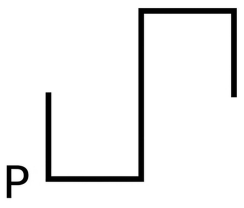
An ant is at the bottom-left corner of a grid (point P) as shown below. It aims to move to the top-right corner of the grid. The ant moves only along the lines marked in the grid such that the current distance to the top-right corner strictly decreases.

Which one of the following is a part of a possible trajectory of the ant during the movement?

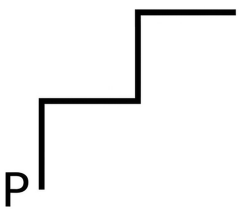
A.



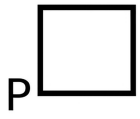
B.



C.



D.



Session : 3 | Forenoon | 06/02/22

Q22.3.1 MCQ | 2022 | 1M

Mr. X speaks _____ Japanese _____ Chinese.

A. neither / or **B.** either / nor

C. neither / nor **D.** also / but

Q22.3.2 MCQ | 2022 | 1M

A sum of money is to be distributed among P, Q, R, and S in the proportion 5:2:4:3, respectively.

If R gets ₹1000 more than S, what is the share of Q (in ₹)?

A. 500 **B.** 1000 **C.** 1500 **D.** 2000

Q22.3.3 MCQ | 2022 | 1M

A trapezium has vertices marked as P, Q, R and S (in that order anticlockwise). The side PQ is parallel to side SR. Further, it is given that, $PQ = 11$ cm, $QR = 4$ cm, $RS = 6$ cm and $SP = 3$ cm.

What is the shortest distance between PQ and SR (in cm)?

A. 1.80 **B.** 2.40 **C.** 4.20 **D.** 5.76

Q22.3.4 MCQ | 2022 | 1M

The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.

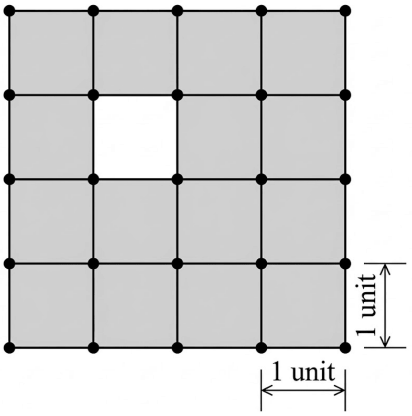


Fig. Q22.3.4

What is the maximum number of squares without a "hole in the interior" that can be formed within the 4×4 grid using the unit squares as building blocks?

A. 15 **B.** 20 **C.** 21 **D.** 26

Q22.3.5 MCQ 2022 1M

An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard posted is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard. If the security guard does not move around the posted location and has a 360° view, which one of the following correctly represents the set of ALL possible locations among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery?

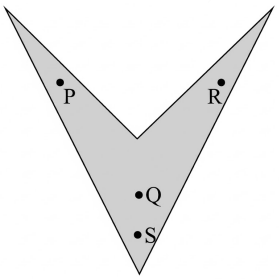


Fig. Q22.3.5

A. P and Q **B.** Q **C.** Q and S **D.** R and S

Q22.3.6 MCQ 2022 2M

Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences.

Which one of the following is the correct logical inference based on the information in the above passage?

A. Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous

B. Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects

C. Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous

D. Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence

Q22.3.7 MCQ 2022 2M

Consider the following inequalities:

(i) $2x - 1 > 7$

(ii) $2x - 9 < 1$

Which one of the following expressions below satisfies the above two inequalities?

A. $x \leq -4$ **B.** $-4 < x \leq 4$ **C.** $4 < x < 5$ **D.** $x \geq 5$

Q22.3.8 MCQ 2022 2M

Four points $P(0, 1)$, $Q(0, -3)$, $R(-2, -1)$, and $S(2, -1)$ represent the vertices of a quadrilateral.

What is the area enclosed by the quadrilateral?

A. 4 **B.** $4\sqrt{2}$ **C.** 8 **D.** $8\sqrt{2}$

Q22.3.9 MCQ 2022 2M

In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below.

Statement of P: R has copied in the exam.
Statement of Q: S has copied in the exam.
Statement of R: P did not copy in the exam.
Statement of S: Only one of us is telling the truth.
Statement of T: R is telling the truth.

The investigating team had authentic information that S never lies.

Based on the information given above, the person who has copied in the exam is

A. R **B.** P **C.** Q **D.** T

Q22.3.10 MCQ 2022 2M

Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively.

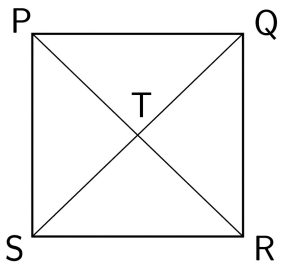


Fig. Q22.3.10

Let X, Y and Z represent the following operations:

X: rotation of the square by 180 degree with respect to the S-Q axis.

Y: rotation of the square by 180 degree with respect to the P-R axis.
Z: rotation of the square by 90 degree clockwise with respect to the axis perpendicular, going into the screen and passing through the point T.

Consider the following three distinct sequences of operation (which are applied in the left to right order):

(1) XYZZ
(2) XY
(3) ZZZZ

Which one of the following statements is correct as per the information provided above?

A. The sequence of operations (1) and (2) are equivalent

B. The sequence of operations (1) and (3) are equivalent

C. The sequence of operations (2) and (3) are equivalent

D. The sequence of operations (1), (2) and (3) are equivalent

Session : 4 | Afternoon | 06/02/22

Q22.4.1 MCQ 2022 1M

Inhaling the smoke from a burning _____ could _____ you quickly

A. tire / tier **B.** tire / tyre

C. tyre / tire **D.** tyre / tier

Q22.4.2 MCQ 2022 1M

A sphere of radius r cm is packed in a box of cubical shape. What should be the minimum volume (in cm^3) of the box that can enclose the sphere?

A. $\frac{r^3}{8}$ **B.** r^3 **C.** $2r^3$ **D.** $8r^3$

Q22.4.3 MCQ 2022 1M

Pipes P and Q can fill a storage tank in full with water in 10 and 6 minutes, respectively. Pipe R draws the water out from the storage tank at a rate of 34 litres per minute. P, Q and R operate at a constant rate.
If it takes one hour to completely empty a full storage tank with all the pipes operating simultaneously, what is the capacity of the storage tank (in litres)?

A. 26.8 **B.** 60.0 **C.** 120.0 **D.** 127.5

Q22.4.4 MCQ 2022 1M

Six persons P, Q, R, S, T and U are sitting around a circular table facing the center not necessarily in the same order. Consider the following statements:

- P sits next to S and T.
- Q sits diametrically opposite to P.
- The shortest distance between S and R is equal to the shortest distance between T and U.

Based on the above statements, Q is a neighbor of:

A. U and S **B.** R and T **C.** R and U **D.** P and S

Q22.4.5 MCQ 2022 1M

A building has several rooms and doors as shown in the top view of the building. The doors are closed initially.
What is the minimum number of doors that need to be opened in order to go from the point P to the point Q?

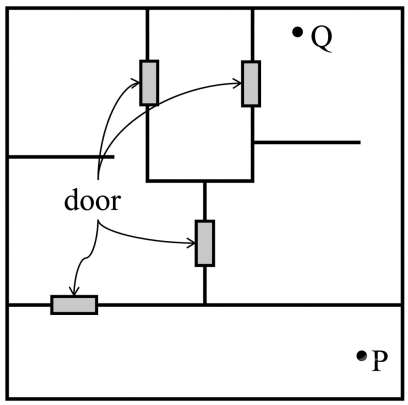


Fig. Q22.4.5

A. 4 **B.** 3 **C.** 2 **D.** 1

Q22.4.6 MCQ 2022 2M

Rice, a versatile and inexpensive source of carbohydrate, is a critical component of diet worldwide. Climate change, causing extreme weather, poses a threat to sustained availability of rice. Scientists are working on developing Green Super Rice (GSR), which is resilient under extreme weather conditions yet gives higher yields sustainably.
Which one of the following is the CORRECT logical inference based on the information given in the above passage?

A. GSR is an alternative to regular rice, but it grows only in an extreme weather

B. GSR may be used in future in response to adverse effects of climate change

C. GSR grows in an extreme weather, but the quantity of produce is lesser than regular rice

D. Regular rice will continue to provide good yields even in extreme weather

Q22.4.7 MCQ 2022 2M

A game consists of spinning an arrow around a stationary disk. When the arrow comes to rest, there are eight equally likely outcomes. It could come to rest in any one of the sectors numbered 1, 2, 3, 4, 5, 6, 7 or 8.
Two such disks are used in a game where their arrows are independently spun.
What is the probability that the sum of the numbers on the resulting sectors upon spinning the two disks is equal to 8 after the arrows come to rest?

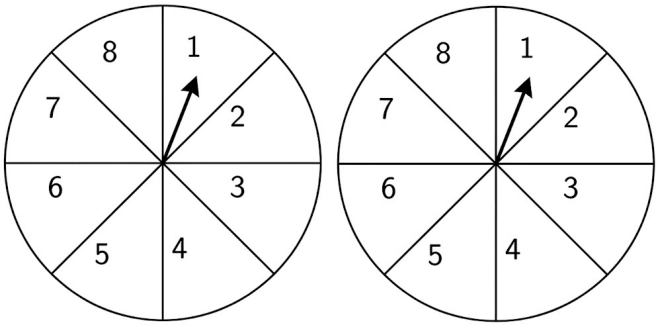


Fig. Q22.4.7

A. $\frac{1}{16}$ **B.** $\frac{5}{64}$ **C.** $\frac{3}{32}$ **D.** $\frac{7}{64}$

Q22.4.8 MCQ 2022 2M

Consider the following inequalities:

(i) $3p - q < 4$
(ii) $3q - p < 12$

Which one of the following expressions below satisfies the above two inequalities?

A. $p + q < 8$ **B.** $p + q = 8$

C. $8 \leq p + q < 16$ **D.** $p + q \geq 16$

Q22.4.9 MCQ 2022 2M

Given below are three statements and four conclusions drawn based on the statements.

Statement 1: Some engineers are writers.
Statement 2: No writer is an actor.
Statement 3: All actors are engineers.

Conclusion I: Some writers are engineers.
Conclusion II: All engineers are actors.
Conclusion III: No actor is a writer.
Conclusion IV: Some actors are writers.

Which one of the following options can be logically inferred?

A. Only conclusion I is correct

B. Only conclusion II and conclusion III are correct

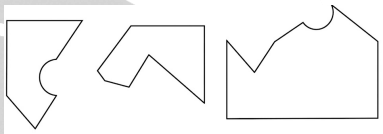
C. Only conclusion I and conclusion III are correct

D. Either conclusion III or conclusion IV is correct

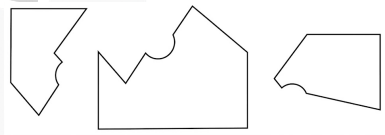
Q22.4.10 MCQ 2022 2M

Which one of the following sets of pieces can be assembled to form a square with a single round hole near the center? Pieces cannot overlap.

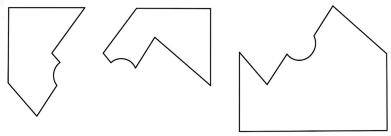
A.



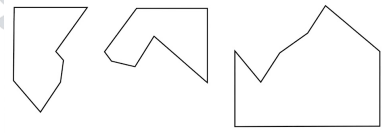
B.



C.



D.



Session : 5 | Forenoon | 12/02/22

Q22.5.1 MCQ 2022 1M

You should _____ when to say _____.

A. no / no **B.** no / know **C.** know / know **D.** know / no

Q22.5.2 MCQ 2022 1M

Two straight lines pass through the origin $(x_0, y_0) = (0, 0)$. One of them passes through the point $(x_1, y_1) = (1, 3)$ and the other passes through the point $(x_2, y_2) = (1, 2)$.
What is the area enclosed between the straight lines in the interval $[0, 1]$ on the x-axis?

A. 0.5 **B.** 1.0 **C.** 1.5 **D.** 2.0

Q22.5.3 MCQ 2022 1M

If
$p : q = 1 : 2$
$q : r = 4 : 3$
$r : s = 4 : 5$
and u is 50% more than s, what is the ratio $p : u$?

A. 2:15 **B.** 16:15 **C.** 1:5 **D.** 16:45

Q22.5.4 MCQ 2022 1M

Given the statements:

- P is the sister of Q.
- Q is the husband of R.
- R is the mother of S.
- T is the husband of P.

Based on the above information, T is _____ of S.

A. the grandfather **B.** an uncle

C. the father **D.** a brother

Q22.5.5 MCQ 2022 1M

In the following diagram, the point R is the center of the circle. The lines PQ and ZV are tangential to the circle. The relation among the areas of the squares, PXWR, RUVZ and SPQT is

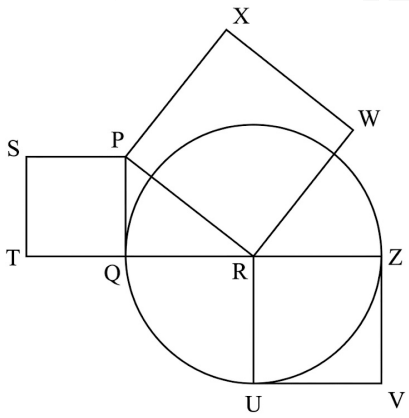


Fig. Q22.5.5

A. Area of SPQT = Area of RUVZ = Area of PXWR

B. Area of SPQT = Area of PXWR − Area of RUVZ

C. Area of PXWR = Area of SPQT − Area of RUVZ

D. Area of PXWR = Area of RUVZ − Area of SPQT

Q22.5.6 MCQ 2022 2M

Healthy eating is a critical component of healthy aging. When should one start eating healthy? It turns out that it is never too early. For example, babies who start eating healthy in the first year are more likely to have better overall health as they get older.
Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Healthy eating is important for those with good health conditions, but not for others

B. Eating healthy can be started at any age, earlier the better

C. Eating healthy and better overall health are more correlated at a young age, but not older age

D. Healthy eating is more important for adults than kids

Q22.5.7 MCQ 2022 2M

P invested ₹5000 per month for 6 months of a year and Q invested ₹x per month for 8 months of the year in a partnership business. The profit is shared in proportion to the total investment made in that year.
If at the end of that investment year, Q receives $\frac{4}{9}$ of the total profit, what is the value of x (in ₹)?

A. 2500 **B.** 3000 **C.** 4687 **D.** 8437

Q22.5.8 MCQ 2022 2M

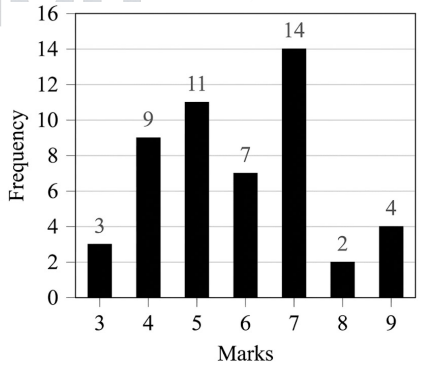


Fig. Q22.5.8

The above frequency chart shows the frequency distribution of marks obtained by a set of students in an exam.
From the data presented above, which one of the following is CORRECT?

A. mean > mode > median **B.** mode > median > mean

C. mode > mean > median **D.** median > mode > mean

Q22.5.9 MCQ | 2022 | 2M

In the square grid shown below, a person standing at P2 position is required to move to P5 position.

The only movement allowed for a step involves, "two moves along one direction followed by one move in a perpendicular direction". The permissible directions for movement are shown as dotted arrows in the right.

For example, a person at a given position **Y** can move only to the positions marked X on the right.

Without occupying any of the shaded squares at the end of each step, the minimum number of steps required to go from P2 to P5 is

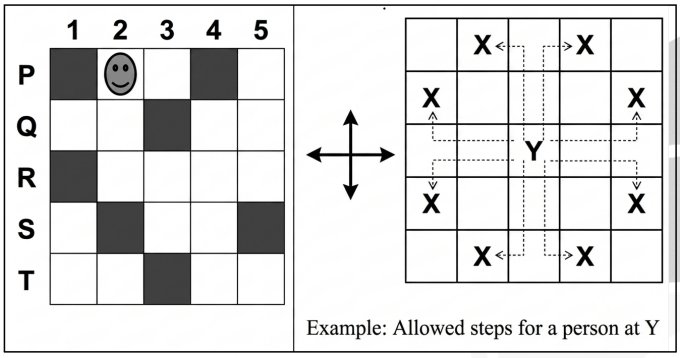


Fig. Q22.5.9

A. 4 **B.** 5 **C.** 6 **D.** 7

Q22.5.10 MCQ | 2022 | 2M

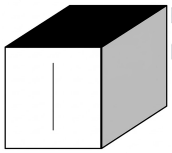
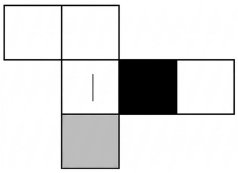
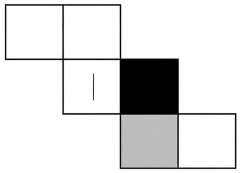
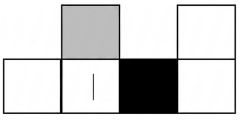


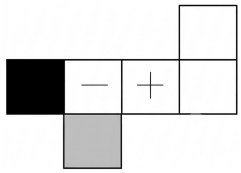
Fig. Q22.5.10

Consider a cube made by folding a single sheet of paper of appropriate shape. The interior faces of the cube are all blank. However, the exterior faces that are not visible in the given view may not be blank. Which one of the following represents a possible unfolding of the cube?

A.


B.


C.


D.


Session : 6 | Afternoon | 12/02/22

Q22.6.1 MCQ | 2022 | 1M

The movie was funny and I _____.

A. could help laughing **B.** couldn't help laughed

C. couldn't help laughing **D.** could helped laughed

Q22.6.2 MCQ | 2022 | 1M

$x : y : z = \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$.

What is the value of $\frac{x + z - y}{y}$?

A. 0.75 **B.** 1.25 **C.** 2.25 **D.** 3.25

Q22.6.3 MCQ | 2022 | 1M

Both the numerator and the denominator of $\frac{3}{4}$ are increased by a positive integer, x, and those of $\frac{15}{17}$ are decreased by the same integer. This operation results in the same value for both the fractions.

What is the value of x?

A. 1 **B.** 2 **C.** 3 **D.** 4

Q22.6.4 MCQ | 2022 | 1M

A survey of 450 students about their subjects of interest resulted in the following outcome:

- 150 students are interested in Mathematics.
- 200 students are interested in Physics.
- 175 students are interested in Chemistry.
- 50 students are interested in Mathematics and Physics.
- 60 students are interested in Physics and Chemistry.
- 40 students are interested in Mathematics and Chemistry.
- 30 students are interested in Mathematics, Physics and Chemistry.
- Remaining students are interested in Humanities.

Based on the above information, the number of students interested in Humanities is

A. 10 **B.** 30 **C.** 40 **D.** 45

Q22.6.5 MCQ 2022 1M

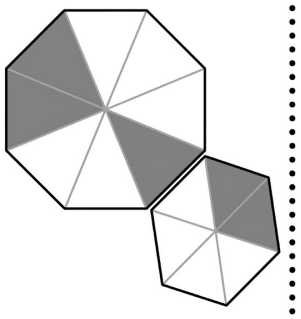
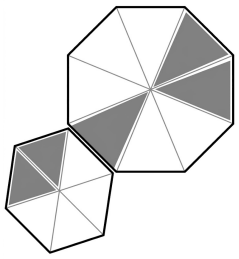
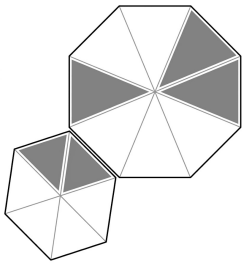
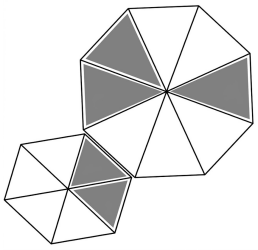


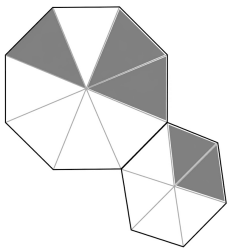
Fig. Q22.6.5

For the picture shown above, which one of the following is the correct picture representing reflection with respect to the mirror shown as the dotted line?

A. 

B. 

C. 

D. 

Q22.6.6 MCQ 2022 2M

In the last few years, several new shopping malls were opened in the city. The total number of visitors in the malls is impressive. However, the total revenue generated through sales in the shops in these malls is generally low. Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Fewer people are visiting the malls but spending more

B. More people are visiting the malls but not spending enough

C. More people are visiting the malls and spending more

D. Fewer people are visiting the malls and not spending enough

Q22.6.7 MCQ 2022 2M

In a partnership business, the monthly investment by three friends for the first six months is in the ratio 3:4:5. After six months, they had to increase their monthly investments by 10%, 15% and 20%, respectively, of their initial monthly investment. The new investment ratio was kept constant for the next six months.
What is the ratio of their shares in the total profit (in the same order) at the end of the year such that the share is proportional to their individual total investment over the year?

A. 22 : 23 : 24

B. 22 : 33 : 50

C. 33 : 46 : 60

D. 63 : 86 : 110

Q22.6.8 MCQ 2022 2M

Consider the following equations of straight lines:
Line L_1: $2x - 3y = 5$;
Line L_2: $3x + 2y = 8$;
Line L_3: $4x - 6y = 5$;
Line L_4: $6x - 9y = 6$.
Which one among the following is the correct statement?

A. L_1 is parallel to L_2 and L_1 is perpendicular to L_3

B. L_2 is parallel to L_4 and L_2 is perpendicular to L_1

C. L_3 is perpendicular to L_4 and L_3 is parallel to L_2

D. L_4 is perpendicular to L_2 and L_4 is parallel to L_3

Q22.6.9 MCQ 2022 2M

Given below are two statements and four conclusions drawn based on the statements.

Statement 1: Some soaps are clean.
Statement 2: All clean objects are wet.

Conclusion I: Some clean objects are soaps.
Conclusion II: No clean object is a soap.
Conclusion III: Some wet objects are soaps.
Conclusion IV: All wet objects are soaps.

A. Only conclusion I is correct

B. Either conclusion I or conclusion II is correct

C. Either conclusion III or conclusion IV is correct

D. Only conclusion I and conclusion III are correct

Q22.6.10 MCQ | 2022 | 2M

An ant walks in a straight line on a plane leaving behind a trace of its movement. The initial position of the ant is at point P facing east.
The ant first turns 72° anticlockwise at P, and then does the following two steps in sequence exactly FIVE times before halting:

1. moves forward for 10 cm.
2. turns 144° clockwise.

The pattern made by the trace left behind by the ant is .

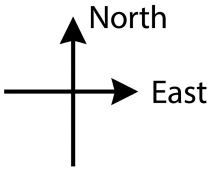
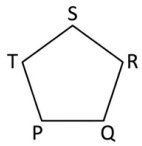
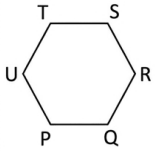


Fig. Q22.6.10

A. 

PQ = QR = RS = ST = TP = 10 cm

B. 

PQ = QR = RS = ST = TU = UP = 10 cm

C.

SQ = QT = TR = RP = PS = 10 cm

D.

SW = WR = RP = PT = TQ = QU = US = 10 cm

Session : 7 | Forenoon | 13/02/22

Q22.7.1 MCQ | 2022 | 1M

After playing _____ hours of tennis, I am feeling _____ tired to walk back.

A. too / too **B.** too / two **C.** two / two **D.** two / too

Q22.7.2 MCQ | 2022 | 1M

The average of the monthly salaries of M, N and S is ₹4000.
The average of the monthly salaries of N, S and P is ₹5000.
The monthly salary of P is ₹6000.
What is the monthly salary of M as a percentage of the monthly salary of P?

A. 50% **B.** 75% **C.** 100% **D.** 125%

Q22.7.3 MCQ | 2022 | 1M

A person travelled 80 km in 6 hours. If the person travelled the first part with a uniform speed of 10 kmph and the remaining part with a uniform speed of 18 kmph.
What percentage of the total distance is travelled at a uniform speed of 10 kmph?

A. 28.25 **B.** 37.25 **C.** 43.75 **D.** 50.00

Q22.7.4 MCQ | 2022 | 1M

Four girls P, Q, R and S are studying languages in a University. P is learning French and Dutch. Q is learning Chinese and Japanese. R is learning Spanish and French. S is learning Dutch and Japanese.
Given that: French is easier than Dutch; Chinese is harder than Japanese; Dutch is easier than Japanese, and Spanish is easier than French.
Based on the above information, which girl is learning the most difficult pair of languages?

A. P **B.** Q **C.** R **D.** S

Q22.7.5 MCQ | 2022 | 1M

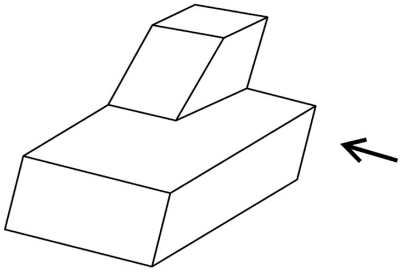
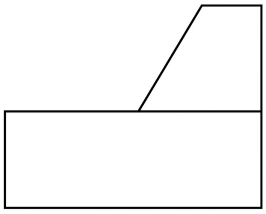
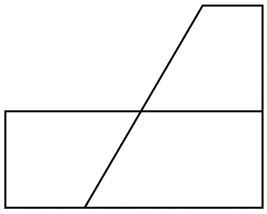
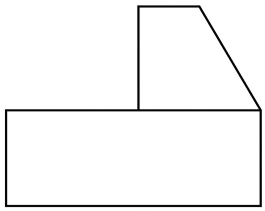


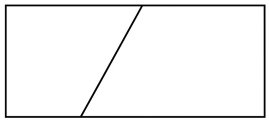
Fig. Q22.7.5

A block with a trapezoidal cross-section is placed over a block with rectangular cross section as shown above. Which one of the following is the correct drawing of the view of the 3D object as viewed in the direction indicated by an arrow in the above figure?

A. 

B. 

C. 

D. 

Q22.7.6 MCQ 2022 2M

Humans are naturally compassionate and honest. In a study using strategically placed wallets that appear "lost", it was found that wallets with money are more likely to be returned than wallets without money. Similarly, wallets that had a key and money are more likely to be returned than wallets with the same amount of money alone. This suggests that the primary reason for this behavior is compassion and empathy.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Wallets with a key are more likely to be returned because people do not care about money

B. Wallets with a key are more likely to be returned because people relate to suffering of others

C. Wallets used in experiments are more likely to be returned than wallets that are really lost

D. Money is always more important than keys

Q22.7.7 MCQ 2022 2M

A rhombus is formed by joining the midpoints of the sides of a unit square.

What is the diameter of the largest circle that can be inscribed within the rhombus?

A. $\frac{1}{\sqrt{2}}$ **B.** $\frac{1}{2\sqrt{2}}$ **C.** $\sqrt{2}$ **D.** $2\sqrt{2}$

Q22.7.8 MCQ 2022 2M

An equilateral triangle, a square and a circle have equal areas.

What is the ratio of the perimeters of the equilateral triangle to square to circle?

A. $3\sqrt{3} : 2 : \sqrt{\pi}$ **B.** $\sqrt{(3\sqrt{3})} : 2 : \sqrt{\pi}$

C. $\sqrt{(3\sqrt{3})} : 4 : 2\sqrt{\pi}$ **D.** $\sqrt{(3\sqrt{3})} : 2 : 2\sqrt{\pi}$

Q22.7.9 MCQ 2022 2M

Given below are three conclusions drawn based on the following three statements.

Statement 1: All teachers are professors.
Statement 2: No professor is a male.
Statement 3: Some males are engineers.

Conclusion I: No engineer is a professor.
Conclusion II: Some engineers are professors.
Conclusion III: No male is a teacher.

Which one of the following options can be logically inferred?

A. Only conclusion III is correct

B. Only conclusion I and conclusion II are correct

C. Only conclusion II and conclusion III are correct

D. Only conclusion I and conclusion III are correct

Q22.7.10 MCQ 2022 2M

In a 12-hour clock that runs correctly, how many times do the second, minute, and hour hands of the clock coincide, in a 12-hour duration from 3 PM in a day to 3 AM the next day?

A. 11 **B.** 12 **C.** 144 **D.** 2

Session : 8 | Afternoon | 13/02/22

Q22.8.1 MCQ 2022 1M

Writing too many things on the _____ while teaching could make the students get _____.

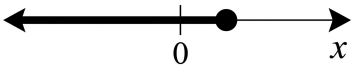
A. bored / board **B.** board / bored

C. board / board **D.** bored / bored

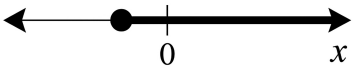
Q22.8.2 MCQ 2022 1M

Which one of the following is a representation (not to scale and in bold) of all values of x satisfying the inequality $2 - 5x \leq -\frac{6x-5}{3}$ on the real number line?

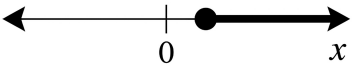
A.



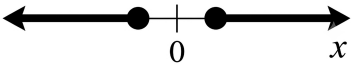
B.



C.



D.



Q22.8.3 MCQ 2022 1M

If $f(x) = 2\ln(\sqrt{e^x})$, what is the area bounded by $f(x)$ for the interval [0, 2] on the x-axis?

A. $\frac{1}{2}$ **B.** 1 **C.** 2 **D.** 4

Q22.8.4 MCQ 2022 1M

A person was born on the fifth Monday of February in a particular year.

Which one of the following statements is correct based on the above information?

A. The 2nd February of that year is a Tuesday

B. There will be five Sundays in the month of February in that year

C. The 1st February of that year is a Sunday

D. All Mondays of February in that year have even dates

Q22.8.5 MCQ 2022 1M

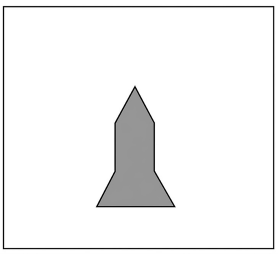
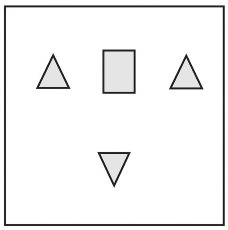


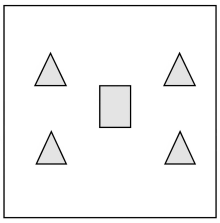
Fig. Q22.8.5

Which one of the groups given below can be assembled to get the shape that is shown above using each piece only once without overlapping with each other? (rotation and translation operations may be used).

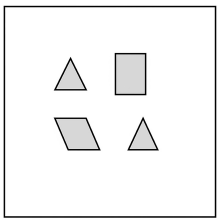
A.



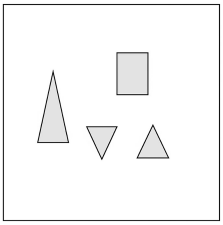
B.



C.



D.



Q22.8.6 MCQ 2022 2M

Fish belonging to species S in the deep sea have skins that are extremely black (ultra-black skin). This helps them not only to avoid predators but also sneakily attack their prey. However, having this extra layer of black pigment results in lower collagen on their skin, making their skin more fragile.

Which one of the following is the CORRECT logical inference based on the information in the above passage?

A. Having ultra-black skin is only advantageous to species S

B. Species S with lower collagen in their skin are at an advantage because it helps them avoid predators

C. Having ultra-black skin has both advantages and disadvantages to species S

D. Having ultra-black skin is only disadvantageous to species S but advantageous only to their predators

Q22.8.7 MCQ 2022 2M

For the past m days, the average daily production at a company was 100 units per day.
If today's production of 180 units changes the average to 110 units per day, what is the value of m?

A. 18 **B.** 10 **C.** 7 **D.** 5

Q22.8.8 MCQ 2022 2M

Consider the following functions for non-zero positive integers, p and q.

$$f(p,q) = \underbrace{p \times p \times p \times \ldots\ldots \cdots \times p}_{q \text{ terms}} = p^q; \quad f(p,1) = p$$

$$g(p,q) = p^{p^{p^{p^{\cdots}}}} \text{ (up to } q \text{ terms)}; \quad g(p,1) = p$$

Which one of the following options is correct based on the above?

A. $f(2,2) = g(2,2)$ **B.** $f(g(2,2),2) < f(2,g(2,2))$

C. $g(2,1) \neq f(2,1)$ **D.** $f(3,2) > g(3,2)$

Q22.8.9 MCQ 2022 2M

Four cities P, Q, R and S are connected through one-way routes as shown in the figure. The travel time between any two connected cities is one hour. The boxes beside each city name describe the starting time of first train of the day and their frequency of operation. For example, from city P, the first trains of the day start at 8 AM with a frequency of 90 minutes to each of R and S. A person does not spend additional time at any city other than the waiting time for the next connecting train.
If the person starts from R at 7 AM and is required to visit S and return to R, what is the minimum time required?

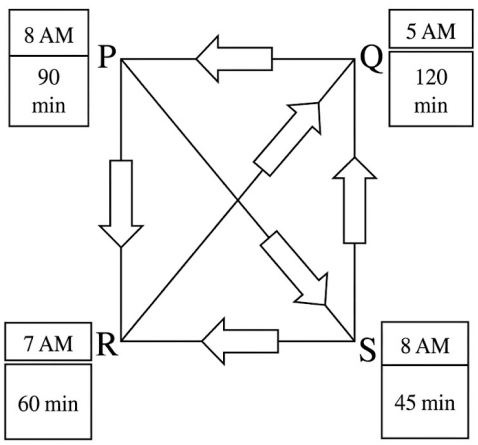


Fig. Q22.8.9

A. 6 hours 30 minutes **B.** 3 hours 45 minutes

C. 4 hours 30 minutes **D.** 5 hours 15 minutes

Q22.8.10 MCQ 2022 2M

Equal sized circular regions are shaded in a square sheet of paper of 1 cm side length. Two cases, case M and case N, are considered as shown in the figures below. In the case M, four circles are shaded in the square sheet and in the case N, nine circles are shaded in the square sheet as shown.
What is the ratio of the areas of unshaded regions of case M to that of case N?

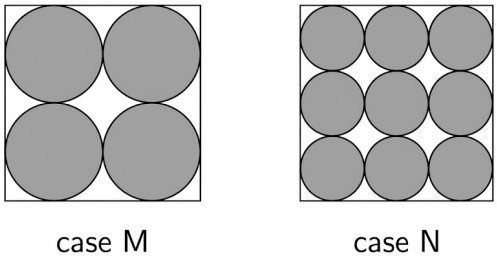


Fig. Q22.8.10

A. 2:3 **B.** 1:1 **C.** 3:2 **D.** 2:1

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q22.1.1 D	Q22.1.2 C	Q22.1.3 D	Q22.1.4 A	Q22.1.5 B	Q22.1.6 B
Q22.1.7 D	Q22.1.8 D	Q22.1.9 B	Q22.1.10 D		
Session : 2					
Q22.2.1 D	Q22.2.2 A	Q22.2.3 A	Q22.2.4 A	Q22.2.5 C	Q22.2.6 C
Q22.2.7 A	Q22.2.8 B	Q22.2.9 A	Q22.2.10 C		
Session : 3					
Q22.3.1 C	Q22.3.2 D	Q22.3.3 B	Q22.3.4 B	Q22.3.5 C	Q22.3.6 D
Q22.3.7 C	Q22.3.8 C	Q22.3.9 B	Q22.3.10 B		
Session : 4					
Q22.4.1 C	Q22.4.2 D	Q22.4.3 C	Q22.4.4 C	Q22.4.5 C	Q22.4.6 B
Q22.4.7 D	Q22.4.8 A	Q22.4.9 C	Q22.4.10 C		
Session : 5					
Q22.5.1 D	Q22.5.2 A	Q22.5.3 D	Q22.5.4 B	Q22.5.5 B	Q22.5.6 B
Q22.5.7 B	Q22.5.8 B	Q22.5.9 B	Q22.5.10 MTA		
Session : 6					
Q22.6.1 C	Q22.6.2 B	Q22.6.3 C	Q22.6.4 D	Q22.6.5 A	Q22.6.6 B
Q22.6.7 D	Q22.6.8 D	Q22.6.9 D	Q22.6.10 C		
Session : 7					
Q22.7.1 D	Q22.7.2 A	Q22.7.3 C	Q22.7.4 B	Q22.7.5 A	Q22.7.6 B
Q22.7.7 A	Q22.7.8 B	Q22.7.9 A	Q22.7.10 MTA		
Session : 8					
Q22.8.1 B	Q22.8.2 C	Q22.8.3 C	Q22.8.4 A	Q22.8.5 B OR C	Q22.8.6 C
Q22.8.7 C	Q22.8.8 A	Q22.8.9 A	Q22.8.10 B		

YEAR

2021

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	IN, CY, PH, AR, BM	10
Session 2	AE, AG, MN, PE, PI, TF, EY	10
Session 3	EE, GG	10
Session 4	EC, ST, MA, XH	10
Session 5	CS1, BT	10
Session 6	CS2	10
Session 7	ME1, CE1, XE	10
Session 8	ME2, CE2, ES, XL	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 06/02/21

Q21.1.1 MCQ 2021 1M

Getting to the top is _____ than staying on top.

A. more easy **B.** much easy **C.** easiest **D.** easier

Q21.1.2 MCQ 2021 1M

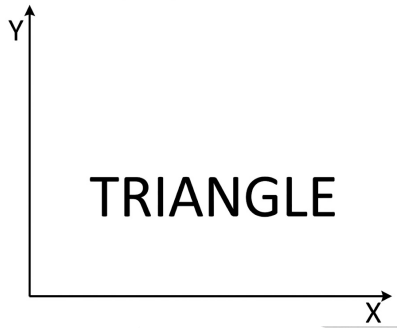


Fig. Q21.1.2

The mirror image of the text about the x-axis is:

A. ⊥RIAИGLE **B.** ⊥RIAИƆLE

C. ⊥RIAИƆLƎ **D.** ⊥RIAИƆLƆ

Q21.1.3 MCQ 2021 1M

In a company, 35% of the employees drink coffee, 40% of the employees drink tea and 10% of the employees drink both tea and coffee. What % of employees drink neither tea nor coffee?

A. 15 **B.** 25 **C.** 35 **D.** 40

Q21.1.4 MCQ 2021 1M

$\oplus$ and $\odot$ are two operators on numbers p and q such that

$p \oplus q = \frac{p^2 + q^2}{pq}$ and $p \odot q = \frac{p^2}{q}$;

If $x \oplus y = 2 \odot 2$, then $x =$

A. $\frac{y}{2}$ **B.** y **C.** $\frac{3y}{2}$ **D.** $2y$

Q21.1.5 MCQ 2021 1M

Four persons P, Q, R and S are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and R cannot sit adjacent to each other. S should be seated to the right of Q. The number of distinct seating arrangements possible is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q21.1.6 MCQ 2021 2M

Statement: Either P marries Q or X marries Y.
Among the options below, the logical NEGATION of the above statement is:

A. P does not marry Q and X marries Y.

B. Neither P marries Q nor X marries Y.

C. X does not marry Y and P marries Q.

D. P marries Q and X marries Y.

Q21.1.7 MCQ 2021 2M

Consider two rectangular sheets, Sheet M and Sheet N of dimensions 6 cm × 4 cm each.
Folding operation 1: The sheet is folded into half by joining the short edges of the current shape.
Folding operation 2: The sheet is folded into half by joining the long edges of the current shape.
Folding operation 1 is carried out on Sheet M three times.
Folding operation 2 is carried out on Sheet N three times.
The ratio of perimeters of the final folded shape of Sheet N to the final folded shape of Sheet M is _____.

A. 13:7 **B.** 3:2 **C.** 7:5 **D.** 5:13

Q21.1.8 MCQ 2021 2M

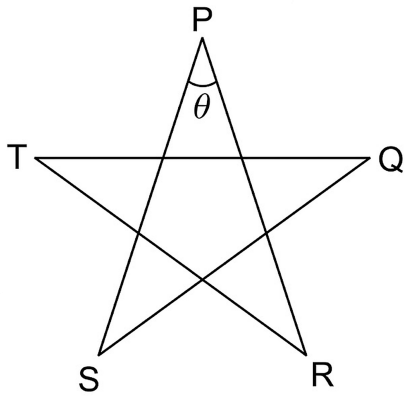


Fig. Q21.1.8

Five line segments of equal lengths, PR, PS, QS, QT and RT are used to form a star as shown in the figure above. The value of θ, in degrees, is _____

A. 36 **B.** 45 **C.** 72 **D.** 108

Q21.1.9 MCQ 2021 2M

A function, λ, is defined by

$$\lambda(p,q) = \begin{cases} (p-q)^2, & \text{if } p \geq q, \\ p+q, & \text{if } p < q. \end{cases}$$

The value of the expression $\frac{\lambda(-(-3+2),(-2+3))}{(-(-2+1))}$ is:

A. -1 **B.** 0 **C.** $\frac{16}{3}$ **D.** 16

Q21.1.10 MCQ 2021 2M

Humans have the ability to construct worlds entirely in their minds, which don't exist in the physical world. So far as we know, no other species possesses this ability. This skill is so important that we have different words to refer to its different flavors, such as imagination, invention and innovation.

Based on the above passage, which one of the following is TRUE?

A. No species possess the ability to construct worlds in their minds.

B. The terms imagination, invention and innovation refer to unrelated skills.

C. We do not know of any species other than humans who possess the ability to construct mental worlds.

D. Imagination, invention and innovation are unrelated to the ability to construct mental worlds.

Session : 2 | Afternoon | 06/02/21

Q21.2.1 MCQ 2021 1M

(i) Arun and Aparna are here.
(ii) Arun and Aparna is here.
(iii) Arun's families is here.
(iv) Arun's family is here.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iv)

C. (ii) and (iv) **D.** (iii) and (iv)

Q21.2.2 MCQ 2021 1M

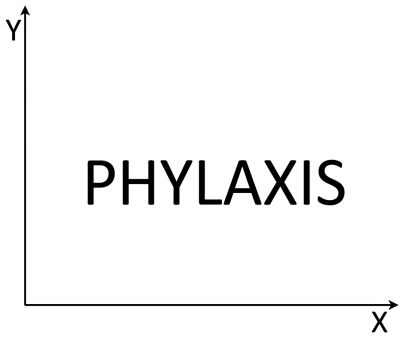


Fig. Q21.2.2

The mirror image of the text PHYLAXIS about the x-axis is

A. PHʎΓ∀XIS **B.** ⅁HʎΓ∀XIS

C. dHʎΓ∀XIS **D.** ⅁HʎΓ∀XIS

Q21.2.3 MCQ 2021 1M

Two identical cube shaped dice each with faces numbered 1 to 6 are rolled simultaneously. The probability that an even number is rolled out on each dice is

A. $\frac{1}{36}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{4}$

Q21.2.4 MCQ 2021 1M

$\oplus$ and $\odot$ are two operators on numbers p and q such that

$$p \odot q = p - q, \quad \text{and} \quad p \oplus q = p \times q$$

Then, $(9 \odot (6 \oplus 7)) \odot (7 \oplus (6 \odot 5)) =$

A. 40 **B.** -26 **C.** -33 **D.** -40

Q21.2.5 MCQ 2021 1M

Four persons P, Q, R and S are to be seated in a row. R should not be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 6 **B.** 9 **C.** 18 **D.** 24

Q21.2.6 MCQ 2021 2M

On a planar field, you travelled 3 units East from a point O. Next you travelled 4 units South to arrive at point P. Then you travelled from P in the North-East direction such that you arrive at a point that is 6 units East of point O. Next, you travelled in the North-West direction, so that you arrive at point Q that is 8 units North of point P.

The distance of point Q to point O, in the same units, should be _____

A. 3 **B.** 4 **C.** 5 **D.** 6

Q21.2.7 MCQ | 2021 | 2M

The author said, "Musicians rehearse before their concerts. Actors rehearse their roles before the opening of a new play. On the other hand, I find it strange that many public speakers think they can just walk on to the stage and start speaking. In my opinion, it is no less important for public speakers to rehearse their talks."
Based on the above passage, which one of the following is TRUE?

A. The author is of the opinion that rehearsing is important for musicians, actors and public speakers.

B. The author is of the opinion that rehearsing is less important for public speakers than for musicians and actors.

C. The author is of the opinion that rehearsing is more important only for musicians than public speakers.

D. The author is of the opinion that rehearsal is more important for actors than musicians.

Q21.2.8 MCQ | 2021 | 2M

1. Some football players play cricket.
2. All cricket players play hockey.

Among the options given below, the statement that logically follows from the two statements 1 and 2 above, is:

A. No football player plays hockey.

B. Some football players play hockey.

C. All football players play hockey.

D. All hockey players play football.

Q21.2.9 MCQ | 2021 | 2M

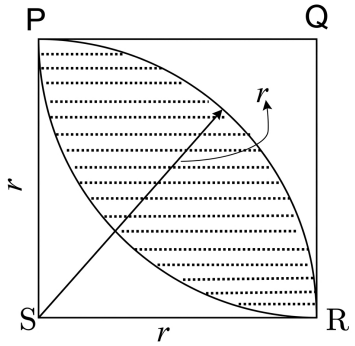


Fig. Q21.2.9

In the figure shown above, PQRS is a square. The shaded portion is formed by the intersection of sectors of circles with radius equal to the side of the square and centers at S and Q.
The probability that any point picked randomly within the square falls in the shaded area is _____

A. $4-\frac{\pi}{2}$ **B.** $\frac{1}{2}$ **C.** $\frac{\pi}{2}-1$ **D.** $\frac{\pi}{4}$

Q21.2.10 MCQ | 2021 | 2M

In an equilateral triangle PQR, side PQ is divided into four equal parts, side QR is divided into six equal parts and side PR is divided into eight equal parts. The length of each subdivided part in cm is an integer. The minimum area of the triangle PQR possible, in cm^2, is

A. 18 **B.** 24 **C.** $48\sqrt{3}$ **D.** $144\sqrt{3}$

Session : 3 | Forenoon | 07/02/21

Q21.3.1 MCQ | 2021 | 1M

The people _____ were at the demonstration were from all sections of society.

A. whose **B.** which **C.** who **D.** whom

Q21.3.2 MCQ | 2021 | 1M

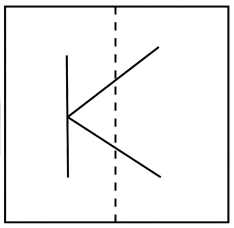
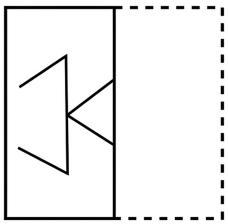
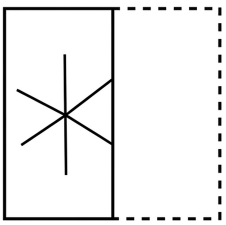
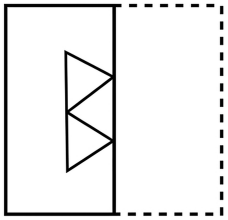


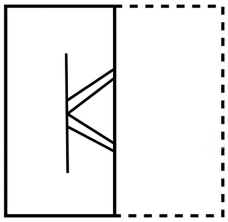
Fig. Q21.3.2

A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

A.


B.


C.


D.


Q21.3.3 MCQ 2021 1M

For a regular polygon having 10 sides, the interior angle between the sides of the polygon, in degrees, is:

A. 396 **B.** 324 **C.** 216 **D.** 144

Q21.3.4 MCQ 2021 1M

Which one of the following numbers is exactly divisible by $(11^{13} + 1)$?

A. $11^{26} + 1$ **B.** $11^{33} + 1$

C. $11^{39} - 1$ **D.** $11^{52} - 1$

Q21.3.5 MCQ 2021 1M

Oasis is to *sand* as *island* is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

A. Stone **B.** Land **C.** Water **D.** Mountain

Q21.3.6 MCQ 2021 2M

The importance of sleep is often overlooked by students when they are preparing for exams. Research has consistently shown that sleep deprivation greatly reduces the ability to recall the material learnt. Hence, cutting down on sleep to study longer hours can be counterproductive.

Which one of the following statements is the CORRECT inference from the above passage?

A. Sleeping well alone is enough to prepare for an exam. Studying has lesser benefit.

B. Students are efficient and are not wrong in thinking that sleep is a waste of time.

C. If a student is extremely well prepared for an exam, he needs little or no sleep.

D. To do well in an exam, adequate sleep must be part of the preparation.

Q21.3.7 MCQ 2021 2M

In the figure shown above, each inside square is formed by joining the midpoints of the sides of the next larger square. The area of the smallest square (shaded) as shown, in cm^2 is:

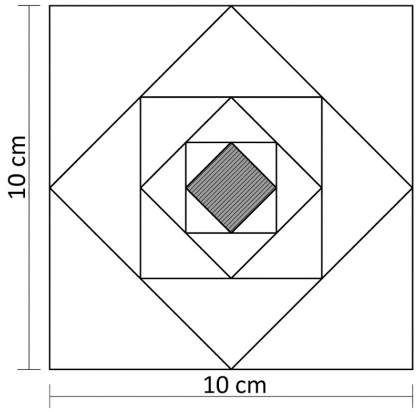


Fig. Q21.3.7

A. 12.50 **B.** 6.25 **C.** 3.125 **D.** 1.5625

Q21.3.8 MCQ 2021 2M

Let X be a continuous random variable denoting the temperature measured. The range of temperature is $[0, 100]$ degree Celsius and let the probability density function of X be $f(x) = 0.01$ for $0 \leq X \leq 100$.

The mean of X is _____

A. 2.5 **B.** 5.0 **C.** 25.0 **D.** 50.0

Q21.3.9 MCQ 2021 2M

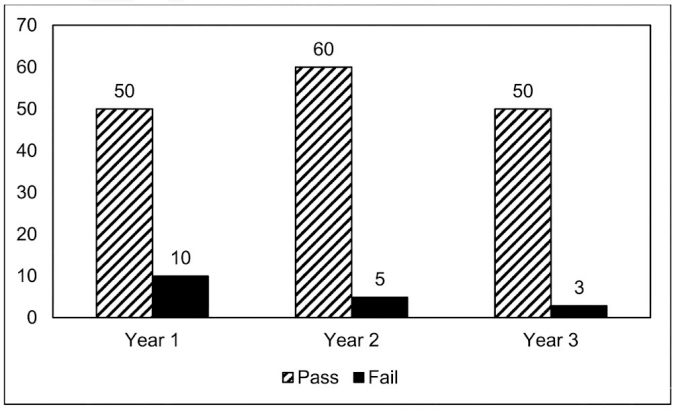


Fig. Q21.3.9

The number of students passing or failing in an exam for a particular subject are presented in the bar chart above. Students who pass the exam cannot appear for the exam again. Students who fail the exam in the first attempt must appear for the exam in the following year. Students always pass the exam in their second attempt.

The number of students who took the exam for the first time in the year 2 and the year 3 respectively, are _____.

A. 65 and 53 **B.** 60 and 50 **C.** 55 and 53 **D.** 55 and 48

Q21.3.10 MCQ 2021 2M

Seven cars P, Q, R, S, T, U and V are parked in a row not necessarily in that order. The cars T and U should be parked next to each other. The cars S and V also should be parked next to each other, whereas P and Q cannot be parked next to each other. Q and S must be parked next to each other. R is parked to the immediate right of V. T is parked to the left of U.
Based on the above statements, the only INCORRECT option given below is:

A. There are two cars parked in between Q and V.

B. Q and R are not parked together.

C. V is the only car parked in between S and R.

D. Car P is parked at the extreme end.

Session : 4 | Afternoon | 07/02/21

Q21.4.1 MCQ 2021 1M

The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?

A. 9,92,500 **B.** 9,95,006 **C.** 10,00,000 **D.** 12,51,506

Q21.4.2 MCQ 2021 1M

p and q are positive integers and $\frac{p}{q} + \frac{q}{p} = 3$.

then, $\frac{p^2}{q^2} + \frac{q^2}{p^2} =$

A. 3 **B.** 7 **C.** 9 **D.** 11

Q21.4.3 MCQ 2021 1M

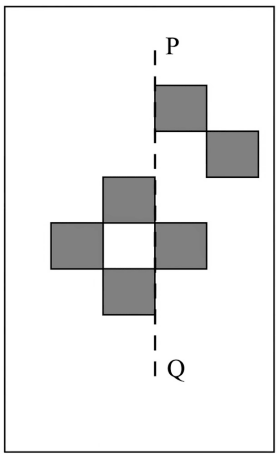


Fig. Q21.4.3

The least number of squares that must be added so that the line P-Q becomes the line of symmetry is ____

A. 4 **B.** 3 **C.** 6 **D.** 7

Q21.4.4 MCQ 2021 1M

Nostalgia is to *anticipation* as _____ is to _____.
Which one of the following options maintains a similar logical relation in the above sentence?

A. Present, past **B.** Future, past

C. Past, future **D.** Future, present

Q21.4.5 MCQ 2021 1M

Consider the following sentences:

(i) I woke up from sleep
(ii) I woked up from sleep.
(iii) I was woken up from sleep.
(iv) I was wokened up from sleep.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (i) and (iv)

Q21.4.6 MCQ 2021 2M

Given below are two statements and two conclusions.

Statement 1: All purple are green.
Statement 2: All black are green.
Conclusion I: Some black are purple.
Conclusion II: No black is purple.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct.

B. Only conclusion II is correct.

C. Either conclusion I or II is correct.

D. Both conclusion I and II are correct.

Q21.4.7 MCQ 2021 2M

Computers are ubiquitous. They are used to improve efficiency in almost all fields from agriculture to space exploration. Artificial intelligence (AI) is currently a hot topic. AI enables computers to learn, given enough training data. For humans, sitting in front of a computer for long hours can lead to health issues.

Which of the following can be deduced from the above passage?

(i) Nowadays, computers are present in almost all places.
(ii) Computers cannot be used for solving problems in engineering.
(iii) For humans, there are both positive and negative effects of using computers.
(iv) Artificial intelligence can be done without data.

A. (ii) and (iii) **B.** (ii) and (iv)

C. (i), (iii) and (iv) **D.** (i) and (iii)

Q21.4.8 MCQ 2021 2M

Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is _____

A. $\frac{\pi}{3}$ **B.** $\frac{2\pi}{3}$ **C.** $\frac{3\pi}{2}$ **D.** 3π

Q21.4.9 MCQ 2021 2M

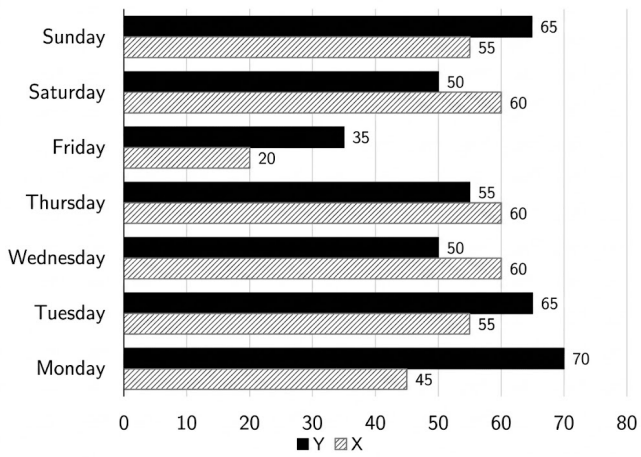


Fig. Q21.4.9

The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.
The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

A. 4 **B.** 5 **C.** 6 **D.** 7

Q21.4.10 MCQ 2021 2M

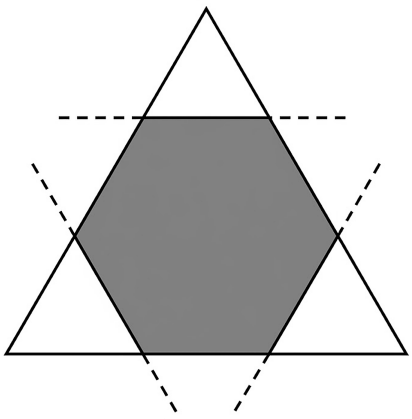


Fig. Q21.4.10

Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above.
The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is

A. 2:3 **B.** 3:4 **C.** 4:5 **D.** 5:6

Session : 5 | Forenoon | 13/02/21

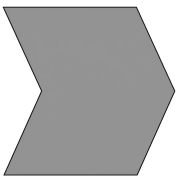
Q21.5.1 MCQ 2021 1M

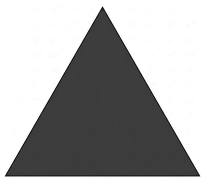
The ratio of boys to girls in a class is 7 to 3.
Among the options below, an acceptable value for the total number of students in the class is:

A. 21 **B.** 37 **C.** 50 **D.** 73

Q21.5.2 MCQ 2021 1M

A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.
Which one of the following is NOT a convex polygon?

A. 

B. 

C.

D.

Q21.5.3 MCQ 2021 1M

Consider the following sentences:

(i) Everybody in the class is prepared for the exam.
(ii) Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the CORRECT observation about the above two sentences?

A. (i) is grammatically correct and (ii) is unambiguous

B. (i) is grammatically incorrect and (ii) is unambiguous

C. (i) is grammatically correct and (ii) is ambiguous

D. (i) is grammatically incorrect and (ii) is ambiguous

Q21.5.4 MCQ 2021 1M

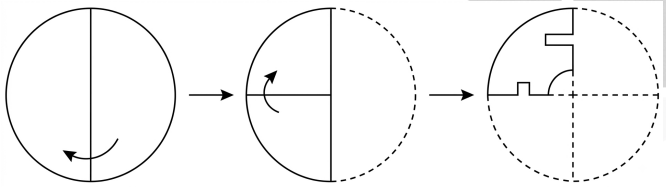
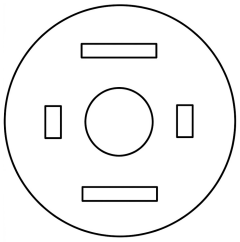
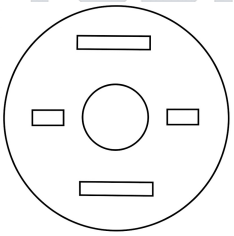
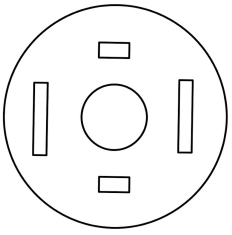


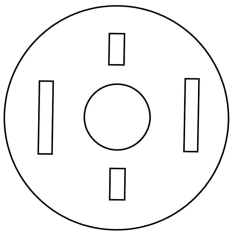
Fig. Q21.5.4

A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

A.


B.


C.


D.


Q21.5.5 MCQ 2021 1M

_____ is to *surgery* as *writer* is to _____.

Which one of the following options maintains a similar logical relation in the above sentence?

A. Plan, outline **B.** Hospital, library

C. Doctor, book **D.** Medicine, grammar

Q21.5.6 MCQ 2021 2M

We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm × 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____

A. $\frac{\pi}{2}$ **B.** $\frac{3}{\pi}$ **C.** $\frac{9}{\pi}$ **D.** 3π

Q21.5.7 MCQ 2021 2M

Items	Cost (₹)	Profit %	Marked Price (₹)
P	5,400	- - -	5,860
Q	- - -	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$\left(\text{Profit \%} = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100\right)$$

The discount on item Q, as a percentage of its marked price, is _____

A. 25 **B.** 12.5 **C.** 10 **D.** 5

Q21.5.8 MCQ 2021 2M

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is

A. 0.3024 **B.** 0.4235 **C.** 0.6976 **D.** 0.8125

Q21.5.9 MCQ 2021 2M

Given below are two statements 1 and 2, and two conclusions I and II

Statement 1: All bacteria are microorganisms.
Statement 2: All pathogens are microorganisms.

Conclusion I: Some pathogens are bacteria.
Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

A. Only conclusion I is correct

B. Only conclusion II is correct

C. Either conclusion I or II is correct

D. Neither conclusion I nor II is correct

Q21.5.10 MCQ 2021 2M

Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality.
Which one of the following statements summarizes the passage?

A. The proposed AOM addresses the core problems that cause obesity.

B. If obesity reduces, poverty will naturally reduce, since obesity causes poverty.

C. AOM are addressing the core problems and are likely to succeed.

D. AOM are addressing the problem superficially.

Session : 6 | Afternoon | 13/02/21

Q21.6.1 MCQ 2021 1M

Gauri said that she can play the keyboard _____ her sister.

A. as well as **B.** as better as

C. as nicest as **D.** as worse as

Q21.6.2 MCQ 2021 1M

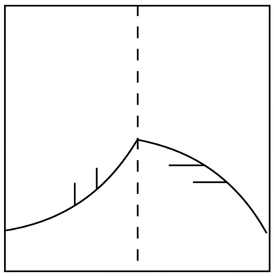
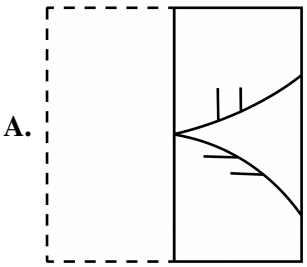
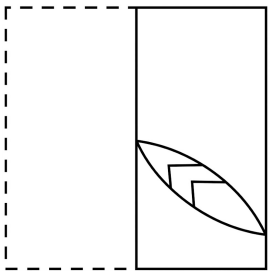
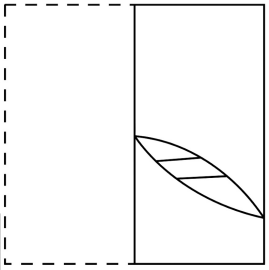


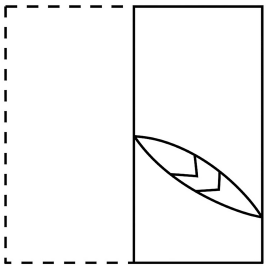
Fig. Q21.6.2

A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

A. 

B. 

C. 

D. 

Q21.6.3 MCQ 2021 1M

If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos\theta =$

A. $\frac{1}{2}$ **B.** $\frac{1}{\sqrt{3}}$ **C.** $\frac{1}{\sqrt{2}}$ **D.** $\frac{\sqrt{3}}{2}$

Q21.6.4 MCQ 2021 1M

If $(x-\frac{1}{2})^2-(x-\frac{3}{2})^2 = x+2$, then the value of x is:

A. 2 **B.** 4 **C.** 6 **D.** 8

Q21.6.5 MCQ 2021 1M

Pen : Write :: Knife : _____
Which of the the following options maintains a similar logical relation to the above?

A. Vegetables **B.** Sharp **C.** Cut **D.** Blunt

Q21.6.6 MCQ 2021 2M

Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.
Which one of the following statements is the CORRECT inference of the above passage?

A. Listening to music has no effect on learning and a positive effect on physical exercise.

B. Listening to music has a clear positive effect both on physical exercise and on learning.

C. Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.

D. Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.

Q21.6.7 MCQ 2021 2M

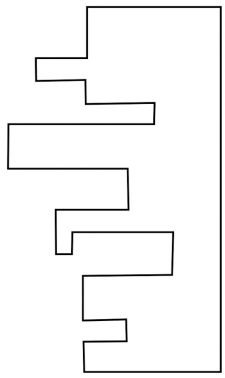
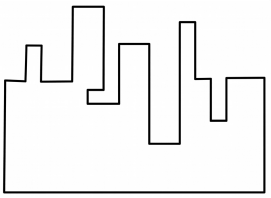
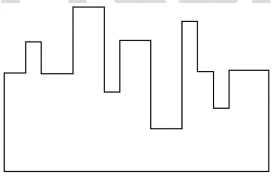
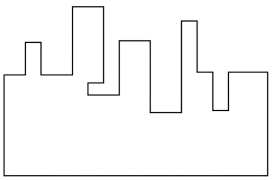


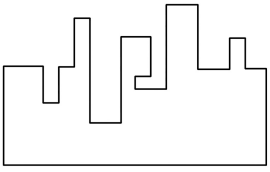
Fig. Q21.6.7

A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

A. 

B. 

C. 

D. 

Q21.6.8 MCQ 2021 2M

The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21.

The total number of students in all the three classes in the beginning was:

A. 22 **B.** 66 **C.** 88 **D.** 110

Q21.6.9 MCQ 2021 2M

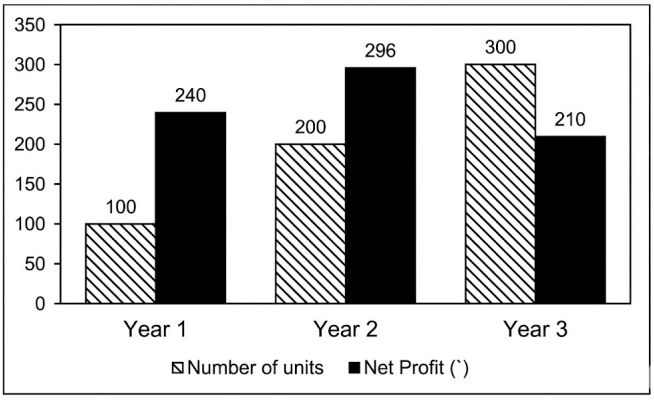


Fig. Q21.6.9

The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

A. 4:3 **B.** 1:1 **C.** 3:4 **D.** 1:2

Q21.6.10 MCQ 2021 2M

Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

Observation I: S is taller than R.
Observation II: Q is the shortest of all.
Observation III: U is taller than only one student.
Observation IV: T is taller than S but is not the tallest.

The number of students that are taller than R is the same as the number of students shorter than _____.

A. T **B.** R **C.** S **D.** P

Session : 7 | Forenoon | 14/02/21

Q21.7.1 MCQ 2021 1M

Consider the following sentences:

(i) After his surgery, Raja hardly could walk.
(ii) After his surgery, Raja could barely walk.
(iii) After his surgery, Raja barely could walk.
(iv) After his surgery, Raja could hardly walk.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (iii) and (iv) **D.** (ii) and (iv)

Q21.7.2 MCQ 2021 1M

Ms. X came out of a building through its front door to find her shadow due to the morning sun falling to her right side with the building to her back. From this, it can be inferred that building is facing _____

A. North **B.** East **C.** West **D.** South

Q21.7.3 MCQ 2021 1M

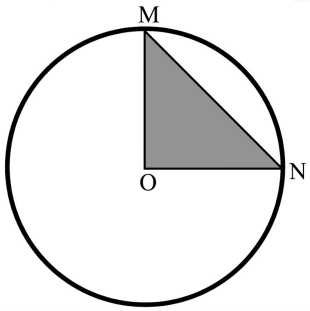


Fig. Q21.7.3

In the above figure, O is the center of the circle and, M and N lie on the circle.
The area of the right triangle MON is 50 cm^2.
What is the area of the circle in cm^2?

A. 2π **B.** 50π **C.** 75π **D.** 100π

Q21.7.4 MCQ 2021 1M

If
" $\oplus$ " means " $-$ ",
" $\otimes$ " means " $\div$ ",
"Δ" means " $+$ ",
"∇" means " $\times$ ",

then, the value of the expression $\Delta 2 \oplus 3\Delta((4 \otimes 2)\nabla 4) =$

A. -1 **B.** -0.5 **C.** 6 **D.** 7

Q21.7.5 MCQ 2021 1M

"The increased consumption of leafy vegetables in the recent months is a clear indication that the people in the state have begun to lead a healthy lifestyle"
Which of the following can be logically inferred from the information presented in the above statement?

A. The people in the state did not consume leafy vegetables earlier.

B. Consumption of leafy vegetables may not be the only indicator of healthy lifestyle.

C. Leading a healthy lifestyle is related to a diet with leafy vegetables.

D. The people in the state have increased awareness of health hazards causing by consumption of junk foods.

Q21.7.6 MCQ 2021 2M

Oxpeckers and rhinos manifest a symbiotic relationship in the wild. The oxpeckers warn the rhinos about approaching poachers, thus possibly saving the lives of the rhinos. Oxpeckers also feed on the parasitic ticks found on rhinos. In the symbiotic relationship described above, the primary benefits for oxpeckers and rhinos respectively are,

A. Oxpeckers get a food source, rhinos have no benefit.

B. Oxpeckers save their habitat from poachers while the rhinos have no benefit.

C. Oxpeckers get a food source, rhinos may be saved from the poachers.

D. Oxpeckers save the lives of poachers, rhinos save their own lives.

Q21.7.7 MCQ 2021 2M

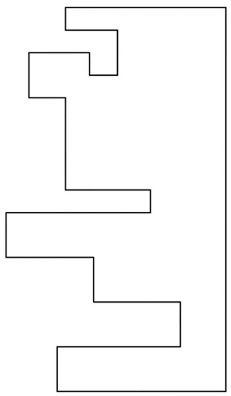
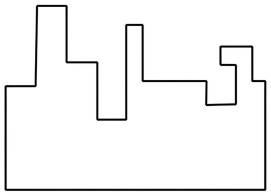
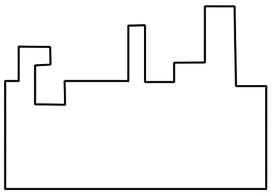
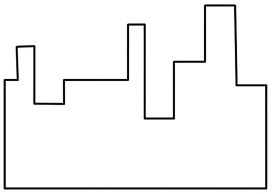


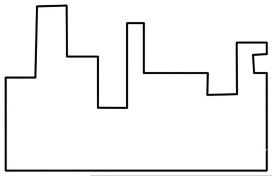
Fig. Q21.7.7

A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

A. 

B. 

C. 

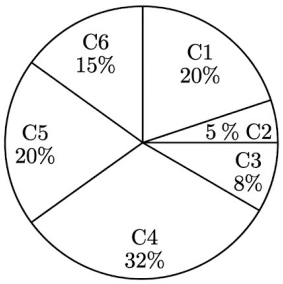
D. 

Q21.7.8 MCQ | 2021 | 2M

The number of hens, ducks and goats in farm P are 65, 91 and 169, respectively. The total number of hens, ducks and goats in a nearby farm Q is 416. The ratio of hens:ducks:goats in farm Q is 5:14:13. All the hens, ducks and goats are sent from farm Q to farm P.
The new ratio of hens:ducks:goats in farm P is _____

A. 5:7:13 **B.** 5:14:13 **C.** 10:21:26 **D.** 21:10:26

Q21.7.9 MCQ | 2021 | 2M



Company	Ratio
C1	3:2
C2	1:4
C3	5:3
C4	2:3
C5	9:1
C6	3:4

Fig. Q21.7.9

The distribution of employees at the rank of executives, across different companies C1, C2, ..., C6 is presented in the chart given above. The ratio of executives with a management degree to those without a management degree in each of these companies is provided in the table above. The total number of executives across all companies is 10,000.
The total number of management degree holders among the executives in companies C2 and C5 together is _____.

A. 225 **B.** 600 **C.** 1900 **D.** 2500

Q21.7.10 MCQ | 2021 | 2M

Five persons P, Q, R, S and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q.
The number of distinct seating arrangements possible is:

A. 4 **B.** 8 **C.** 10 **D.** 16

Session : 8 | Afternoon | 14/02/21

Q21.8.1 MCQ | 2021 | 1M

Five persons P, Q, R, S and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be seated at either end of the row. P should not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

A. 2 **B.** 3 **C.** 4 **D.** 5

Q21.8.2 MCQ | 2021 | 1M

Consider the following sentences:

(i) The number of candidates who appear for the GATE examination is staggering.
(ii) A number of candidates from my class are appearing for the GATE examination.
(iii) The number of candidates who appear for the GATE examination are staggering.
(iv) A number of candidates from my class is appearing for the GATE examination.

Which of the above sentences are grammatically CORRECT?

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (ii) and (iv)

Q21.8.3 MCQ | 2021 | 1M

A digital watch X beeps every 30 seconds while watch Y beeps every 32 seconds. They beeped together at 10 AM. The immediate next time that they will beep together is ___

A. 10.08 AM **B.** 10.42 AM

C. 11.00 AM **D.** 10.00 PM

Q21.8.4 MCQ 2021 1M

If $\oplus \div \odot = 2$; $\oplus \div \Delta = 3$; $\odot + \Delta = 5$; $\Delta \times \otimes = 10$, Then, the value of $(\otimes - \oplus)^2$, is:

A. 0 **B.** 1 **C.** 4 **D.** 16

Q21.8.5 MCQ 2021 1M

The front door of Mr. X's house faces East. Mr. X leaves the house, walking 50 m straight from the back door that is situated directly opposite to the front door. He then turns to his right, walks for another 50 m and stops. The direction of the point Mr. X is now located at with respect to the starting point is _____

A. South-East **B.** North-East **C.** West **D.** North-West

Q21.8.6 MCQ 2021 2M

Given below are two statements 1 and 2, and two conclusions I and II.

Statement 1: All entrepreneurs are wealthy.
Statement 2: All wealthy are risk seekers.

Conclusion I: All risk seekers are wealthy.
Conclusion II: Only some entrepreneurs are risk seekers.

Based on the above statements and conclusions, which one of the following options is CORRECT?

A. Only conclusion I is correct

B. Only conclusion II is correct

C. Neither conclusion I nor II is correct

D. Both conclusions I and II are correct

Q21.8.7 MCQ 2021 2M

A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is ____

A. 3/16 **B.** 45/236 **C.** 1/4 **D.** 3/4

Q21.8.8 MCQ 2021 2M

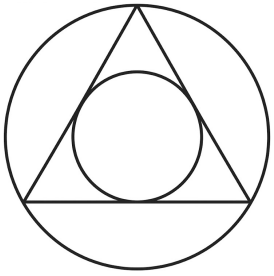


Fig. Q21.8.8

The ratio of the area of the inscribed circle to the area of the circumscribed circle of an equilateral triangle is _____

A. 1/8 **B.** 1/6 **C.** 1/4 **D.** 1/2

Q21.8.9 MCQ 2021 2M

Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. The area of each face of the final folded shape, in square units, equal to

A. 1/4 **B.** 1/8 **C.** 1/16 **D.** 1/32

Q21.8.10 MCQ 2021 2M

The world is going through the worst pandemic in the past hundred years. The air travel industry is facing a crisis, as the resulting quarantine requirement for travelers led to weak demand.
In relation to the first sentence above, what does the second sentence do?

A. Restates an idea from the first sentence.

B. Second sentence entirely contradicts the first sentence.

C. The two statements are unrelated.

D. States an effect of the first sentence.

PrepFusion

Answer Key

Session : 1					
Q21.1.1 D	Q21.1.2 B	Q21.1.3 C	Q21.1.4 B	Q21.1.5 C	Q21.1.6 B
Q21.1.7 A	Q21.1.8 A	Q21.1.9 B	Q21.1.10 C		
Session : 2					
Q21.2.1 B	Q21.2.2 B	Q21.2.3 D	Q21.2.4 D	Q21.2.5 C	Q21.2.6 C
Q21.2.7 A	Q21.2.8 B	Q21.2.9 C	Q21.2.10 D		
Session : 3					
Q21.3.1 C	Q21.3.2 C	Q21.3.3 D	Q21.3.4 D	Q21.3.5 C	Q21.3.6 D
Q21.3.7 C	Q21.3.8 D	Q21.3.9 D	Q21.3.10 A		
Session : 4					
Q21.4.1 C	Q21.4.2 B	Q21.4.3 C	Q21.4.4 C	Q21.4.5 B	Q21.4.6 C
Q21.4.7 D	Q21.4.8 A	Q21.4.9 C	Q21.4.10 A		
Session : 5					
Q21.5.1 C	Q21.5.2 A	Q21.5.3 C	Q21.5.4 A	Q21.5.5 C	Q21.5.6 C
Q21.5.7 C	Q21.5.8 C	Q21.5.9 C OR D	Q21.5.10 D		
Session : 6					
Q21.6.1 A	Q21.6.2 B	Q21.6.3 B	Q21.6.4 B	Q21.6.5 C	Q21.6.6 C
Q21.6.7 A	Q21.6.8 C	Q21.6.9 A	Q21.6.10 C		
Session : 7					
Q21.7.1 D	Q21.7.2 D	Q21.7.3 D	Q21.7.4 D	Q21.7.5 C	Q21.7.6 C
Q21.7.7 A	Q21.7.8 C	Q21.7.9 C	Q21.7.10 D		
Session : 8					
Q21.8.1 B	Q21.8.2 A	Q21.8.3 A	Q21.8.4 B	Q21.8.5 D	Q21.8.6 C
Q21.8.7 B	Q21.8.8 C	Q21.8.9 B	Q21.8.10 D		

YEAR

2020

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	IN, ME1, MT, PE, PH	10
Session 2	CY, ME2, PI	10
Session 3	AR, BM, BT, CH, MA, MN, ST, XE, XL	10
Session 4	AE, AG, EC, GG	10
Session 5	EE, EY, TF	10
Session 6	CS	10
Session 7	CE1	10
Session 8	CE2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 01/02/20

Q20.1.1 MCQ 2020 1M

He is known for his unscrupulous ways. He always sheds _____ tears to deceive people.

A. fox's **B.** crocodile's **C.** crocodile **D.** fox

Q20.1.2 MCQ 2020 1M

Jofra Archer, the England fast bowler, is _____ than accurate.

A. more fast **B.** faster **C.** less fast **D.** more faster

Q20.1.3 MCQ 2020 1M

Select the word that fits the analogy:
Build : Building :: Grow : _____

A. Grown **B.** Grew **C.** Growth **D.** Growed

Q20.1.4 MCQ 2020 1M

I do not think you know the case well enough to have opinions. Having said that, I agree with your other point. What does the phrase "having said that" mean in the given text?

A. as opposed to what I have said

B. despite what I have said

C. in addition to what I have said

D. contrary to what I have said

Q20.1.5 MCQ 2020 1M

Define $[x]$ as the greatest integer less than or equal to x, for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is _____.

A. 1 **B.** 3 **C.** 4 **D.** 6

Q20.1.6 MCQ 2020 2M

Crowd funding deals with mobilisation of funds for a project from a large number of people, who would be willing to invest smaller amounts through web-based platforms in the project.
Based on the above paragraph, which of the following is correct about crowd funding?

A. Funds raised through unwilling contributions on web-based platforms.

B. Funds raised through large contributions on web-based platforms.

C. Funds raised through coerced contributions on web-based platforms.

D. Funds raised through voluntary contributions on web-based platforms.

Q20.1.7 MCQ 2020 2M

P, Q, R and S are to be uniquely coded using α and β. If P is coded as $\alpha\alpha$ and Q as $\alpha\beta$, then R and S, respectively, can be coded as _____.

A. $\beta\alpha$ and $\alpha\beta$ **B.** $\beta\beta$ and $\alpha\alpha$

C. $\alpha\beta$ and $\beta\beta$ **D.** $\beta\alpha$ and $\beta\beta$

Q20.1.8 MCQ 2020 2M

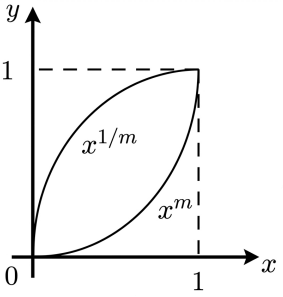
The sum of the first n terms in the sequence 8, 88, 888, 8888, . . . is _____.

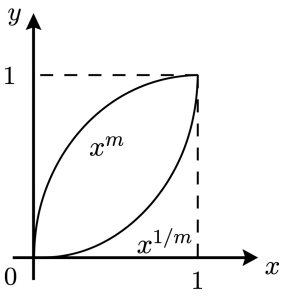
A. $\frac{81}{80}(10^n - 1) + \frac{9}{8}n$ **B.** $\frac{81}{80}(10^n - 1) - \frac{9}{8}n$

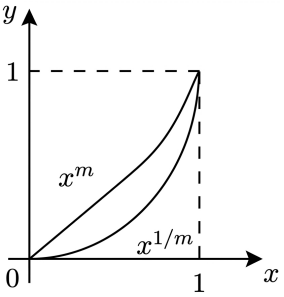
C. $\frac{80}{81}(10^n - 1) + \frac{8}{9}n$ **D.** $\frac{80}{81}(10^n - 1) - \frac{8}{9}n$

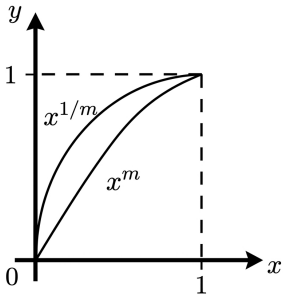
Q20.1.9 MCQ 2020 2M

Select the graph that schematically represents BOTH $y = x^m$ and $y = x^{1/m}$ properly in the interval $0 \le x \le 1$, for integer values of m, where $m > 1$.

A.


B.


C.


D.


Q20.1.10 MCQ 2020 2M

The bar graph shows the data of the students who appeared and passed in an examination for four schools P, Q, R and S. The average of success rates (in percentage) of these four schools is _____.

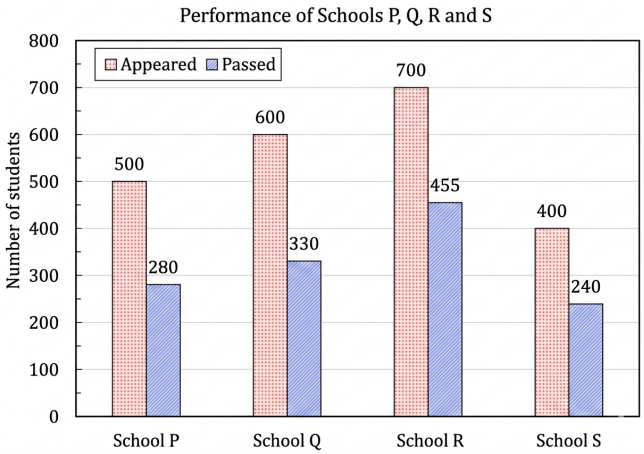


Fig. Q20.1.10

A. 58.5 % **B.** 58.8 % **C.** 59.0 % **D.** 59.3 %

Session : 2 | Afternoon | 01/02/20

Q20.2.1 MCQ 2020 1M

While I agree _____ his proposal this time, I do not often agree _____ him.

A. to, with **B.** with, to **C.** with, with **D.** to, to

Q20.2.2 MCQ 2020 1M

The recent measures to improve the output would _____ the level of production to our satisfaction.

A. increase **B.** decrease **C.** speed **D.** equalise

Q20.2.3 MCQ 2020 1M

Select the word that fits the analogy:
White: Whitening : : Light: _____

A. Lightning **B.** Lightening

C. Lighting **D.** Enlightening

Q20.2.4 MCQ 2020 1M

In one of the greatest innings ever seen in 142 years of Test history, Ben Stokes upped the tempo in a five-and-a-half hour long stay of 219 balls including 11 fours and 8 sixes that saw him finish on a 135 not out as England squared the five-match series.
Based on their connotations in the given passage, which one of the following meanings DOES NOT match?

A. upped = increased **B.** squared = lost

C. tempo = enthusiasm **D.** saw = resulted in

Q20.2.5 MCQ 2020 1M

There are five levels $\{P, Q, R, S, T\}$ in a linear supply chain before a product reaches customers, as shown in the figure.

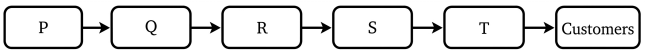


Fig. Q20.2.5

At each of the five levels, the price of the product is increased by 25%. If the product is produced at level P at the cost of Rs. 120 per unit, what is the price paid (in rupees) by the customers?

A. 187.50 **B.** 234.38 **C.** 292.96 **D.** 366.21

Q20.2.6 MCQ 2020 2M

Climate change and resilience deal with two aspects – reduction of sources of non-renewable energy resources and reducing vulnerability of climate change aspects. The terms 'mitigation' and 'adaptation' are used to refer to these aspects, respectively.
Which of the following assertions is best supported by the above information?

A. Mitigation deals with consequences of climate change

B. Adaptation deals with causes of climate change

C. Mitigation deals with actions taken to reduce the use of fossil fuels

D. Adaptation deals with actions taken to combat greenhouse gas emissions

Q20.2.7 MCQ 2020 2M

Find the missing element in the following figure.

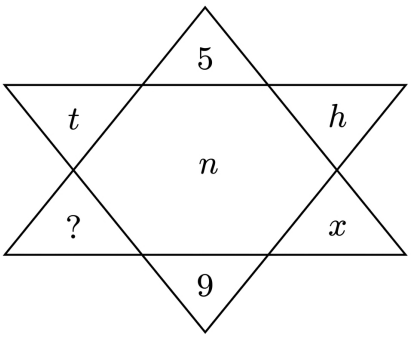


Fig. Q20.2.7

A. d **B.** e **C.** w **D.** y

Q20.2.8 MCQ 2020 2M

It was estimated that 52 men can complete a strip in a newly constructed highway connecting cities P and Q in 10 days. Due to an emergency, 12 men were sent to another project. How many number of days, more than the original estimate, will be required to complete the strip?

A. 3 days **B.** 5 days

C. 10 days **D.** 13 days

Q20.2.9 MCQ 2020 2M

An engineer measures THREE quantities X, Y and Z in an experiment. She finds that they follow a relationship that is represented in the figure below: (the product of X and Y linearly varies with Z)

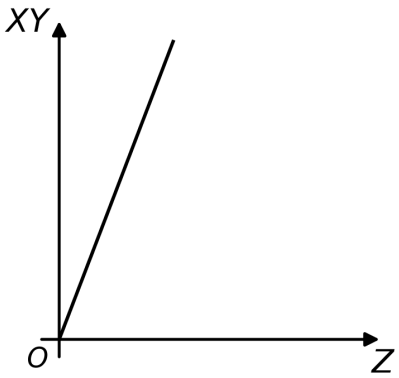


Fig. Q20.2.9

Then, which of the following statements is FALSE?

A. For fixed Z; X is proportional to Y

B. For fixed Y; X is proportional to Z

C. For fixed X; Z is proportional to Y

D. XY/Z is constant

Q20.2.10 MCQ 2020 2M

The two pie-charts given below show the data of total students and only girls registered in different streams in a university. If the total number of students registered in the university is 5000, and the total number of the registered girls is 1500; then, the ratio of boys enrolled in Arts to the girls enrolled in Management is _____.

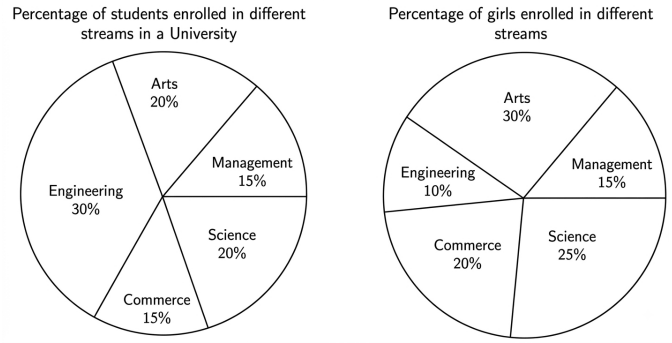


Fig. Q20.2.10

A. 2 : 1 **B.** 9 : 22 **C.** 11 : 9 **D.** 22 : 9

Session : 3 | Forenoon | 02/02/20

Q20.3.1 MCQ 2020 1M

Rajiv Gandhi Khel Ratna Award was conferred ____ Mary Kom, a six-time world champion in boxing, recently in a ceremony ____ the Rashtrapati Bhawan (the President's official residence) in New Delhi.

A. with, at **B.** on, in **C.** on, at **D.** to, at

Q20.3.2 MCQ 2020 1M

Despite a string of poor performances, the chances of K. L. Rahul's selection in the team are _____.

A. slim **B.** bright **C.** obvious **D.** uncertain

Q20.3.3 MCQ 2020 1M

Select the word that fits the analogy:
Cover : Uncover :: Associate : _____

A. Unassociate **B.** Inassociate

C. Misassociate **D.** Dissociate

Q20.3.4 MCQ 2020 1M

Hit by floods, the Kharif (summer sown) crops in various parts of the country have been affected. Officials believe that the loss in production of the kharif crops can be recovered in the output of the rabi (winter sown) crops so that the country can achieve its food-grain production target of 291 million tons in the crop year 2019-20 (July-June). They are hopeful that good rains in July-August will help the soil retain moisture for a longer period, helping winter sown crops such as wheat and pulses during the November-February period.
Which of the following statements can be inferred from the given passage?

A. Officials want the food-grain production target to be met by the November-February period.

B. Officials feel that the food-grain production target cannot be met due to floods.

C. Officials declared that the food-grain production target will be met due to good rains.

D. Officials hope that the food-grain production target will be met due to good rabi produce.

Q20.3.5 MCQ | 2020 | 1M

The difference between the sum of the first $2n$ natural numbers and the sum of the first n odd natural numbers is ___.

A. $n^2 - n$ **B.** $n^2 + n$ **C.** $2n^2 - n$ **D.** $2n^2 + n$

Q20.3.6 MCQ | 2020 | 2M

Repo rate is the rate at which Reserve Bank of India (RBI) lends commercial banks, and reverse repo rate is the rate at which RBI borrows money from commercial banks.
Which of the following statements can be inferred from the above passage?

A. Decrease in repo rate will increase cost of borrowing and decrease lending by commercial banks.

B. Increase in repo rate will decrease cost of borrowing and increase lending by commercial banks.

C. Increase in repo rate will decrease cost of borrowing and decrease lending by commercial banks.

D. Decrease in repo rate will decrease cost of borrowing and increase lending by commercial banks.

Q20.3.7 MCQ | 2020 | 2M

P, Q, R, S, T, U, V, and W are seated around a circular table.

I. S is seated opposite to W.
II. U is seated at the second place to the right of R.
III. T is seated at the third place to the left of R.
IV. V is a neighbour of S.

Which of the following must be true?

A. P is a neighbour of R.

B. Q is a neighbour of R.

C. P is not seated opposite to Q.

D. R is the left neighbour of S.

Q20.3.8 MCQ | 2020 | 2M

The distance between Delhi and Agra is 233 km. A car P started travelling from Delhi to Agra and another car Q started from Agra to Delhi along the same road 1 hour after the car P started. The two cars crossed each other 75 minutes after the car Q started. Both cars were travelling at constant speed. The speed of car P was 10 km/hr more than the speed of car Q. How many kilometers the car Q had travelled when the cars crossed each other?

A. 66.6 **B.** 75.2 **C.** 88.2 **D.** 116.5

Q20.3.9 MCQ | 2020 | 2M

For a matrix $M = [m_{ij}]; i, j = 1, 2, 3, 4$, the diagonal elements are all zero and $m_{ij} = -m_{ji}$.
The minimum number of elements required to fully specify the matrix is ____.

A. 0 **B.** 6 **C.** 12 **D.** 16

Q20.3.10 MCQ | 2020 | 2M

The profit shares of two companies P and Q are shown in the figure. If the two companies have invested a fixed and equal amount every year, then the ratio of the total revenue of company P to the total revenue of company Q, during 2013 - 2018 is _____.

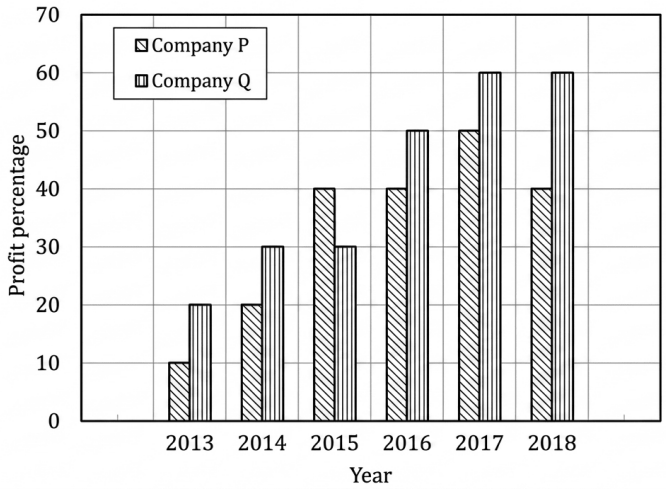


Fig. Q20.3.10

A. 15 : 17 **B.** 16 : 17 **C.** 17 : 15 **D.** 17 : 16

Session : 4 | Afternoon | 02/02/20

Q20.4.1 MCQ 2020 1M

The untimely loss of life is a cause of serious global concern as thousands of people get killed _____ accidents every year while many other die _____ diseases like cardio vascular disease, cancer, etc.

A. in, of **B.** from, of **C.** during, from **D.** from, from

Q20.4.2 MCQ 2020 1M

He was not only accused of theft _____ of conspiracy.

A. rather **B.** but also **C.** but even **D.** rather than

Q20.4.3 MCQ 2020 1M

Select the word that fits the analogy:
Explicit: Implicit :: Express: _____

A. Impress **B.** Repress **C.** Compress **D.** Suppress

Q20.4.4 MCQ 2020 1M

The Canadian constitution requires that equal importance be given to English and French. Last year, Air Canada lost a lawsuit, and had to pay a six-figure fine to a French-speaking couple after they filed complaints about formal in-flight announcements in English lasting 15 seconds, as opposed to informal 5 second messages in French.
The French-speaking couple were upset at _____.

A. the in-flight announcements being made in English.

B. the English announcements being clearer than the French ones.

C. the English announcements being longer than the French ones.

D. equal importance being given to English and French.

Q20.4.5 MCQ 2020 1M

A superadditive function $f(\cdot)$ satisfies the following property

$$f(x_1 + x_2) \geq f(x_1) + f(x_2)$$

Which of the following functions is a superadditive function for $x > 1$?

A. e^x **B.** $\sqrt{x}$ **C.** $1/x$ **D.** e^{-x}

Q20.4.6 MCQ 2020 2M

The global financial crisis in 2008 is considered to be the most serious world-wide financial crisis, which started with the sub-prime lending crisis in USA in 2007. The sub-prime lending crisis led to the banking crisis in 2008 with the collapse of Lehman Brothers in 2008. The sub-prime lending refers to the provision of loans to those borrowers who may have difficulties in repaying loans, and it arises because of excess liquidity following the East Asian crisis. Which one of the following sequences shows the correct precedence as per the given passage?

A. East Asian crisis→ subprime lending crisis → banking crisis→ global financial crisis.

B. Subprime lending crisis→ global financial crisis→ banking crisis→ East Asian crisis.

C. Banking crisis→ subprime lending crisis→ global financial crisis → East Asian crisis.

D. Global financial crisis → East Asian crisis → banking crisis → subprime lending crisis.

Q20.4.7 MCQ 2020 2M

It is quarter past three in your watch. The angle between the hour hand and the minute hand is _____.

A. 0° **B.** 7.5° **C.** 15° **D.** 22.5°

Q20.4.8 MCQ 2020 2M

A circle with centre O is shown in the figure. A rectangle PQRS of maximum possible area is inscribed in the circle. If the radius of the circle is a, then the area of the shaded portion is _____.

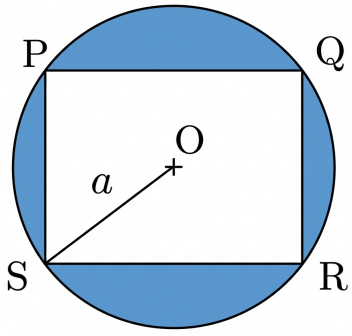


Fig. Q20.4.8

A. $\pi a^2 - a^2$ **B.** $\pi a^2 - \sqrt{2}a^2$

C. $\pi a^2 - 2a^2$ **D.** $\pi a^2 - 3a^2$

Q20.4.9 MCQ 2020 2M

a, b, c are real numbers. The quadratic equation $ax^2 - bx + c = 0$ has equal roots, which is β, then

A. $\beta = b/a$ **B.** $\beta^2 = ac$

C. $\beta^3 = bc/(2a^2)$ **D.** $b^2 \neq 4ac$

Q20.4.10 MCQ 2020 2M

The following figure shows the data of students enrolled in 5 years (2014 to 2018) for two schools P and Q. During this period, the ratio of the average number of the students enrolled in school P to the average of the difference of the number of students enrolled in schools P and Q is

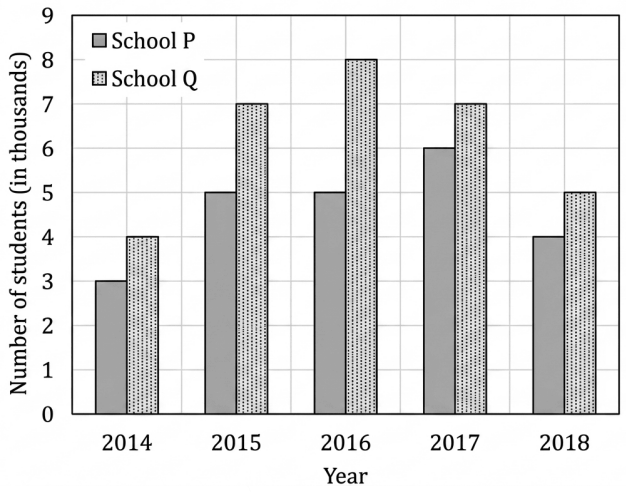


Fig. Q20.4.10

A. 8 : 23 **B.** 23 : 8 **C.** 23 : 31 **D.** 31 : 23

Session : 5 | Forenoon | 08/02/20

Q20.5.1 MCQ 2020 1M

This book, including all its chapters, ____ interesting. The students as well as the instructor ____ in agreement about it.

A. is, was **B.** are, are **C.** is, are **D.** were, was

Q20.5.2 MCQ 2020 1M

People were prohibited _____ their vehicles near the entrance of the main administrative building.

A. to park **B.** from parking

C. parking **D.** to have parked

Q20.5.3 MCQ 2020 1M

Select the word that fits the analogy:
Do : Undo :: Trust : _____

A. Entrust **B.** Intrust **C.** Distrust **D.** Untrust

Q20.5.4 MCQ 2020 1M

Stock markets _____ at the news of the coup.

A. poised **B.** plunged **C.** plugged **D.** probed

Q20.5.5 MCQ 2020 1M

If P, Q, R, S are four individuals, how many teams of size exceeding one can be formed, with Q as a member?

A. 5 **B.** 6 **C.** 7 **D.** 8

Q20.5.6 MCQ 2020 2M

Non-performing Assets (NPAs) of a bank in India is defined as an asset, which remains unpaid by a borrower for a certain period of time in terms of interest, principal, or both. Reserve Bank of India (RBI) has changed the definition of NPA thrice during 1993-2004, in terms of the holding period of loans. The holding period was reduced by one quarter each time. In 1993, the holding period was four quarters (360 days).
Based on the above paragraph, the holding period of loans in 2004 after the third revision was _____ days.

A. 45 **B.** 90 **C.** 135 **D.** 180

Q20.5.7 MCQ 2020 2M

Select the next element of the series: Z, WV, RQP, ____

A. LKJI **B.** JIHG **C.** KJIH **D.** NMLK

Q20.5.8 MCQ 2020 2M

In four-digit integer numbers from 1001 to 9999, the digit group "37" (in the same sequence) appears ____ times.

A. 270 **B.** 279 **C.** 280 **D.** 299

Q20.5.9 MCQ 2020 2M

Given a semicircle with O as the centre, as shown in the figure, the ratio $\frac{\overline{AC} + \overline{CB}}{\overline{AB}}$ is ____, where $\overline{AC}$, $\overline{CB}$ and $\overline{AB}$ are chords.

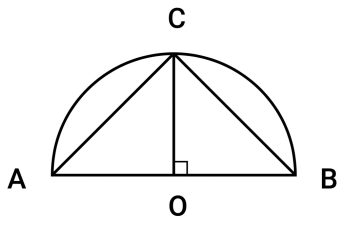


Fig. Q20.5.9

A. $\sqrt{2}$ **B.** $\sqrt{3}$ **C.** 2 **D.** 3

Q20.5.10 MCQ | 2020 | 2M

The revenue and expenditure of four different companies P, Q, R and S in 2015 are shown in the figure. If the revenue of company Q in 2015 was 20% more than that in 2014, and company Q had earned a profit of 10% on expenditure in 2014, then its expenditure (in million rupees) in 2014 was _____.

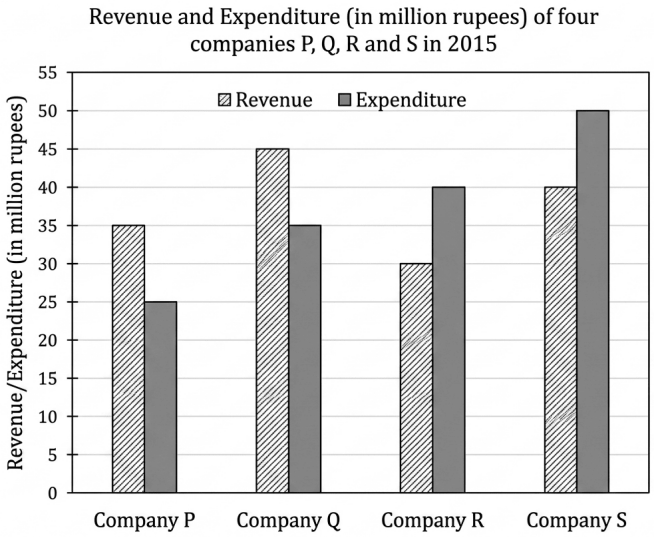


Fig. Q20.5.10

A. 32.7 **B.** 33.7 **C.** 34.1 **D.** 35.1

Session : 6 | Afternoon | 08/02/20

Q20.6.1 MCQ | 2020 | 1M

Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks.

A. during, for **B.** for, since **C.** for, in **D.** within, for

Q20.6.2 MCQ | 2020 | 1M

His knowledge of the subject was excellent but his classroom performance was _____.

A. extremely poor **B.** good

C. desirable **D.** praiseworthy

Q20.6.3 MCQ | 2020 | 1M

Select the word that fits the analogy:
Cook : Cook :: Fly : _____

A. Flyer **B.** Flying **C.** Flew **D.** Flighter

Q20.6.4 MCQ | 2020 | 1M

The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.
Which one of the following statements can be inferred from the given passage?

A. Nation-states do not have environmental concerns.

B. Nation-states are responsible for providing fresh water to billions of people.

C. Billions of people are responsible for man-made emissions.

D. Billions of people are affected by melting glaciers.

Q20.6.5 MCQ 2020 1M

There are multiple routes to reach from node 1 to node 2, as shown in the network.

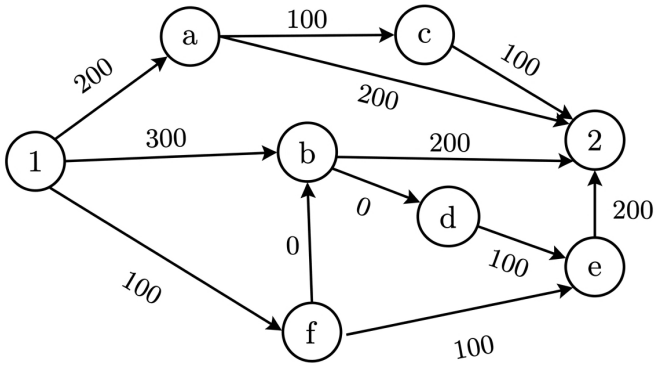


Fig. Q20.6.5

The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

A. 1-a-c-2 **B.** 1-f-b-2 **C.** 1-b-2 **D.** 1-f-e-2

Q20.6.6 MCQ 2020 2M

Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs). Which one of the following statements can be inferred from the given passage?

A. GST is imposed on the production of goods and services.

B. GST includes all indirect taxes.

C. GST does not have a component specific to UT.

D. GST is imposed at the point of usage of goods and services.

Q20.6.7 MCQ 2020 2M

If $P = 3$, $R = 27$, $T = 243$, then $Q + S =$ _____.

A. 40 **B.** 80 **C.** 90 **D.** 110

Q20.6.8 MCQ 2020 2M

The figure below shows an annular ring with outer and inner radii as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is _____.

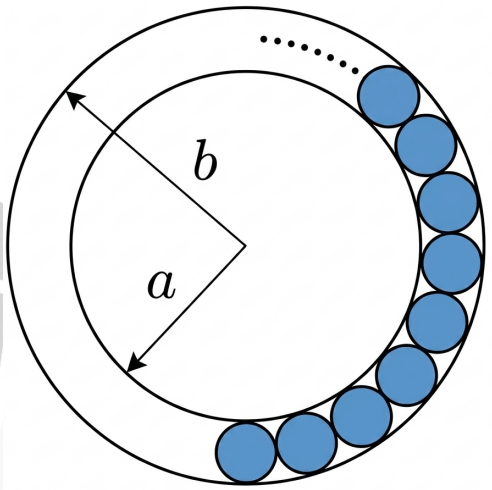


Fig. Q20.6.8

A. $\pi\left[(b^2 - a^2) - \frac{n}{4}(b-a)^2\right]$

B. $\pi\left[(b^2 - a^2) - n(b-a)^2\right]$

C. $\pi\left[(b^2 - a^2) + \frac{n}{4}(b-a)^2\right]$

D. $\pi\left[(b^2 - a^2) + n(b-a)^2\right]$

Q20.6.9 MCQ 2020 2M

Two straight lines are drawn perpendicular to each other in the X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha + \beta$ is _____.

A. 60° **B.** 90° **C.** 120° **D.** 180°

Q20.6.10 MCQ 2020 2M

The total revenue of a company during 2014-2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014-2018 is _____.

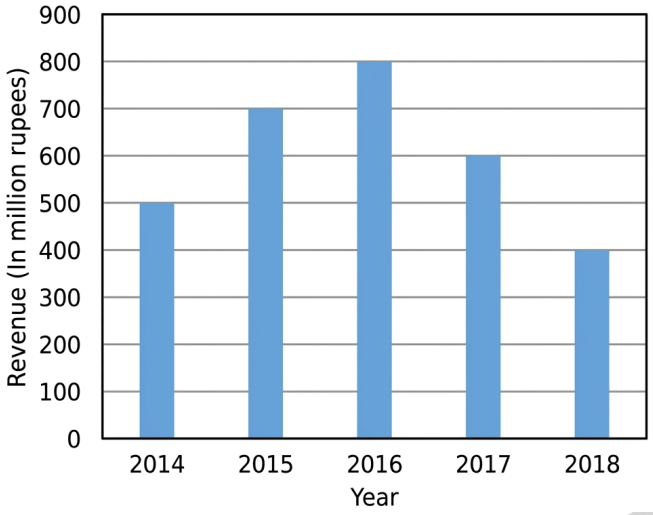


Fig. Q20.6.10

A. 16.67% profit **B.** 16.67% loss

C. 20% profit **D.** 20% loss

Session : 7 | Forenoon | 09/02/20

Q20.7.1 MCQ 2020 1M

It is a common criticism that most of the academicians live in their _____, so, they are not aware of the real life challenges.

A. homes **B.** ivory towers

C. glass palaces **D.** big flats

Q20.7.2 MCQ 2020 1M

His hunger for reading is insatiable. He reads indiscriminately. He is most certainly a/an _____ reader.

A. all-round **B.** precocious **C.** voracious **D.** wise

Q20.7.3 MCQ 2020 1M

Select the word that fits the analogy:
Fuse : Fusion :: Use : _____

A. Usage **B.** User **C.** Uses **D.** Usion

Q20.7.4 MCQ 2020 1M

If $0, 1, 2, \ldots, 7, 8, 9$ are coded as O, P, Q, . . . , V, W, X, then 45 will be coded as _____.

A. TS **B.** ST **C.** SS **D.** SU

Q20.7.5 MCQ 2020 1M

The sum of two positive numbers is 100. After subtracting 5 from each number, the product of the resulting numbers is 0. One of the original numbers is _____.

A. 80 **B.** 85 **C.** 90 **D.** 95

Q20.7.6 MCQ 2020 2M

The American psychologist Howard Gardner expounds that human intelligence can be sub-categorised into multiple kinds, in such a way that individuals differ with respect to their relative competence in each kind. Based on this theory, modern educationists insist on prescribing multi-dimensional curriculum and evaluation parameters that enable development and assessment of multiple intelligences. Which of the following statements can be inferred from the given text?

A. Howard Gardner insists that the teaching curriculum and evaluation needs to be multi-dimensional.

B. Howard Gardner wants to develop and assess the theory of multiple intelligences.

C. Modern educationists want to develop and assess the theory of multiple intelligences.

D. Modern educationists insist that the teaching curriculum and evaluation needs to be multi-dimensional.

Q20.7.7 MCQ 2020 2M

Five friends P, Q, R, S and T went camping. At night, they had to sleep in a row inside the tent. P, Q and T refused to sleep next to R since he snored loudly. P and S wanted to avoid Q as he usually hugged people in sleep.
Assuming everyone was satisfied with the sleeping arrangements, what is the order in which they slept?

A. RSPTQ **B.** SPRTQ **C.** QRSPT **D.** QTSPR

Q20.7.8 MCQ 2020 2M

Insert seven numbers between 2 and 34, such that the resulting sequence including 2 and 34 is an arithmetic progression. The sum of these inserted seven numbers is_____.

A. 120 **B.** 124 **C.** 126 **D.** 130

Q20.7.9 MCQ 2020 2M

The unit's place in 26591749^{110016} is _____.

A. 1 **B.** 3 **C.** 6 **D.** 9

Q20.7.10 MCQ 2020 2M

The total expenditure of a family, on different activities in a month, is shown in the pie-chart. The extra money spent on education as compared to transport (in percent) is _____.

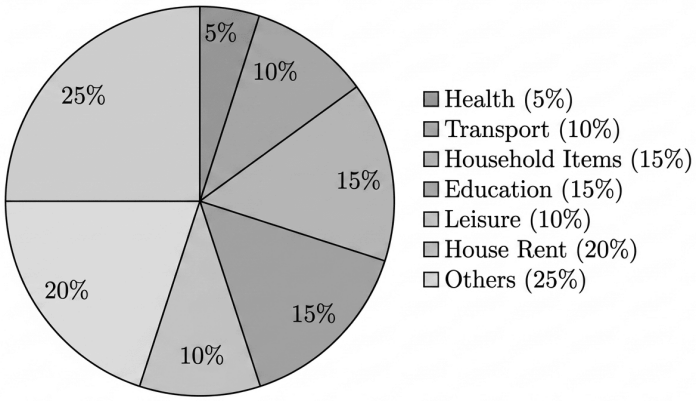


Fig. Q20.7.10

A. 5 **B.** 33.3 **C.** 50 **D.** 100

Session : 8 | Afternoon | 09/02/20

Q20.8.1 MCQ 2020 1M

Rescue teams deployed _____ disaster hit areas combat _____ a lot of difficulties to save the people.

A. with, at **B.** in, with **C.** with, with **D.** to, to

Q20.8.2 MCQ 2020 1M

Select the most appropriate word that can replace the underlined word without changing the meaning of the sentence.
Now-a-days, most children have a tendency to belittle the legitimate concerns of their parents.

A. disparage **B.** applaud **C.** reduce **D.** begrudge

Q20.8.3 MCQ 2020 1M

Select the word that fits the analogy:
Partial : Impartial :: Popular : _____

A. Impopular **B.** Dispopular

C. Mispopular **D.** Unpopular

Q20.8.4 MCQ 2020 1M

After the inauguration of the new building, the Head of the Department (HoD) collated faculty preferences for office space. *P* wanted a room adjacent to the lab. *Q* wanted to be close to the lift. *R* wanted a view of the playground and *S* wanted a corner office.
Assuming that everyone was satisfied, which among the following shows a possible allocation?

A.

	PLAYGROUND			
ROAD	HoD	Q	R	S
	P			LIFT
	LAB			

B.

	PLAYGROUND			
ROAD	S	R	P	HoD
	Q			LIFT
	LAB			

C.

	PLAYGROUND			
ROAD	S	R	HoD	Q
	P			LIFT
	LAB			

D.

	PLAYGROUND			
ROAD	HoD	S	R	Q
	P			LIFT
	LAB			

Q20.8.5 MCQ 2020 1M

If $f(x) = x^2$ for each $x \in (-\infty, \infty)$, then $\frac{f(f(f(x)))}{f(x)}$ is equal to _____.

A. $f(x)$ **B.** $(f(x))^2$ **C.** $(f(x))^3$ **D.** $(f(x))^4$

Q20.8.6 MCQ 2020 2M

Nominal interest rate is defined as the amount paid by the borrower to the lender for using the borrowed amount for a specific period of time. Real interest rate calculated on the basis of actual value (inflation-adjusted), is approximately equal to the difference between nominal rate and expected rate of inflation in the economy.
Which of the following assertions is best supported by the above information?

A. Under high inflation, real interest rate is low and borrowers get benefited.

B. Under low inflation, real interest rate is high and borrowers get benefited.

C. Under high inflation, real interest rate is low and lenders get benefited.

D. Under low inflation, real interest rate is low and borrowers get benefited.

Q20.8.7 MCQ 2020 2M

For the year 2019, which of the previous year's calendar can be used?

A. 2011 **B.** 2012 **C.** 2013 **D.** 2014

Q20.8.8 MCQ 2020 2M

The ratio of 'the sum of the odd positive integers from 1 to 100' to 'the sum of the even positive integers from 150 to 200' is _____.

A. 45 : 95 **B.** 1 : 2 **C.** 50 : 91 **D.** 1 : 1

Q20.8.9 MCQ 2020 2M

In a school of 1000 students, 300 students play chess and 600 students play football. If 50 students play both chess and football, the number of students who play neither is _____.

A. 200 **B.** 150 **C.** 100 **D.** 50

Q20.8.10 MCQ 2020 2M

The monthly distribution of 9 Watt LED bulbs sold by two firms X and Y from January to June 2018 is shown in the pie-chart and the corresponding table. If the total number of LED bulbs sold by two firms during April–June 2018 is 50000, then the number of LED bulbs sold by firm Y during April–June 2018 is _____.

Month	Ratio of LED bulbs sold by two firms (X : Y)
January	7 : 8
February	2 : 3
March	2 : 1
April	3 : 2
May	1 : 4
June	9 : 11

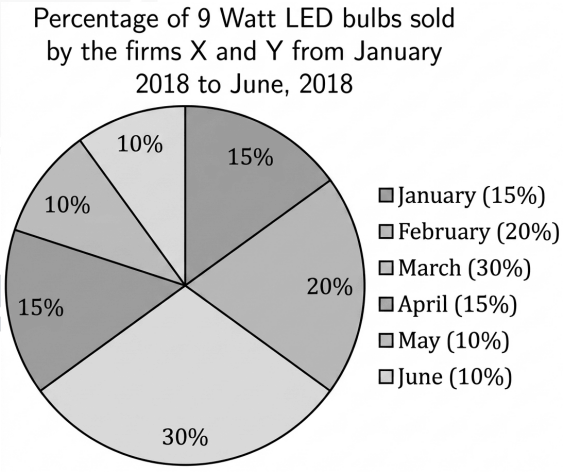


Fig. Q20.8.10

A. 11250 **B.** 9750 **C.** 8750 **D.** 8250

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q20.1.1 C	Q20.1.2 A	Q20.1.3 C	Q20.1.4 B	Q20.1.5 D	Q20.1.6 D
Q20.1.7 D	Q20.1.8 D	Q20.1.9 A	Q20.1.10 C		
Session : 2					
Q20.2.1 A	Q20.2.2 A	Q20.2.3 B	Q20.2.4 B	Q20.2.5 D	Q20.2.6 C
Q20.2.7 A	Q20.2.8 A	Q20.2.9 A	Q20.2.10 D		
Session : 3					
Q20.3.1 C	Q20.3.2 B	Q20.3.3 D	Q20.3.4 D	Q20.3.5 B	Q20.3.6 D
Q20.3.7 C	Q20.3.8 B	Q20.3.9 B	Q20.3.10 B		
Session : 4					
Q20.4.1 A	Q20.4.2 B	Q20.4.3 B	Q20.4.4 C	Q20.4.5 A	Q20.4.6 A
Q20.4.7 B	Q20.4.8 C	Q20.4.9 C	Q20.4.10 B		
Session : 5					
Q20.5.1 C	Q20.5.2 B	Q20.5.3 C	Q20.5.4 B	Q20.5.5 C	Q20.5.6 B
Q20.5.7 C	Q20.5.8 C	Q20.5.9 A	Q20.5.10 C		
Session : 6					
Q20.6.1 D	Q20.6.2 A	Q20.6.3 A	Q20.6.4 D	Q20.6.5 B	Q20.6.6 D
Q20.6.7 C	Q20.6.8 A	Q20.6.9 B	Q20.6.10 C		
Session : 7					
Q20.7.1 B	Q20.7.2 C	Q20.7.3 A	Q20.7.4 B	Q20.7.5 D	Q20.7.6 D
Q20.7.7 A	Q20.7.8 C	Q20.7.9 A	Q20.7.10 C		
Session : 8					
Q20.8.1 B	Q20.8.2 A	Q20.8.3 D	Q20.8.4 C	Q20.8.5 C	Q20.8.6 A
Q20.8.7 C	Q20.8.8 C	Q20.8.9 B	Q20.8.10 MTA		

YEAR

2019

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CY,ME-1,MN	10
Session 2	ME-2,XE,XL,TF	10
Session 3	CS,CH	10
Session 4	AE,AG,AR,BT,EY,GG,IN,MA,MT,PE,PH,PI,ST	10
Session 5	EC	10
Session 6	EE	10
Session 7	CE-1	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 02/02/19

Q19.1.1 MCQ 2019 1M

John Thomas, an _____ writer, passed away in 2018.

A. imminent **B.** prominent **C.** eminent **D.** dominant

Q19.1.2 MCQ 2019 1M

_____ I permitted him to leave, I wouldn't have had any problem with him being absent, _____ I?

A. Had, wouldn't **B.** Have, would

C. Had, would **D.** Have, wouldn't

Q19.1.3 MCQ 2019 1M

A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

A. 3.75 hours **B.** 4 hours and 15 mins

C. 8.5 hours **D.** 7.5 hours

Q19.1.4 MCQ 2019 1M

The sum and product of two integers are 26 and 165 respectively. The difference between these two integers is

A. 2 **B.** 3 **C.** 4 **D.** 6

Q19.1.5 MCQ 2019 1M

The minister avoided any mention of the issue of women's reservation in the private sector. He was accused of _____ the issue.

A. collaring **B.** skirting **C.** tying **D.** belting

Q19.1.6 MCQ 2019 2M

Under a certain legal system, prisoners are allowed to make one statement. If their statement turns out to be true then they are hanged. If the statement turns out to be false then they are shot. One prisoner made a statement and the judge had no option but to set him free. Which one of the following could be that statement?

A. I did not commit the crime

B. I committed the crime

C. I will be shot

D. You committed the crime

Q19.1.7 MCQ 2019 2M

A person divided an amount of Rs. 100,000 into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs. 120 less. Find the ratio between his investments in the two schemes.

A. 9 : 16 **B.** 11 : 14 **C.** 37 : 63 **D.** 47 : 53

Q19.1.8 MCQ 2019 2M

Congo was named by Europeans. Congo's dictator Mobuto later changed the name of the country and the river to Zaire with the objective of Africanising names of persons and spaces. However, the name Zaire was a Portuguese alteration of *Nzadi o Nzere*, a local African term meaning 'River that swallows Rivers'. Zaire was the Portuguese name for the Congo river in the 16th and 17th centuries. Which one of the following statements can be inferred from the paragraph above?

A. Mobuto was not entirely successful in Africanising the name of his country

B. The term *Nzadi o Nzere* was of Portuguese origin

C. Mobuto's desire to Africanise names was prevented by the Portuguese

D. As a dictator Mobuto ordered the Portuguese to alter the name of the river to Zaire

Q19.1.9 MCQ 2019 2M

A firm hires employees at five different skill levels P, Q, R, S, T. The shares of employment at these skill levels of total employment in 2010 is given in the pie chart as shown. There were a total of 600 employees in 2010 and the total employment increased by 15% from 2010 to 2016. The total employment at skill levels P, Q and R remained unchanged during this period. If the employment at skill level S increased by 40% from 2010 to 2016, how many employees were there at skill level T in 2016?

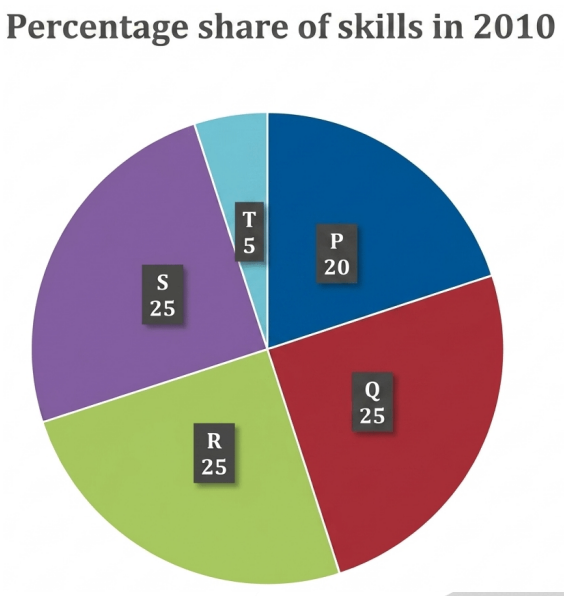


Fig. Q19.1.9

A. 30 **B.** 35 **C.** 60 **D.** 72

Q19.1.10 MCQ | 2019 | 2M

M and N had four children P, Q, R and S. Of them, only P and R were married. They had children X and Y respectively. If Y is a legitimate child of W, which one of the following statements is necessarily FALSE?

A. M is the grandmother of Y **B.** R is the father of Y

C. W is the wife of R **D.** W is the wife of P

Session : 2 | Afternoon | 02/02/19

Q19.2.1 MCQ | 2019 | 1M

Once the team of analysts identify the problem, we _____ in a better position to comment on the issue.
Which one of the following choices CANNOT fill the given blank?

A. will be **B.** were to be

C. are going to be **D.** might be

Q19.2.2 MCQ | 2019 | 1M

A final examination is the _____ of a series of evaluations that a student has to go through.

A. culmination **B.** consultation

C. desperation **D.** insinuation

Q19.2.3 MCQ | 2019 | 1M

If IMHO = JNIP; IDK = JEL; and SO = TP, then IDC = _____.

A. JDE **B.** JED **C.** JDC **D.** JCD

Q19.2.4 MCQ | 2019 | 1M

The product of three integers X, Y and Z is 192. Z is equal to 4 and P is equal to the average of X and Y. What is the minimum possible value of P?

A. 6 **B.** 7 **C.** 8 **D.** 9.5

Q19.2.5 MCQ | 2019 | 1M

Are there enough seats here? There are _____ people here than I expected.

A. many **B.** most **C.** least **D.** more

Q19.2.6 MCQ | 2019 | 2M

Fiscal deficit was 4% of the GDP in 2015 and that increased to 5% in 2016. If the GDP increased by 10% from 2015 to 2016, the percentage increase in the actual fiscal deficit is _____.

A. 37.50 **B.** 35.70 **C.** 25.00 **D.** 10.00

Q19.2.7 MCQ | 2019 | 2M

Two pipes P and Q can fill a tank in 6 hours and 9 hours respectively, while a third pipe R can empty the tank in 12 hours. Initially, P and R are open for 4 hours. Then P is closed and Q is opened. After 6 more hours R is closed. The total time taken to fill the tank (in hours) is _____.

A. 13.50 **B.** 14.50 **C.** 15.50 **D.** 16.50

Q19.2.8 MCQ | 2019 | 2M

While teaching a creative writing class in India, I was surprised at receiving stories from the students that were all set in distant places: in the American West with cowboys and in Manhattan penthouses with clinking ice cubes. This was, till an eminent Caribbean writer gave the writers in the once-colonised countries the confidence to see the shabby lives around them as worthy of being "told".
The writer of this passage is surprised by the creative writing assignments of his students, because _____.

A. Some of the students had written stories set in foreign places

B. None of the students had written stories set in India

C. None of the students had written about ice cubes and cowboys

D. Some of the students had written about ice cubes and cowboys

Q19.2.9 MCQ 2019 2M

Mola is a digital platform for taxis in a city. It offers three types of rides – Pool, Mini and Prime. The table below presents the number of rides for the past four months. The platform earns one US dollar per ride. What is the percentage share of revenue contributed by Prime to the total revenues of Mola, for the entire duration?

Type	Month			
	January	February	March	April
Pool	170	320	215	190
Mini	110	220	180	70
Prime	75	180	120	90

A. 16.24 **B.** 23.97 **C.** 25.86 **D.** 38.74

Q19.2.10 MCQ 2019 2M

X is an online media provider. By offering unlimited and exclusive online content at attractive prices for a loyalty membership, X is almost forcing its customers towards its loyalty membership. If its loyalty membership continues to grow at its current rate, within the next eight years more households will be watching X than cable television. Which one of the following statements can be inferred from the above paragraph?

A. Most households that subscribe to X's loyalty membership discontinue watching cable television

B. Non-members prefer to watch cable television

C. Cable television operators don't subscribe to X's loyalty membership

D. The X is cancelling accounts of non-members

Session : 3 | Forenoon | 03/02/19

Q19.3.1 MCQ 2019 1M

The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

A. break down **B.** break **C.** breaks down **D.** breaks

Q19.3.2 MCQ 2019 1M

The search engine's business model _____ around the fulcrum of trust.

A. revolves **B.** plays **C.** sinks **D.** bursts

Q19.3.3 MCQ 2019 1M

Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

A. 1 **B.** 2 **C.** 3 **D.** 6

Q19.3.4 MCQ 2019 1M

Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs. _____.

A. 666 **B.** 3000 **C.** 6000 **D.** 12000

Q19.3.5 MCQ 2019 1M

A court is to a judge as _____ is to a teacher.

A. a student **B.** a punishment

C. a syllabus **D.** a school

Q19.3.6 MCQ 2019 2M

The police arrested four criminals – P, Q, R and S. The criminals knew each other. They made the following statements:

P says "Q committed the crime."
Q says "S committed the crime."
R says "I did not do it."
S says "What Q said about me is false."

Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

A. P **B.** R **C.** S **D.** Q

Q19.3.7 MCQ 2019 2M

In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.

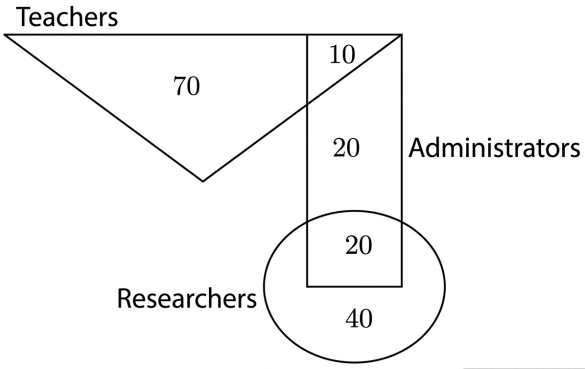


Fig. Q19.3.7

A. 0 to 15 **B.** 16 to 30 **C.** 31 to 45 **D.** 46 to 60

Q19.3.8 MCQ 2019 2M

"A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net."
Which one of the following statements can be inferred from the given passage?

A. Beggars are lazy people who beg because they are unwilling to work

B. Beggars are created because of the lack of social welfare schemes

C. Begging is an offence that has to be dealt with firmly

D. Begging has to be banned because it adversely affects the welfare of the state

Q19.3.9 MCQ 2019 2M

In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is

A. 1000 **B.** 975 **C.** 900 **D.** 225

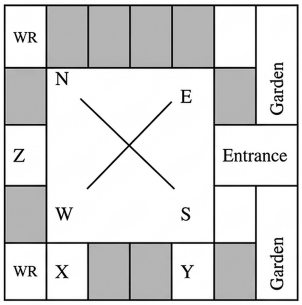
Q19.3.10 MCQ 2019 2M

Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

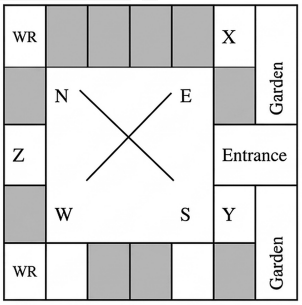
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.
Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.
Request by Z: I believe in Vaastu and so want to stay in the South-west wing.

The shaded rooms are already occupied. WR is washroom.

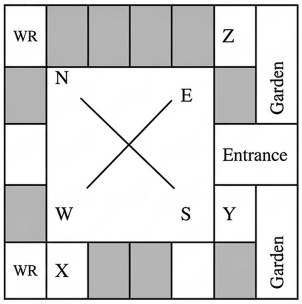
A.



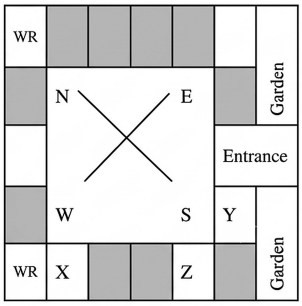
B.



C.



D.



Session : 4 | Afternoon | 03/02/19

Q19.4.1 MCQ 2019 1M

The fishermen, _____ the flood victims owed their lives, were rewarded by the government.

A. whom **B.** to which **C.** to whom **D.** that

Q19.4.2 MCQ 2019 1M

Some students were not involved in the strike.
If the above statement is true, which of the following conclusions is/are logically necessary?

1. Some who were involved in the strike were students.
2. No student was involved in the strike.
3. At least one student was involved in the strike.
4. Some who were not involved in the strike were students.

A. 1 and 2 **B.** 3 **C.** 4 **D.** 2 and 3

Q19.4.3 MCQ 2019 1M

The radius as well as the height of a circular cone increases by 10%. The percentage increase in its volume is_____.

A. 17.1 **B.** 21.0 **C.** 33.1 **D.** 72.8

Q19.4.4 MCQ 2019 1M

Five numbers 10, 7, 5, 4 and 2 are to be arranged in a sequence from left to right following the directions given below:

1. No two odd or even numbers are next to each other.
2. The second number from the left is exactly half of the left-most number.
3. The middle number is exactly twice the right-most number.

Which is the second number from the right?

A. 2 **B.** 4 **C.** 7 **D.** 10

Q19.4.5 MCQ 2019 1M

Until Iran came along, India had never been _____ in kabaddi.

A. defeated **B.** defeating **C.** defeat **D.** defeatist

Q19.4.6 MCQ 2019 2M

Since the last one year, after a 125 basis point reduction in repo rate by the Reserve Bank of India, banking institutions have been making a demand to reduce interest rates on small saving schemes. Finally, the government announced yesterday a reduction in interest rates on small saving schemes to bring them on par with fixed deposit interest rates.
Which one of the following statements can be inferred from the given passage?

A. Whenever the Reserve Bank of India reduces the repo rate, the interest rates on small saving schemes are also reduced

B. Interest rates on small saving schemes are always maintained on par with fixed deposit interest rates

C. The government sometimes takes into consideration the demands of banking institutions before reducing the interest rates on small saving schemes

D. A reduction in interest rates on small saving schemes follow only after a reduction in repo rate by the Reserve Bank of India

Q19.4.7 MCQ 2019 2M

In a country of 1400 million population, 70% own mobile phones. Among the mobile phone owners, only 294 million access the Internet. Among these Internet users, only half buy goods from e-commerce portals. What is the percentage of these buyers in the country?

A. 10.50 **B.** 14.70 **C.** 15.00 **D.** 50.00

Q19.4.8 MCQ 2019 2M

The nomenclature of Hindustani music has changed over the centuries. Since the medieval period *dhrupad* styles were identified as *baanis*. Terms like *gayaki* and *baaj* were used to refer to vocal and instrumental styles, respectively. With the institutionalization of music education the term *gharana* became acceptable. *Gharana* originally referred to hereditary musicians from a particular lineage, including disciples and grand disciples.
Which one of the following pairings is NOT correct?

A. *dhrupad*, *baani* **B.** *gayaki*, vocal

C. *baaj*, institution **D.** *gharana*, lineage

Q19.4.9 MCQ 2019 2M

Two trains started at 7AM from the same point. The first train travelled north at a speed of 80 km/h and the second train travelled south at a speed of 100 km/h. The time at which they were 540 km apart is ____ AM.

A. 9 **B.** 10 **C.** 11 **D.** 11.30

Q19.4.10 MCQ 2019 2M

"I read somewhere that in ancient times the prestige of a kingdom depended upon the number of taxes that it was able to levy on its people. It was very much like the prestige of a head-hunter in his own community."
Based on the paragraph above, the prestige of a head-hunter depended upon _____

A. the prestige of the kingdom

B. the prestige of the heads

C. the number of taxes he could levy

D. the number of heads he could gather

Session : 5 | Forenoon | 09/02/19

Q19.5.1 MCQ 2019 1M

The strategies that the company _____ to sell its products _____ house-to-house marketing.

A. use, includes **B.** uses, include

C. used, includes **D.** uses, including

Q19.5.2 MCQ 2019 1M

The boat arrived _____ dawn.

A. in **B.** at **C.** on **D.** under

Q19.5.3 MCQ 2019 1M

It would take one machine 4 hours to complete a production order and another machine 2 hours to complete the same order. If both machines work simultaneously at their respective constant rates, the time taken to complete the same order is _____ hours.

A. 2/3 **B.** 3/4 **C.** 4/3 **D.** 7/3

Q19.5.4 MCQ 2019 1M

Five different books (P, Q, R, S, T) are to be arranged on a shelf. The books R and S are to be arranged first and second, respectively from the right side of the shelf. The number of different orders in which P, Q and T may be arranged is _____.

A. 2 **B.** 6 **C.** 12 **D.** 120

Q19.5.5 MCQ 2019 1M

When he did not come home, she _____ him lying dead on the roadside somewhere.

A. concluded **B.** looked **C.** notice **D.** pictured

Q19.5.6 MCQ 2019 2M

Four people are standing in a line facing you. They are Rahul, Mathew, Seema and Lohit. One is an engineer, one is a doctor, one a teacher and another a dancer. You are told that:

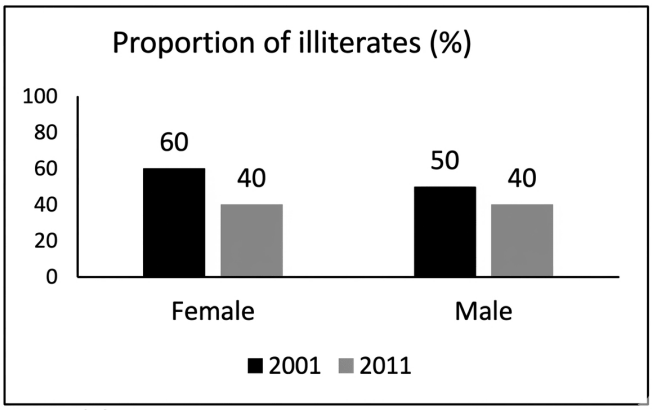
1. Mathew is not standing next to Seema
2. There are two people standing between Lohit and the engineer
3. Rahul is not a doctor
4. The teacher and the dancer are standing next to each other
5. Seema is turning to her right to speak to the doctor standing next to her

Who among them is an engineer?

A. Seema **B.** Lohit **C.** Rahul **D.** Mathew

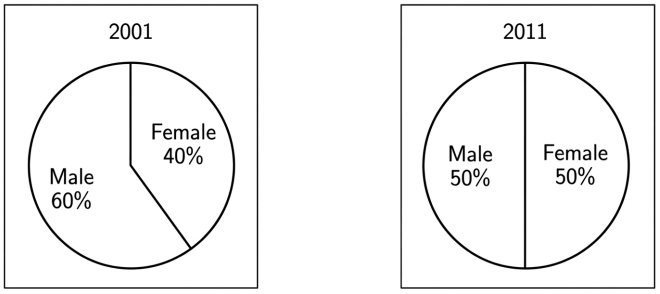
Q19.5.7 MCQ 2019 2M

The bar graph in Panel (a) shows the proportion of male and female illiterates in 2001 and 2011. The proportions of males and females in 2001 and 2011 are given in Panel (b) and (c), respectively. The total population did not change during this period.
The percentage increase in the total number of literates from 2001 to 2011 is _____.



Panel (a)

Fig. Q19.5.7



Panel (b) Panel (c)

Fig. Q19.5.7

A. 30.43 **B.** 33.43 **C.** 34.43 **D.** 35.43

Q19.5.8 MCQ 2019 2M

"Indian history was written by British historians – extremely well documented and researched, but not always impartial. History had to serve its purpose: Everything was made subservient to the glory of the Union Jack. Latter-day Indian scholars presented a contrary picture."
From the text above, we can infer that:
Indian history written by British historians _____

A. was well documented and not researched but was always biased

B. was not well documented and researched and was always biased

C. was well documented and researched but was sometimes biased

D. was not well documented and researched and was sometimes biased

Q19.5.9 MCQ 2019 2M

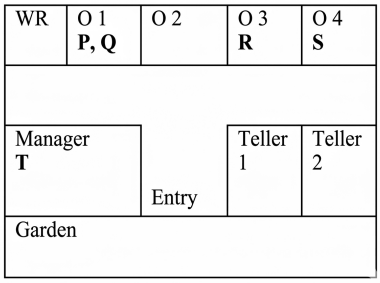
Two design consultants, P and Q, started working from 8 AM for a client. The client budgeted a total of USD 3000 for the consultants. P stopped working when the hour hand moved by 210 degrees on the clock. Q stopped working when the hour hand moved by 240 degrees. P took two tea breaks of 15 minutes each during her shift, but took no lunch break. Q took only one lunch break for 20 minutes, but no tea breaks. The market rate for consultants is USD 200 per hour and breaks are not paid. After paying the consultants, the client shall have USD _____ remaining in the budget.

A. 000.00 **B.** 166.67 **C.** 300.00 **D.** 433.33

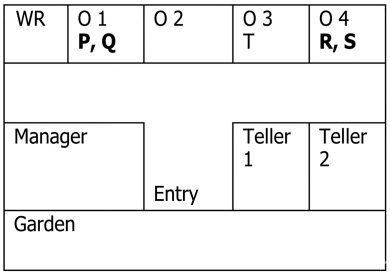
Q19.5.10 MCQ 2019 2M

Five people P, Q, R, S and T work in a bank. P and Q don't like each other but have to share an office till T gets a promotion and moves to the big office next to the garden. R, who is currently sharing an office with T wants to move to the adjacent office with S, the handsome new intern. Given the floor plan, what is the current location of Q, R and T? (O = Office, WR = Washroom)

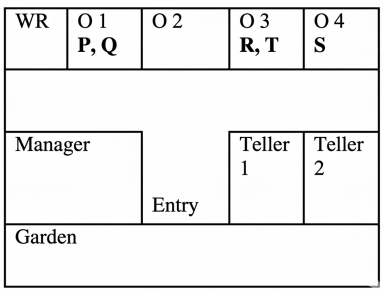
A.



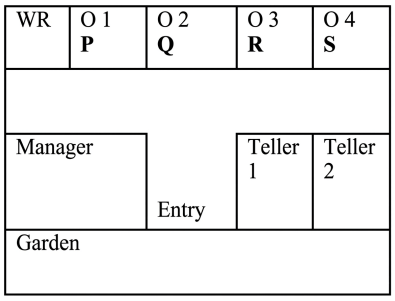
B.



C.



D.



Session : 6 | Afternoon | 09/02/19

Q19.6.1 MCQ 2019 1M

I am not sure if the bus that has been booked will be able to _____ all the students.

A. sit **B.** deteriorate **C.** fill **D.** accommodate

Q19.6.2 MCQ 2019 1M

The passengers were angry _____ the airline staff about the delay.

A. on **B.** about **C.** with **D.** towards

Q19.6.3 MCQ 2019 1M

The missing number in the given sequence 343, 1331, ____, 4913 is

A. 3375 **B.** 2744 **C.** 2197 **D.** 4096

Q19.6.4 MCQ 2019 1M

It takes two hours for a person X to mow the lawn. Y can mow the same lawn in four hours. How long (in minutes) will it take X and Y, if they work together to mow the lawn?

A. 60 **B.** 80 **C.** 90 **D.** 120

Q19.6.5 MCQ 2019 1M

Newspapers are a constant source of delight and recreation for me. The ____ trouble is that I read ____ many of them.

A. even, quite **B.** even, too **C.** only, quite **D.** only, too

Q19.6.6 MCQ 2019 2M

How many integers are there between 100 and 1000 all of whose digits are even?

A. 60 **B.** 80 **C.** 100 **D.** 90

Q19.6.7 MCQ 2019 2M

The ratio of the number of boys and girls who participated in an examination is 4:3. The total percentage of candidates who passed the examination is 80 and the percentage of girls who passed is 90. The percentage of boys who passed is _____.

A. 55.50 **B.** 72.50 **C.** 80.50 **D.** 90.00

Q19.6.8 MCQ 2019 2M

An award-winning study by a group of researchers suggests that men are as prone to buying on impulse as women but women feel more guilty about shopping.
Which one of the following statements can be inferred from the given text?

A. Some men and women indulge in buying on impulse

B. All men and women indulge in buying on impulse

C. Few men and women indulge in buying on impulse

D. Many men and women indulge in buying on impulse

Q19.6.9 MCQ 2019 2M

Given two sets $X = \{1, 2, 3\}$ and $Y = \{2, 3, 4\}$, we construct a set Z of all possible fractions where the numerators belong to set X and the denominators belong to set Y. The product of elements having minimum and maximum values in the set Z is _____.

A. 1/12 **B.** 1/8 **C.** 1/6 **D.** 3/8

Q19.6.10 MCQ 2019 2M

Consider five people – Mita, Ganga, Rekha, Lakshmi and Sana. Ganga is taller than both Rekha and Lakshmi. Lakshmi is taller than Sana. Mita is taller than Ganga.
Which of the following conclusions are true?

1. Lakshmi is taller than Rekha
2. Rekha is shorter than Mita
3. Rekha is taller than Sana
4. Sana is shorter than Ganga

A. 1 and 3 **B.** 3 only **C.** 2 and 4 **D.** 1 only

Session : 7 | Forenoon | 10/02/19

Q19.7.1 MCQ 2019 1M

The lecture was attended by quite _____ students, so the hall was not very _____.

A. a few, quite **B.** few, quiet

C. a few, quiet **D.** few, quite

Q19.7.2 MCQ 2019 1M

They have come a long way in _____ trust among the users.

A. creating **B.** created **C.** creation **D.** create

Q19.7.3 MCQ 2019 1M

On a horizontal ground, the base of a straight ladder is 6 m away from the base of a vertical pole. The ladder makes an angle of 45° to the horizontal. If the ladder is resting at a point located at one-fifth of the height of the pole from the bottom, the height of the pole is ____ meters.

A. 15 **B.** 25 **C.** 30 **D.** 35

Q19.7.4 MCQ 2019 1M

If E = 10; J = 20; O = 30; and T = 40, what will be P + E + S + T?

A. 51 **B.** 82 **C.** 120 **D.** 164

Q19.7.5 MCQ 2019 1M

The CEO's decision to quit was as shocking to the Board as it was to _____.

A. I **B.** me **C.** my **D.** myself

Q19.7.6 MCQ 2019 2M

The new cotton technology, Bollgard-II, with herbicide tolerant traits has developed into a thriving business in India. However, the commercial use of this technology is not legal in India. Notwithstanding that, reports indicate that the herbicide tolerant Bt cotton had been purchased by farmers at an average of Rs 200 more than the control price of ordinary cotton, and planted in 15% of the cotton growing area in the 2017 Kharif season.
Which one of the following statements can be inferred from the given passage?

A. Farmers want to access the new technology if India benefits from it

B. Farmers want to access the new technology even if it is not legal

C. Farmers want to access the new technology for experimental purposes

D. Farmers want to access the new technology by paying high price

Q19.7.7 MCQ 2019 2M

In a sports academy of 300 people, 105 play only cricket, 70 play only hockey, 50 play only football, 25 play both cricket and hockey, 15 play both hockey and football and 30 play both cricket and football. The rest of them play all three sports. What is the percentage of people who play at least two sports?

A. 23.30 **B.** 25.00 **C.** 28.00 **D.** 50.00

Q19.7.8 MCQ 2019 2M

"The increasing interest in tribal characters might be a mere coincidence, but the timing is of interest. None of this, though, is to say that the tribal hero has arrived in Hindi cinema, or that the new crop of characters represents the acceptance of the tribal character in the industry. The films and characters are too few to be described as a pattern."
What does the word 'arrived' mean in the paragraph above?

A. reached a terminus **B.** came to a conclusion

C. attained a status **D.** went to a place

Q19.7.9 MCQ 2019 2M

A square has sides 5 cm smaller than the sides of a second square. The area of the larger square is four times the area of the smaller square. The side of the larger square is ___ cm.

A. 18.50 **B.** 15.10 **C.** 10.00 **D.** 8.50

Q19.7.10 MCQ 2019 2M

P, Q, R, S and T are related and belong to the same family. P is the brother of S. Q is the wife of P. R and T are the children of the siblings P and S respectively. Which one of the following statements is necessarily FALSE?

A. S is the aunt of R **B.** S is the aunt of T

C. S is the sister-in-law of Q **D.** S is the brother of P

Session : 8 | Afternoon | 10/02/19

Q19.8.1 MCQ 2019 1M

Daytime temperatures in Delhi can _____ 40°C.

A. get **B.** stand **C.** reach **D.** peak

Q19.8.2 MCQ 2019 1M

The growth rate of ABC Motors in 2017 was the same _____ XYZ Motors in 2016.

A. as off **B.** as those of **C.** as that off **D.** as that of

Q19.8.3 MCQ 2019 1M

Suresh wanted to lay a new carpet in his new mansion with an area of 70×55 sq. mts. However an area of 550 sq. mts. had to be left out for flower pots. If the cost of carpet is Rs. 50 per sq. mts., how much money (in Rs.) will be spent by Suresh for the carpet now?

A. Rs. 1,65,000 **B.** Rs. 1,92,500

C. Rs. 2,75,000 **D.** Rs. 1,27,500

Q19.8.4 MCQ 2019 1M

A retaining wall with measurements $30\,m \times 12\,m \times 6\,m$ was constructed with bricks of dimensions $8\,cm \times 6\,cm \times 6\,cm$. If 60% of the wall consists of bricks, the number of bricks used for the construction is ____ lakhs.

A. 30 **B.** 40 **C.** 45 **D.** 75

Q19.8.5 MCQ 2019 1M

Hima Das was ____ only Indian athlete to win ____ gold for India.

A. the, many **B.** the, a **C.** an, a **D.** an, the

Q19.8.6 MCQ 2019 2M

Mohan, the manager, wants his four workers to work in pairs. No pair should work for more than 5 hours. Ram and John have worked together for 5 hours. Krishna and Amir have worked as a team for 2 hours. Krishna does not want to work with Ram. Whom should Mohan allot to work with John, if he wants all the workers to continue working?

A. Amir **B.** Krishna **C.** Ram **D.** None of the three

Q19.8.7 MCQ 2019 2M

Population of state X increased by x% and the population of state Y increased by y% from 2001 to 2011. Assume that x is greater than y. Let P be the ratio of the population of state X to state Y in a given year. The percentage increase in P from 2001 to 2011 is _____.

A. $\frac{x}{y}$ **B.** $x - y$ **C.** $\frac{100(x-y)}{100+x}$ **D.** $\frac{100(x-y)}{100+y}$

Q19.8.8 MCQ 2019 2M

The Newspaper reports that over 500 hectares of tribal land spread across 28 tribal settlements in Mohimitampuram forest division have already been "alienated". A top forest official said, "First the tribals are duped out of their land holdings. Second, the families thus rendered landless are often forced to encroach further into the forests".
On the basis of the information available in the paragraph, _____ is/are responsible for duping the tribals.

A. forest officials **B.** landless families

C. The Newspaper **D.** it cannot be inferred who

Q19.8.9 MCQ 2019 2M

An oil tank can be filled by pipe X in 5 hours and pipe Y in 4 hours, each pump working on its own. When the oil tank is full and the drainage hole is open, the oil is drained in 20 hours. If initially the tank was empty and someone started the two pumps together but left the drainage hole open, how many hours will it take for the tank to be filled? (Assume that the rate of drainage is independent of the Head)

A. 1.50 **B.** 2.00 **C.** 2.50 **D.** 4.00

Q19.8.10 MCQ 2019 2M

"Popular Hindi fiction, despite – or perhaps because of – its wide reach, often does not appear in our cinema. As ideals that viewers are meant to look up to rather than identify with, Hindi film protagonists usually read books of aspirational value: textbooks, English books, or high value literature."
Which one of the following CANNOT be inferred from the paragraph above?

A. Though popular Hindi fiction has wide reach, it often does not appear in the movies

B. Protagonists in Hindi movies, being ideals for viewers, read only books of aspirational value

C. Textbooks, English books or high literature have aspirational value, but not popular Hindi fiction

D. People do not look up to writers of textbooks, English books, or high value literature

PrepFusion

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q19.1.1 C	Q19.1.2 C	Q19.1.3 D	Q19.1.4 C	Q19.1.5 B	Q19.1.6 C
Q19.1.7 D	Q19.1.8 A	Q19.1.9 C	Q19.1.10 D		
Session : 2					
Q19.2.1 B	Q19.2.2 A	Q19.2.3 B	Q19.2.4 MTA	Q19.2.5 D	Q19.2.6 A
Q19.2.7 B	Q19.2.8 B	Q19.2.9 B	Q19.2.10 A		
Session : 3					
Q19.3.1 C	Q19.3.2 A	Q19.3.3 B	Q19.3.4 C	Q19.3.5 D	Q19.3.6 B
Q19.3.7 C	Q19.3.8 B	Q19.3.9 C	Q19.3.10 D		
Session : 4					
Q19.4.1 C	Q19.4.2 C	Q19.4.3 C	Q19.4.4 C	Q19.4.5 A	Q19.4.6 C
Q19.4.7 A	Q19.4.8 C	Q19.4.9 B	Q19.4.10 D		
Session : 5					
Q19.5.1 B	Q19.5.2 B	Q19.5.3 C	Q19.5.4 B	Q19.5.5 D	Q19.5.6 D
Q19.5.7 A	Q19.5.8 C	Q19.5.9 B	Q19.5.10 C		
Session : 6					
Q19.6.1 D	Q19.6.2 C	Q19.6.3 C	Q19.6.4 B	Q19.6.5 D	Q19.6.6 C
Q19.6.7 B	Q19.6.8 A	Q19.6.9 D	Q19.6.10 C		
Session : 7					
Q19.7.1 C	Q19.7.2 A	Q19.7.3 C	Q19.7.4 C	Q19.7.5 B	Q19.7.6 B
Q19.7.7 B	Q19.7.8 C	Q19.7.9 C	Q19.7.10 B		
Session : 8					
Q19.8.1 C	Q19.8.2 D	Q19.8.3 A	Q19.8.4 C	Q19.8.5 B	Q19.8.6 B
Q19.8.7 D	Q19.8.8 D	Q19.8.9 C	Q19.8.10 B or D		

YEAR

2018

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, EY, PE, XE, XL	10
Session 2	ME-2, AE, MA ,PI	10
Session 3	CS, MN	10
Session 4	AG, AR, BT, CH, CY, GG, IN, MT, PH, TF	10
Session 5	EC	10
Session 6	EE	10
Session 7	CE-1	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 03/02/18

Q18.1.1 MCQ 2018 1M

"Going by the _____ that many hands make light work, the school _____ involved all the students in the task."
The words that best fill the blanks in the above sentence are

A. principle, principal **B.** principal, principle

C. principle, principle **D.** principal, principal

Q18.1.2 MCQ 2018 1M

"Her _____ should not be confused with miserliness; she is ever willing to assist those in need."
The word that best fills the blank in the above sentence is

A. cleanliness **B.** punctuality

C. frugality **D.** greatness

Q18.1.3 MCQ 2018 1M

Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys?

A. 1 **B.** 7 **C.** 100 **D.** 700

Q18.1.4 MCQ 2018 1M

A rectangle becomes a square when its length and breadth are reduced by 10 m and 5 m, respectively. During this process, the rectangle loses 650 m^2 of area. What is the area of the original rectangle in square meters?

A. 1125 **B.** 2250 **C.** 2924 **D.** 4500

Q18.1.5 MCQ 2018 1M

A number consists of two digits. The sum of the digits is 9. If 45 is subtracted from the number, its digits are interchanged. What is the number?

A. 63 **B.** 72 **C.** 81 **D.** 90

Q18.1.6 MCQ 2018 2M

For integers a, b and c, what would be the minimum and maximum values respectively of $a+b+c$ if $\log|a|+\log|b|+\log|c| = 0$?

A. -3 and 3 **B.** -1 and 1 **C.** -1 and 3 **D.** 1 and 3

Q18.1.7 MCQ 2018 2M

Given that a and b are integers and $a + a^2 b^3$ is odd, which one of the following statements is correct?

A. a and b are both odd **B.** a and b are both even

C. a is even and b is odd **D.** a is odd and b is even

Q18.1.8 MCQ 2018 2M

From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while travelling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same direction as the train. What is the length of the train and that of the platform in meters, respectively?

A. 210 and 140 **B.** 162.5 and 187.5

C. 245 and 130 **D.** 175 and 200

Q18.1.9 MCQ 2018 2M

Which of the following functions describe the graph shown in the below figure?

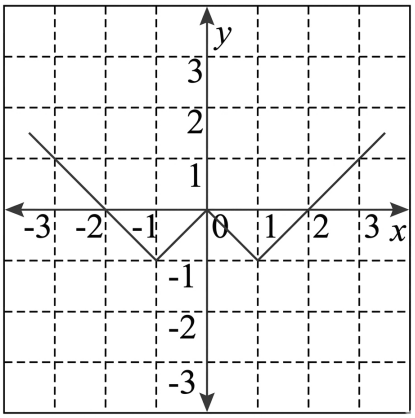


Fig. Q18.1.9

A. $y = \big||x| + 1\big| - 2$ **B.** $y = \big||x| - 1\big| - 1$

C. $y = \big||x| + 1\big| - 1$ **D.** $y = \big||x - 1| - 1\big|$

Q18.1.10 MCQ 2018 2M

Consider the following three statements:

(i) Some roses are red.
(ii) All red flowers fade quickly.
(iii) Some roses fade quickly.

Which of the following statements can be logically inferred from the above statements?

A. If (i) is true and (ii) is false, then (iii) is false.

B. If (i) is true and (ii) is false, then (iii) is true.

C. If (i) and (ii) are true, then (iii) is true.

D. If (i) and (ii) are false, then (iii) is false.

Session : 2 | Afternoon | 03/02/18

Q18.2.1 MCQ 2018 1M

"The dress _____ her so well that they all immediately _____ her on her appearance."
The words that best fill the blanks in the above sentence are

A. complemented, complemented

B. complimented, complemented

C. complimented, complimented

D. complemented, complimented

Q18.2.2 MCQ 2018 1M

"The judge's standing in the legal community, though shaken by false allegations of wrongdoing, remained ____."
The word that best fills the blank in the above sentence is

A. undiminished **B.** damaged

C. illegal **D.** uncertain

Q18.2.3 MCQ 2018 1M

Find the missing group of letters in the following series:
BC, FGH, LMNO, _____

A. UVWXY **B.** TUVWX **C.** STUVW **D.** RSTUV

Q18.2.4 MCQ 2018 1M

The perimeters of a circle, a square and an equilateral triangle are equal.
Which one of the following statements is true?

A. The circle has the largest area.

B. The square has the largest area.

C. The equilateral triangle has the largest area.

D. All the three shapes have the same area.

Q18.2.5 MCQ 2018 1M

The value of the expression $\frac{1}{1+\log_u vw} + \frac{1}{1+\log_v wu} + \frac{1}{1+\log_w uv}$ is _____.

A. -1 **B.** 0 **C.** 1 **D.** 3

Q18.2.6 MCQ 2018 2M

Forty students watched films A, B and C over a week. Each student watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C.
How many students watched all three films?

A. 0 **B.** 2 **C.** 4 **D.** 8

Q18.2.7 MCQ 2018 2M

A wire would enclose an area of 1936 m^2, if it is bent into a square. The wire is cut into two pieces. The longer piece is thrice as long as the shorter piece. The long and the short pieces are bent into a square and a circle, respectively. Which of the following choices is closest to the sum of the areas enclosed by the two pieces in square meters?

A. 1096 **B.** 1111 **C.** 1243 **D.** 2486

Q18.2.8 MCQ 2018 2M

A contract is to be completed in 52 days and 125 identical robots were employed, each operational for 7 hours a day. After 39 days, five-seventh of the work was completed. How many additional robots would be required to complete the work on time, if each robot is now operational for 8 hours a day?

A. 50 **B.** 89 **C.** 146 **D.** 175

Q18.2.9 MCQ 2018 2M

A house has a number which needs to be identified. The following three statements are given that can help in identifying the house number.

(i) If the house number is a multiple of 3, then it is a number from 50 to 59.
(ii) If the house number is NOT a multiple of 4, then it is a number from 60 to 69.
(iii) If the house number is NOT a multiple of 6, then it is a number from 70 to 79.

What is the house number?

A. 54 **B.** 65 **C.** 66 **D.** 76

Q18.2.10 MCQ 2018 2M

An unbiased coin is tossed six times in a row and four different such trials are conducted. One trial implies six tosses of the coin. If H stands for head and T stands for tail, the following are the observations from the four trials:
(1) HTHTTH (2) TTHHHT (3) HTTHHT
(4) HHHHT ___ ___
Which statement describing the last two coin tosses of the fourth trial has the highest probability of being correct?

A. Two T will occur.

B. One H and one T will occur.

C. Two H will occur.

D. One H will be followed by one T.

Session : 3 | Forenoon | 04/02/18

Q18.3.1 MCQ 2018 1M

"From where are they bringing their books? _____ bringing _____ books from _____."
The words that best fill the blanks in the above sentence are

A. Their, they're, there **B.** They're, their, there

C. There, their, they're **D.** They're, there, there

Q18.3.2 MCQ 2018 1M

"A _____ investigation can sometimes yield new facts, but typically organized ones are more successful."
The word that best fills the blank in the above sentence is

A. meandering **B.** timely

C. consistent **D.** systematic

Q18.3.3 MCQ 2018 1M

The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

A. πd **B.** πd^2 **C.** $\frac{1}{4}\pi d^2$ **D.** $\frac{1}{2}\pi d$

Q18.3.4 MCQ 2018 1M

What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

A. 3047 **B.** 6047 **C.** 7987 **D.** 63847

Q18.3.5 MCQ 2018 1M

What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, ____, 0

A. 2880 **B.** 1440 **C.** 720 **D.** 0

Q18.3.6 MCQ 2018 2M

In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $8/9^{th}$ of the eligible men and $2/3^{rd}$ of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

A. 1,50,000 **B.** 2,00,000 **C.** 1,75,000 **D.** 1,51,000

Q18.3.7 MCQ 2018 2M

If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}$, $q^{-y} = \frac{1}{r}$, $r^{-z} = \frac{1}{p}$, what is the value of the product xyz?

A. -1 **B.** $\frac{1}{pqr}$ **C.** 1 **D.** pqr

Q18.3.8 MCQ 2018 2M

In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

A. 2:3 **B.** 1:1 **C.** 3:2 **D.** 2:1

Q18.3.9 MCQ 2018 2M

In the figure below, $\angle DEC + \angle BFC$ is equal to _____.

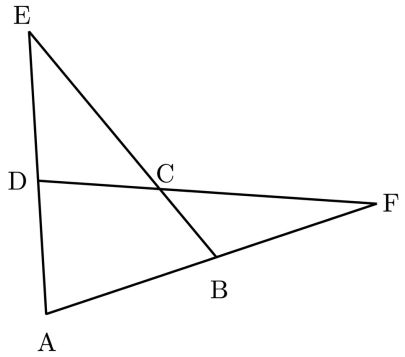


Fig. Q18.3.9

A. $\angle BCD - \angle BAD$ **B.** $\angle BAD + \angle BCF$

C. $\angle BAD + \angle BCD$ **D.** $\angle CBA + \angle ADC$

Q18.3.10 MCQ 2018 2M

A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

A. Three green faces and four red faces.

B. Four green faces and three red faces.

C. Five green faces and two red faces.

D. Six green faces and one red face.

Session : 4 | Afternoon | 04/02/18

Q18.4.1 MCQ 2018 1M

"When she fell down the _____, she received many _____ but little help."
The words that best fill the blanks in the above sentence are

A. stairs, stares **B.** stairs, stairs

C. stares, stairs **D.** stares, stares

Q18.4.2 MCQ 2018 1M

"In spite of being warned repeatedly, he failed to correct his _____ behaviour."
The word that best fills the blank in the above sentence is

A. rational **B.** reasonable **C.** errant **D.** good

Q18.4.3 MCQ 2018 1M

For $0 \le x \le 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.

A. $\left(0, \frac{\pi}{2}\right)$ **B.** $\left(\frac{\pi}{2}, \pi\right)$ **C.** $\left(\pi, \frac{3\pi}{2}\right)$ **D.** $\left(\frac{3\pi}{2}, 2\pi\right)$

Q18.4.4 MCQ 2018 1M

The area of an equilateral triangle is $\sqrt{3}$. What is the perimeter of the triangle?

A. 2 **B.** 4 **C.** 6 **D.** 8

Q18.4.5 MCQ 2018 1M

Arrange the following three-dimensional objects in the descending order of their volumes:

(i) A cuboid with dimensions 10 cm, 8 cm and 6 cm
(ii) A cube of side 8 cm
(iii) A cylinder with base radius 7 cm and height 7 cm
(iv) A sphere of radius 7 cm

A. (i), (ii), (iii), (iv) **B.** (ii), (i), (iv), (iii)

C. (iii), (ii), (i), (iv) **D.** (iv), (iii), (ii), (i)

Q18.4.6 MCQ 2018 2M

An automobile travels from city A to city B and returns to city A by the same route. The speed of the vehicle during the onward and return journeys were constant at 60 km/h and 90 km/h, respectively. What is the average speed in km/h for the entire journey?

A. 72 **B.** 73 **C.** 74 **D.** 75

Q18.4.7 MCQ 2018 2M

A set of 4 parallel lines intersect with another set of 5 parallel lines. How many parallelograms are formed?

A. 20 **B.** 48 **C.** 60 **D.** 72

Q18.4.8 MCQ 2018 2M

To pass a test, a candidate needs to answer at least 2 out of 3 questions correctly. A total of 6,30,000 candidates appeared for the test. Question A was correctly answered by 3,30,000 candidates. Question B was answered correctly by 2,50,000 candidates. Question C was answered correctly by 2,60,000 candidates. Both questions A and B were answered correctly by 1,00,000 candidates. Both questions B and C were answered correctly by 90,000 candidates. Both questions A and C were answered correctly by 80,000 candidates. If the number of students answering all questions correctly is the same as the number answering none, how many candidates failed to clear the test?

A. 30,000 **B.** 2,70,000 **C.** 3,90,000 **D.** 4,20,000

Q18.4.9 MCQ 2018 2M

If $x^2 + x - 1 = 0$ what is the value of $x^4 + \frac{1}{x^4}$?

A. 1 **B.** 5 **C.** 7 **D.** 9

Q18.4.10 MCQ 2018 2M

In a detailed study of annual crow births in India, it was found that there was relatively no growth during the period 2002 to 2004 and a sudden spike from 2004 to 2005. In another unrelated study, it was found that the revenue from cracker sales in India which remained fairly flat from 2002 to 2004, saw a sudden spike in 2005 before declining again in 2006. The solid line in the graph below refers to annual sale of crackers and the dashed line refers to the annual crow

births in India. Choose the most appropriate inference from the above data.

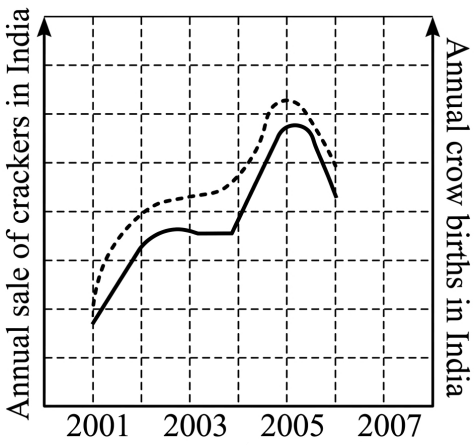


Fig. Q18.4.10

A. There is a strong correlation between crow birth and cracker sales.

B. Cracker usage increases crow birth rate.

C. If cracker sale declines, crow birth will decline.

D. Increased birth rate of crows will cause an increase in the sale of crackers.

Session : 5 | Forenoon | 10/02/18

Q18.5.1 MCQ 2018 1M

"By giving him the last ____ of the cake, you will ensure lasting ____ in our house today."
The words that best fill the blanks in the above sentence are

A. peas, piece **B.** piece, peace

C. peace, piece **D.** peace, peas

Q18.5.2 MCQ 2018 1M

"Even though there is a vast scope for its _____, tourism has remained a/an _____ area."
The words that best fill the blanks in the above sentence are

A. improvement, neglected **B.** rejection, approved

C. fame, glum **D.** interest, disinterested

Q18.5.3 MCQ 2018 1M

If the number 715■423 is divisible by 3 (■ denotes the missing digit in the thousandths place), then the smallest whole number in the place of ■ is ____.

A. 0 **B.** 2 **C.** 5 **D.** 6

Q18.5.4 MCQ 2018 1M

What is the value of $1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \cdots$?

A. 2 **B.** $\frac{7}{4}$ **C.** $\frac{3}{2}$ **D.** $\frac{4}{3}$

Q18.5.5 MCQ 2018 1M

A 1.5 m tall person is standing at a distance of 3 m from a lamp post. The light from the lamp at the top of the post casts her shadow. The length of the shadow is twice her height. What is the height of the lamp post in meters?

A. 1.5 **B.** 3 **C.** 4.5 **D.** 6

Q18.5.6 MCQ 2018 2M

Leila aspires to buy a car worth Rs. 10,00,000 after 5 years. What is the minimum amount in Rupees that she should deposit now in a bank which offers 10% annual rate of interest, if the interest was compounded annually?

A. 5,00,000 **B.** 6,21,000 **C.** 6,66,667 **D.** 7,50,000

Q18.5.7 MCQ 2018 2M

Two alloys A and B contain gold and copper in the ratios of 2:3 and 3:7 by mass, respectively. Equal masses of alloys A and B are melted to make an alloy C. The ratio of gold to copper in alloy C is ____.

A. 5:10 **B.** 7:13 **C.** 6:11 **D.** 9:13

Q18.5.8 MCQ 2018 2M

The Cricket Board has long recognized John's potential as a leader of the team. However, his on-field temper has always been a matter of concern for them since his junior days. While this aggression has filled stadia with die-hard fans, it has taken a toll on his own batting. Until recently, it appeared that he found it difficult to convert his aggression into big scores. Over the past three seasons though, that picture of John has been replaced by a cerebral, calculative and successful batsman-captain. After many years, it appears that the team has finally found a complete captain.
Which of the following statements can be logically inferred from the above paragraph?

(i) Even as a junior cricketer, John was considered a good captain.
(ii) Finding a complete captain is a challenge.
(iii) Fans and the Cricket Board have differing views on what they want in a captain.
(iv) Over the past three seasons John has accumulated big scores.

A. (i), (ii) and (iii) only **B.** (iii) and (iv) only

C. (ii) and (iv) only **D.** (i), (ii), (iii) and (iv)

Q18.5.9 MCQ 2018 2M

A cab was involved in a hit and run accident at night. You are given the following data about the cabs in the city and the accident.

(i) 85% of cabs in the city are green and the remaining cabs are blue.
(ii) A witness identified the cab involved in the accident as blue.
(iii) It is known that a witness can correctly identify the cab colour only 80% of the time.

Which of the following options is closest to the probability that the accident was caused by a blue cab?

A. 12% **B.** 15% **C.** 41% **D.** 80%

Q18.5.10 MCQ 2018 2M

A coastal region with unparalleled beauty is home to many species of animals. It is dotted with coral reefs and unspoilt white sandy beaches. It has remained inaccessible to tourists due to poor connectivity and lack of accommodation. A company has spotted the opportunity and is planning to develop a luxury resort with helicopter service to the nearest major city airport. Environmentalists are upset that this would lead to the region becoming crowded and polluted like any other major beach resorts.
Which one of the following statements can be logically inferred from the information given in the above paragraph?

A. The culture and tradition of the local people will be influenced by the tourists.

B. The region will become crowded and polluted due to tourism.

C. The coral reefs are on the decline and could soon vanish.

D. Helicopter connectivity would lead to an increase in tourists coming to the region.

Session : 6 | Afternoon | 10/02/18

Q18.6.1 MCQ 2018 1M

"Since you have gone off the _____, the _____ sand is likely to damage the car."
The words that best fill the blanks in the above sentence are

A. course, coarse **B.** course, course

C. coarse, course **D.** coarse, coarse

Q18.6.2 MCQ 2018 1M

"A common misconception among writers is that sentence structure mirrors thought; the more _____ the structure, the more complicated the ideas."
The word that best fills the blank in the above sentence is

A. detailed **B.** simple **C.** clear **D.** convoluted

Q18.6.3 MCQ 2018 1M

The three roots of the equation $f(x) = 0$ are $x = \{-2, 0, 3\}$. What are the three values of x for which $f(x - 3) = 0$?

A. $-5, -3, 0$ **B.** $-2, 0, 3$ **C.** $0, 6, 8$ **D.** $1, 3, 6$

Q18.6.4 MCQ 2018 1M

For what values of k given below is $\frac{(k+2)^2}{k-3}$ an integer?

A. 4, 8, 18 **B.** 4, 10, 16 **C.** 4, 8, 28 **D.** 8, 26, 28

Q18.6.5 MCQ 2018 1M

Functions $F(a, b)$ and $G(a, b)$ are defined as follows: $F(a, b) = (a - b)^2$ and $G(a, b) = |a - b|$, where $|x|$ represents the absolute value of x.
What would be the value of $G(F(1, 3), G(1, 3))$?

A. 2 **B.** 4 **C.** 6 **D.** 36

Q18.6.6 MCQ 2018 2M

An e-mail password must contain three characters. The password has to contain one numeral from 0 to 9, one upper case and one lower case character from the English alphabet. How many distinct passwords are possible?

A. 6,760 **B.** 13,520 **C.** 40,560 **D.** 1,05,456

Q18.6.7 MCQ 2018 2M

In a certain code, AMCF is written as EQGJ and NKUF is written as ROYJ. How will DHLP be written in that code?

A. RSTN **B.** TLPH **C.** HLPT **D.** XSVR

Q18.6.8 MCQ 2018 2M

A class of twelve children has two more boys than girls. A group of three children are randomly picked from this class to accompany the teacher on a field trip. What is the probability that the group accompanying the teacher contains more girls than boys?

A. 0 **B.** $\frac{325}{864}$ **C.** $\frac{525}{864}$ **D.** $\frac{5}{12}$

Q18.6.9 MCQ 2018 2M

A designer uses marbles of four different colours for his designs. The cost of each marble is the same, irrespective of the colour. The table below shows the percentage of marbles of each colour used in the current design. The cost of each marble increased by 25%. Therefore, the designer decided to reduce equal numbers of marbles of each colour to keep the total cost unchanged. What is the percentage of blue marbles in the new design?

Blue	Black	Red	Yellow
40%	25%	20%	15%

A. 35.75 **B.** 40.25 **C.** 43.75 **D.** 46.25

Q18.6.10 MCQ 2018 2M

P, Q, R and S crossed a lake in a boat that can hold a maximum of two persons, with only one set of oars. The following additional facts are available.

(i) The boat held two persons on each of the three forward trips across the lake and one person on each of the two return trips.
(ii) P is unable to row when someone else is in the boat.
(iii) Q is unable to row with anyone else except R.
(iv) Each person rowed for at least one trip.
(v) Only one person can row during a trip.

Who rowed twice?

A. P **B.** Q **C.** R **D.** S

Session : 7 | Forenoon | 11/02/18

Q18.7.1 MCQ 2018 1M

"The driver applied the _____ as soon as she approached the hotel where she wanted to take a _____."
The words that best fill the blanks in the above sentence are

A. brake, break **B.** break, break

C. brake, brake **D.** break, brake

Q18.7.2 MCQ 2018 1M

"It is no surprise that every society has had codes of behaviour; however, the nature of these codes is often _____."
The word that best fills the blank in the above sentence is

A. unpredictable **B.** simple **C.** expected **D.** strict

Q18.7.3 MCQ 2018 1M

Hema's age is 5 years more than twice Hari's age. Suresh's age is 13 years less than 10 times Hari's age. If Suresh is 3 times as old as Hema, how old is Hema?

A. 14 **B.** 17 **C.** 18 **D.** 19

Q18.7.4 MCQ 2018 1M

Tower A is 90 m tall and tower B is 140 m tall. They are 100 m apart. A horizontal skywalk connects the floors at 70 m in both the towers. If a taut rope connects the top of tower A to the bottom of tower B, at what distance (in metres) from tower A will the rope intersect the skywalk?

A. 22.22 **B.** 50 **C.** 57.87 **D.** 77.78

Q18.7.5 MCQ 2018 1M

The temperature T in a room varies as a function of the outside temperature T_0 and the number of persons in the room p, according to the relation $T = K(\Theta p + T_0)$, where Θ and K are constants. What would be the value of Θ given the following data?

T_0	p	T
25	2	32.4
30	5	42.0

A. 0.8 **B.** 1.0 **C.** 2.0 **D.** 10.0

Q18.7.6 MCQ 2018 2M

A fruit seller sold a basket of fruits at 12.5% loss. Had he sold it for Rs. 108 more, he would have made a 10% gain. What is the loss in Rupees incurred by the fruit seller?

A. 48 **B.** 52 **C.** 60 **D.** 108

Q18.7.7 MCQ 2018 2M

The price of a wire made of a superalloy material is proportional to the square of its length. The price of 10 m length of the wire is Rs. 1600. What would be the total price (in Rs.) of two wires of lengths 4 m and 6 m?

A. 768 **B.** 832 **C.** 1440 **D.** 1600

Q18.7.8 MCQ 2018 2M

Which of the following function(s) is an accurate description of the graph for the range(s) indicated?

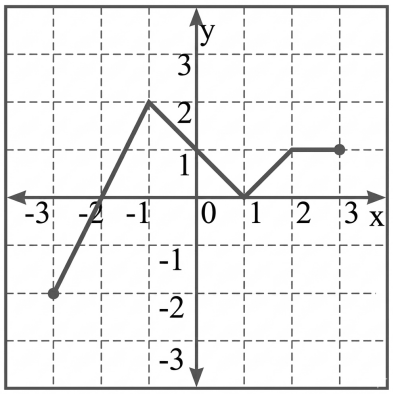


Fig. Q18.7.8

(i) $y = 2x + 4$ for $-3 \le x \le -1$
(ii) $y = |x - 1|$ for $-1 \le x \le 2$
(iii) $y = \big||x| - 1\big|$ for $-1 \le x \le 2$
(iv) $y = 1$ for $2 \le x \le 3$

A. (i), (ii) and (iii) only **B.** (i), (ii) and (iv) only

C. (i) and (iv) only **D.** (ii) and (iv) only

Q18.7.9 MCQ 2018 2M

Consider a sequence of numbers $a_1, a_2, a_3, \ldots, a_n$ where

$$a_n = \frac{1}{n} - \frac{1}{n+2}, \quad \text{for each integer } n > 0.$$

What is the sum of the first 50 terms?

A. $\left(1 + \frac{1}{2}\right) - \frac{1}{50}$ **B.** $\left(1 + \frac{1}{2}\right) + \frac{1}{50}$

C. $\left(1 + \frac{1}{2}\right) - \left(\frac{1}{51} + \frac{1}{52}\right)$ **D.** $1 - \left(\frac{1}{51} + \frac{1}{52}\right)$

Q18.7.10 MCQ 2018 2M

Each of the letters arranged as below represents a unique integer from 1 to 9. The letters are positioned in the figure such that $(A \times B \times C)$, $(B \times G \times E)$ and $(D \times E \times F)$ are equal. Which integer among the following choices cannot be represented by the letters A, B, C, D, E, F or G?

A		D
B	G	E
C		F

Fig. Q18.7.10

A. 4 **B.** 5 **C.** 6 **D.** 9

Session : 8 | Afternoon | 11/02/18

Q18.8.1 MCQ 2018 1M

"His face _____ with joy when the solution of the puzzle was _____ to him."
The words that best fill the blanks in the above sentence are

A. shone, shown **B.** shone, shone

C. shown, shone **D.** shown, shown

Q18.8.2 MCQ 2018 1M

"Although it does contain some pioneering ideas, one would hardly characterize the work as _____."
The word that best fills the blank in the above sentence is

A. innovative **B.** simple **C.** dull **D.** boring

Q18.8.3 MCQ 2018 1M

$\underbrace{a + a + a + \cdots + a}_{n \text{ times}} = a^2 b$ and $\underbrace{b + b + b + \cdots + b}_{m \text{ times}} = ab^2$,

where a, b, n and m are natural numbers.
What is the value of

$$\left(\underbrace{m + m + m + \cdots + m}_{n \text{ times}}\right)\left(\underbrace{n + n + n + \cdots + n}_{m \text{ times}}\right)?$$

A. $2a^2b^2$ **B.** a^4b^4 **C.** $ab(a + b)$ **D.** $a^2 + b^2$

Q18.8.4 MCQ 2018 1M

A three-member committee has to be formed from a group of 9 people. How many such distinct committees can be formed?

A. 27 **B.** 72 **C.** 81 **D.** 84

Q18.8.5 MCQ 2018 1M

For non-negative integers, a, b, c, what would be the value of $a + b + c$ if

$$\log a + \log b + \log c = 0?$$

A. 3 **B.** 1 **C.** 0 **D.** -1

Q18.8.6 MCQ 2018 2M

In manufacturing industries, loss is usually taken to be proportional to the square of the deviation from a target. If the loss is Rs. 4900 for a deviation of 7 units, what would be the loss in Rupees for a deviation of 4 units from the target?

A. 400 **B.** 1200 **C.** 1600 **D.** 2800

Q18.8.7 MCQ 2018 2M

A faulty wall clock is known to gain 15 minutes every 24 hours. It is synchronized to the correct time at 9 AM on 11th July. What will be the correct time to the nearest minute when the clock shows 2 PM on 15th July of the same year?

A. 12:45 PM **B.** 12:58 PM **C.** 1:00 PM **D.** 2:00 PM

Q18.8.8 MCQ 2018 2M

The annual average rainfall in a tropical city is 1000 mm. On a particular rainy day (24-hour period), the cumulative rainfall experienced by the city is shown in the graph. Over the 24-hour period, 50% of the rainfall falling on a rooftop, which had an obstruction-free area of 50 m^2, was harvested into a tank. What is the total volume of water collected in the tank in liters?

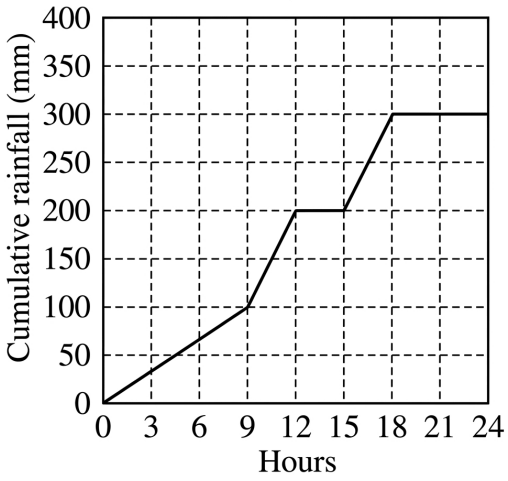


Fig. Q18.8.8

A. 25,000 **B.** 18,749 **C.** 7,500 **D.** 3,125

Q18.8.9 MCQ 2018 2M

Given that $\frac{\log P}{y-z} = \frac{\log Q}{z-x} = \frac{\log R}{x-y} = 10$ for $x \neq y \neq z$, what is the value of the product PQR?

A. 0 **B.** 1 **C.** xyz **D.** 10^{xyz}

Q18.8.10 MCQ 2018 2M

Each of the letters in the figure below represents a unique integer from 1 to 9. The letters are positioned in the figure such that each of (A+B+C), (C+D+E), (E+F+G) and (G+H+K) is equal to 13. Which integer does E represent?

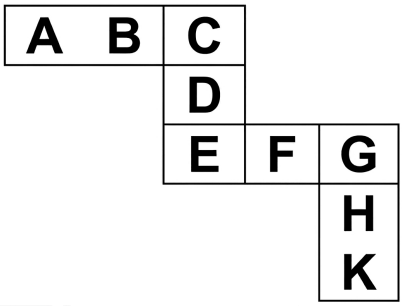


Fig. Q18.8.10

A. 1 **B.** 4 **C.** 6 **D.** 7

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Session : 1					
Q18.1.1 A	Q18.1.2 C	Q18.1.3 B	Q18.1.4 B	Q18.1.5 B	Q18.1.6 A
Q18.1.7 D	Q18.1.8 D	Q18.1.9 B	Q18.1.10 C		
Session : 2					
Q18.2.1 D	Q18.2.2 A	Q18.2.3 B	Q18.2.4 A	Q18.2.5 C	Q18.2.6 C
Q18.2.7 C	Q18.2.8 MTA	Q18.2.9 D	Q18.2.10 B		
Session : 3					
Q18.3.1 B	Q18.3.2 A	Q18.3.3 D	Q18.3.4 C	Q18.3.5 B	Q18.3.6 B
Q18.3.7 C	Q18.3.8 B	Q18.3.9 A	Q18.3.10 C		
Session : 4					
Q18.4.1 A	Q18.4.2 C	Q18.4.3 B	Q18.4.4 C	Q18.4.5 D	Q18.4.6 A
Q18.4.7 C	Q18.4.8 D	Q18.4.9 C	Q18.4.10 A		
Session : 5					
Q18.5.1 B	Q18.5.2 A	Q18.5.3 B	Q18.5.4 D	Q18.5.5 B	Q18.5.6 B
Q18.5.7 B	Q18.5.8 C	Q18.5.9 C	Q18.5.10 D		
Session : 6					
Q18.6.1 A	Q18.6.2 D	Q18.6.3 D	Q18.6.4 C	Q18.6.5 A	Q18.6.6 C
Q18.6.7 C	Q18.6.8 MTA	Q18.6.9 C	Q18.6.10 C		
Session : 7					
Q18.7.1 A	Q18.7.2 A	Q18.7.3 D	Q18.7.4 MTA	Q18.7.5 B	Q18.7.6 C
Q18.7.7 B	Q18.7.8 B	Q18.7.9 C	Q18.7.10 B		
Session : 8					
Q18.8.1 A	Q18.8.2 A	Q18.8.3 B	Q18.8.4 D	Q18.8.5 A	Q18.8.6 C
Q18.8.7 B	Q18.8.8 C	Q18.8.9 B	Q18.8.10 B		

YEAR

2017

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, AR, TF	10
Session 2	ME-2, AG, GG	10
Session 3	EC-1, BT, CY, EY, PI	10
Session 4	EC-2, AE, MA, MT, PE, PH	10
Session 5	CS-1, EE-1	10
Session 6	CS-2, EE-2	10
Session 7	CE-1, CH, MN	10
Session 8	CE-2, IN, XE, XL	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 80 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 04/02/17

Q17.1.1 MCQ 2017 1M

He was one of my best _____ and I felt his loss _____.

A. friend, keenly **B.** friends, keen

C. friend, keener **D.** friends, keenly

Q17.1.2 MCQ 2017 1M

As the two speakers became increasingly agitated, the debate became _____.

A. lukewarm **B.** poetic **C.** forgiving **D.** heated

Q17.1.3 MCQ 2017 1M

A right-angled cone (with base radius 5 cm and height 12 cm), as shown in the figure below, is rolled on the ground keeping the point P fixed until the point Q (at the base of the cone, as shown) touches the ground again.

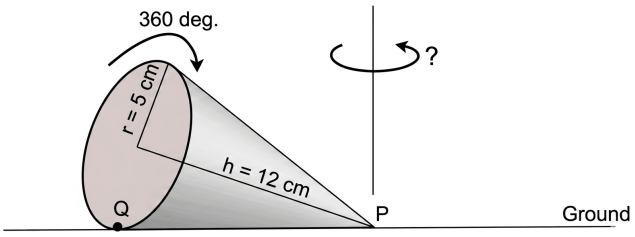


Fig. Q17.1.3

By what angle (in radians) does the cone travel?

A. $\frac{5\pi}{12}$ **B.** $\frac{5\pi}{24}$ **C.** $\frac{24\pi}{5}$ **D.** $\frac{10\pi}{13}$

Q17.1.4 MCQ 2017 1M

In a company with 100 employees, 45 earn Rs. 20,000 per month, 25 earn Rs. 30,000, 20 earn Rs. 40,000, 8 earn Rs. 60,000, and 2 earn Rs. 1,50,000. The median of the salaries is

A. Rs. 20,000 **B.** Rs. 30,000

C. Rs. 32,300 **D.** Rs. 40,000

Q17.1.5 MCQ 2017 1M

P, Q, and R talk about S's car collection. P states that S has at least 3 cars. Q believes that S has less than 3 cars. R indicates that to his knowledge, S has at least one car. Only one of P, Q and R is right. The number of cars owned by S is

A. 0 **B.** 1 **C.** 3 **D.** Cannot be determined

Q17.1.6 MCQ 2017 2M

"Here, throughout the early 1820s, Stuart continued to fight his losing battle to allow his sepoys to wear their caste-marks and their own choice of facial hair on parade, being again reprimanded by the commander-in-chief. He retort that 'A stronger instance than this of European prejudice with relation to this country has never come under my observations' had no effect on his superiors."

According to this paragraph, which of the statements below is most accurate?

A. Stuart's commander-in-chief was moved by this demonstration of his prejudice.

B. The Europeans were accommodating of the sepoys' desire to wear their caste-marks.

C. Stuart's 'losing battle' refers to his inability to succeed in enabling sepoys to wear caste-marks.

D. The commander-in-chief was exempt from the European prejudice that dictated how the sepoys were to dress.

Q17.1.7 MCQ 2017 2M

What is the sum of the missing digits in the subtraction problem below?

$$\begin{array}{r} 5____ \\ -\,48_89 \\ \hline 1111 \end{array}$$

A. 8 **B.** 10 **C.** 11 **D.** Cannot be determined

Q17.1.8 MCQ 2017 2M

Let S_1 be the plane figure consisting of the points (x, y) given by the inequalities $|x - 1| \leq 2$ and $|y + 2| \leq 3$. Let S_2 be the plane figure given by the inequalities $x - y \geq -2$, $y \geq 1$, and $x \leq 3$. Let S be the union of S_1 and S_2. The area of S is

A. 26 **B.** 28 **C.** 32 **D.** 34

Q17.1.9 MCQ 2017 2M

Two very famous sportsmen Mark and Steve happened to be brothers, and played for country K. Mark teased James, an opponent from country E, "There is no way you are good enough to play for your country." James replied, "Maybe not, but at least I am the best player in my own family."

Which one of the following can be inferred from this conversation?

A. Mark was known to play better than James

B. Steve was known to play better than Mark

C. James and Steve were good friends

D. James played better than Steve

Q17.1.10 MCQ 2017 2M

The growth of bacteria (lactobacillus) in milk leads to curd formation. A minimum bacterial population density of 0.8 (in suitable units) is needed to form curd. In the graph below, the population density of lactobacillus in 1 litre of milk is plotted as a function of time, at two different temperatures, 25 °C and 37 °C.

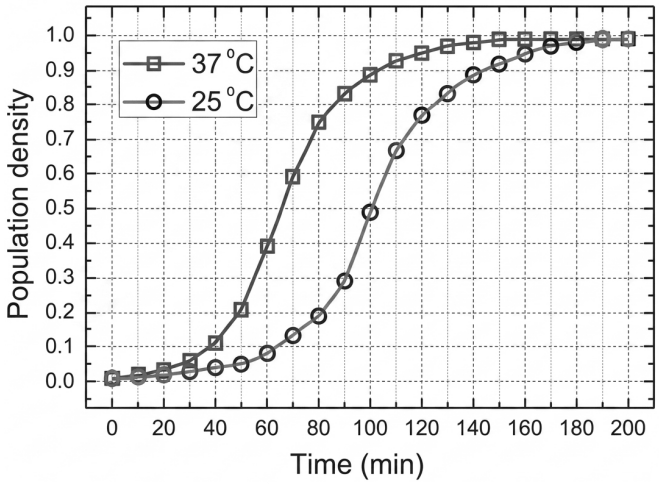


Fig. Q17.1.10

Consider the following statements based on the data shown above:

i. The growth in bacterial population stops earlier at 37 °C as compared to 25 °C

ii. The time taken for curd formation at 25 °C is twice the time taken at 37 °C

Which one of the following options is correct?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 2 | Afternoon | 04/02/17

Q17.2.1 MCQ 2017 1M

The ways in which this game can be played _____ potentially infinite.

A. is **B.** is being **C.** are **D.** are being

Q17.2.2 MCQ 2017 1M

If you choose plan P, you will have to _____ plan Q, as these two are mutually _____.

A. forgo, exclusive **B.** forget, inclusive

C. accept, exhaustive **D.** adopt, intrusive

Q17.2.3 MCQ 2017 1M

If a and b are integers and $a - b$ is even, which of the following must always be even?

A. ab **B.** $a^2 + b^2 + 1$ **C.** $a^2 + b + 1$ **D.** $ab - b$

Q17.2.4 MCQ 2017 1M

A couple has 2 children. The probability that both children are boys if the older one is a boy is

A. 1/4 **B.** 1/3 **C.** 1/2 **D.** 1

Q17.2.5 MCQ 2017 1M

P looks at Q while Q looks at R. P is married, R is not. The number of pairs of people in which a married person is looking at an unmarried person is

A. 0 **B.** 1 **C.** 2 **D.** Cannot be determined

Q17.2.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters."
Which of the following is closest in meaning to 'cleaving'?

A. deteriorating **B.** arguing **C.** departing **D.** splitting

Q17.2.7 MCQ 2017 2M

X bullocks and Y tractors take 8 days to plough a field. If we halve the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

A. 30 **B.** 35 **C.** 40 **D.** 45

Q17.2.8 MCQ 2017 2M

There are 4 women P, Q, R, S, and 5 men V, W, X, Y, Z in a group. We are required to form pairs each consisting of one woman and one man. P is not to be paired with Z, and Y must necessarily be paired with someone. In how many ways can 4 such pairs be formed?

A. 74 **B.** 76 **C.** 78 **D.** 80

Q17.2.9 MCQ 2017 2M

All people in a certain island are either 'Knights' or 'Knaves' and each person knows every other person's identity.
Knights NEVER lie, and knaves ALWAYS lie.
P says "Both of us are knights".
Q says "None of us are knaves".
Which one of the following can be logically inferred from the above?

A. Both P and Q are knights

B. P is a knight; Q is a knave

C. Both P and Q are knaves

D. The identities of P, Q cannot be determined

Q17.2.10 MCQ 2017 2M

In the graph below, the concentration of a particular pollutant in a lake is plotted over (alternate) days of a month in winter (average temperature 10 °C) and a month in summer (average temperature 30 °C).

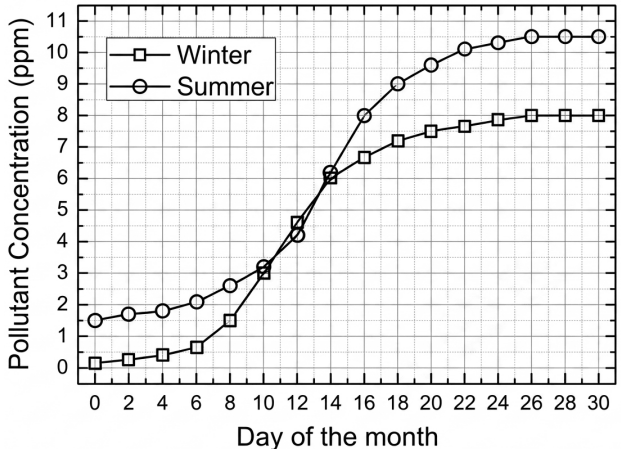


Fig. Q17.2.10

Consider the following statements based on the data shown above:

i. Over the given months, the difference between the maximum and the minimum pollutant concentrations is the same in both winter and summer.
ii. There are at least four days in the summer month such that the pollutant concentrations on those days are within 1 ppm of the pollutant concentrations on the corresponding days in the winter month.

Which one of the following options is correct?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 3 | Forenoon | 05/02/17

Q17.3.1 MCQ 2017 1M

She has a sharp tongue and it can occasionally turn___

A. hurtful **B.** left **C.** methodical **D.** vital

Q17.3.2 MCQ 2017 1M

I _____ made arrangements had I _____ informed earlier.

A. could have, been **B.** would have, being

C. had, have **D.** had been, been

Q17.3.3 MCQ 2017 1M

In the summer, water consumption is known to decrease overall by 25%. A Water Board official states that in the summer household consumption decreases by 20%, while other consumption increases by 70%.
Which of the following statements is correct?

A. The ratio of household to other consumption is 8/17

B. The ratio of household to other consumption is 1/17

C. The ratio of household to other consumption is 17/8

D. There are errors in the official's statement.

Q17.3.4 MCQ 2017 1M

40% of deaths on city roads may be attributed to drunken driving. The number of degrees needed to represent this as a slice of a pie chart is

A. 120 **B.** 144 **C.** 160 **D.** 212

Q17.3.5 MCQ 2017 1M

Some tables are shelves. Some shelves are chairs. All chairs are benches. Which of the following conclusions can be deduced from the preceding sentences?

i. At least one bench is a table
ii. At least one shelf is a bench
iii. At least one chair is a table
iv. All benches are chairs

A. Only i **B.** Only ii **C.** Only ii and iii **D.** Only iv

Q17.3.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters".
Here, the word 'antagonistic' is closest in meaning to

A. impartial **B.** argumentative **C.** separated **D.** hostile

Q17.3.7 MCQ 2017 2M

S, T, U, V, W, X, Y, and Z are seated around a circular table. T's neighbours are Y and V. Z is seated third to the left of T and second to the right of S. U's neighbours are S and Y, and T and W are not seated opposite each other. Who is third to the left of V?

A. X **B.** W **C.** U **D.** T

Q17.3.8 MCQ 2017 2M

Trucks (10 m long) and cars (5 m long) go on a single lane bridge. There must be a gap of at least 20 m after each truck and a gap of at least 15 m after each car. Trucks and cars travel at a speed of 36 km/h. If cars and trucks go alternately, what is the maximum number of vehicles that can use the bridge in one hour?

A. 1440 **B.** 1200 **C.** 720 **D.** 600

Q17.3.9 MCQ 2017 2M

There are 3 Indians and 3 Chinese in a group of 6 people. How many subgroups of this group can we choose so that every subgroup has at least one Indian?

A. 56 **B.** 52 **C.** 48 **D.** 44

Q17.3.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.

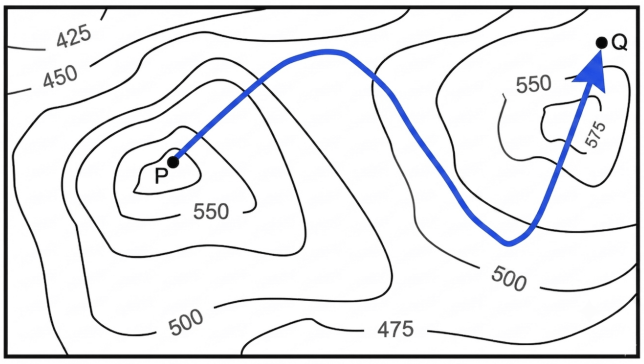


Fig. Q17.3.10

The path from P to Q is best described by

A. Up-Down-Up-Down **B.** Down-Up-Down-Up

C. Down-Up-Down **D.** Up-Down-Up

Session : 4 | Afternoon | 05/02/17

Q17.4.1 MCQ 2017 1M

The ninth and the tenth of this month are Monday and Tuesday _____.

A. figuratively **B.** retrospectively

C. respectively **D.** rightfully

Q17.4.2 MCQ 2017 1M

It is _____ to read this year's textbook _____ the last year's.

A. easier, than **B.** most easy, than

C. easier, from **D.** easiest, from

Q17.4.3 MCQ 2017 1M

A rule states that in order to drink beer, one must be over 18 years old. In a bar, there are 4 people. P is 16 years old, Q is 25 years old, R is drinking milkshake and S is drinking a beer. What must be checked to ensure that the rule is being followed?

A. Only P's drink

B. Only P's drink and S's age

C. Only S's age

D. Only P's drink, Q's drink and S's age

Q17.4.4 MCQ 2017 1M

Fatima starts from point P, goes North for 3 km, and then East for 4 km to reach point Q. She then turns to face point P and goes 15 km in that direction. She then goes North for 6 km. How far is she from point P, and in which direction should she go to reach point P?

A. 8 km, East **B.** 12 km, North

C. 6 km, East **D.** 10 km, North

Q17.4.5 MCQ 2017 1M

500 students are taking one or more courses out of Chemistry, Physics, and Mathematics. Registration records indicate course enrolment as follows: Chemistry (329), Physics (186), Mathematics (295), Chemistry and Physics (83), Chemistry and Mathematics (217), and Physics and Mathematics (63). How many students are taking all 3 subjects?

A. 37 **B.** 43 **C.** 47 **D.** 53

Q17.4.6 MCQ 2017 2M

"If you are looking for a history of India, or for an account of the rise and fall of the British Raj, or for the reason of the cleaving of the subcontinent into two mutually antagonistic parts and the effects this mutilation will have in the respective sections, and ultimately on Asia, you will not find it in these pages; for though I have spent a lifetime in the country, I lived too near the seat of events, and was too intimately associated with the actors, to get the perspective needed for the impartial recording of these matters."
Which of the following statements best reflects the author's opinion?

A. An intimate association does not allow for the necessary perspective.

B. Matters are recorded with an impartial perspective.

C. An intimate association offers an impartial perspective.

D. Actors are typically associated with the impartial recording of matters.

Q17.4.7 MCQ 2017 2M

Each of P, Q, R, S, W, X, Y and Z has been married at most once. X and Y are married and have two children P and Q. Z is the grandfather of the daughter S of P. Further, Z and W are married and are parents of R.
Which one of the following must necessarily be FALSE?

A. X is the mother-in-law of R

B. P and R are not married to each other

C. P is a son of X and Y

D. Q cannot be married to R

Q17.4.8 MCQ 2017 2M

1200 men and 500 women can build a bridge in 2 weeks. 900 men and 250 women will take 3 weeks to build the same bridge. How many men will be needed to build the bridge in one week?

A. 3000 **B.** 3300 **C.** 3600 **D.** 3900

Q17.4.9 MCQ 2017 2M

The number of 3-digit numbers such that the digit 1 is never to the immediate right of 2 is

A. 781 **B.** 791 **C.** 881 **D.** 891

Q17.4.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot.

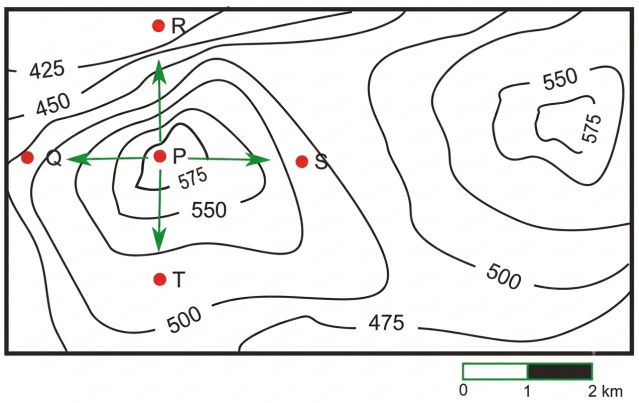


Fig. Q17.4.10

Which of the following is the steepest path leaving from P?

A. P to Q **B.** P to R **C.** P to S **D.** P to T

Session : 5 | Forenoon | 11/02/17

Q17.5.1 MCQ 2017 1M

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

A. was wishing **B.** is wishing **C.** wished **D.** had wished

Q17.5.2 MCQ 2017 1M

Research in the workplace reveals that people work for many reasons _____.

A. money beside **B.** beside money

C. money besides **D.** besides money

Q17.5.3 MCQ 2017 1M

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

A. Rahul and Murali **B.** Srinivas and Arul

C. Srinivas and Murali **D.** Srinivas and Rahul

Q17.5.4 MCQ 2017 1M

Find the smallest number y such that $y \times 162$ is a perfect cube.

A. 24 **B.** 27 **C.** 32 **D.** 36

Q17.5.5 MCQ 2017 1M

The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

A. 0.3^k **B.** 0.6^k **C.** 0.7^k **D.** 0.9^k

Q17.5.6 MCQ 2017 2M

"The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history." Which of the following statements best reflects the author's opinion?

A. Nationalists are highly imaginative.

B. History is viewed through the filter of nationalism.

C. Our colonial past never happened.

D. Nationalism has to be both adequately and properly imagined.

Q17.5.7 MCQ 2017 2M

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

A. 2 **B.** 3 **C.** 4 **D.** Cannot be determined

Q17.5.8 MCQ 2017 2M

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

A. the maximum of x and y **B.** the minimum of x and y

C. 1 **D.** none of the above

Q17.5.9 MCQ 2017 2M

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

A. 21 **B.** 18 **C.** 16 **D.** 14

Q17.5.10 MCQ 2017 2M

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?

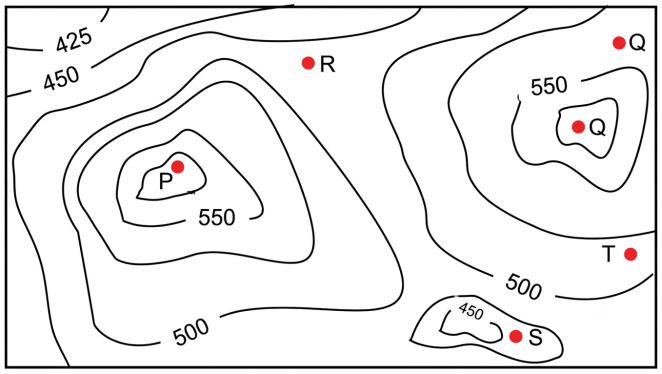


Fig. Q17.5.10

A. P, Q **B.** P, Q, T **C.** R, S, T **D.** Q, R, S

Session : 6 | Afternoon | 11/02/17

Q17.6.1 MCQ 2017 1M

Choose the option with words that are not synonyms.

A. aversion, dislike **B.** luminous, radiant

C. plunder, loot **D.** yielding, resistant

Q17.6.2 MCQ 2017 1M

Saturn is _____ to be seen on a clear night with the naked eye.

A. enough bright **B.** bright enough

C. as enough bright **D.** bright as enough

Q17.6.3 MCQ 2017 1M

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

A. V **B.** W **C.** X **D.** Y

Q17.6.4 MCQ 2017 1M

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

A. 12 **B.** 15 **C.** 18 **D.** 19

Q17.6.5 MCQ 2017 1M

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

A. 1/5 **B.** 7/30 **C.** 1/4 **D.** 4/15

Q17.6.6 MCQ 2017 2M

"We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms." The author's belief that ideology is not as important as literature is revealed by the word:

A. 'culture' **B.** 'seemingly'

C. 'urgent' **D.** 'political'

Q17.6.7 MCQ 2017 2M

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

A. The box labelled 'Apples'

B. The box labelled 'Apples and Oranges'

C. The box labelled 'Oranges'

D. Cannot be determined

Q17.6.8 MCQ 2017 2M

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

A. 90 digits **B.** 91 digits **C.** 92 digits **D.** 93 digits

Q17.6.9 MCQ 2017 2M

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q17.6.10 MCQ 2017 2M

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

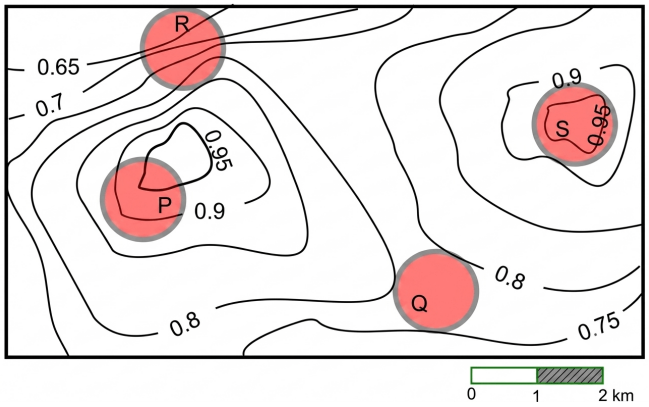


Fig. Q17.6.10

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

A. P **B.** Q **C.** R **D.** S

Session : 7 | Forenoon | 12/02/17

Q17.7.1 MCQ 2017 1M

The bacteria in milk are destroyed when it _____ heated to 80 degree Celsius.

A. would be **B.** will be **C.** is **D.** was

Q17.7.2 MCQ 2017 1M

_____ with someone else's email account is now a very serious offence.

A. Involving **B.** Assisting **C.** Tampering **D.** Incubating

Q17.7.3 MCQ 2017 1M

Consider the following sentences:
All benches are beds. No bed is a bulb. Some bulbs are lamps.
Which of the following can be inferred?

i. Some beds are lamps.
ii. Some lamps are beds.

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Q17.7.4 MCQ 2017 1M

If the radius of a right circular cone is increased by 50%, its volume increases by

A. 75% **B.** 100% **C.** 125% **D.** 237.5%

Q17.7.5 MCQ 2017 1M

The following sequence of numbers is arranged in increasing order: 1, x, x, x, y, y, 9, 16, 18. Given that the mean and median are equal, and are also equal to twice the mode, the value of y is

A. 5 **B.** 6 **C.** 7 **D.** 8

Q17.7.6 MCQ 2017 2M

The old concert hall was demolished because of fears that the foundation would be affected by the construction of the new metro line in the area. Modern technology for underground metro construction tried to mitigate the impact of pressurized air pockets created by the excavation of large amounts of soil. But even with these safeguards, it was feared that the soil below the concert hall would not be stable.
From this, one can infer that

A. the foundations of old buildings create pressurized air pockets underground, which are difficult to handle during metro construction.

B. metro construction has to be done carefully considering its impact on the foundations of existing buildings.

C. old buildings in an area form an impossible hurdle to metro construction in that area.

D. pressurized air can be used to excavate large amounts of soil from underground areas.

Q17.7.7 MCQ 2017 2M

Students applying for hostel rooms are allotted rooms in order of seniority. Students already staying in a room will move if they get a room in their preferred list. Preferences of lower ranked applicants are ignored during allocation. Given the data below, which room will Ajit stay in?

Names	Student seniority	Current room	Room preference list
Amar	1	P	R, S, Q
Akbar	2	None	R, S
Anthony	3	Q	P
Ajit	4	S	Q, P, R

A. P **B.** Q **C.** R **D.** S

Q17.7.8 MCQ 2017 2M

The last digit of $(2171)^7 + (2172)^9 + (2173)^{11} + (2174)^{13}$ is

A. 2 **B.** 4 **C.** 6 **D.** 8

Q17.7.9 MCQ 2017 2M

Two machines M1 and M2 are able to execute any of four jobs P, Q, R and S. The machines can perform one job on one object at a time. Jobs P, Q, R and S take 30 minutes, 20 minutes, 60 minutes and 15 minutes each respectively. There are 10 objects each requiring exactly 1 job. Job P is to be performed on 2 objects, Job Q on 3 objects, Job R on 1 object and Job S on 4 objects. What is the minimum time needed to complete all the jobs?

A. 2 hours **B.** 2.5 hours **C.** 3 hours **D.** 3.5 hours

Q17.7.10 MCQ 2017 2M

The bar graph below shows the output of five carpenters over one month, each of whom made different items of furniture: chairs, tables, and beds.

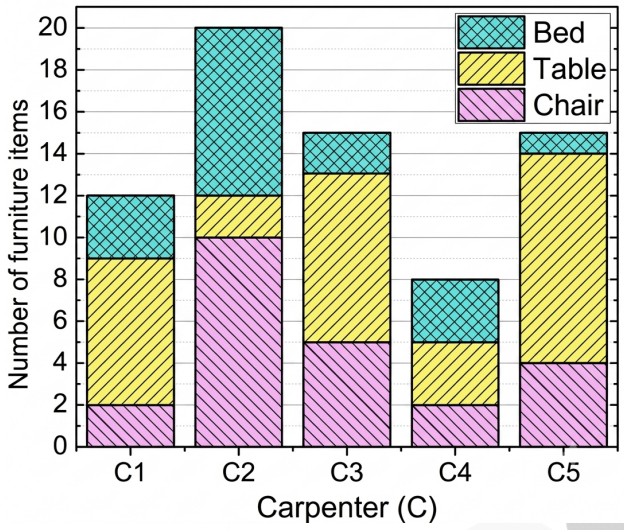


Fig. Q17.7.10

Consider the following statements:

i. The number of beds made by carpenter C2 is exactly the same as the number of tables made by carpenter C3.

ii. The total number of chairs made by all carpenters is less than the total number of tables.

Which one of the following is true?

A. Only i **B.** Only ii

C. Both i and ii **D.** Neither i nor ii

Session : 8 | Afternoon | 12/02/17

Q17.8.1 MCQ 2017 1M

The event would have been successful if you _____ able to come.

A. are **B.** had been **C.** have been **D.** would have been

Q17.8.2 MCQ 2017 1M

There was no doubt that their work was thorough.

Which of the words below is closest in meaning to the underlined word above?

A. pretty **B.** complete **C.** sloppy **D.** haphazard

Q17.8.3 MCQ 2017 1M

Four cards lie on a table. Each card has a number printed on one side and a colour on the other. The faces visible on the cards are 2, 3, red, and blue.

Proposition: If a card has an even value on one side, then its opposite face is red.

The cards which MUST be turned over to verify the above proposition are

A. 2, red **B.** 2, 3, red **C.** 2, blue **D.** 2, red, blue

Q17.8.4 MCQ 2017 1M

What is the value of x when $81\times\left(\frac{16}{25}\right)^{x+2}\div\left(\frac{2}{5}\right)^{2x+4}=144$?

A. 1 **B.** -1 **C.** -2 **D.** Cannot be determined

Q17.8.5 MCQ 2017 1M

Two dice are thrown simultaneously. The probability that the product of the numbers appearing on the top faces of the dice is a perfect square is

A. 1/9 **B.** 2/9 **C.** 1/3 **D.** 4/9

Q17.8.6 MCQ 2017 2M

Bhaichung was observing the pattern of people entering and leaving a car service centre. There was a single window where customers were being served. He saw that people inevitably came out of the centre in the order that they went in. However, the time they spent inside seemed to vary a lot: some people came out in a matter of minutes while for others it took much longer.

From this, what can one conclude?

A. The centre operates on a first-come-first-served basis, but with variable service times depending on specific customer needs.

B. Customers were served in an arbitrary order, since they took varying amounts of time for service completion in the centre.

C. Since some people came out within a few minutes of entering the centre, the system is likely to operate on a last-come-first-served basis.

D. Entering the centre early ensured that one would have shorter service times and most people attempted to do this.

Q17.8.7 MCQ 2017 2M

A map shows the elevations of Darjeeling, Gangtok, Kalimpong, Pelling, and Siliguri. Kalimpong is at a lower elevation than Gangtok. Pelling is at a lower elevation than Gangtok. Pelling is at a higher elevation than Siliguri. Darjeeling is at a higher elevation than Gangtok.
Which of the following statements can be inferred from the paragraph above?

i. Pelling is at a higher elevation than Kalimpong
ii. Kalimpong is at a lower elevation than Darjeeling
iii. Kalimpong is at a lower elevation than Pelling
iv. Siliguri is at a lower elevation than Gangtok

A. Only ii
B. Only ii and iii
C. Only ii and iv
D. Only iii and iv

Q17.8.8 MCQ 2017 2M

P, Q, R, S, T and U are seated around a circular table. R is seated two places to the right of Q. P is seated three places to the left of R. S is seated opposite U. If P and U now switch seats, which of the following must necessarily be true?

A. P is immediately to the right of R

B. T is immediately to the left of P

C. T is immediately to the left of P or P is immediately to the right of Q

D. U is immediately to the right of R or P is immediately to the left of T

Q17.8.9 MCQ 2017 2M

Budhan covers a distance of 19 km in 2 hours by cycling one fourth of the time and walking the rest. The next day he cycles (at the same speed as before) for half the time and walks the rest (at the same speed as before) and covers 26 km in 2 hours. The speed in km/h at which Budhan walks is

A. 1 **B.** 4 **C.** 5 **D.** 6

Q17.8.10 MCQ 2017 2M

The points in the graph below represent the halts of a lift for durations of 1 minute, over a period of 1 hour.

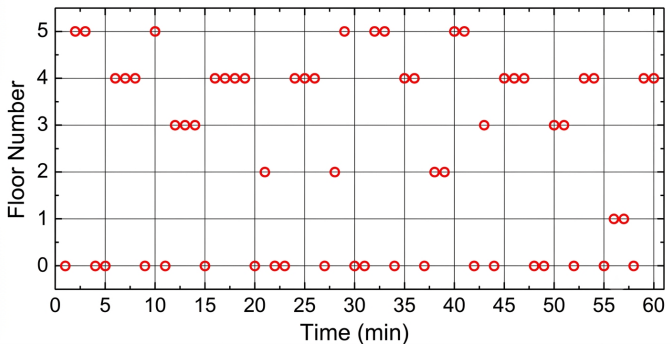


Fig. Q17.8.10

Which of the following statements are correct?

i. The elevator never moves directly from any non-ground floor to another non-ground floor over the one hour period
ii. The elevator stays on the fourth floor for the longest duration over the one hour period

A. Only i
B. Only ii
C. Both i and ii
D. Neither i nor ii

Answer Key

Session : 1					
Q17.1.1 D	Q17.1.2 D	Q17.1.3 D	Q17.1.4 B	Q17.1.5 A	Q17.1.6 C
Q17.1.7 D	Q17.1.8 C	Q17.1.9 B	Q17.1.10 A		
Session : 2					
Q17.2.1 C	Q17.2.2 A	Q17.2.3 D	Q17.2.4 C	Q17.2.5 B	Q17.2.6 D
Q17.2.7 A	Q17.2.8 C	Q17.2.9 D	Q17.2.10 B		
Session : 3					
Q17.3.1 A	Q17.3.2 A	Q17.3.3 D	Q17.3.4 B	Q17.3.5 B	Q17.3.6 D
Q17.3.7 A	Q17.3.8 A	Q17.3.9 A	Q17.3.10 C		
Session : 4					
Q17.4.1 C	Q17.4.2 A	Q17.4.3 B	Q17.4.4 A	Q17.4.5 D	Q17.4.6 A
Q17.4.7 B	Q17.4.8 C	Q17.4.9 C	Q17.4.10 B		
Session : 5					
Q17.5.1 C	Q17.5.2 D	Q17.5.3 C	Q17.5.4 D	Q17.5.5 C	Q17.5.6 B
Q17.5.7 A	Q17.5.8 B	Q17.5.9 D	Q17.5.10 C		
Session : 6					
Q17.6.1 D	Q17.6.2 B	Q17.6.3 A	Q17.6.4 B	Q17.6.5 D	Q17.6.6 B
Q17.6.7 B	Q17.6.8 A	Q17.6.9 C	Q17.6.10 C		
Session : 7					
Q17.7.1 C	Q17.7.2 C	Q17.7.3 D	Q17.7.4 C	Q17.7.5 D	Q17.7.6 B
Q17.7.7 B	Q17.7.8 B	Q17.7.9 A	Q17.7.10 C		
Session : 8					
Q17.8.1 B	Q17.8.2 B	Q17.8.3 C	Q17.8.4 B	Q17.8.5 B	Q17.8.6 A
Q17.8.7 C	Q17.8.8 C	Q17.8.9 D	Q17.8.10 D		

YEAR

2016

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	ME-1, EC-1	10
Session 2	ME-2, BT, CH, GG, MN, PH	10
Session 3	ME-3, EC-2	10
Session 4	EC-3, AR, CY, IN, MA, PE	10
Session 5	CS-1, CE-1	10
Session 6	CS-2, EE-1	10
Session 7	CE-2, AG, EY, MT, PI	10
Session 8	EE-2, AE, TF, XL , XE	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 77 **MSQ:** 1 **NAT:** 2

Session : 1 | Forenoon | 30/01/16

Q16.1.1 MCQ 2016 1M

Which of the following is **CORRECT** with respect to grammar and usage?
Mount Everest is _____.

A. the highest peak in the world

B. highest peak in the world

C. one of highest peak in the world

D. one of the highest peak in the world

Q16.1.2 MCQ 2016 1M

The policeman asked the victim of a theft, "What did you _____?"

A. loose **B.** lose **C.** loss **D.** louse

Q16.1.3 MCQ 2016 1M

Despite the new medicine's _____ in treating diabetes, it is not _____ widely.

A. effectiveness — prescribed

B. availability — used

C. prescription — available

D. acceptance — proscribed

Q16.1.4 MCQ 2016 1M

In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

A. 2029198 **B.** 2467482 **C.** 2789080 **D.** 3577422

Q16.1.5 MCQ 2016 1M

Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

A. 3.00 **B.** 4.99 **C.** 6.02 **D.** 7.01

Q16.1.6 MCQ 2016 2M

A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?

A. 15 **B.** 33 **C.** 35 **D.** 37

Q16.1.7 MCQ 2016 2M

In a world filled with uncertainty, he was glad to have many good friends. He had always assisted them in times of need and was confident that they would reciprocate. However, the events of the last week proved him wrong.
Which of the following inference(s) is/are logically valid and can be inferred from the above passage?

(i) His friends were always asking him to help them.
(ii) He felt that when in need of help, his friends would let him down.
(iii) He was sure that his friends would help him when in need.
(iv) His friends did not help him last week.

A. (i) and (ii) **B.** (iii) and (iv)

C. (iii) only **D.** (iv) only

Q16.1.8 MCQ 2016 2M

Leela is older than her cousin Pavithra. Pavithra's brother Shiva is older than Leela. When Pavithra and Shiva are visiting Leela, all three like to play chess. Pavithra wins more often than Leela does.
Which one of the following statements must be **TRUE** based on the above?

A. When Shiva plays chess with Leela and Pavithra, he often loses.

B. Leela is the oldest of the three.

C. Shiva is a better chess player than Pavithra.

D. Pavithra is the youngest of the three.

Q16.1.9 MCQ 2016 2M

If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$, the value of abc is _____.

A. $(rqs)^{-1}$ **B.** 0 **C.** 1 **D.** $r + q + s$

Q16.1.10 MCQ 2016 2M

P, **Q**, **R** and **S** are working on a project. **Q** can finish the task in 25 days, working alone for 12 hours a day. **R** can finish the task in 50 days, working alone for 12 hours per day. **Q** worked 12 hours a day but took sick leave in the beginning for two days. **R** worked 18 hours a day on all days. What is the ratio of work done by **Q** and **R** after 7 days from the start of the project?

A. 10:11 **B.** 11:10 **C.** 20:21 **D.** 21:20

Session : 2 | Afternoon | 30/01/16

Q16.2.1 MCQ 2016 1M

The volume of a sphere of diameter 1 unit is _____ than the volume of a cube of side 1 unit.

A. least **B.** less **C.** lesser **D.** low

Q16.2.2 MCQ 2016 1M

The unruly crowd demanded that the accused be _____ without trial.

A. hanged **B.** hanging **C.** hankering **D.** hung

Q16.2.3 MSQ 2016 1M

Choose the statement(s) where the underlined word is used correctly:

(i) A prone is a dried plum.
(ii) He was lying prone on the floor.
(iii) People who eat a lot of fat are prone to heart disease.

A. (i) and (iii) only **B.** (iii) only

C. (i) and (ii) only **D.** (ii) and (iii) only

Q16.2.4 MCQ 2016 1M

Fact: If it rains, then the field is wet.
Read the following statements:

(i) It rains
(ii) The field is not wet
(iii) The field is wet
(iv) It did not rain

Which one of the options given below is **NOT** logically possible, based on the given fact?

A. If (iii), then (iv). **B.** If (i), then (iii).

C. If (i), then (ii). **D.** If (ii), then (iv).

Q16.2.5 MCQ 2016 1M

A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is _____.

A. 1.43 **B.** 2.06 **C.** 2.68 **D.** 2.88

Q16.2.6 MCQ 2016 2M

Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution,
Which of the following statements will have the highest probability of being **TRUE**?

A. No student in group **Q** scored less marks than any student in group **P**.

B. No student in group **P** scored less marks than any student in group **Q**.

C. Most students of group **Q** scored marks in a narrower range than students in group **P**.

D. The median of the marks of group **P** is 100.

Q16.2.7 MCQ 2016 2M

A smart city integrates all modes of transport, uses clean energy and promotes sustainable use of resources. It also uses technology to ensure safety and security of the city, something which critics argue, will lead to a surveillance state.
Which of the following can be logically inferred from the above paragraph?

(i) All smart cities encourage the formation of surveillance states.
(ii) Surveillance is an integral part of a smart city.
(iii) Sustainability and surveillance go hand in hand in a smart city.
(iv) There is a perception that smart cities promote surveillance.

A. (i) and (iv) only **B.** (ii) and (iii) only

C. (iv) only **D.** (i) only

Q16.2.8 MCQ 2016 2M

Find the missing sequence in the letter series.

$B,\ FH,\ LNP,\ ____$

A. SUWY **B.** TUVW **C.** TVXZ **D.** TWXZ

Q16.2.9 MCQ 2016 2M

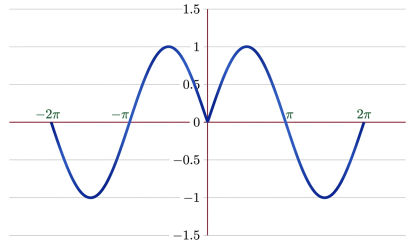
The binary operation $\square$ is defined as $a \square b = ab + (a + b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x = a$, for any a, is _____.

A. 0 **B.** 1 **C.** 2 **D.** 10

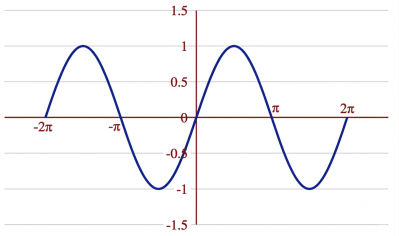
Q16.2.10 MCQ 2016 2M

Which of the following curves represents the function $y = \ln\left(\left|e^{[|\sin(|x|)|]}\right|\right)$ for $|x| < 2\pi$? Here, x represents the abscissa and y represents the ordinate.

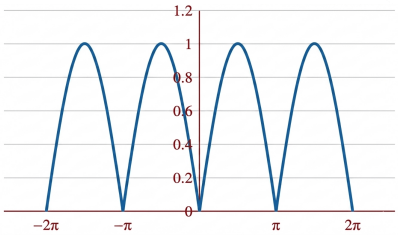
A.



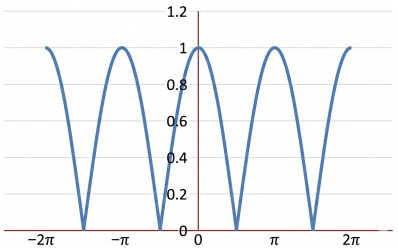
B.



C.



D.



Session : 3 | Forenoon | 31/01/16

Q16.3.1 MCQ 2016 1M

Based on the given statements, select the appropriate option with respect to grammar and usage.

Statements:

(i) The height of Mr. **X** is 6 feet.

(ii) The height of Mr. **Y** is 5 feet.

A. Mr. **X** is longer than Mr. **Y**.

B. Mr. **X** is more elongated than Mr. **Y**.

C. Mr. **X** is taller than Mr. **Y**.

D. Mr. **X** is lengthier than Mr. **Y**.

Q16.3.2 MCQ 2016 1M

The students _____ the teacher on teachers' day for twenty years of dedicated teaching.

A. facilitated **B.** felicitated **C.** fantasized **D.** facillitated

Q16.3.3 MCQ 2016 1M

After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the rest is history. What does the underlined phrase mean in this context?

A. history will rest in peace

B. rest is recorded in history books

C. rest is well known

D. rest is archaic

Q16.3.4 MCQ 2016 1M

Given $(9 \text{ inches})^{\frac{1}{2}} = (0.25 \text{ yards})^{\frac{1}{2}}$, which one of the following statements is **TRUE**?

A. 3 inches = 0.5 yards **B.** 9 inches = 1.5 yards

C. 9 inches = 0.25 yards **D.** 81 inches = 0.0625 yards

Q16.3.5 MCQ 2016 1M

S, **M**, **E** and **F** are working in shifts in a team to finish a project. **M** works with twice the efficiency of others but for half as many days as **E** worked. **S** and **M** have 6 hour shifts in a day, whereas **E** and **F** have 12 hour shifts. What is the ratio of contribution of **M** to contribution of **E** in the project?

A. 1:1 **B.** 1:2 **C.** 1:4 **D.** 2:1

Q16.3.6 MCQ 2016 2M

The Venn diagram shows the preference of the student population for leisure activities.

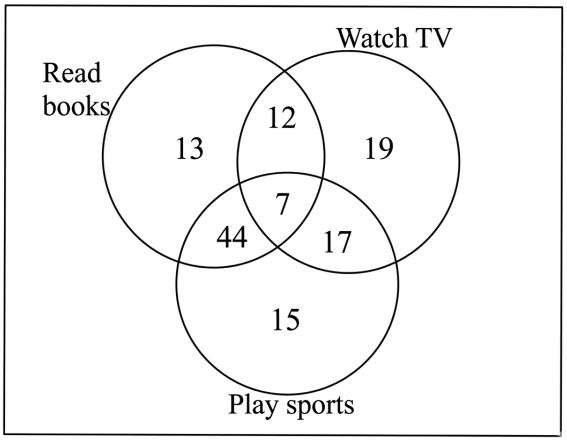


Fig. Q16.3.6

From the data given, the number of students who like to read books or play sports is _____.

A. 44 **B.** 51 **C.** 79 **D.** 108

Q16.3.7 MCQ 2016 2M

Social science disciplines were in existence in an amorphous form until the colonial period when they were institutionalized. In varying degrees, they were intended to further the colonial interest. In the time of globalization and the economic rise of postcolonial countries like India, conventional ways of knowledge production have become obsolete.

Which of the following can be logically inferred from the above statements?

(i) Social science disciplines have become obsolete.
(ii) Social science disciplines had a pre-colonial origin.
(iii) Social science disciplines always promote colonialism.
(iv) Social science must maintain disciplinary boundaries.

A. (ii) only **B.** (i) and (iii) only

C. (ii) and (iv) only **D.** (iii) and (iv) only

Q16.3.8 MCQ 2016 2M

Two and a quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show 1:30. What is the actual current time shown by the clock?

A. 8:15 **B.** 11:15 **C.** 12:15 **D.** 12:45

Q16.3.9 MCQ 2016 2M

M and **N** start from the same location. **M** travels 10 km East and then 10 km North-East. **N** travels 5 km South and then 4 km South-East. What is the shortest distance (in km) between **M** and **N** at the end of their travel?

A. 18.60 **B.** 22.50 **C.** 20.61 **D.** 25.00

Q16.3.10 MCQ 2016 2M

A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a **MINIMUM**?

A. 30 **B.** 40 **C.** 120 **D.** 180

Session : 4 | Afternoon | 31/01/16

Q16.4.1 MCQ 2016 1M

An apple costs Rs. 10. An onion costs Rs. 8.

Select the most suitable sentence with respect to grammar and usage.

A. The price of an apple is greater than an onion.

B. The price of an apple is more than onion.

C. The price of an apple is greater than that of an onion.

D. Apples are more costlier than onions.

Q16.4.2 MCQ 2016 1M

The Buddha said, "Holding on to anger is like grasping a hot coal with the intent of throwing it at someone else; you are the one who gets burnt."

Select the word below which is closest in meaning to the word underlined above.

A. burning **B.** igniting **C.** clutching **D.** flinging

Q16.4.3 MCQ 2016 1M

M has a son **Q** and a daughter **R**. He has no other children. **E** is the mother of **P** and daughter-in-law of **M**. How is **P** related to **M**?

A. **P** is the son-in-law of **M**.

B. **P** is the grandchild of **M**.

C. **P** is the daughter-in law of **M**.

D. **P** is the grandfather of **M**.

Q16.4.4 MCQ 2016 1M

The number that least fits this set:
(324, 441, 97 and 64) is _____.

A. 324 **B.** 441 **C.** 97 **D.** 64

Q16.4.5 MCQ 2016 1M

It takes 10 s and 15 s, respectively, for two trains travelling at different constant speeds to completely pass a telegraph post. The length of the first train is 120 m and that of the second train is 150 m. The magnitude of the difference in the speeds of the two trains (in m/s) is _____.

A. 2.0 **B.** 10.0 **C.** 12.0 **D.** 22.0

Q16.4.6 MCQ 2016 2M

The velocity V of a vehicle along a straight line is measured in m/s and plotted as shown with respect to time in seconds. At the end of the 7 seconds, how much will the odometer reading increase by (in m)?

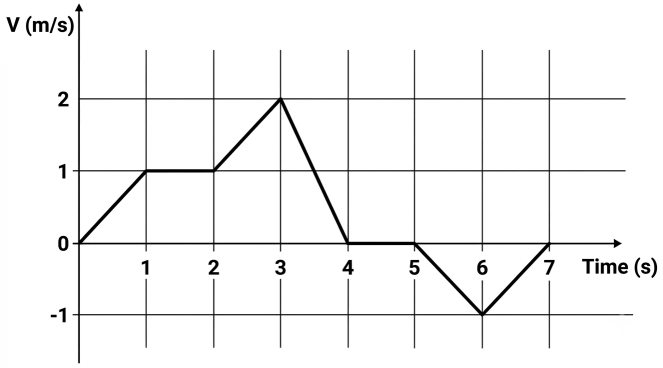


Fig. Q16.4.6

A. 0 **B.** 3 **C.** 4 **D.** 5

Q16.4.7 MCQ 2016 2M

The overwhelming number of people infected with rabies in India has been flagged by the World Health Organization as a source of concern. It is estimated that inoculating 70% of pets and stray dogs against rabies can lead to a significant reduction in the number of people infected with rabies.
Which of the following can be logically inferred from the above sentences?

A. The number of people in India infected with rabies is high.

B. The number of people in other parts of the world who are infected with rabies is low.

C. Rabies can be eradicated in India by vaccinating 70% of stray dogs.

D. Stray dogs are the main source of rabies worldwide.

Q16.4.8 MCQ 2016 2M

A flat is shared by four first year undergraduate students. They agreed to allow the oldest of them to enjoy some extra space in the flat. Manu is two months older than Sravan, who is three months younger than Trideep. Pavan is one month older than Sravan. Who should occupy the extra space in the flat?

A. Manu **B.** Sravan **C.** Trideep **D.** Pavan

Q16.4.9 MCQ 2016 2M

Find the area bounded by the lines $3x+2y = 14$, $2x-3y = 5$ in the first quadrant.

A. 14.95 **B.** 15.25 **C.** 15.70 **D.** 20.35

Q16.4.10 MCQ 2016 2M

A straight line is fit to a data set $(\ln x,\ y)$. This line intercepts the abscissa at $\ln x = 0.1$ and has a slope of -0.02. What is the value of y at $x = 5$ from the fit?

A. −0.030 **B.** −0.014 **C.** 0.014 **D.** 0.030

Session : 5 | Forenoon | 06/02/16

Q16.5.1 MCQ 2016 1M

Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

A. I will not leave the place until the minister does not meet me.

B. I will not leave the place until the minister doesn't meet me.

C. I will not leave the place until the minister meet me.

D. I will not leave the place until the minister meets me.

Q16.5.2 MCQ 2016 1M

A rewording of something written or spoken is a _____.

A. paraphrase **B.** paradox **C.** paradigm **D.** paraffin

Q16.5.3 MCQ 2016 1M

Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.

A. figurative **B.** collateral **C.** literal **D.** figurine

Q16.5.4 MCQ 2016 1M

If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

A. zentaga **B.** tagafer **C.** tagazen **D.** relffer

Q16.5.5 MCQ 2016 1M

A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

A. 56 **B.** 64 **C.** 72 **D.** 96

Q16.5.6 MCQ 2016 2M

A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

A. Elegance **B.** Executive **C.** Smooth **D.** Soft

Q16.5.7 MCQ 2016 2M

Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.
Which of the following can be logically inferred from the above sentences?

A. India is a country of exactly seventeen languages.

B. Linguistic pluralism is the only indicator of a nation's diversity.

C. Indian currency notes have sufficient space for all the Indian languages.

D. Linguistic pluralism is strong evidence of India's diversity.

Q16.5.8 MCQ 2016 2M

Consider the following statements relating to the level of poker play of four players **P**, **Q**, **R** and **S**.

I. **P** always beats **Q**
II. **R** always beats **S**
III. **S** loses to **P** only sometimes
IV. **R** always loses to **Q**

Which of the following can be logically inferred from the above statements?

(i) **P** is likely to beat all the three other players
(ii) **S** is the absolute worst player in the set

A. (i) only **B.** (ii) only

C. (i) and (ii) **D.** neither (i) nor (ii)

Q16.5.9 MCQ 2016 2M

If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

A. $(x^3 + 8)$ **B.** $(x - 1)$ **C.** $(2x - 5)$ **D.** $(x + 1)$

Q16.5.10 MCQ 2016 2M

In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

A. 40.00 **B.** 46.02 **C.** 60.01 **D.** 92.02

Session : 6 | Afternoon | 06/02/16

Q16.6.1 MCQ 2016 1M

The man who is now Municipal Commissioner worked as _____.

A. the security guard at a university

B. a security guard at the university

C. a security guard at university

D. the security guard at the university

Q16.6.2 MCQ 2016 1M

Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.
Choose the option which is closest in meaning to the underlined phrase in the above sentence.

A. put up with **B.** put in with

C. put down to **D.** put up against

Q16.6.3 MCQ 2016 1M

Find the odd one in the following group of words.
mock, deride, praise, jeer

A. mock **B.** deride **C.** praise **D.** jeer

Q16.6.4 MCQ 2016 1M

Pick the odd one from the following options.

A. CADBE **B.** JHKIL **C.** XVYWZ **D.** ONPMQ

Q16.6.5 MCQ 2016 1M

In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

A. n^4 **B.** 4^n **C.** 2^{2n-1} **D.** 4^{n-1}

Q16.6.6 MCQ 2016 2M

Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

A. 35 **B.** 45 **C.** 65 **D.** 90

Q16.6.7 MCQ 2016 2M

Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.
Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

(i) The author believes that computers are not good for us.
(ii) Mobile computers and the internet are both intended inventions.

A. (i) only **B.** (ii) only

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q16.6.8 MCQ 2016 2M

All hill-stations have a lake.
Ooty has two lakes.
Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

(i) Ooty is not a hill-station.
(ii) No hill-station can have more than one lake.

A. (i) only **B.** (ii) only

C. both (i) and (ii) **D.** neither (i) nor (ii)

Q16.6.9 MCQ 2016 2M

In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?

Fig. Q16.6.9

A. 21 **B.** 27 **C.** 30 **D.** 36

Q16.6.10 MCQ 2016 2M

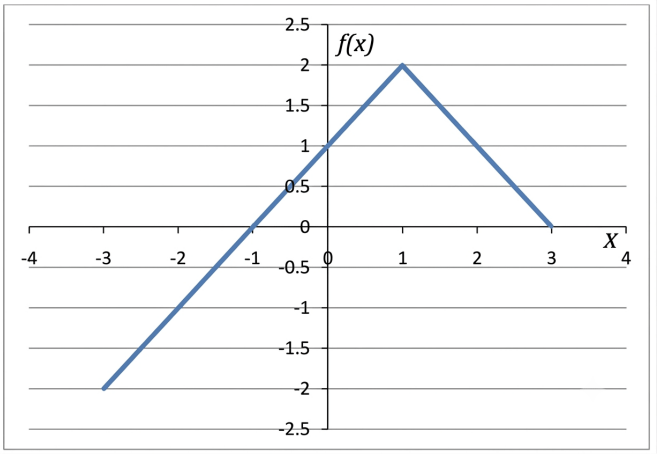


Fig. Q16.6.10

Choose the correct expression for $f(x)$ given in the graph.

A. $f(x) = 1 - |x - 1|$ **B.** $f(x) = 1 + |x - 1|$

C. $f(x) = 2 - |x - 1|$ **D.** $f(x) = 2 + |x - 1|$

Session : 7 | Forenoon | 07/02/16

Q16.7.1 MCQ 2016 1M

If I were you, I _____ that laptop. It's much too expensive.

A. won't buy **B.** shan't buy

C. wouldn't buy **D.** would buy

Q16.7.2 MCQ 2016 1M

He turned a deaf ear to my request.
What does the underlined phrasal verb mean?

A. ignored **B.** appreciated **C.** twisted **D.** returned

Q16.7.3 MCQ 2016 1M

Choose the most appropriate set of words from the options given below to complete the following sentence.
_____ _____ is a will, _____ is a way.

A. Wear, there, their **B.** Were, their, there

C. Where, there, there **D.** Where, their, their

Q16.7.4 MCQ 2016 1M

(x% of y) + (y% of x) is equivalent to _____.

A. 2% of xy **B.** 2% of $(xy/100)$

C. xy% of 100 **D.** 100% of xy

Q16.7.5 MCQ 2016 1M

The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.

A. 39 **B.** 57 **C.** 66 **D.** 93

Q16.7.6 MCQ 2016 2M

Two finance companies, P and Q, declared fixed annual rates of interest on the amounts invested with them. The rates of interest offered by these companies may differ from year to year. Year-wise annual rates of interest offered by these companies are shown by the line graph provided below.

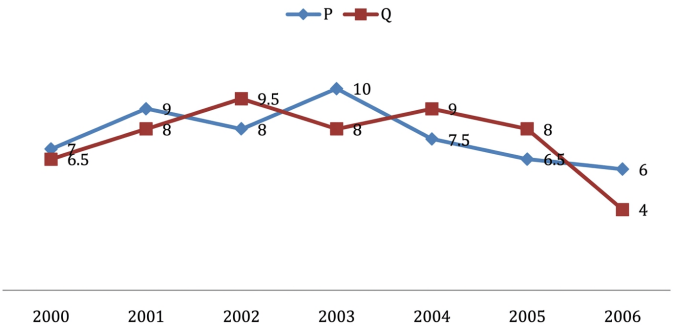


Fig. Q16.7.6

If the amounts invested in the companies, P and Q, in 2006 are in the ratio 8:9, then the amounts received after one year as interests from companies P and Q would be in the ratio:

A. 2:3 **B.** 3:4 **C.** 6:7 **D.** 4:3

Q16.7.7 MCQ 2016 2M

Today, we consider Ashoka as a great ruler because of the copious evidence he left behind in the form of stone carved edicts. Historians tend to correlate greatness of a king at his time with the availability of evidence today.
Which of the following can be logically inferred from the above sentences?

A. Emperors who do not leave significant sculpted evidence are completely forgotten.

B. Ashoka produced stone carved edicts to ensure that later historians will respect him.

C. Statues of kings are a reminder of their greatness.

D. A king's greatness, as we know him today, is interpreted by historians.

Q16.7.8 MCQ 2016 2M

Fact 1: Humans are mammals.
Fact 2: Some humans are engineers.
Fact 3: Engineers build houses.

If the above statements are facts, which of the following can be logically inferred?

I. All mammals build houses.
II. Engineers are mammals.
III. Some humans are not engineers.

A. II only **B.** III only

C. I, II and III **D.** I only

Q16.7.9 MCQ 2016 2M

A square pyramid has a base perimeter x, and the slant height is half of the perimeter. What is the lateral surface area of the pyramid?

A. x^2 **B.** $0.75\,x^2$ **C.** $0.50\,x^2$ **D.** $0.25\,x^2$

Q16.7.10 MCQ 2016 2M

Ananth takes 6 hours and Bharath takes 4 hours to read a book. Both started reading copies of the book at the same time. After how many hours is the number of pages **to be** read by Ananth, twice that **to be** read by Bharath? Assume Ananth and Bharath read all the pages with constant pace.

A. 1 **B.** 2 **C.** 3 **D.** 4

Session : 8 | Afternoon | 07/02/16

Q16.8.1 MCQ 2016 1M

The chairman requested the aggrieved shareholders to _____ him.

A. bare with **B.** bore with

C. bear with **D.** bare

Q16.8.2 MCQ 2016 1M

Identify the correct spelling out of the given options:

A. Managable **B.** Manageable

C. Mangaeble **D.** Managible

Q16.8.3 MCQ 2016 1M

Pick the odd one out in the following:
13, 23, 33, 43, 53

A. 23 **B.** 33 **C.** 43 **D.** 53

Q16.8.4 MCQ 2016 1M

R2D2 is a robot. R2D2 can repair aeroplanes. No other robot can repair aeroplanes.
Which of the following can be logically inferred from the above statements?

A. R2D2 is a robot which can only repair aeroplanes.

B. R2D2 is the only robot which can repair aeroplanes.

C. R2D2 is a robot which can repair only aeroplanes.

D. Only R2D2 is a robot.

Q16.8.5 MCQ 2016 1M

If $|9y - 6| = 3$, then $y^2 - 4y/3$ is _____.

A. 0 **B.** $+1/3$ **C.** $-1/3$ **D.** undefined

Q16.8.6 NAT 2016 2M

The following graph represents the installed capacity for cement production (in tonnes) and the actual production (in tonnes) of nine cement plants of a cement company. Capacity utilization of a plant is defined as ratio of actual production of cement to installed capacity. A plant with installed capacity of at least 200 tonnes is called a large plant and a plant with lesser capacity is called a small plant. The difference between total production of large plants and small plants, in tonnes is .

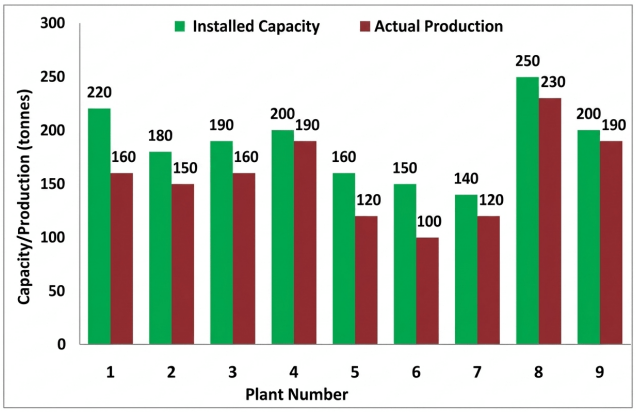


Fig. Q16.8.6

Q16.8.7 MCQ 2016 2M

A poll of students appearing for masters in engineering indicated that 60% of the students believed that mechanical engineering is a profession unsuitable for women. A research study on women with masters or higher degrees in mechanical engineering found that 99% of such women were successful in their professions.
Which of the following can be logically inferred from the above paragraph?

A. Many students have misconceptions regarding various engineering disciplines.

B. Men with advanced degrees in mechanical engineering believe women are well suited to be mechanical engineers.

C. Mechanical engineering is a profession well suited for women with masters or higher degrees in mechanical engineering.

D. The number of women pursuing higher degrees in mechanical engineering is small.

Q16.8.8 MCQ 2016 2M

Sourya committee had proposed the establishment of Sourya Institutes of Technology (SITs) in line with Indian Institutes of Technology (IITs) to cater to the technological and industrial needs of a developing country.
Which of the following can be logically inferred from the above sentence?

Based on the proposal,

(i) In the initial years, SIT students will get degrees from IIT.
(ii) SITs will have a distinct national objective.
(iii) SIT like institutions can only be established in consultation with IIT.
(iv) SITs will serve technological needs of a developing country.

A. (iii) and (iv) only. **B.** (i) and (iv) only.

C. (ii) and (iv) only. **D.** (ii) and (iii) only.

Q16.8.9 MCQ 2016 2M

Shaquille O' Neal is a 60% career free throw shooter, meaning that he successfully makes 60 free throws out of 100 attempts on average. What is the probability that he will successfully make exactly 6 free throws in 10 attempts?

A. 0.2508 **B.** 0.2816 **C.** 0.2934 **D.** 0.6000

Q16.8.10 NAT 2016 2M

The numeral in the units position of $211^{870} + 146^{127} \times 3^{424}$ is _____.

Answer Key

Session : 1					
Q16.1.1 A	Q16.1.2 B	Q16.1.3 A	Q16.1.4 A	Q16.1.5 C	Q16.1.6 C
Q16.1.7 B	Q16.1.8 D	Q16.1.9 C	Q16.1.10 C		
Session : 2					
Q16.2.1 B	Q16.2.2 A	Q16.2.3 D	Q16.2.4 C	Q16.2.5 B	Q16.2.6 C
Q16.2.7 C	Q16.2.8 C	Q16.2.9 A	Q16.2.10 C		
Session : 3					
Q16.3.1 C	Q16.3.2 B	Q16.3.3 C	Q16.3.4 C	Q16.3.5 B	Q16.3.6 D
Q16.3.7 A	Q16.3.8 D	Q16.3.9 C	Q16.3.10 B		
Session : 4					
Q16.4.1 C	Q16.4.2 C	Q16.4.3 B	Q16.4.4 C or D	Q16.4.5 A	Q16.4.6 D
Q16.4.7 A	Q16.4.8 C	Q16.4.9 B	Q16.4.10 A		
Session : 5					
Q16.5.1 D	Q16.5.2 A	Q16.5.3 A	Q16.5.4 C	Q16.5.5 D	Q16.5.6 B
Q16.5.7 D	Q16.5.8 D	Q16.5.9 B	Q16.5.10 B		
Session : 6					
Q16.6.1 B	Q16.6.2 A	Q16.6.3 C	Q16.6.4 D	Q16.6.5 B	Q16.6.6 A
Q16.6.7 D	Q16.6.8 D	Q16.6.9 C	Q16.6.10 C		
Session : 7					
Q16.7.1 C	Q16.7.2 A	Q16.7.3 C	Q16.7.4 A	Q16.7.5 A	Q16.7.6 D
Q16.7.7 D	Q16.7.8 B	Q16.7.9 D	Q16.7.10 C		
Session : 8					
Q16.8.1 C	Q16.8.2 B	Q16.8.3 B	Q16.8.4 B	Q16.8.5 C	Q16.8.6 120.00
Q16.8.7 C	Q16.8.8 C	Q16.8.9 A	Q16.8.10 7.00		

YEAR

2015

8 Sessions · 80 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AE, AR, BT, CH, CY, EC-1, EY, XE, XL	10
Session 2	EC-2, ME-1	10
Session 3	EC-3, ME-2	10
Session 4	AE, GG, IN, MA, ME-3, MN, MT, PH, PI, TF	10
Session 5	CS-1, EE-1	10
Session 6	CS-2, EE-2	10
Session 7	CE-1, CS-3	10
Session 8	CE-2	10

PrepFusion

Year Snapshot

Total Questions: 80 **MCQ:** 72 **MSQ:** 0 **NAT:** 8

Session : 1 | Forenoon | 31/01/15

Q15.1.1 MCQ 2015 1M

Choose the most appropriate word from the options given below to complete the following sentence.
The principal presented the chief guest with a _____, as token of appreciation.

A. momento **B.** memento **C.** momentum **D.** moment

Q15.1.2 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Frogs _____.

A. croak **B.** roar **C.** hiss **D.** patter

Q15.1.3 MCQ 2015 1M

Choose the word most similar in meaning to the given word:
Educe

A. Exert **B.** Educate **C.** Extract **D.** Extend

Q15.1.4 MCQ 2015 1M

Operators $\square$, $\diamond$ and $\rightarrow$ are defined by:

$$a\square b = \frac{a-b}{a+b}; a\diamond b = \frac{a+b}{a-b}; a \rightarrow b = ab.$$

Find the value of $(66\square 6) \rightarrow (66\diamond 6)$.

A. -2 **B.** -1 **C.** 1 **D.** 2

Q15.1.5 MCQ 2015 1M

If $\log_x(5/7) = -1/3$, then the value of x is

A. 343/125 **B.** 125/343 **C.** -25/49 **D.** -49/25

Q15.1.6 MCQ 2015 2M

The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underlined part. Following the requirements of the standard written English, select the answer that produces the most effective sentence.
Tuberculosis, together with its effects, ranks one of the leading causes of death in India.

A. ranks as one of the leading causes of death

B. rank as one of the leading causes of death

C. has the rank of one of the leading causes of death

D. are one of the leading causes of death

Q15.1.7 MCQ 2015 2M

Read the following paragraph and choose the correct statement.
Climate change has reduced human security and threatened human well being. An ignored reality of human progress is that human security largely depends upon environmental security. But on the contrary, human progress seems contradictory to environmental security. To keep up both at the required level is a challenge to be addressed by one and all. One of the ways to curb the climate change may be suitable scientific innovations, while the other may be the Gandhian perspective on small scale progress with focus on sustainability.

A. Human progress and security are positively associated with environmental security.

B. Human progress is contradictory to environmental security.

C. Human security is contradictory to environmental security.

D. Human progress depends upon environmental security.

Q15.1.8 NAT 2015 2M

Fill in the missing value

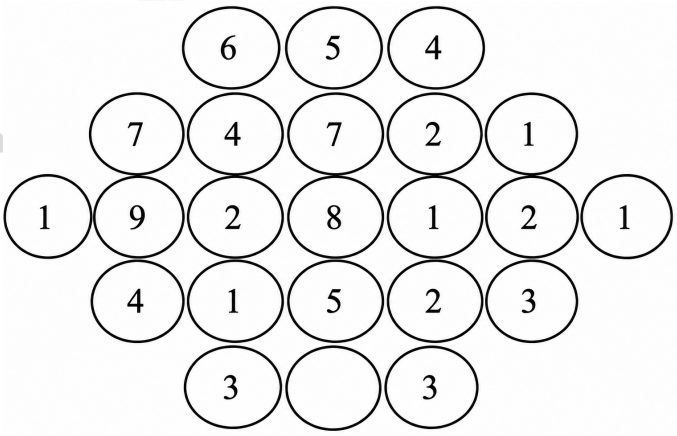


Fig. Q15.1.8

Q15.1.9 MCQ 2015 2M

A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible.

A. 1 : 4 **B.** 1 : 3 **C.** 1 : 2 **D.** 2 : 3

Q15.1.10 MCQ 2015 2M

Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Humpty Dumpty always falls while having lunch

B. Humpty Dumpty does not fall sometimes while having lunch

C. Humpty Dumpty never falls during dinner

D. When Humpty Dumpty does not sit on the wall, the wall does not break

Session : 2 | Afternoon | 31/01/15

Q15.2.1 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Dhoni, as well as the other team members of Indian team, _____ present on the occasion.

A. were **B.** was **C.** has **D.** have

Q15.2.2 MCQ 2015 1M

Choose the word most similar in meaning to the given word:
Awkward

A. Inept **B.** Graceful **C.** Suitable **D.** Dreadful

Q15.2.3 MCQ 2015 1M

What is the adverb for the given word below?
Misogynous

A. Misogynousness **B.** Misogynity

C. Misogynously **D.** Misogynous

Q15.2.4 MCQ 2015 1M

An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O, and P, in that order. The cumulative distances travelled and the corresponding electricity consumption are shown in the Table below:

Stretch	Cumulative distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

A. M **B.** N **C.** O **D.** P

Q15.2.5 MCQ 2015 1M

Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $\frac{1}{6}$ and that of Ramesh is $\frac{1}{8}$. What is the probability that only one of them will be selected?

A. $\frac{47}{48}$ **B.** $\frac{1}{4}$ **C.** $\frac{13}{48}$ **D.** $\frac{35}{48}$

Q15.2.6 MCQ 2015 2M

In the following sentence certain parts are underlined and marked P, Q, and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error.
The student corrected all the errors (P) that the instructor marked (Q) on the answer book (R).

A. P **B.** Q **C.** R **D.** No Error

Q15.2.7 MCQ 2015 2M

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:
I. All film stars are playback singers.
II. All film directors are film stars.

Conclusions:
I. All film directors are playback singers.
II. Some film stars are film directors.

A. Only conclusion I follows.

B. Only conclusion II follows.

C. Neither conclusion I nor II follows.

D. Both conclusions I and II follow.

Q15.2.8 NAT 2015 2M

A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5 metre per leap respectively, what distance in metres will the tiger have to run before it catches the deer?

Q15.2.9 MCQ 2015 2M

If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval

A. $[1, \frac{2}{3}]$ **B.** $[-\frac{1}{2}, 1]$ **C.** $[-1, \frac{1}{2}]$ **D.** $[2, -4]$

Q15.2.10 MCQ 2015 2M

Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future.

Which of the statement(s) below is/are logically valid and can be inferred from the above statements?

(i) The artists expected funding for the arts to increase this year.
(ii) The Chief Minister was receptive to the idea of increasing funding for the arts.
(iii) The Chief Minister is a prominent artist.
(iv) Schools are giving less importance to arts education nowadays.

A. (iii) and (iv) **B.** (i) and (iv)

C. (i), (ii) and (iv) **D.** (i) and (iii)

Session : 3 | Forenoon | 01/02/15

Q15.3.1 MCQ 2015 1M

Choose the most appropriate word from the options given below to complete the following sentence.

If the athlete had wanted to come first in the race, he _____ several hours every day.

A. should practise **B.** should have practised

C. practised **D.** should be practising

Q15.3.2 MCQ 2015 1M

Choose the most suitable one word substitute for the following expression:

Connotation of a road or way

A. Pertinacious **B.** Viaticum

C. Clandestine **D.** Ravenous

Q15.3.3 MCQ 2015 1M

Choose the correct verb to fill in the blank below:

Let us _____.

A. introvert **B.** alternate **C.** atheist **D.** altruist

Q15.3.4 MCQ 2015 1M

Find the missing sequence in the letter series below:

A, CD, GHI, ?, UVWXY

A. LMN **B.** MNO **C.** MNOP **D.** NOPQ

Q15.3.5 MCQ 2015 1M

If $x > y > 1$, which of the following must be true?

i. $\ln x > \ln y$ iii. $y^x > x^y$

ii. $e^x > e^y$ iv. $\cos x > \cos y$

A. (i) and (ii) **B.** (i) and (iii)

C. (iii) and (iv) **D.** (ii) and (iv)

Q15.3.6 MCQ 2015 2M

Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English.

A. It would remain between you and me.

B. It would remain between I and you.

C. It would remain between you and I.

D. It would remain with me.

Q15.3.7 MCQ 2015 2M

In the following question, the first and the last sentence of the passage are in order and numbered 1 and 6. The rest of the passage is split into 4 parts and numbered as 2, 3, 4, and 5. These 4 parts are not arranged in proper order. Read the sentences and arrange them in a logical sequence to make a passage and choose the correct sequence from the given options.

1. On Diwali, the family rises early in the morning.
2. The whole family, including the young and the old enjoy doing this.
3. Children let off fireworks later in the night with their friends.
4. At sunset, the lamps are lit and the family performs various rituals.

5. Father, mother, and children visit relatives and exchange gifts and sweets.
6. Houses look so pretty with lighted lamps all around.

A. 2,5,3,4 **B.** 5,2,4,3 **C.** 3,5,4,2 **D.** 4,5,2,3

Q15.3.8 NAT 2015 2M

From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is _____.

Q15.3.9 MCQ 2015 2M

$\log \tan 1^\circ + \log \tan 2^\circ + \ldots\ldots + \log \tan 89^\circ$ is. . .

A. 1 **B.** $\frac{1}{\sqrt{2}}$ **C.** 0 **D.** -1

Q15.3.10 MCQ 2015 2M

Ms. X will be in Bagdogra from 01/05/2014 to 20/05/2014 and from 22/05/2014 to 31/05/2014. On the morning of 21/05/2014, she will reach Kochi via Mumbai.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Ms. X will be in Kochi for one day, only in May.

B. Ms. X will be in Kochi for only one day in May.

C. Ms. X will be only in Kochi for one day in May.

D. Only Ms. X will be in Kochi for one day in May.

Session : 4 | Afternoon | 01/02/15

Q15.4.1 MCQ 2015 1M

Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:

Apparent lifelessness _____ dormant life.

A. harbours **B.** leads to **C.** supports **D.** affects

Q15.4.2 MCQ 2015 1M

Fill in the blank with the correct idiom/phrase.

That boy from the town was a _____ in the sleepy village.

A. dog out of herd **B.** sheep from the heap

C. fish out of water **D.** bird from the flock

Q15.4.3 MCQ 2015 1M

Choose the statement where underlined word is used correctly.

A. When the teacher eludes to different authors, he is being elusive.

B. When the thief keeps eluding the police, he is being elusive.

C. Matters that are difficult to understand, identify or remember are allusive.

D. Mirages can be allusive, but a better way to express them is illusory.

Q15.4.4 MCQ 2015 1M

Tanya is older than Eric.
Cliff is older than Tanya.
Eric is older than Cliff.

If the first two statements are true, then the third statement is:

A. True **B.** False **C.** Uncertain **D.** Data insufficient

Q15.4.5 MCQ 2015 1M

Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

A. 20 **B.** 10 **C.** 8 **D.** 5

Q15.4.6 MCQ 2015 2M

Select the appropriate option in place of underlined part of the sentence.
Increased productivity necessary reflects greater efforts made by the employees.

A. Increase in productivity necessary

B. Increase productivity is necessary

C. Increase in productivity necessarily

D. No improvement required

Q15.4.7 MCQ 2015 2M

Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statements:

I. No manager is a leader.
II. All leaders are executives.

Conclusions:

I. No manager is an executive.

II. No executive is a manager.

A. Only conclusion I follows.

B. Only conclusion II follows.

C. Neither conclusion I nor II follows.

D. Both conclusions I and II follow.

Q15.4.8 NAT 2015 2M

In the given figure angle Q is a right angle, $PS : QS = 3 : 1$, $RT : QT = 5 : 2$ and $PU : UR = 1 : 1$. If area of triangle QTS is 20 cm^2, then the area of triangle PQR in cm^2 is _____.

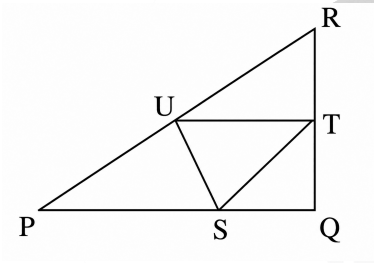


Fig. Q15.4.8

Q15.4.9 MCQ 2015 2M

Right triangle PQR is to be constructed in the xy-plane so that the right angle is at P and line PR is parallel to the x-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequalities: $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties?

A. 110 **B.** 1,100 **C.** 9,900 **D.** 10,000

Q15.4.10 MCQ 2015 2M

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE?

A. X and Y are not independent

B. Y and Z are dependent

C. Y and Z are independent

D. X and Z are independent

Session : 5 | Forenoon | 07/02/15

Q15.5.1 MCQ 2015 1M

Didn't you buy _____ when you went shopping?

A. any paper **B.** much paper

C. no paper **D.** a few paper

Q15.5.2 MCQ 2015 1M

Which of the following options is the closest in meaning to the sentence below?

She enjoyed herself immensely at the party.

A. She had a terrible time at the party

B. She had a horrible time at the party

C. She had a terrific time at the party

D. She had a terrifying time at the party

Q15.5.3 MCQ 2015 1M

Which one of the following combinations is incorrect?

A. Acquiescence – Submission

B. Wheedle – Roundabout

C. Flippancy – Lightness

D. Profligate – Extravagant

Q15.5.4 MCQ 2015 1M

Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

I. Each step is 3/4 foot high.

II. Each step is 1 foot wide.

A. Statement I alone is sufficient, but statement II alone is not sufficient.

B. Statement II alone is sufficient, but statement I alone is not sufficient.

C. Both statements are sufficient, but neither statement alone is sufficient.

D. Statement I and II together are not sufficient.

Q15.5.5 MCQ 2015 1M

Given Set $A = \{2, 3, 4, 5\}$ and Set $B = \{11, 12, 13, 14, 15\}$, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

A. 0.20 **B.** 0.25 **C.** 0.30 **D.** 0.33

Q15.5.6 MCQ 2015 2M

Select the alternative meaning of the underlined part of the sentence.

The chain snatchers took to their heels when the police party arrived.

A. took shelter in a thick jungle

B. open indiscriminate fire

C. took to flight

D. unconditionally surrendered

Q15.5.7 MCQ 2015 2M

The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

I. The water supply authority should impose a partial cut in supply to tackle the situation.

II. The government should appeal to all the residents through mass media for minimal use of water.

III. The government should ban the water supply in lower areas.

A. Statements I and II follow.

B. Statements I and III follow.

C. Statements II and III follow.

D. All statements follow.

Q15.5.8 NAT 2015 2M

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the Civil department and the female students in the Mechanical department?

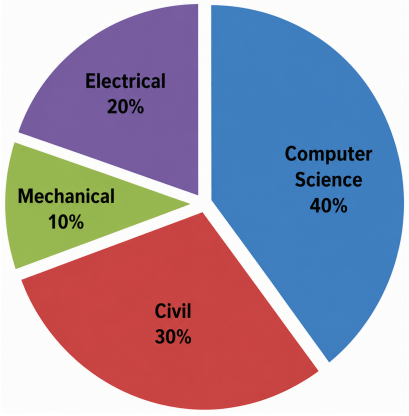


Fig. Q15.5.8

Q15.5.9 MCQ 2015 2M

The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

I. $p + m + c = 27/20$

II. $p + m + c = 13/20$

III. $(p) \cdot (m) \cdot (c) = 1/10$

A. Only relation I is true.

B. Only relation II is true.

C. Relations II and III are true.

D. Relations I and III are true.

Q15.5.10 MCQ 2015 2M

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

A. 2.290 **B.** 2.970 **C.** 6.795 **D.** 8.795

Session : 6 | Afternoon | 07/02/15

Q15.6.1 MCQ 2015 1M

We _____ our friend's birthday and we _____ how to make it up to him.

A. completely forgot — don't just know

B. forgot completely — don't just know

C. completely forgot — just don't know

D. forgot completely — just don't know

Q15.6.2 MCQ 2015 1M

Choose the statement where underlined word is used correctly.

A. The industrialist had a personnel jet.

B. I write my experience in my personnel diary.

C. All personnel are being given the day off.

D. Being religious is a personnel aspect.

Q15.6.3 MCQ 2015 1M

A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

A. fabric **B.** textile **C.** fibre **D.** apparel

Q15.6.4 MCQ 2015 1M

Based on the given statements, select the most appropriate option to solve the given question.

What will be the total weight of 10 poles each of same weight?

Statements:

I. One fourth of the weight of a pole is 5 Kg.

II. The total weight of these poles is 160 kg more than the total weight of two poles.

A. Statement I alone is not sufficient.

B. Statement II alone is not sufficient.

C. Either I or II alone is sufficient.

D. Both statements I and II together are not sufficient.

Q15.6.5 MCQ 2015 1M

Consider a function $f(x) = 1 - |x|$ on $-1 \le x \le 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

A. $0, -1$ **B.** $-1, 0$ **C.** $0, 1$ **D.** $-1, 2$

Q15.6.6 MCQ 2015 2M

Out of the following four sentences, select the most suitable sentence with respect to grammar and usage:

A. Since the report lacked needed information, it was of no use to them.

B. The report was useless to them because there were no needed information in it.

C. Since the report did not contain the needed information, it was not real useful to them.

D. Since the report lacked needed information, it would not had been useful to them.

Q15.6.7 MCQ 2015 2M

In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60°$. What is the length of PS?

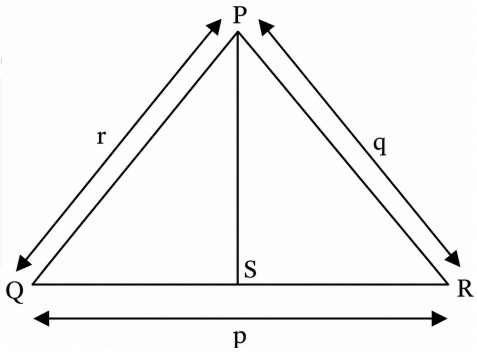


Fig. Q15.6.7

A. $\frac{(q+r)}{qr}$ **B.** $\frac{qr}{(q+r)}$

C. $\sqrt{(q^2+r^2)}$ **D.** $\frac{(q+r)^2}{qr}$

Q15.6.8 NAT 2015 2M

For distinct integers p, q, r, s, define the functions f, g, and h as follows:

$$f(p,q,r,s) = \max(p,q,r,s).$$

$$g(p,q,r,s) = \min(p,q,r,s).$$

The function $h(p,q,r,s)$ is the remainder when the larger of the two products pq and rs is divided by the smaller (for the unequal nonzero products used here). Products of the function values on the same arguments are denoted by $fg = f \times g$ and $fgh = f \times g \times h$. The same maximum/minimum operations also apply to two arguments. What is the value of $fg(h(2,5,7,3),4,6,8)$?

Q15.6.9 MCQ 2015 2M

If the list of letters, P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. $2P, 2R, 2S, 2T, 2U$
II. $P-3, R-3, S-3, T-3, U-3$
III. P^2, R^2, S^2, T^2, U^2

A. I only **B.** I and II **C.** II and III **D.** I and III

Q15.6.10 MCQ 2015 2M

Four branches of a company are located at M, N, O, and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km?

A. 5.34 **B.** 6.74 **C.** 28.5 **D.** 45.49

Session : 7 | Forenoon | 08/02/15

Q15.7.1 MCQ 2015 1M

Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.

A. related **B.** extraneous **C.** outside **D.** useful

Q15.7.2 MCQ 2015 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:

Children : Pediatrician

A. Adult : Orthopaedist **B.** Females : Gynaecologist

C. Kidney : Nephrologist **D.** Skin : Dermatologist

Q15.7.3 MCQ 2015 1M

The Tamil version of ____ John Abraham-starrer *Madras Cafe* ____ cleared by the Censor Board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu ____ this Friday.

A. Mr., was, found, on **B.** a, was, found, at

C. the, was, found, on **D.** a, being, find at

Q15.7.4 MCQ 2015 1M

If ROAD is written as URDG, then SWAN should be written as:

A. VXDQ **B.** VZDQ **C.** VZDP **D.** UXDQ

Q15.7.5 MCQ 2015 1M

A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

A. 59 **B.** 45 **C.** 43 **D.** 35

Q15.7.6 MCQ 2015 2M

Alexander turned his attention towards India, since he had conquered Persia.
Which one of the statements below is logically valid and can be inferred from the above sentence?

A. Alexander would not have turned his attention towards India had he not conquered Persia.

B. Alexander was not ready to rest on his laurels, and wanted to march to India.

C. Alexander was completely in control of his army and could command it to move towards India.

D. Since Alexander's kingdoms extended to Indian borders after the conquest of Persia, he was keen to move further.

Q15.7.7 MCQ 2015 2M

Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed this problem, success at the highest level will continue to elude him.
Which of the statement(s) below is/are logically valid and can be inferred from the above passage?

(i) He was already a successful batsman at the highest level.
(ii) He has to improve his temperament in order to become a great batsman.
(iii) He failed to make many of his good starts count.
(iv) Improving his technical skills will guarantee success.

A. (iii) and (iv) **B.** (ii) and (iii)

C. (i), (ii) and (iii) **D.** (iv) only

Q15.7.8 NAT 2015 2M

The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?

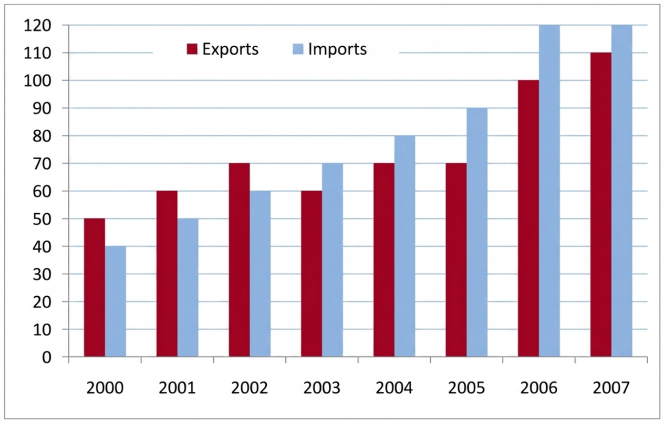


Fig. Q15.7.8

Q15.7.9 MCQ | 2015 | 2M

Choose the most appropriate equation for the function drawn as a thick line, in the plot below.

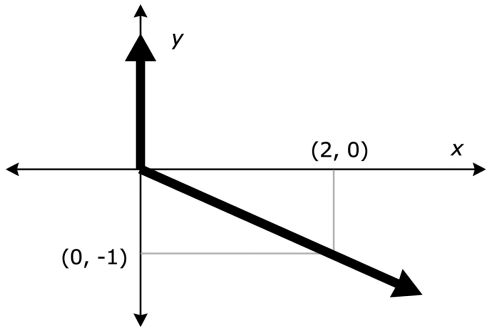


Fig. Q15.7.9

A. $x = y - |y|$ **B.** $x = -(y - |y|)$

C. $x = y + |y|$ **D.** $x = -(y + |y|)$

Q15.7.10 MCQ | 2015 | 2M

The head of a newly formed government desires to appoint five of the six selected members P, Q, R, S, T, and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.
Which is the valid distribution of portfolios?

A. P-Home, Q-Power, R-Defense, S-Telecom, T-Finance

B. R-Home, S-Power, P-Defense, Q-Telecom, T-Finance

C. P-Home, Q-Power, T-Defense, S-Telecom, U-Finance

D. Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

Session : 8 | Afternoon | 08/02/15

Q15.8.1 MCQ | 2015 | 1M

Choose the most appropriate word from the options given below to complete the following sentence.
The official answered _____ that the complaints of the citizen would be looked into.

A. respectably **B.** respectfully

C. reputably **D.** respectively

Q15.8.2 MCQ | 2015 | 1M

Choose the statement where underlined word is used correctly.

A. The minister insured the victims that everything would be all right.

B. He ensured that the company will not have to bear any loss.

C. The actor got himself ensured against any accident.

D. The teacher insured students of good results.

Q15.8.3 MCQ | 2015 | 1M

Which word is not a synonym for the word *vernacular*?

A. regional **B.** indigenous **C.** indigent **D.** colloquial

Q15.8.4 MCQ | 2015 | 1M

Mr. Vivek walks 6 meters North-east, then turns and walks 6 meters South-east, both at 60 degrees to east. He further moves 2 meters South and 4 meters West. What is the straight distance in metres between the point he started from and the point he finally reached?

A. $2\sqrt{2}$ **B.** 2 **C.** $\sqrt{2}$ **D.** $1/\sqrt{2}$

Q15.8.5 MCQ | 2015 | 1M

Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king?

A. $4/52$ **B.** $2/50$

C. $(1/52) \times (1/52)$ **D.** $(1/52) \times (1/51) \times (1/50)$

Q15.8.6 MCQ | 2015 | 2M

The word similar in meaning to 'dreary' is

A. cheerful **B.** dreamy **C.** hard **D.** dismal

Q15.8.7 MCQ 2015 2M

The given question is followed by two statements; select the most appropriate option that solves the question.
Capacity of a solution tank A is 70% of the capacity of tank B. How many gallons of solution are in tank A and tank B?

Statements:

I. Tank A is 80% full and tank B is 40% full.
II. Tank A if full contains 14,000 gallons of solution.

A. Statement I alone is sufficient.

B. Statement II alone is sufficient.

C. Either statement I or II alone is sufficient.

D. Both the statements I and II together are sufficient.

Q15.8.8 NAT 2015 2M

How many four digit numbers can be formed with the 10 digits $0, 1, 2, \ldots, 9$ if no number can start with 0 and if repetitions are not allowed?

Q15.8.9 MCQ 2015 2M

Read the following table giving sales data of five types of batteries for years 2006 to 2012:

Year	Type I	Type II	Type III	Type IV	Type V
2006	75	144	114	102	108
2007	90	126	102	84	126
2008	96	114	75	105	135
2009	105	90	150	90	75
2010	90	75	135	75	90
2011	105	60	165	45	120
2012	115	85	160	100	145

Out of the following, which type of battery achieved highest growth between the years 2006 and 2012?

A. Type V **B.** Type III

C. Type II **D.** Type I

Q15.8.10 MCQ 2015 2M

There are 16 teachers who can teach Thermodynamics (TD), 11 who can teach Electrical Sciences (ES), and 5 who can teach both TD and Engineering Mechanics (EM). There are a total of 40 teachers. 6 cannot teach any of the three subjects, i.e. EM, ES or TD. 6 can teach only ES. 4 can teach all three subjects, i.e. EM, ES and TD. 4 can teach ES and TD. How many can teach both ES and EM but not TD?

A. 1 **B.** 2 **C.** 3 **D.** 4

Answer Key

Session : 1					
Q15.1.1 B	Q15.1.2 A	Q15.1.3 C	Q15.1.4 C	Q15.1.5 A	Q15.1.6 A
Q15.1.7 B	Q15.1.8 3	Q15.1.9 C	Q15.1.10 B		
Session : 2					
Q15.2.1 B	Q15.2.2 A	Q15.2.3 C	Q15.2.4 D	Q15.2.5 B	Q15.2.6 B
Q15.2.7 D	Q15.2.8 800.00	Q15.2.9 B	Q15.2.10 B		
Session : 3					
Q15.3.1 B	Q15.3.2 B	Q15.3.3 B	Q15.3.4 C	Q15.3.5 A	Q15.3.6 A
Q15.3.7 B	Q15.3.8 1.9 to 2.2	Q15.3.9 C	Q15.3.10 B		
Session : 4					
Q15.4.1 A	Q15.4.2 C	Q15.4.3 B	Q15.4.4 B	Q15.4.5 B	Q15.4.6 C
Q15.4.7 C	Q15.4.8 280.00	Q15.4.9 C	Q15.4.10 B		
Session : 5					
Q15.5.1 A	Q15.5.2 C	Q15.5.3 B	Q15.5.4 A	Q15.5.5 A	Q15.5.6 C
Q15.5.7 A	Q15.5.8 32.00	Q15.5.9 D	Q15.5.10 C		
Session : 6					
Q15.6.1 C	Q15.6.2 C	Q15.6.3 D	Q15.6.4 C	Q15.6.5 C	Q15.6.6 A
Q15.6.7 B	Q15.6.8 8.00	Q15.6.9 B	Q15.6.10 A		
Session : 7					
Q15.7.1 B	Q15.7.2 B	Q15.7.3 C	Q15.7.4 B	Q15.7.5 C	Q15.7.6 A
Q15.7.7 B	Q15.7.8 2006.00	Q15.7.9 B	Q15.7.10 B		
Session : 8					
Q15.8.1 B	Q15.8.2 B	Q15.8.3 C	Q15.8.4 A	Q15.8.5 B	Q15.8.6 D
Q15.8.7 D	Q15.8.8 4536	Q15.8.9 D	Q15.8.10 A		

YEAR

2014

9 Sessions · 90 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	CE, CH, XE, AR, AE, MA, MT	10
Session 2	XL, BT, CE, CH, PI, TF, IN	10
Session 3	EC, ME	10
Session 4	EC, ME	10
Session 5	EC, ME	10
Session 6	EC, ME	10
Session 7	CS, EE	10
Session 8	CS, EE	10
Session 9	CS	10

PrepFusion

Year Snapshot

Total Questions: 90 **MCQ:** 71 **MSQ:** 0 **NAT:** 19

Session : 1 | Forenoon | 02/02/14

Q14.1.1 MCQ 2014 1M

A student is required to demonstrate a high level of comprehension of the subject, especially in the social sciences.
The word closest in meaning to comprehension is

A. understanding **B.** meaning

C. concentration **D.** stability

Q14.1.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
One of his biggest _____ was his ability to forgive.

A. vice **B.** virtues **C.** choices **D.** strength

Q14.1.3 MCQ 2014 1M

Rajan was not happy that Sajan decided to do the project on his own. On observing his unhappiness, Sajan explained to Rajan that he preferred to work independently.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Rajan has decided to work only in a group.

B. Rajan and Sajan were formed into a group against their wishes.

C. Sajan had decided to give in to Rajan's request to work with him.

D. Rajan had believed that Sajan and he would be working together.

Q14.1.4 MCQ 2014 1M

If $y = 5x^2 + 3$, then the tangent at $x = 0, y = 3$

A. passes through $x = 0, y = 0$ **B.** has a slope of $+1$

C. is parallel to the x-axis **D.** has a slope of -1

Q14.1.5 NAT 2014 1M

A foundry has a fixed daily cost of Rs 50,000 whenever it operates and a variable cost of Rs $800Q$, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes?

Q14.1.6 MCQ 2014 1M

Find the odd one in the following group: ALRVX, EPVZB, ITZDF, OYEIK

A. ALRVX **B.** EPVZB **C.** ITZDF **D.** OYEIK

Q14.1.7 MCQ 2014 2M

Anuj, Bhola, Chandan, Dilip, Eswar and Faisal live on different floors in a six-storeyed building (the ground floor is numbered 1, the floor above it 2, and so on). Anuj lives on an even-numbered floor. Bhola does not live on an odd numbered floor. Chandan does not live on any of the floors below Faisal's floor. Dilip does not live on floor number 2. Eswar does not live on a floor immediately above or immediately below Bhola. Faisal lives three floors above Dilip. Which of the following floor-person combinations is correct?

	Anuj	Bhola	Chandan	Dilip	Eswar	Faisal
(A)	6	2	5	1	3	4
(B)	2	6	5	1	3	4
(C)	4	2	6	3	1	5
(D)	2	4	6	1	3	5

Q14.1.8 NAT 2014 2M

The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3 : 4 : 5 : 6. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

Q14.1.9 MCQ 2014 2M

One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft?

A. 3.0 **B.** 2.5 **C.** 1.5 **D.** 1.25

Q14.1.10 MCQ 2014 2M

The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure. Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)

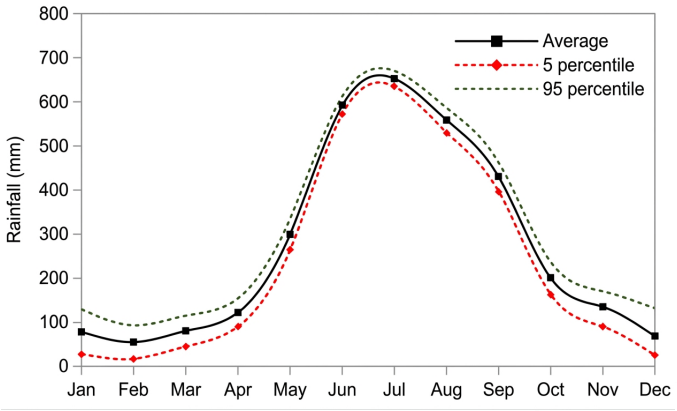


Fig. Q14.1.10

(i) On average, it rains more in July than in December
(ii) Every year, the amount of rainfall in August is more than that in January
(iii) July rainfall can be estimated with better confidence than February rainfall
(iv) In August, there is at least 500 mm of rainfall

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (iii) and (iv)

Session : 2 | Afternoon | 02/02/14

Q14.2.1 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
A person suffering from Alzheimer's disease _____ short-term memory loss.

A. experienced **B.** has experienced

C. is experiencing **D.** experiences

Q14.2.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
_____ is the key to their happiness; they are satisfied with what they have.

A. Contentment **B.** Ambition

C. Perseverance **D.** Hunger

Q14.2.3 MCQ 2014 1M

Which of the following options is the closest in meaning to the sentence below?
"As a woman, I have no country."

A. Women have no country.

B. Women are not citizens of any country.

C. Women's solidarity knows no national boundaries.

D. Women of all countries have equal legal rights.

Q14.2.4 NAT 2014 1M

In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is _____ years.

Q14.2.5 MCQ 2014 1M

The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

A. 3-4 years **B.** 4-5 years **C.** 5-6 years **D.** 6-7 years

Q14.2.6 MCQ 2014 2M

In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

A. Statement 1 by itself determines the eldest child.

B. Statement 2 by itself determines the eldest child.

C. Statements 1 and 2 are both required to determine the eldest child.

D. Statements 1 and 2 are not sufficient to determine the eldest child.

Q14.2.7 MCQ 2014 2M

Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era.

When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis.
From *"BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier*
The main point of the paragraph is:

A. The twenty-first century is a digital world

B. Big data is obsessed with exactness

C. Exactitude is not critical in dealing with big data

D. Sparse data leads to a bias in the analysis

Q14.2.8 MCQ | 2014 | 2M

The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?

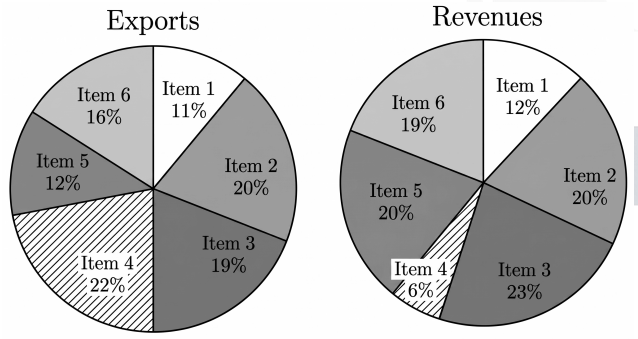


Fig. Q14.2.8

A. 1:2 **B.** 2:1 **C.** 1:4 **D.** 4:1

Q14.2.9 MCQ | 2014 | 2M

X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?

A. 1 **B.** $\sqrt{2}$ **C.** $\sqrt{3}$ **D.** 2

Q14.2.10 NAT | 2014 | 2M

10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____

Session : 3 | Forenoon | 15/02/14

Q14.3.1 MCQ | 2014 | 1M

Choose the most appropriate phrase from the options given below to complete the following sentence
The aircraft _____ take off as soon as its flight plan was filed.

A. is allowed to **B.** will be allowed to

C. was allowed to **D.** has been allowed to

Q14.3.2 MCQ | 2014 | 1M

Read the statements:

All women are entrepreneurs.
Some women are doctors.

Which of the following conclusions can be logically inferred from the above statements?

A. All women are doctors

B. All doctors are entrepreneurs

C. All entrepreneurs are women

D. Some entrepreneurs are doctors

Q14.3.3 MCQ | 2014 | 1M

Choose the most appropriate word from the options given below to complete the following sentence.
Many ancient cultures attributed disease to supernatural causes. However, modern science has largely helped _____ such notions.

A. impel **B.** dispel **C.** propel **D.** repel

Q14.3.4 MCQ | 2014 | 1M

The statistics of runs scored in a series by four batsmen are provided in the following table. Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

A. K **B.** L **C.** M **D.** N

Q14.3.5 NAT 2014 1M

What is the next number in the series?

12 35 81 173 357 _____

Q14.3.6 MCQ 2014 2M

Find the odd one from the following group:

W,E,K,O I,Q,W,A F,N,T,X N,V,B,D

A. W,E,K,O **B.** I,Q,W,A **C.** F,N,T,X **D.** N,V,B,D

Q14.3.7 MCQ 2014 2M

For submitting tax returns, all resident males with annual income below Rs 10 lakh should fill up Form P and all resident females with income below Rs 8 lakh should fill up Form Q. All people with incomes above Rs 10 lakh should fill up Form R, except non residents with income above Rs 15 lakhs, who should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

A. a resident male with annual income Rs 9 lakh

B. a resident female with annual income Rs 9 lakh

C. a non-resident male with annual income Rs 16 lakh

D. a non-resident female with annual income Rs 16 lakh

Q14.3.8 NAT 2014 2M

A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

Q14.3.9 MCQ 2014 2M

The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit 1/5th of the exports?

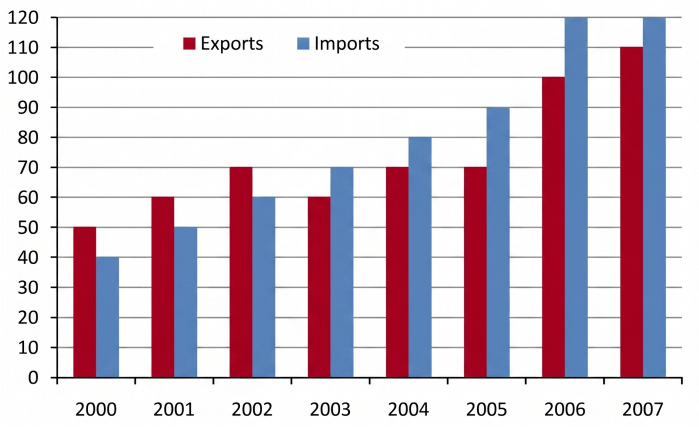


Fig. Q14.3.9

A. 2005 **B.** 2004 **C.** 2007 **D.** 2006

Q14.3.10 MCQ 2014 2M

You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

A. 1/4 **B.** 1/3 **C.** 1/2 **D.** 2/3

Session : 4 | Afternoon | 15/02/14

Q14.4.1 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.

Communication and interpersonal skills are _____ important in their own ways.

A. each **B.** both **C.** all **D.** either

Q14.4.2 MCQ 2014 1M

Which of the options given below best completes the following sentence?

She will feel much better if she _____.

A. will get some rest **B.** gets some rest

C. will be getting some rest **D.** is getting some rest

Q14.4.3 MCQ 2014 1M

Choose the most appropriate pair of words from the options given below to complete the following sentence.
She could not _____ the thought of _____ the election to her bitter rival.

A. bear, loosing **B.** bare, loosing

C. bear, losing **D.** bare, losing

Q14.4.4 MCQ 2014 1M

A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: 1 → 0.167; 2 → 0.167; 3 → 0.152; 4 → 0.166; 5 → 0.168; 6 → 0.180. We call this die

A. irregular **B.** biased **C.** Gaussian **D.** insufficient

Q14.4.5 NAT 2014 1M

Fill in the missing number in the series.
2 3 6 15 _____ 157.5 630

Q14.4.6 MCQ 2014 2M

Find the odd one in the following group

Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X

A. Q,W,Z,B **B.** B,H,K,M **C.** W,C,G,J **D.** M,S,V,X

Q14.4.7 MCQ 2014 2M

Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct?

A. The number of red lights is equal to the number of blue lights

B. The number of green lights is equal to the number of yellow lights

C. The sum of the red and green lights is equal to the sum of the yellow and blue lights

D. The sum of the red and blue lights is equal to the sum of the green and yellow lights

Q14.4.8 NAT 2014 2M

The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number?

Q14.4.9 MCQ 2014 2M

The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?

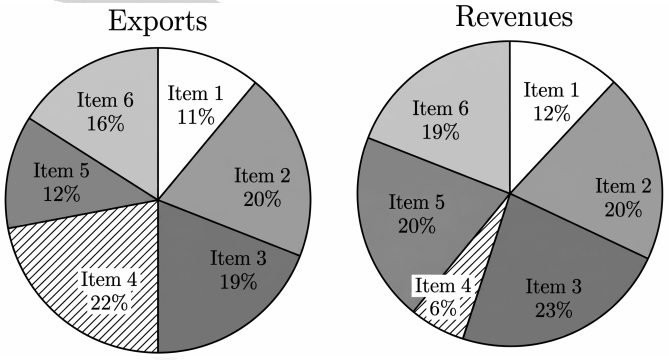


Fig. Q14.4.9

A. Item 2 **B.** Item 3 **C.** Item 6 **D.** Item 5

Q14.4.10 MCQ 2014 2M

It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes?

A. 4 times the draining rate

B. 3 times the draining rate

C. 2.5 times the draining rate

D. 2 times the draining rate

Session : 5 | Forenoon | 16/02/14

Q14.5.1 MCQ 2014 1M

"India is a country of rich heritage and cultural diversity." Which one of the following facts best supports the claim made in the above sentence?

A. India is a union of 28 states and 7 union territories.

B. India has a population of over 1.1 billion.

C. India is home to 22 official languages and thousands of dialects.

D. The Indian cricket team draws players from over ten states.

Q14.5.2 MCQ 2014 1M

The value of one U.S. dollar is 65 Indian Rupees today, compared to 60 last year. The Indian Rupee has _____.

A. depressed **B.** depreciated

C. appreciated **D.** stabilized

Q14.5.3 MCQ 2014 1M

'Advice' is _____.

A. a verb **B.** a noun

C. an adjective **D.** both a verb and a noun

Q14.5.4 NAT 2014 1M

The next term in the series 81, 54, 36, 24, _____ is

Q14.5.5 MCQ 2014 1M

In which of the following options will the expression $P < M$ be definitely true?

A. $M < R > P > S$ **B.** $M > S < P < F$

C. $Q < M < F = P$ **D.** $P = A < R < M$

Q14.5.6 MCQ 2014 2M

Find the next term in the sequence: 7G, 11K, 13M, ___

A. 15Q **B.** 17Q **C.** 15P **D.** 17P

Q14.5.7 MCQ 2014 2M

The multi-level hierarchical pie chart shows the population of animals in a reserve forest. The correct conclusions from this information are:

Fig. Q14.5.7

(i) Butterflies are birds
(ii) There are more tigers in this forest than red ants
(iii) All reptiles in this forest are either snakes or crocodiles
(iv) Elephants are the largest mammals in this forest

A. (i) and (ii) only **B.** (i), (ii), (iii) and (iv)

C. (i), (iii) and (iv) only **D.** (i), (ii) and (iii) only

Q14.5.8 NAT 2014 2M

A man can row at 8 km per hour in still water. If it takes him thrice as long to row upstream, as to row downstream, then find the stream velocity in km per hour.

Q14.5.9 NAT 2014 2M

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. If the company registered a profit of Rs. 10 lakhs in 2012, at what price (in Rs.) was each air purifier sold?

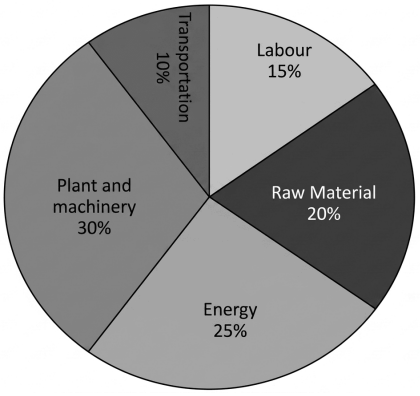


Fig. Q14.5.9

Q14.5.10 NAT 2014 2M

A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch is accepted is

Session : 6 | Afternoon | 16/02/14

Q14.6.1 MCQ 2014 1M

Which of the following options is the closest in meaning to the word underlined in the sentence below?
In a democracy, everybody has the freedom to disagree with the government.

A. dissent **B.** descent **C.** decent **D.** decadent

Q14.6.2 MCQ 2014 1M

After the discussion, Tom said to me, 'Please revert!'. He expects me to _____.

A. retract **B.** get back to him

C. move in reverse **D.** retreat

Q14.6.3 MCQ 2014 1M

While receiving the award, the scientist said, "I feel vindicated". Which of the following is closest in meaning to the word "vindicated"?

A. punished **B.** substantiated

C. appreciated **D.** chastened

Q14.6.4 MCQ 2014 1M

Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

A. $2^{n-m}P$ **B.** $2^{m-n}P$ **C.** $2(n-m)P$ **D.** $2(m-n)P$

Q14.6.5 NAT 2014 1M

In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

Q14.6.6 MCQ 2014 2M

Find the next term in the sequence: 13M, 17Q, 19S, ___

A. 21W **B.** 21V **C.** 23W **D.** 23V

Q14.6.7 MCQ 2014 2M

If 'KCLFTSB' stands for 'best of luck' and 'SHSWDG' stands for 'good wishes', which of the following indicates 'ace the exam'?

A. MCHTX **B.** MXHTC

C. XMHCT **D.** XMHTC

Q14.6.8 MCQ 2014 2M

Industrial consumption of power doubled from 2000-2001 to 2010-2011. Find the annual rate of increase in percent assuming it to be uniform over the years.

A. 5.6 **B.** 7.2 **C.** 10.0 **D.** 12.2

Q14.6.9 NAT 2014 2M

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013?

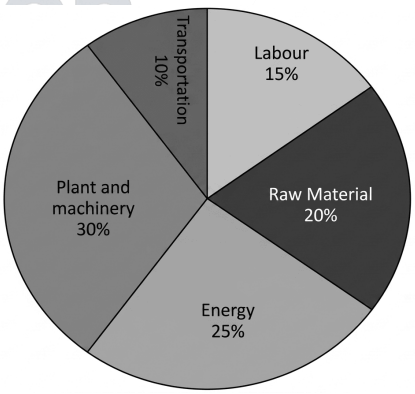


Fig. Q14.6.9

Q14.6.10 MCQ 2014 2M

A five digit number is formed using the digits 1,3,5,7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

A. 6666660 **B.** 6666600 **C.** 6666666 **D.** 6666606

Session : 7 | Afternoon | 01/03/14

Q14.7.1 MCQ 2014 1M

Which of the following options is the closest in meaning to the phrase underlined in the sentence below?
It is fascinating to see life forms cope with varied environmental conditions.

A. adopt to **B.** adapt to **C.** adept in **D.** accept with

Q14.7.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____.

A. superb **B.** medium **C.** mediocre **D.** exhilarating

Q14.7.3 MCQ 2014 1M

In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?

A. He wants all the money

B. He will return the money

C. He will assume final responsibility

D. He will resist all enquiries

Q14.7.4 NAT 2014 1M

If $(z + 1/z)^2 = 98$, compute $(z^2 + 1/z^2)$.

Q14.7.5 MCQ 2014 1M

The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b|x| + c = 0$ has

A. no roots **B.** 2 real roots

C. 3 real roots **D.** 4 real roots

Q14.7.6 MCQ 2014 2M

The Palghat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. The exact reasons for the formation of this gap are not clear. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures.
What can be inferred from this passage?

A. The Palghat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala

B. The regions in Tamil Nadu and Kerala that are near the Palghat Gap are low-lying

C. The low terrain of the Palghat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala

D. Higher summer temperatures result in higher rainfall near the Palghat Gap area

Q14.7.7 MCQ 2014 2M

Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.
On which of the following assumptions does the statement above rely?

A. Strategies are now available for eliminating psychiatric illnesses

B. Certain psychiatric illnesses have a genetic basis

C. All human diseases can be traced back to genes and how they are expressed

D. In the future, genetics will become the only relevant field for identifying psychiatric illnesses

Q14.7.8 NAT 2014 2M

Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____.

Q14.7.9 NAT 2014 2M

In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

Q14.7.10 NAT 2014 2M

When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines? _____

Session : 8 | Forenoon | 02/03/14

Q14.8.1 MCQ 2014 1M

Choose the most appropriate phrase from the options given below to complete the following sentence.
India is a post-colonial country because

A. it was a former British colony

B. Indian Information Technology professionals have colonized the world

C. India does not follow any colonial practices

D. India has helped other countries gain freedom

Q14.8.2 MCQ 2014 1M

Who _____ was coming to see us this evening?

A. you said **B.** did you say

C. did you say that **D.** had you said

Q14.8.3 MCQ 2014 1M

Match the columns.

Column 1	Column 2
1) eradicate	P) misrepresent
2) distort	Q) soak completely
3) saturate	R) use
4) utilize	S) destroy utterly

A. 1:S, 2:P, 3:Q, 4:R **B.** 1:P, 2:Q, 3:R, 4:S

C. 1:Q, 2:R, 3:S, 4:P **D.** 1:S, 2:P, 3:R, 4:Q

Q14.8.4 MCQ 2014 1M

What is the average of all multiples of 10 from 2 to 198?

A. 90 **B.** 100 **C.** 110 **D.** 120

Q14.8.5 MCQ 2014 1M

The value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$ is

A. 3.464 **B.** 3.932 **C.** 4.000 **D.** 4.444

Q14.8.6 MCQ 2014 2M

The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

A. Kaliningrad was historically Russian in its ethnic make up

B. Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia

C. Koenigsberg was renamed Kaliningrad, as that was its original Russian name

D. Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

Q14.8.7 MCQ 2014 2M

The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.
Which one of the following statements, if true, does not contradict this conclusion?

A. A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent

B. More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency

C. Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test

D. The number of people with malarial fever (also contracted from mosquito bites) has increased this year

Q14.8.8 MCQ 2014 2M

If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

A. 2, 4 **B.** 2, 14 **C.** 4, 52 **D.** 14, 52

Q14.8.9 NAT 2014 2M

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

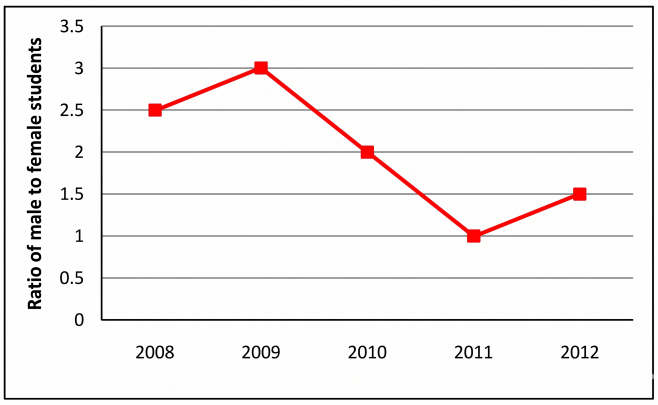


Fig. Q14.8.9

Q14.8.10 MCQ 2014 2M

At what time between 6 a.m. and 7 a.m. will the minute hand and hour hand of a clock make an angle closest to 60°?

A. 6:22 a.m. **B.** 6:27 a.m. **C.** 6:38 a.m. **D.** 6:45 a.m.

Session : 9 | Afternoon | 02/03/14

Q14.9.1 MCQ 2014 1M

While trying to collect (I) an envelope from under the table, (II) Mr. X fell down (III) and was losing consciousness. (IV)

Which one of the above underlined parts of the sentence is NOT appropriate?

A. I **B.** II **C.** III **D.** IV

Q14.9.2 MCQ 2014 1M

If she _____ how to calibrate the instrument, she _____ done the experiment.

A. knows, will have

B. knew, had

C. had known, could have

D. should have known, would have

Q14.9.3 MCQ 2014 1M

Choose the word that is opposite in meaning to the word "coherent".

A. sticky **B.** well-connected

C. rambling **D.** friendly

Q14.9.4 MCQ 2014 1M

Which number does not belong in the series below?
2, 5, 10, 17, 26, 37, 50, 64

A. 17 **B.** 37 **C.** 64 **D.** 26

Q14.9.5 MCQ 2014 1M

The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

A. 1.34 **B.** 1.74 **C.** 3.02 **D.** 3.91

Q14.9.6 MCQ 2014 2M

A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

A. Students should come at 9.00 a.m. and parents should come at 10.00 a.m.

B. Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.

C. Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.

D. Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

Q14.9.7 MCQ | 2014 | 2M

By the beginning of the 20^{th} century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. However, the clinching evidence was provided by experimental measurements of the position of a star which was directly behind our sun. Which of the following inference(s) may be drawn from the above passage?

(i) Our understanding of the universe changes based on the positions of stars
(ii) Paradigm shifts usually occur at the beginning of centuries
(iii) Stars are important objects in the universe
(iv) Experimental evidence was important in confirming this paradigm shift

A. (i), (ii) and (iv) **B.** (iii) only

C. (i) and (iv) **D.** (iv) only

Q14.9.8 MCQ | 2014 | 2M

The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/ USD to Rs. 60/ USD. India's GDP in USD during the period 2012-2013.

A. increased by 5%

B. decreased by 20%

C. decreased by 11%

D. decreased by 13%

Q14.9.9 MCQ | 2014 | 2M

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?

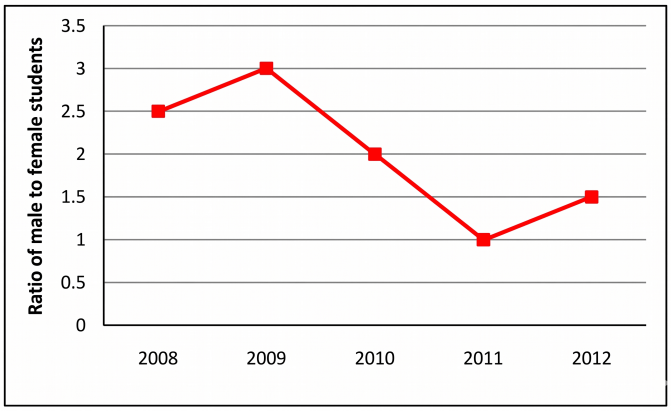


Fig. Q14.9.9

A. 1:1 **B.** 2:1 **C.** 1.5:1 **D.** 2.5:1

Q14.9.10 MCQ | 2014 | 2M

Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

A. 1634 **B.** 1737 **C.** 3142 **D.** 3162

Answer Key

Session : 1					
Q14.1.1 A	Q14.1.2 B	Q14.1.3 D	Q14.1.4 C	Q14.1.5 1300	Q14.1.6 D
Q14.1.7 B	Q14.1.8 180	Q14.1.9 D	Q14.1.10 B		
Session : 2					
Q14.2.1 D	Q14.2.2 A	Q14.2.3 C	Q14.2.4 25	Q14.2.5 A	Q14.2.6 A
Q14.2.7 C	Q14.2.8 D	Q14.2.9 C	Q14.2.10 0.48-0.49		
Session : 3					
Q14.3.1 C	Q14.3.2 D	Q14.3.3 B	Q14.3.4 A	Q14.3.5 725	Q14.3.6 D
Q14.3.7 B	Q14.3.8 560	Q14.3.9 D	Q14.3.10 B		
Session : 4					
Q14.4.1 B	Q14.4.2 B	Q14.4.3 C	Q14.4.4 B	Q14.4.5 45	Q14.4.6 C
Q14.4.7 D	Q14.4.8 163	Q14.4.9 D	Q14.4.10 A		
Session : 5					
Q14.5.1 C	Q14.5.2 B	Q14.5.3 B	Q14.5.4 16	Q14.5.5 D	Q14.5.6 B
Q14.5.7 D	Q14.5.8 4	Q14.5.9 20000	Q14.5.10 0.80 - 0.82		
Session : 6					
Q14.6.1 A	Q14.6.2 B	Q14.6.3 B	Q14.6.4 A	Q14.6.5 495	Q14.6.6 C
Q14.6.7 B	Q14.6.8 B	Q14.6.9 22	Q14.6.10 B		
Session : 7					
Q14.7.1 B	Q14.7.2 C	Q14.7.3 C	Q14.7.4 96	Q14.7.5 D	Q14.7.6 C
Q14.7.7 B	Q14.7.8 850	Q14.7.9 48	Q14.7.10 6		
Session : 8					
Q14.8.1 A	Q14.8.2 B	Q14.8.3 A	Q14.8.4 B	Q14.8.5 C	Q14.8.6 B
Q14.8.7 D	Q14.8.8 D	Q14.8.9 140	Q14.8.10 A		
Session : 9					
Q14.9.1 D	Q14.9.2 C	Q14.9.3 C	Q14.9.4 C	Q14.9.5 C	Q14.9.6 B
Q14.9.7 D	Q14.9.8 D	Q14.9.9 C	Q14.9.10 C		

YEAR

2013

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, CE, GG, MA, MT, PH, TF	10
Session 2	AE, AG, BT, CH, CY, MN, XE, XL	10
Session 3	CS, ME, PI	10
Session 4	EC, EE, IN	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 20/01/13

Q13.1.1 MCQ 2013 1M

A number is as much greater than 75 as it is smaller than 117. The number is:

A. 91 **B.** 93 **C.** 89 **D.** 96

Q13.1.2 MCQ 2013 1M

The professor (I) ordered to (II) the students to go (III) out of the class. (IV)

Which of the above underlined parts of the sentence is grammatically incorrect?

A. I **B.** II **C.** III **D.** IV

Q13.1.3 MCQ 2013 1M

Which of the following options is the closest in meaning to the word given below:

Primeval

A. Modern **B.** Historic **C.** Primitive **D.** Antique

Q13.1.4 MCQ 2013 1M

Friendship, no matter how _____ it is, has its limitations.

A. cordial **B.** intimate **C.** secret **D.** pleasant

Q13.1.5 MCQ 2013 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:

Medicine: Health

A. Science: Experiment **B.** Wealth: Peace

C. Education: Knowledge **D.** Money: Happiness

Q13.1.6 MCQ 2013 2M

X and Y are two positive real numbers such that $2X+Y \leq 6$ and $X+2Y \leq 8$. For which of the following values of (X, Y) the function $f(X, Y) = 3X+6Y$ will give maximum value?

A. $(4/3, 10/3)$ **B.** $(8/3, 20/3)$

C. $(8/3, 10/3)$ **D.** $(4/3, 20/3)$

Q13.1.7 MCQ 2013 2M

If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is:

A. 2, 1/3 **B.** 1/2, 3 **C.** 3/2, 9 **D.** 2/3, 9

Q13.1.8 MCQ 2013 2M

Following table provides figures (in rupees) on annual expenditure of a firm for two years - 2010 and 2011.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the following two categories have registered increase by same percentage?

A. Raw material and Salary & wages

B. Salary & wages and Advertising

C. Power & fuel and Advertising

D. Raw material and Research & Development

Q13.1.9 MCQ 2013 2M

A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

A. 5 **B.** 10 **C.** 15 **D.** 30

Q13.1.10 MCQ 2013 2M

Abhishek is elder to Savar.
Savar is younger to Anshul.

Which of the given conclusions is logically valid and is inferred from the above statements?

A. Abhishek is elder to Anshul

B. Anshul is elder to Abhishek

C. Abhishek and Anshul are of the same age

D. No conclusion follows

Session : 2 | Afternoon | 20/01/13

Q13.2.1 MCQ 2013 1M

If $3 \le X \le 5$ and $8 \le Y \le 11$ then which of the following options is TRUE?

A. $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{5}$ **B.** $\frac{3}{11} \le \frac{X}{Y} \le \frac{5}{8}$

C. $\frac{3}{11} \le \frac{X}{Y} \le \frac{8}{5}$ **D.** $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{11}$

Q13.2.2 MCQ 2013 1M

The Headmaster _____ to speak to you.
Which of the following options is incorrect to complete the above sentence?

A. is wanting **B.** wants **C.** want **D.** was wanting

Q13.2.3 MCQ 2013 1M

Mahatama Gandhi was known for his humility as

A. he played an important role in humiliating exit of British from India.

B. he worked for humanitarian causes.

C. he displayed modesty in his interactions.

D. he was a fine human being.

Q13.2.4 MCQ 2024 1M

All engineering students (I) should learn mechanics, (II) mathematics and (III) how to do computation. (IV)

Which of the above underlined parts of the sentence is not appropriate?

A. I **B.** II **C.** III **D.** IV

Q13.2.5 MCQ 2013 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:
water: pipe::

A. cart: road **B.** electricity: wire

C. sea: beach **D.** music: instrument

Q13.2.6 MCQ 2013 2M

Velocity of an object fired directly in upward direction is given by $V = 80 - 32t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

A. (1, 3/2) **B.** (1/2, 1) **C.** (1/2, 3/2) **D.** (1, 3)

Q13.2.7 MCQ 2013 2M

In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

A. 0.35 **B.** 0.45 **C.** 0.5 **D.** 0.4

Q13.2.8 MCQ 2013 2M

Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

A. USA and Japan **B.** USA and Australia

C. England and France **D.** Japan and Australia

Q13.2.9 MCQ 2013 2M

If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

A. 30 **B.** -30 **C.** -42 **D.** 42

Q13.2.10 MCQ 2013 2M

All professors are researchers
Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments:

A. All scientists are researchers

B. All professors are scientists

C. Some researchers are scientists

D. No conclusion follows

Session : 3 | Forenoon | 10/02/13

Q13.3.1 MCQ 2013 1M

Which one of the following options is the closest in meaning to the word given below?

Nadir

A. Highest **B.** Lowest **C.** Medium **D.** Integration

Q13.3.2 MCQ 2013 1M

Complete the sentence:

Universalism is to particularism as diffuseness is to _____.

A. specificity **B.** neutrality

C. generality **D.** adaptation

Q13.3.3 MCQ 2013 1M

What will be the maximum sum of $44, 42, 40, \ldots$?

A. 502 **B.** 504 **C.** 506 **D.** 500

Q13.3.4 MCQ 2013 1M

Were you a bird, you _____ in the sky.

A. would fly **B.** shall fly

C. should fly **D.** shall have flown

Q13.3.5 MCQ 2013 1M

Choose the grammatically **INCORRECT** sentence:

A. He is of Asian origin.

B. They belonged to Africa.

C. She is an European.

D. They migrated from India to Australia.

Q13.3.6 MCQ 2013 2M

Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{80}+\sqrt{81}}$$

A. 7 **B.** 8 **C.** 9 **D.** 10

Q13.3.7 MCQ 2013 2M

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

A. 13/90 **B.** 12/90 **C.** 78/90 **D.** 77/90

Q13.3.8 MCQ 2013 2M

After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

A. Failure is the pillar of success.

B. Honesty is the best policy.

C. Life begins and ends with adventures.

D. No adversity justifies giving up hope.

Q13.3.9 MCQ 2013 2M

A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

A. 36 **B.** 30 **C.** 24 **D.** 18

Q13.3.10 MCQ 2013 2M

The current erection cost of a structure is $Rs.\ 13,200$. If the labour wages per day increase by $1/5$ of the current wages and the working hours decrease by $1/24$ of the current period, then the new cost of erection in $Rs.$ is

A. 16,500 **B.** 15,180 **C.** 11,000 **D.** 10,120

Session : 4 | Afternoon | 10/02/13

Q13.4.1 MCQ 2013 1M

They were requested not to **quarrel** with others.
Which one of the following options is the closest in meaning to the word **quarrel**?

A. make out **B.** call out **C.** dig out **D.** fall out

Q13.4.2 MCQ 2013 1M

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in °C on Thursday was

A. 40 **B.** 43 **C.** 46 **D.** 49

Q13.4.3 MCQ 2013 1M

Complete the sentence:
Dare _____ mistakes.

A. commit **B.** to commit
C. committed **D.** committing

Q13.4.4 MCQ 2013 1M

Choose the grammatically **CORRECT** sentence:

A. Two and two add four.

B. Two and two become four.

C. Two and two are four.

D. Two and two make four.

Q13.4.5 MCQ 2013 1M

Statement: You can always give me a ring whenever you need.
Which one of the following is the best inference from the above statement?

A. Because I have a nice caller tune.

B. Because I have a better telephone facility.

C. Because a friend in need is a friend indeed.

D. Because you need not pay towards the telephone bills when you give me a ring.

Q13.4.6 MCQ 2013 2M

A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

A. 30 **B.** 36 **C.** 40 **D.** 24

Q13.4.7 MCQ 2013 2M

Find the sum to n terms of the series $10 + 84 + 734 + \ldots$

A. $\frac{9(9^n + 1)}{10} + 1$ **B.** $\frac{9(9^n - 1)}{8} + 1$

C. $\frac{9(9^n - 1)}{8} + n$ **D.** $\frac{9(9^n - 1)}{8} + n^2$

Q13.4.8 MCQ 2013 2M

Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists, and so on.
Which one of the following is the best inference from the above statement?

A. The emergence of nationalism in colonial India led to our Independence.

B. Nationalism in India emerged in the context of colonialism.

C. Nationalism in India is homogeneous.

D. Nationalism in India is heterogeneous.

Q13.4.9 MCQ 2013 2M

The set of values of p for which the roots of the equation $3x^2 + 2x + p(p - 1) = 0$ are of opposite sign is

A. $(-\infty, 0)$ **B.** $(0, 1)$ **C.** $(1, \infty)$ **D.** $(0, \infty)$

Q13.4.10 MCQ 2013 2M

What is the chance that a leap year, selected at random, will contain 53 Saturdays?

A. 2/7 **B.** 3/7 **C.** 1/7 **D.** 5/7

Answer Key

Session : 1					
Q13.1.1 D	Q13.1.2 B	Q13.1.3 C	Q13.1.4 B	Q13.1.5 C	Q13.1.6 A
Q13.1.7 B	Q13.1.8 D	Q13.1.9 A	Q13.1.10 D		
Session : 2					
Q13.2.1 B	Q13.2.2 C	Q13.2.3 C	Q13.2.4 D	Q13.2.5 B	Q13.2.6 C
Q13.2.7 C	Q13.2.8 C	Q13.2.9 B	Q13.2.10 C		
Session : 3					
Q13.3.1 B	Q13.3.2 A	Q13.3.3 C	Q13.3.4 A	Q13.3.5 C	Q13.3.6 B
Q13.3.7 D	Q13.3.8 D	Q13.3.9 C	Q13.3.10 B		
Session : 4					
Q13.4.1 D	Q13.4.2 C	Q13.4.3 B	Q13.4.4 D	Q13.4.5 C	Q13.4.6 C
Q13.4.7 D	Q13.4.8 D	Q13.4.9 B	Q13.4.10 A		

PrepFusion

YEAR

2012

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, GG, TF	10
Session 2	AE, AG, MN	10
Session 3	BT, CE, CH, CS, ME, PH, PI	10
Session 4	CY, EC, EE, IN, MA, MT, XE, XL	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 29/01/12

Q12.1.1 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Pacify

A. Excite **B.** Soothe **C.** Deplete **D.** Tire

Q12.1.2 MCQ 2012 1M

Choose the most appropriate pair of words from the options given below to complete the following sentence:

The high level of _____ of the questions in the test was _____ by an increase in the period of time allotted for answering them.

A. difficulty, compensated **B.** exactitude, magnified

C. aptitude, decreased **D.** attitude, mitigated

Q12.1.3 MCQ 2012 1M

Choose the grammatically **CORRECT** sentence:

A. He laid in bed till 8 o'clock in the morning.

B. He layed in bed till 8 o'clock in the morning.

C. He lain in bed till 8 o'clock in the morning.

D. He lay in bed till 8 o'clock in the morning.

Q12.1.4 MCQ 2012 1M

Which one of the parts (A, B, C, D) in the sentence contains an **ERROR**?

No sooner had the doctor seen the results of the blood test, than he suggested the patient to see the specialist.

A. no sooner had **B.** results of the blood test

C. suggested the patient **D.** see the specialist

Q12.1.5 MCQ 2012 1M

Ten teams participate in a tournament. Every team plays each of the other teams twice. The total number of matches to be played is

A. 20 **B.** 45 **C.** 60 **D.** 90

Q12.1.6 MCQ 2012 2M

A value of x that satisfies the equation $log\ x + log\ (x - 7) = log\ (x + 11) + log\ 2$ is

A. 1 **B.** 2 **C.** 7 **D.** 11

Q12.1.7 MCQ 2012 2M

Let $f(x) = x - [x]$, where $x \geq 0$ and $[x]$ is the greatest integer not larger than x. Then $f(x)$ is a

A. monotonically increasing function

B. monotonically decreasing function

C. linearly increasing function between two integers

D. linearly decreasing function between two integers

Q12.1.8 MCQ 2012 2M

Ravi is taller than Arun but shorter than Iqbal. Sam is shorter than Ravi. Mohan is shorter than Arun. Balu is taller than Mohan and Sam. The tallest person can be

A. Mohan **B.** Ravi **C.** Balu **D.** Arun

Q12.1.9 MCQ 2012 2M

A smuggler has 10 capsules in which five are filled with narcotic drugs and the rest contain the original medicine. All the 10 capsules are mixed in a single box, from which the customs officials picked two capsules at random and tested for the presence of narcotic drugs. The probability that the smuggler will be caught is

A. 0.50 **B.** 0.67 **C.** 0.78 **D.** 0.82

Q12.1.10 MCQ 2012 2M

The documents expose the cynicism of the government officials – and yet as the media website reflects, not a single newspaper has reported on their existence.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?

A. Nobody other than the government officials knew about the existence of the documents.

B. Newspapers did report about the documents but nobody cared.

C. Media reports did not show the existence of the documents.

D. The documents reveal the attitude of the government officials.

Session : 2 | Afternoon | 29/01/12

Q12.2.1 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
I _____ to have bought a diamond ring.

A. have a liking **B.** should have liked

C. would like **D.** may like

Q12.2.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
Food prices _____ again this month.

A. have raised **B.** have been raising

C. have been rising **D.** have arose

Q12.2.3 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
The administrators went on to implement yet another unreasonable measure, arguing that the measures were already _____ and one more would hardly make a difference.

A. reflective **B.** utopian **C.** luxuriant **D.** unpopular

Q12.2.4 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
To those of us who had always thought him timid, his ____ came as a surprise.

A. intrepidity **B.** inevitability **C.** inability **D.** inertness

Q12.2.5 MCQ 2012 1M

The arithmetic mean of five different natural numbers is 12. The largest possible value among the numbers is

A. 12 **B.** 40 **C.** 50 **D.** 60

Q12.2.6 MCQ 2012 2M

Two policemen, A and B, fire once each at the same time at an escaping convict. The probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

A. 0.14 **B.** 0.22 **C.** 0.33 **D.** 0.40

Q12.2.7 MCQ 2012 2M

The total runs scored by four cricketers P, Q, R, and S in years 2009 and 2010 are given in the following table:

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

A. P **B.** Q **C.** R **D.** S

Q12.2.8 MCQ 2012 2M

If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

A. sum of squares of two natural numbers

B. sum of cubes of two natural numbers

C. sum of square roots of two natural numbers

D. sum of cube roots of two natural numbers

Q12.2.9 MCQ 2012 2M

Two points $(4, p)$ and $(0, q)$ lie on a straight line having a slope of $3/4$. The value of $(p - q)$ is

A. -3 **B.** 0 **C.** 3 **D.** 4

Q12.2.10 MCQ 2012 2M

In the early nineteenth century, theories of social evolution were inspired less by Biology than by the conviction of social scientists that there was a growing improvement in social institutions. Progress was taken for granted and social scientists attempted to discover its laws and phases.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?

Social scientists

A. did not question that progress was a fact.

B. did not approve of Biology.

C. framed the laws of progress.

D. emphasized Biology over Social Sciences.

Session : 3 | Forenoon | 12/02/12

Q12.3.1 MCQ 2012 1M

The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹50 per unit. The number of units to be produced by the firm such that the profit is maximized is

A. 5 **B.** 10 **C.** 15 **D.** 25

Q12.3.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

Despite several _____ the mission succeeded in its attempt to resolve the conflict.

A. attempts **B.** setbacks

C. meetings **D.** delegations

Q12.3.3 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Mitigate

A. Diminish **B.** Divulge **C.** Dedicate **D.** Denote

Q12.3.4 MCQ 2012 1M

Choose the grammatically **INCORRECT** sentence:

A. They gave us the money back less the service charges of Three Hundred rupees.

B. This country's expenditure is not less than that of Bangladesh.

C. The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.

D. This country's expenditure on educational reforms is very less.

Q12.3.5 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one _____ was hurt in the stampede.

A. that **B.** which **C.** who **D.** whom

Q12.3.6 MCQ 2012 2M

Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.

Which one of the following is the best inference from the above advertisement?

A. Gender-discriminatory

B. Xenophobic

C. Not designed to make the post attractive

D. Not gender-discriminatory

Q12.3.7 MCQ 2012 2M

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters **B.** 10 meters **C.** 12 meters **D.** 14 meters

Q12.3.8 MCQ 2012 2M

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q12.3.9 MCQ 2012 2M

Which of the following assertions are **CORRECT**?

P: Adding 7 to each entry in a list adds 7 to the mean of the list

Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list

R: Doubling each entry in a list doubles the mean of the list

S: Doubling each entry in a list leaves the standard deviation of the list unchanged

A. P, Q **B.** Q, R **C.** P, R **D.** R, S

Q12.3.10 MCQ 2012 2M

Given the sequence of terms, AD, CG, FK, JP, the next term is

A. OV **B.** OW **C.** PV **D.** PW

Session : 4 | Afternoon | 12/02/12

Q12.4.1 MCQ 2012 1M

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$

A. 2.23 **B.** 4.33 **C.** 11.37 **D.** 27.64

Q12.4.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

If the tired soldier wanted to lie down, he _____ the mattress out on the balcony.

A. should take **B.** shall take

C. should have taken **D.** will have taken

Q12.4.3 MCQ 2012 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Given the seriousness of the situation that he had to face, his _____ was impressive.

A. beggary **B.** nomenclature

C. jealousy **D.** nonchalance

Q12.4.4 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Latitude

A. Eligibility **B.** Freedom

C. Coercion **D.** Meticulousness

Q12.4.5 MCQ 2012 1M

One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT**?

I requested that he should be given the driving test today instead of tomorrow.

A. requested that **B.** should be given

C. the driving test **D.** instead of tomorrow

Q12.4.6 MCQ 2012 2M

One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them.

Which one of the following statements best sums up the meaning of the above passage?

A. Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.

B. The legions were treated inhumanly as if the men were animals.

C. Discipline was the armies' inheritance from their seniors.

D. The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

Q12.4.7 MCQ 2012 2M

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

A. 5 **B.** 6 **C.** 9 **D.** 10

Q12.4.8 MCQ 2012 2M

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

A. 2 **B.** 3 **C.** 4 **D.** 8

Q12.4.9 MCQ 2012 2M

The data given in the following table summarizes the monthly budget of an average household.

Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

The approximate percentage of the monthly budget **NOT** spent on savings is

A. 10% **B.** 14% **C.** 81% **D.** 86%

Q12.4.10 MCQ 2012 2M

A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

A. 1/4 **B.** 1/16 **C.** 7/16 **D.** 9/16

PrepFusion

Answer Key

Session : 1					
Q12.1.1 B	Q12.1.2 A	Q12.1.3 D	Q12.1.4 C	Q12.1.5 D	Q12.1.6 D
Q12.1.7 C	Q12.1.8 C	Q12.1.9 C	Q12.1.10 D		
Session : 2					
Q12.2.1 C	Q12.2.2 C	Q12.2.3 D	Q12.2.4 A	Q12.2.5 C	Q12.2.6 A
Q12.2.7 B	Q12.2.8 A	Q12.2.9 C	Q12.2.10 A		
Session : 3					
Q12.3.1 A	Q12.3.2 B	Q12.3.3 A	Q12.3.4 D	Q12.3.5 Marks to All	Q12.3.6 D
Q12.3.7 B	Q12.3.8 B	Q12.3.9 C	Q12.3.10 A		
Session : 4					
Q12.4.1 D	Q12.4.2 A	Q12.4.3 D	Q12.4.4 B	Q12.4.5 B	Q12.4.6 A
Q12.4.7 A	Q12.4.8 A	Q12.4.9 D	Q12.4.10 C		

PrepFusion

YEAR

2011

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	GG, TF	10
Session 2	AE, MN	10
Session 3	AR, BT, CE, CH, CS, ME, PH, PI	10
Session 4	AG, CY, EC, EE, IN, MA, MT, XE, XL	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 30/01/11

Q11.1.1 MCQ 2011 1M

Choose the most appropriate word or phrase from the options given below to complete the following sentence.

The environmentalists hope _____ the lake to its pristine condition.

A. in restoring **B.** in the restoration of

C. to restore **D.** restoring

Q11.1.2 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Polemical

A. imitative **B.** conciliatory **C.** truthful **D.** ideological

Q11.1.3 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Despite the mixture's _____ nature, we found that by lowering its temperature in the laboratory we could dramatically reduce its tendency to vaporize.

A. acerbic **B.** resilient **C.** volatile **D.** heterogeneous

Q11.1.4 MCQ 2011 1M

If m students require a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in

A. 100 days **B.** $m/100$ days **C.** $100/m$ days **D.** m days

Q11.1.5 MCQ 2011 1M

Choose the most appropriate words from the options given below to complete the following sentence:

Because she had a reputation for _____, we were surprised and pleased when she greeted us so _____.

A. insolence . . . . . irately

B. insouciance . . . . . curtly

C. graciousness . . . . . amiably

D. querulousness . . . . . affably

Q11.1.6 MCQ 2011 2M

The number of solutions for the following system of inequalities is

$$X_1 \geq 0$$
$$X_2 \geq 0$$
$$X_1 + X_2 \leq 10$$
$$2X_1 + 2X_2 \geq 22$$

A. 0 **B.** infinite **C.** 1 **D.** 2

Q11.1.7 MCQ 2011 2M

In a class of 300 students in an M.Tech programme, each student is required to take at least one subject from the following three:

M600: Advanced Engineering Mathematics
C600: Computational Methods for Engineers
E600: Experimental Techniques for Engineers

The registration data for the M.Tech class shows that 100 students have taken M600, 200 students have taken C600, and 60 students have taken E600. What is the maximum possible number of students in the class who have taken all the above three subjects?

A. 20 **B.** 30 **C.** 40 **D.** 50

Q11.1.8 MCQ 2011 2M

Three sisters (R, S, and T) received a total of 24 toys during Christmas. The toys were initially divided among them in a certain proportion. Subsequently, R gave some toys to S which doubled the share of S. Then S in turn gave some of her toys to T, which doubled T's share. Next, some of T's toys were given to R, which doubled the number of toys that R currently had. As a result of all such exchanges, the three sisters were left with equal number of toys. How many toys did R have originally?

A. 8 **B.** 9 **C.** 11 **D.** 12

Q11.1.9 MCQ 2011 2M

The quality of services delivered by a company consists of six factors as shown below in the radar diagram. The dots in the figure indicate the score for each factor on a scale of 0 to 10. The standardized coefficient for each factor is given in the parentheses. The contribution of each factor to the overall service quality is directly proportional to the factor score and its standardized coefficient.

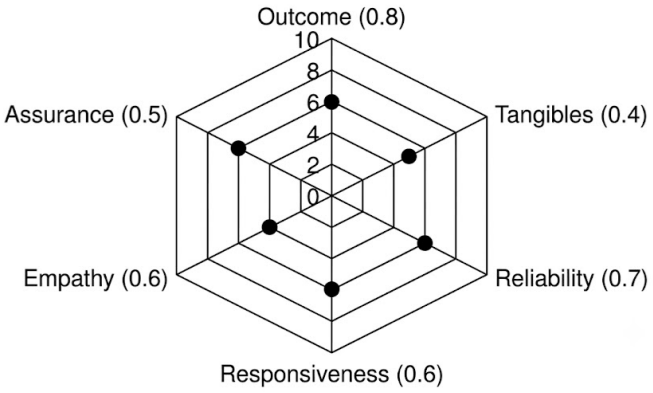


Fig. Q11.1.9

The lowest contribution among all the above factors to the overall quality of services delivered by the company is

A. 10% **B.** 20% **C.** 24% **D.** 40%

Q11.1.10 MCQ 2011 2M

In order to develop to full potential, a baby needs to be physically able to respond to the environment.

It can be inferred from the passage that

A. Full physical potential is needed in order for a baby to be able to respond to the environment.

B. It is necessary for a baby to be able to physically respond to the environment for it to develop its full potential.

C. Response to the environment of physically able babies needs to be developed to its full potential.

D. A physically able baby needs to develop its full potential in order to respond to its environment.

Session : 2 | Afternoon | 30/01/11

Q11.2.1 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Deference

A. aversion **B.** resignation

C. suspicion **D.** contempt

Q11.2.2 MCQ 2011 1M

Choose the most appropriate word(s) from the options given below to complete the following sentence:

We lost confidence in him because he never _____ the grandiose promises he had made.

A. delivered **B.** delivered on **C.** forgot **D.** reneged on

Q11.2.3 MCQ 2011 1M

Choose the word or phrase that best completes the sentence below.

_____ in the frozen wastes of Arctic takes special equipment.

A. To survive **B.** Surviving

C. Survival **D.** That survival

Q11.2.4 MCQ 2011 1M

In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

A. 4 **B.** 12 **C.** 64 **D.** 81

Q11.2.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

The _____ of evidence was on the side of the plaintiff since all but one witness testified that his story was correct.

A. paucity **B.** propensity

C. preponderance **D.** accuracy

Q11.2.6 MCQ 2011 2M

If $(2y + 1)/(y + 2) < 1$, then which of the following alternatives gives the **CORRECT** range of y?

A. $-2 < y < 2$ **B.** $-2 < y < 1$

C. $-3 < y < 1$ **D.** $-4 < y < 1$

Q11.2.7 MCQ 2011 2M

A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as $(4, 3)$. In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as $(3, 2)$. Based on the above information, the roots of the correct quadratic equation are

A. (-3, 4) **B.** (3, -4) **C.** (6, 1) **D.** (4, 2)

Q11.2.8 MCQ 2011 2M

L, M and N are waiting in a queue meant for children to enter the zoo. There are 5 children between L and M, and 8 children between M and N. If there are 3 children ahead of N and 21 children behind L, then what is the minimum number of children in the queue?

A. 28 **B.** 27 **C.** 41 **D.** 40

Q11.2.9 MCQ 2011 2M

Four archers P, Q, R and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bulls' eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.

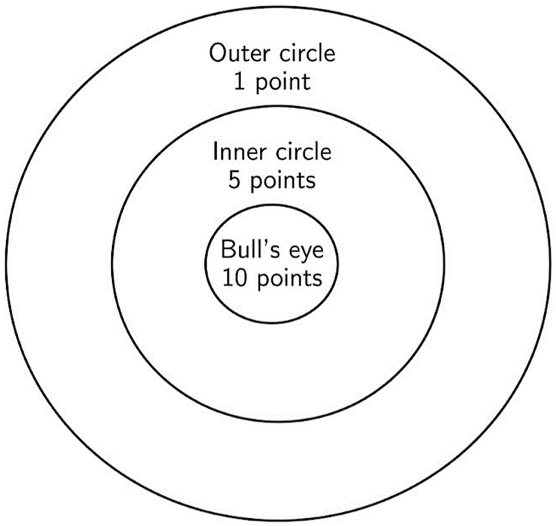


Fig. Q11.2.9

The final scores received by the players during the tournament are listed in the table below.

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively

A. P and S **B.** Q and R **C.** Q and Q **D.** R and Q

Q11.2.10 MCQ 2013 2M

Nimbus clouds are dark and ragged, stratus clouds appear dull in colour and cover the entire sky. Cirrus clouds are thin and delicate, whereas cumulus clouds look like cotton balls.

It can be inferred from the passage that

A. A cumulus cloud on the ground is called fog

B. It is easy to predict the weather by studying clouds

C. Clouds are generally of very different shapes, sizes and mass

D. There are four basic cloud types: stratus, nimbus, cumulus and cirrus

Session : 3 | Forenoon | 13/02/11

Q11.3.1 MCQ 2011 1M

Choose the most appropriate word(s) from the options given below to complete the following sentence.

I contemplated _____ Singapore for my vacation but decided against it.

A. to visit **B.** having to visit

C. visiting **D.** for a visit

Q11.3.2 MCQ 2011 1M

If $\log(P) = (1/2)\log(Q) = (1/3)\log(R)$, then which of the following options is **TRUE**?

A. $P^2 = Q^3R^2$ **B.** $Q^2 = PR$

C. $Q^2 = R^3P$ **D.** $R = P^2Q^2$

Q11.3.3 MCQ 2011 1M

Which of the following options is the closest in the meaning to the word below:

Inexplicable

A. Incomprehensible **B.** Indelible

C. Inextricable **D.** Infallible

Q11.3.4 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Amalgamate

A. merge **B.** split **C.** collect **D.** separate

Q11.3.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.

A. hyperbolic **B.** restrained

C. argumentative **D.** indifferent

Q11.3.6 MCQ 2011 2M

The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = 100/q$. How many units should be produced to minimize the total cost ($V + F$)?

A. 5 **B.** 4 **C.** 7 **D.** 6

Q11.3.7 MCQ 2011 2M

P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:

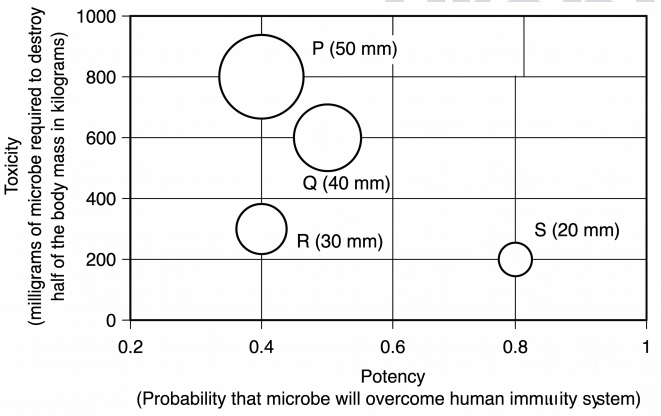


Fig. Q11.3.7

A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

A. P **B.** Q **C.** R **D.** S

Q11.3.8 MCQ 2011 2M

Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage which topic would not be included in a unit on bereavement?

A. how to write a letter of condolence

B. what emotional stages are passed through in the healing process

C. what the leading causes of death are

D. how to give support to a grieving friend

Q11.3.9 MCQ 2011 2M

A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

A. 7.58 litres **B.** 7.84 litres **C.** 7 litres **D.** 7.29 litres

Q11.3.10 MCQ 2011 2M

A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

A. 4 **B.** 5 **C.** 6 **D.** 7

Session : 4 | Afternoon | 13/02/11

Q11.4.1 MCQ 2011 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair:

Gladiator : Arena

A. dancer : stage **B.** commuter : train

C. teacher : classroom **D.** lawyer : courtroom

Q11.4.2 MCQ 2011 1M

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and the rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

A. 100 **B.** 110 **C.** 90 **D.** 95

Q11.4.3 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

It was her view that the country's problems had been _____ by foreign technocrats, so that to invite them to come back would be counter-productive.

A. identified **B.** ascertained

C. exacerbated **D.** analysed

Q11.4.4 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Frequency

A. periodicity **B.** rarity **C.** gradualness **D.** persistency

Q11.4.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which _____ treatments are unsatisfactory.

A. similar **B.** most **C.** uncommon **D.** available

Q11.4.6 MCQ 2011 2M

The horse has played a little known but very important role in the field of medicine. Horses were injected with toxins of diseases until their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way.

It can be inferred from the passage, that horses were

A. given immunity to diseases

B. generally quite immune to diseases

C. given medicines to fight toxins

D. given diphtheria and tetanus serums

Q11.4.7 MCQ 2011 2M

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.

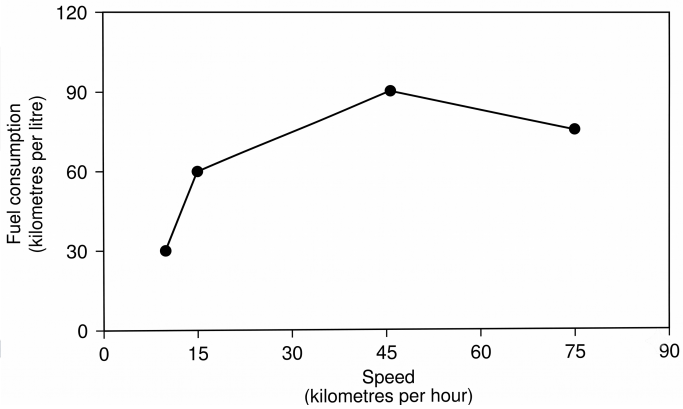


Fig. Q11.4.7

The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

A. P **B.** Q **C.** R **D.** S

Q11.4.8 MCQ 2011 2M

Three friends, R, S and T shared toffee from a bowl. R took $1/3^{rd}$ of the toffees, but returned four to the bowl. S took $1/4^{th}$ of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

A. 38 **B.** 31 **C.** 48 **D.** 41

Q11.4.9 MCQ 2011 2M

Given that $f(y) = |y|/y$, and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is

A. 0 **B.** -1 **C.** 1 **D.** 2

Q11.4.10 MCQ 2011 2M

The sum of n terms of the series $4 + 44 + 444 + \ldots$ is

A. $(4/81)[10^{n+1} - 9n - 1]$

B. $(4/81)[10^{n-1} - 9n - 1]$

C. $(4/81)[10^{n+1} - 9n - 10]$

D. $(4/81)[10^{n} - 9n - 10]$

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Answer Key

Session : 1					
Q11.1.1 C	Q11.1.2 B	Q11.1.3 C	Q11.1.4 D	Q11.1.5 D	Q11.1.6 A
Q11.1.7 B	Q11.1.8 C	Q11.1.9 A	Q11.1.10 B		
Session : 2					
Q11.2.1 D	Q11.2.2 B	Q11.2.3 A	Q11.2.4 C	Q11.2.5 C	Q11.2.6 B
Q11.2.7 C	Q11.2.8 A	Q11.2.9 B	Q11.2.10 D		
Session : 3					
Q11.3.1 C	Q11.3.2 B	Q11.3.3 A	Q11.3.4 B	Q11.3.5 B	Q11.3.6 A
Q11.3.7 D	Q11.3.8 C	Q11.3.9 D	Q11.3.10 C		
Session : 4					
Q11.4.1 D	Q11.4.2 A	Q11.4.3 C	Q11.4.4 B	Q11.4.5 D	Q11.4.6 A
Q11.4.7 B	Q11.4.8 C	Q11.4.9 D	Q11.4.10 C		

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YEAR

2010

3 Sessions · 30 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	TF	10
Session 2	MN	10
Session 3	EC, EE, IN, CS, ME, CE	10

Year Snapshot

Total Questions: 30 **MCQ:** 30 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 07/02/10

Q10.1.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:
Ephemeral

A. effeminate **B.** ghostlike **C.** soft **D.** short-lived

Q10.1.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.
Erudition : Scholar

A. steadfast : mercurial **B.** competence : strict

C. skill : craftsman **D.** nurse : doctor

Q10.1.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
The two child norm with _____ for the violators will have significant implications for our demographic profile.

A. disincentives **B.** incitements

C. restrictions **D.** restraints

Q10.1.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
There is no fixed relation between food and famine; famines can occur with or without substantial _____ in food output.

A. aberration **B.** weakening

C. decline **D.** deterioration

Q10.1.5 MCQ 2010 1M

Consider the function $f(x) = \max(7 - x, x + 3)$. In which range does f take its minimum value?

A. $-6 \le x < -2$ **B.** $-2 \le x < 2$

C. $2 \le x < 6$ **D.** $6 \le x < 10$

Q10.1.6 MCQ 2010 2M

It has taken fifty six long and frustrating years to turn bronze into gold for India's Olympics aspirations. Beijing 2008 marks a defining moment in India's Olympic history. From Delhi to Beijing is a long journey but one that our Olympians have undertaken with courage.

Which of the following statements best sums up the meaning of the above passage?

A. India's participation in Olympics has been frustrating.

B. Beijing Olympics was a landmark in India's Olympic history.

C. Our Olympians have undertaken a long journey to Beijing.

D. India's bronze medal turned into gold at Beijing.

Q10.1.7 MCQ 2010 2M

Consider the series

$$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{8} + \frac{1}{9} - \frac{1}{16} + \frac{1}{32} + \frac{1}{27} - \frac{1}{64} + \ldots.$$

The sum of the infinite series above is:

A. ∞ **B.** $\frac{5}{6}$ **C.** $\frac{1}{2}$ **D.** 0

Q10.1.8 MCQ 2010 2M

A gathering of 50 linguists discovered that 4 knew Kannada, Telugu and Tamil, 7 knew only Telugu and Tamil, 5 knew only Kannada and Tamil, 6 knew only Telugu and Kannada. If the number of linguists who knew Tamil is 24 and those who knew Kannada is also 24, how many linguists knew only Telugu?

A. 9 **B.** 10 **C.** 11 **D.** 8

Q10.1.9 MCQ 2010 2M

A tank has 100 litres of water. At the end of every hour the following two operations are performed in sequence: i) water equal to $m\%$ of the current contents of the tank is added to the tank, ii) water equal to $n\%$ of the current contents of the tank is removed from the tank. At the end of 5 hours the tank contains exactly 100 litres of water. The relation between m and n is:

A. m=n **B.** m>n

C. m<n **D.** None of the previous

Q10.1.10 MCQ 2010 2M

A student is answering a multiple choice examination with 65 questions with a marking scheme as follows: i) 1 mark for each correct answer, ii) $-\frac{1}{4}$ for a wrong answer, iii) $-\frac{1}{8}$ for a question that has not been attempted. If the student gets 37 marks in the test then the least possible number of questions the student has **NOT** answered is:

A. 6 **B.** 5 **C.** 7 **D.** 4

Session : 2 | Afternoon | 07/02/10

Q10.2.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:

Exhort

A. urge **B.** condemn **C.** restrain **D.** scold

Q10.2.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Preamble : Constitution

A. amendment : law **B.** prologue : play

C. episode : serial **D.** plot : story

Q10.2.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

The committee wrote a _____ report, extolling only the strengths of the proposal.

A. reasonable **B.** supportive **C.** biased **D.** fragmented

Q10.2.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

If the country has to achieve real prosperity, it is _____ that the fruits of progress reach all, and in equal measure.

A. inevitable **B.** contingent **C.** oblivious **D.** imperative

Q10.2.5 MCQ 2010 1M

A person invests Rs. 1000 at 10% annual compound interest for 2 years. At the end of two years the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is:

A. 1776 **B.** 1760 **C.** 1920 **D.** 1936

Q10.2.6 MCQ 2010 2M

The ban on smoking in designated public places can save a large number of people from the well known effects of environmental tobacco smoke. Passive smoking seriously impairs respiratory health. The ban rightly seeks to protect non-smokers from its ill effects.

Which of the following statements best sums up the meaning of the above passage?

A. Effects of environmental tobacco are well known.

B. The ban on smoking in public places protects the non smokers.

C. Passive smoking is bad for health.

D. The ban on smoking in public places excludes passive smoking.

Q10.2.7 MCQ 2010 2M

Given the sequence A, B, B, C, C, C, D, D, D, D, . . . etc., that is one A, two Bs, three Cs, four Ds, five Es and so on, the 240^{th} letter in the sequence will be:

A. V **B.** U **C.** T **D.** W

Q10.2.8 MCQ 2010 2M

Consider the set of integers $\{1, 2, 3, \ldots, 5000\}$. The number of integers that is divisible by neither 3 nor 4 is:

A. 1668 **B.** 2084 **C.** 2500 **D.** 2916

Q10.2.9 MCQ 2010 2M

A positive integer m in base 10 when represented in base 2 has the representation p and in base 3 has the representation q. We get $p - q = 990$ where the subtraction is done in base 10. Which of the following is necessarily true:

A. $m \geq 14$ **B.** $9 \leq m \leq 13$ **C.** $6 \leq m \leq 8$ **D.** $m < 6$

Q10.2.10 MCQ 2010 2M

Given the following four functions $f_1(n) = n^{100}$, $f_2(n) = (1.2)^n$, $f_3(n) = 2^{n/2}$, $f_4(n) = 3^{n/3}$ which function will have the largest value for sufficiently large values of n (i.e. $n \to \infty$)?

A. f_4 **B.** f_3 **C.** f_2 **D.** f_1

Session : 3 | Forenoon | 14/02/10

Q10.3.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:
Circuitous

A. cyclic **B.** indirect **C.** confusing **D.** crooked

Q10.3.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.
Unemployed : Worker

A. fallow : land **B.** unaware : sleeper

C. wit : jester **D.** renovated : house

Q10.3.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
If we manage to _____ our natural resources, we would leave a better planet for our children.

A. uphold **B.** restrain **C.** cherish **D.** conserve

Q10.3.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

His rather casual remarks on politics _____ his lack of seriousness about the subject.

A. masked **B.** belied **C.** betrayed **D.** suppressed

Q10.3.5 MCQ 2010 1M

25 persons are in a room. 15 of them play hockey. 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

A. 2 **B.** 17 **C.** 13 **D.** 3

Q10.3.6 MCQ 2010 2M

Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage?

A. Modern warfare has resulted in civil strife.

B. Chemical agents are useful in modern warfare.

C. Use of chemical agents in warfare would be undesirable.

D. People in military establishments like to use chemical agents in war.

Q10.3.7 MCQ 2010 2M

If 137 + 276 = 435 how much is 731 + 672?

A. 534 **B.** 1403 **C.** 1623 **D.** 1513

Q10.3.8 MCQ 2010 2M

5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

A. 20 days **B.** 18 days **C.** 16 days **D.** 15 days

Q10.3.9 MCQ 2010 2M

Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

A. 50 **B.** 51 **C.** 52 **D.** 54

Q10.3.10 MCQ 2010 2M

Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age.
ii. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
iii. There are no twins.

In what order were they born (oldest first)?

A. HSIG **B.** SGHI **C.** IGSH **D.** IHSG

Answer Key

Session : 1					
Q10.1.1 D	Q10.1.2 C	Q10.1.3 A	Q10.1.4 C	Q10.1.5 B	Q10.1.6 B
Q10.1.7 B	Q10.1.8 C	Q10.1.9 B	Q10.1.10 D		
Session : 2					
Q10.2.1 A	Q10.2.2 B	Q10.2.3 C	Q10.2.4 D	Q10.2.5 D	Q10.2.6 B
Q10.2.7 A	Q10.2.8 C	Q10.2.9 A	Q10.2.10 A		
Session : 3					
Q10.3.1 B	Q10.3.2 A	Q10.3.3 D	Q10.3.4 C	Q10.3.5 D	Q10.3.6 C
Q10.3.7 C	Q10.3.8 D	Q10.3.9 B	Q10.3.10 C		

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